IKEA Store Bengaluru

IKEA India Pvt. Ltd.

Technical Specifications
Architecture

29th March 2019

Notice

This document and its contents have been prepared and are intended solely for IKEA India Pvt. Ltd. information and use in relation to IKEA Store at Bengaluru

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Change Log

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R1	Protection Wall	Removable Protection added. Reinforced wall Nomenclature updated.

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ATK-ARC-001 – General

1. General

1.1. Introduction

1.1.1. This Section specifies the General Requirements & provides the summary of the applicable guidelines & standards as per IKEA Engineering & Construction Manual, IKEA Standards & References, as required in the completion of works.

1.2. Terminology

1.2.1. Employer: IKEA (India) Pvt. Ltd. or Client

1.2.2. Engineer: Client's representative

1.2.3. Contractor: General Contractor

1.2.4. Manufacturer: Manufacturer, Supplier or Installer

1.3. Related Documents

1.3.1. This section to be read in conjunction with Employer's requirements and general provision of contract, including General, Supplementary and Special Conditions of Contract Documents.

1.4. Codes and Standards

1.4.1. All materials, workmanship and testing shall be in accordance with the following standards & codes, in their latest revisions. This is not exhaustive and contractor shall verify the listing and supplement with the latest applicable standards & codes.

1.4.2. Comply with the requirements of the reference standards noted herein, except where more stringent requirements are listed herein or otherwise required by the Contract Documents.

1.4.3. Where materials, finishes, fittings, equipment and building components that comply with standards other than those included in these specifications are proposed, the Contractor shall adequately demonstrate that such items are of an equal quality and standard and are fit for their intended purpose, through submission of internationally recognized and authorized test methods, technical and certification reports (in compliance with assessment/evaluation criteria based on standards already included in these specifications), to the approval of engineer, prior to use of such items

1.4.4. Conform to the requirements of the following Standards and Codes of Practice where applicable:

1.4.4.1. National Building Code (NBC) of India 2016 Volume 1 & 2.

1.4.4.2. Compliance to all relevant Indian Standards & Codes (BIS), International Standards & Practices, such as BS EN, ASTM, DIN, NFPA, etc.

ATK-ARC-002 – Ceramic Tile

1. General

1.1. Introduction

1.1.1. This section specifies supply and installation of the Ceramic tiling, for the floor, skirting and wall as shown on the Tender drawings & documents, as required in completion of the work.

1.2. Related Documents

1.2.1. This section to be read in conjunction with Employer's requirements and general provision of contract, including General, Supplementary and Special Conditions of Contract Documents.

1.3. Codes and Standards

1.3.1. All materials, workmanship and testing shall be in accordance with the following standards & codes, in their latest revisions. This is not exhaustive, and contractor shall verify the listing and supplement with the latest applicable standards & codes.

1.3.2. Comply with the requirements of the reference standards noted herein, except where more stringent requirements are listed herein or otherwise required by the Contract Documents.

1.3.3. Where materials, finishes, fittings, equipment and building components that comply with standards other than those included in these specifications are proposed, the Contractor shall adequately demonstrate that such items are of an equal quality and standard and are fit for their intended purpose, through submission of internationally recognized and authorized test methods, technical and certification reports (in compliance with assessment/evaluation criteria based on standards already included in these specifications), to the approval of engineer, prior to use of such items

1.3.4. Conform to the requirements of the following Standards and Codes of Practice where applicable:

1.3.4.1. National Building Code (NBC) of India 2016 Volume 1 & 2.

1.3.4.2. IS Code 15477: 2004 Specification for adhesives for use with ceramic tiles and mosaics.

1.3.4.3. IS Code 2333: 1992 Specification for plaster of Paris for ceramic industry (second revision).

1.3.4.4. IS Code 4457: 2007 Specification for ceramic unglazed vitreous acid resisting tile (second revision).

1.3.4.5. IS Code 13630 Ceramic tiles - Methods of test, sampling and basis for acceptance.

1.3.4.5.1. (Part 1): 2006 Determination of dimensions and surface quality (first revision).

1.3.4.5.2. (Part 2): 2006 Determination of water absorption and bulk density (first revision).

1.3.4.5.3. (Part 3): 2006 Determination of moisture expansion using boiling water - Unglazed tiles (first revision).

1.3.4.5.4. (Part 4): 2006 Determination of linear thermal expansion (first revision).

1.3.4.5.5. (Part 5): 2006 Determination of resistance to thermal shock (first revision).

1.3.4.5.6. (Part 6): 2006 Determination of modulus of rupture and breaking strength (first revision).

1.3.4.5.7. (Part 7): 2006 Determination of chemical resistance - Unglazed tiles (first revision).

1.3.4.5.8. (Part 8): 2006 Determination of chemical resistance - Glazed tiles (first revision).

1.3.4.5.9. (Part 9): 2006 Determination of crazing resistance - Glazed tiles (first revision).

1.3.4.5.10. (Part 10): 2006 Determination of frost resistance (first revision).

1.3.4.5.11. (Part 11): 2006 Determination of resistance to surface abrasion - Glazed tiles (first revision).

1.3.4.5.12. (Part 12): 2006 Determination of resistance to deep abrasion - Unglazed tiles (first revision).

1.3.4.5.13. (Part 13): 2006 Determination of scratch hardness of surface according to Mohs (first revision).

1.3.4.5.14. (Part 14): 2006 Determination of impact resistance by measurement of coefficient of restitution.

1.3.4.5.15. (Part 15): 2006 Ceramic tiles - Sampling and basis for acceptance

1.3.4.5.16. IS Code 13712: 2006 Ceramic tiles - Definitions, classifications, characteristics and marking (first revision).

1.3.4.5.17. 15622: 2006 Specification for pressed ceramic tiles.

1.3.4.6. NFPA 101 Life Safety Code.

1.4. Submittals

1.4.1. Technical Data

1.4.1.1. Comprehensive technical data viz., technical literature, data sheets, product catalogue & brochure, installation instructions, etc. to be provided for Engineer’s approval.

1.4.2. Shop Drawings

1.4.2.1. Shop drawings shall be submitted to Engineer for approval, prior to start of work at site.

1.4.2.2. Shop drawings shall include all details w.r.t. the material & related product’s installation, to the scale & detailing as instructed by the Engineer.

1.4.2.3. Shop drawings shall be developed as per actual measurements from the site & must depict the same in the drawings.

1.4.2.4. Shop drawings must comply to BS 1192:2007+A2:2016 Collaborative production of architectural, engineering and construction information & Code of practice.

1.4.3. Samples

1.4.3.1. Samples for Initial Selection: For each type of tile and grout indicated. Include Samples of accessories involving colour selection.

1.4.3.2. Samples for Verification:

1.4.3.2.1. Full-size units of each type and composition of tile and for each colour and finish required.

1.4.3.2.2. Assembled samples with grouted joints for each type and composition of tile and for each colour and finish required, at least 200 mm & 300mm square mounted on rigid panel. Use grout of type and in colour or colours approved for completed work.

1.4.3.2.3. Full-size units of each type of trim and accessory for each colour and finish required.

1.4.3.2.4. Metal edge strips in 150-mm lengths.

1.4.4. Method Statements

1.4.4.1. Shall include method of construction/installation, Health and Safety management, testing methods, required relative humidity and temperature level and control systems, possible risk associated with the execution works and precautions to be considered, site control system and any other additional requirements as advised by the Engineer for the execution of works.

1.4.4.2. Method statement shall be approved by the Engineer prior to commencement of the works.

1.4.5. Installer Qualifications:

1.4.5.1. Licensed installer specialized in ceramic flooring installation shall be employed for the work.

1.4.6. Material/Product Certificates:

1.4.6.1. Comprehensive tests performed by a qualified independent testing agency certifying that the product compiles with specified performance requirements.

1.4.7. Maintenance Data:

1.4.7.1. Details of manufacturer's recommended maintenance procedures for each type of product/material to be included in the operation and maintenance manual.

1.4.8. Spare Materials/Parts

1.4.8.1. Provide the following extra materials that match installed products, packaged for storage with suitable labels for identification.

1.4.8.2. 2% or 50m^2 minimum for each type of ceramic flooring (product, pattern or colour)

1.5. Warranty / Guarantee

1.5.1. As per the Employer's requirements and general provision of contract, including General, Supplementary and Special Conditions of Contract Documents.

1.6. Performance Requirements

1.6.1. All finishes, materials and components shall comply with requirements of NFPA 101 Life Safety Code, chapter 10 – Interior Finish, Contents and Furnishings with respect to classification of materials, flame spread, smoke generation and other requirements.

1.7. Environment & Sustainability

1.7.1. Recycling Strategy document for the material or product shall be submitted to the Engineer for approval.

1.7.2. LEED Requirements:

1.7.2.1. All the specifications in this document are for USGBC LEED BD+C submission and are applicable to all materials & procedures followed for the project. The specification as stated here should be adhered to the word. In case of a dispute LEED guidelines shall be followed. Brochures, product specification sheets and/or certificates as specified in this document should be submitted along with the material procurement. All contractors, manufactures must submit documents, records and other submittal related to the different credits as mentioned in the specification.

1.7.2.2. Manufacturer product conforming to ISO 14044 or Environmental Product Declarations which conform to ISO 14025, 14040, 14044, and EN 15804 or ISO 21930 and have at least a cradle to gate scope. (If applicable)**.**

1.7.2.3. Manufacturer Material Ingredient Reporting by any of the listed below documents (If applicable):

1.7.2.3.1. Health Product Declaration. The end use product has a published, complete Health Product Declaration with full disclosure of known hazards in compliance with the Health Product Declaration Open Standard.

1.7.2.3.2. Cradle to Cradle Certificate. The end use product has been certified at the Cradle to Cradle v2 Basic level or Cradle to Cradle v3 Bronze level.

1.7.2.3.3. REACH Optimization. End use products and materials that do not contain substances that meet REACH criteria for substances of very high concern.

1.7.2.4. Manufacturer Raw Material Source and Extraction Reporting by any of the listed below activity (If applicable):

1.7.2.4.1. Third-party verified corporate sustainability reports (CSR) which include environmental impacts of extraction operations and activities associated with the manufacturer’s product and the product’s supply chain**.**

1.7.2.4.2. Self-declared reports from manufacturers that have publicly released a report from their raw material suppliers which include raw material supplier extraction locations.

1.7.2.5. Tile Adhesive should meet VOC level as per LEED Tender Documents. Manufacturer must confirm same by providing lab test report.

1.8. Manufacturer

1.8.1. Manufacturer to be accredited and registered under BS EN ISO 9001.

1.8.2. List of preferred or acceptable Manufacturers attached herewith as Annexure 1.

2. Products

2.1. Materials

2.1.1. Material/Product Specification as per Standard attached herewith as Annexure A.01.

2.1.1.1. Floor Tile:

2.1.1.1.1. Size: 300mm x 300mm

2.1.1.1.2. Thickness: 10mm

2.1.1.2. Skirting Tile:

2.1.1.2.1. Size: 300mm x 100mm

2.1.1.2.2. Thickness: 8mm

2.1.1.3. Wall Tile:

2.1.1.3.1. Size: 300mm x 300mm

2.1.1.3.2. Thickness: 8mm

2.2. Colours

2.2.1. Colours as per Standard attached herewith as Annexure B.01.

2.2.1.1. Floor Tile:

2.2.1.1.1. Colour: Grey NCS S 3000-N

2.2.1.2. Skirting Tile:

2.2.1.2.1. Colour: Grey NCS S 3000-N

2.2.1.3. Wall Tile:

2.2.1.3.1. Colour: White NCS S 0500-N

2.2.1.3.2. Colour: Black (IKEA Restaurant/Kitchen)

3. Execution

3.1. Examination

3.1.1. Examine substrates, areas, and conditions where tile will be installed, with Installer present, for compliance with requirements for installation tolerances and other conditions affecting performance of installed tile.

3.1.1.1. Verify that substrates for setting tile are firm; dry; clean; free of oil, waxy films, and curing compounds; and within flatness tolerances required by referenced BS 5385 Series of tile installation standards for installations indicated.

3.1.1.2. Verify that installation of grounds, anchors, recessed frames, electrical and mechanical units of work, and similar items located in or behind tile has been completed before installing tile.

3.1.1.3. Verify that joints and cracks in tile substrates are coordinated with tile joint locations; if not coordinated, adjust joint locations in consultation with Engineer.

3.1.2. Proceed with installation only after unsatisfactory conditions have been corrected.

3.2. Preparation

3.2.1. Remove coatings, including curing compounds and other substances that contain soap, wax, oil, or silicone, that are incompatible with tile-setting materials.

3.2.2. Provide concrete substrates for tile on floors & walls installed with thin-set mortar that comply with flatness tolerances as specified in BS 5385 of tile installation standards.

3.2.3. Fill cracks, holes, and depressions with trowel able levelling and patching compound per tile-setting material manufacturer's written instructions. Use product specifically recommended by tile-setting material manufacturer. Remove protrusions, bumps, and ridges by sanding or grinding.

3.2.4. Blending: For tile exhibiting colour variations within ranges selected during Sample submittals, verify that tile has been factory blended and packaged so tile units taken from one package show same range of colours as those taken from other packages and match approved Samples. If not, factory blended, either return to manufacturer or blend tiles at Project site before installing.

3.3. Installation

3.3.1. Ceramic Tile Installation shall be carried out as per Manufacturer’s instructions, subject to approval by Engineer.

3.3.2. Installation works shall comply to BS 5385 Wall and floor tiling. Code of practice for tiling and mosaics in specific conditions + AMD 10824.

3.3.3. A proprietary transition strip shall be used between tiled areas and concrete floor.

3.4. Workmanship

3.4.1. Thoroughly clean the screed or backing of grease, oil, wax, paint, dirt or dust and fill all cracks, minor holes and services with a suitable filler recommended by the tile or sheeting manufacturer.

3.4.2. Apply adhesive in a thin film and spread evenly with a notched or plain trowel as recommended by the manufacturer. If a notched trowel is used the notches shall be of the size recommended and shall be renewed periodically to ensure the correct spread of adhesive.

3.4.3. Closely butt joints and cut accurately at right angles to the tiles to an agreed design and with only the minimum necessary jointing. Relay any areas insufficiently bonded or showing an uneven surface.

3.5. Cleaning and Protection

3.5.1. After laying, thoroughly clean the tiling with water and a detergent recommended by the manufacturer, subject to Engineer’s approval. Apply two coats of non-slip, matt emulsion polish. Protect the floors with approved covering.

4. Quality Control

4.1. Delivery, Storage and Handling

4.1.1. All building materials shall be stored on the building site in such a way as to prevent deterioration or the loss or impairment of their structural and other essential properties. Comply to National Building Code (NBC) of India 2016 Volume 1; Part 7 Construction Management, Practices and Safety.

4.2. Source Limitation

4.2.1. All materials & related products shall be from the same manufacturer to ensure material/product integrity & uniformity of colour, texture & quality.

4.3. Mock-up

4.3.1. Build mock-ups for each type of Ceramic flooring indicated including all accessories to verify selections made under sample submittals and to demonstrate aesthetic effects and set quality standards for materials and execution.

4.3.2. Size: Minimum 9 Sq.M of typical poured-in place flooring and base condition, in locations selected by Engineer.

4.3.3. Obtain Engineer's approval of design and patterns.

4.3.4. The Contractor shall provide a mock-up of one complete male and female toilet block and obtain Engineer’s approval for mock-up approval prior to installation of remaining areas.

5. Testing and Commissioning

5.1. Inspection

5.1.1. Inspection upon completion of work, all ceramic tile areas shall be carefully tested and examined for faults in application, shall be corrected immediately.

ATK-ARC-003 – Painting

1. General

1.1. Introduction

1.1.1. This section specifies supply and application of paint systems on the substrates, as shown on the Tender drawings & documents, as required in completion of the work.

1.1.1.1. Concrete

1.1.1.2. Masonry (Block Wall)

1.1.1.3. Gypsum Plasterboard Partition

1.1.1.4. Concrete Floors

1.1.2. Unless specifically designated to be unfinished or unpainted, paint all surfaces exposed to view which will not be covered by other finishes or finish materials when construction is completed, to Engineer's approval.

1.2. Related Documents

1.2.1. This section to be read in conjunction with Employer’s requirements and general provision of contract, including General, Supplementary and Special Conditions of Contract Documents

1.3. Codes and Standards

1.3.1. All materials, workmanship and testing shall be in accordance with the following standards & codes, in their latest revisions. This is not exhaustive and contractor shall verify the listing and supplement with the latest applicable standards & codes.

1.3.2. Comply with the requirements of the reference standards noted herein, except where more stringent requirements are listed herein or otherwise required by the Contract Documents.

1.3.3. Where materials, finishes, fittings, equipment and building components that comply with standards other than those included in these specifications are proposed, the Contractor shall adequately demonstrate that such items are of an equal quality and standard and are fit for their intended purpose, through submission of globally recognized and authorized test methods, technical and certification reports (in compliance with assessment or evaluation criteria based on standards already included in these specifications), to the approval of engineer, prior to use of such items.

1.3.4. Conform to the requirements of the following Standards and Codes of Practice where applicable.

1.3.4.1. National Building Code (NBC) of India 2016 Volume 1 & 2.

1.3.4.2. IS Codes for Water Based Paints and Pigments:

1.3.4.2.1. IS Code 427: 2013 Specification for distemper, dry, colour as required (third revision)

1.3.4.2.2. IS Code 428: 2013 Specification for washable distemper (third revision)

1.3.4.2.3. IS Code 5410: 2013 Specification for cement paint (second revision)

1.3.4.2.4. IS Code 15489: 2013 Specification for paint, plastic emulsion (first revision)

1.3.4.3. IS Codes for Ready Mixed Paints, Enamels and Powder Coatings:

1.3.4.3.1. IS Code 101 Methods of sampling and test for paints, varnishes and related products

1.3.4.3.2. (Part 1/Sec 1) 1986: Test on liquid paints (general and physical), Section 1 Sampling (third revision)

1.3.4.3.3. (Part 1/Sec 2) 1987: Test on liquid paints (general and physical), Section 2 Preliminary examination and preparation of samples for testing (third revision)

1.3.4.3.4. (Part 1/Sec 3) 1986: Test on liquid paints (general and physical), Section 3 Preparation of panels (third revision)

1.3.4.3.5. (Part 1/Sec 4) 1987: Test on liquid paints (general and physical), Section 4 Brushing test (third revision)

1.3.4.3.6. (Part 1/Sec 5) 1989: Test on liquid paints (general and physical), Section 5 Consistency (third revision)

1.3.4.3.7. (Part 1/Sec 6) 1987: Test on liquid paints (general and physical), Section 6 Flash point (third revision)

1.3.4.3.8. (Part 1/Sec 7) 1987: Test on liquid paints (general and physical), Section 7 Mass per 10 litres (third revision)

1.3.4.3.9. (Part 2/Sec 1) 1988: Test on liquid paints (Chemical examination), Section 1 Water content (third revision)

1.3.4.3.10. (Part 2/Sec 2) 1986: Test on liquid paints (Chemical examination), Section 2 Volatile matter (third revision)

1.3.4.3.11. (Part 3/Sec 1) 1986: Tests on paint film formation, Section 1 Drying time (third revision)

1.3.4.3.12. (Part 3/Sec 2) 1989: Tests on paint film formation, Section 2 Film thickness (third revision)

1.3.4.3.13. (Part 3/Sec 4) 1987: Tests on paint film formation, Section 4 Finish (third revision)

1.3.4.3.14. (Part 3/Sec 5) 1987: Tests on paint film formation, Section 5 Fineness of grind (third revision)

1.3.4.3.15. (Part 4/Sec 1) 1988: Optical test, Section 1 Opacity (third revision)

1.3.4.3.16. (Part 4/Sec 2) 1989: Optical test, Section 2 Colour (third revision)

1.3.4.3.17. (Part 4/Sec 3) 1988: Optical test, Section 3 Light fastness test (third revision)

1.3.4.3.18. (Part 4/Sec 4) 1988: Optical test, Section 4 Gloss (third revision)

1.3.4.3.19. (Part 5/Sec 1) 1988: Mechanical test on paint films, Section 1 Hardness tests (third revision)

1.3.4.3.20. (Part 5/Sec 2) 1988: Mechanical test on paint films, 1988 Section 2 Flexibility and adhesion (third revision)

1.3.4.3.21. (Part 5/Sec 3) 1999: Mechanical test on paint films, Section 3 Impact resistance (fourth revision)

1.3.4.3.22. (Part 5/Sec 4) 1988: Mechanical test on paint films, Section 4 Print free test (third revision)

1.3.4.3.23. (Part 6/Sec 1) 1988: Durability tests, Section 1 Resistance to humidity under conditions of condensation (third revision)

1.3.4.3.24. (Part 6/Sec 2) 1989: Durability tests, Section 2 Keeping properties (third revision)

1.3.4.3.25. (Part 6/Sec 3) 1990: Durability tests, Section 3 Moisture vapour permeability (third revision)

1.3.4.3.26. (Part 6/Sec 4) 1991: Durability tests, Section 4 Degradation of coatings (pictorial aids for evaluation) (third revision)

1.3.4.3.27. (Part 6/Sec 5) 1997: Durability tests, Section 5 Accelerated weathering test

1.3.4.3.28. (Part 7/Sec 1) 1989: Environmental tests on paint films, Section 1 Resistance to water (third revision)

1.3.4.3.29. (Part 7/Sec 2) 1989: Environmental tests on paint films, Section 2 Resistance to liquids (third revision)

1.3.4.3.30. (Part 7/Sec 3) 1990: Environmental tests on paint films, Section 3 Resistance to heat (third revision)

1.3.4.3.31. (Part 7/Sec 4) 1990: Environmental tests on paint films, Section 4 Resistance to bleeding of pigments (third revision)

1.3.4.3.32. (Part 8/Sec 1) 1989: Tests for pigments and other solids, Section 1 Residue on sieve (third revision)

1.3.4.3.33. (Part 8/Sec 2) 1990: Tests for pigments and other solids, Section 2 Pigments and non-volatile matter (third revision)

1.3.4.3.34. (Part 8/Sec 3) 1993: Tests for pigments and other solids, Section 3 Ash content (third revision)

1.3.4.3.35. (Part 8/Sec 4) 2015: Tests for pigments and other solids, Section 4 Phthalic anhydride (fourth revision)

1.3.4.3.36. (Part 8/Sec 5) 1993: Tests for pigments and other solids, Section 5 Lead restriction test (third revision)

1.3.4.3.37. (Part 8/Sec 6) 1993: Tests for pigments and other solids, Section 6 Volume solids

1.3.4.3.38. (Part 9/Sec 1) 1993: Tests for lacquers and varnish, Section 1 Acid value (third revision)

1.3.4.3.39. (Part 9/Sec 2) 1993: Tests for lacquers and varnish, Section 2 Rosin test (third revision)

1.3.4.3.40. IS Code 104: 1979 Specification for ready mixed paint, brushing, zinc chrome, priming (second revision)

1.3.4.3.41. IS Code 109: 1968 Specification for ready mixed paint, brushing, priming, plaster to Indian Standard Colours No. 361 and 631 (first revision)

1.3.4.3.42. IS Code 133: 2004 Specification for enamel, interior (a) undercoating, (b) finishing (fourth revision)

1.3.4.3.43. IS Code 133 (Part 1): Specification for enamel, interior 2013 (a) undercoating, (b) finishing: Part 1 For household and decorative purposes (fifth revision)

1.3.4.3.44. IS Code 137: 1965 Specification for ready mixed paint, brushing, matt or egg-shell flat, finishing, interior, to Indian Standard colour, as required (revised)

1.3.4.3.45. IS Code 158: 2015 Specification for ready mixed paint, brushing, bituminous, black, lead- free, acid, alkali, and heat resisting (fourth revision)

1.3.4.3.46. IS Code 164: 1981 Specification for ready mixed paint for road marking (second revision)

1.3.4.3.47. IS Code 168: 1993 Specification for ready mixed paint, air-drying semi-glossy/matt, for general purposes (third revision)

1.3.4.3.48. IS Code 341: 1973 Specification for black Japan, Types A, B and C (first revision)

1.3.4.3.49. IS Code 2074: 1992 Specification for ready mixed paint, air drying red oxide-zinc chrome, priming (second revision)

1.3.4.3.50. IS Code 2075: 2000 Specification for ready mixed paint, stoving, red oxide-zinc chrome, priming (second revision)

1.3.4.3.51. IS Code 2339: 2013 Specification for aluminium paint for general purposes (first revision)

1.3.4.3.52. IS Code 2932 (Part 1) 2013: Specification for enamel, synthetic, exterior (a) undercoating, (b) finishing: Part 1 For household and decorative applications (fourth revision)

1.3.4.3.53. IS Code 2933 (Part 1) 2013: Specification for enamel, exterior: (a) undercoating (b) finishing: Part 1 for domestic and decorative applications

1.3.4.3.54. IS Code 3536: 1999 Specification for ready mixed paint, brushing, wood primer (first revision)

1.3.4.3.55. IS Code 3537: 1966 Specification for ready mixed paint, finishing, interior for general purposes, to Indian Standard Colours No. 101, 216, 217, 219, 275, 281, 352, 353, 358 to 361, 363, 364, 388, 410, 442, 444, 628, 631, 632, 634, 693, 697, white and black.

1.3.4.3.56. IS Code 3539: 1966 Specification for ready mixed paint, undercoating, for use under oil finishes, to Indian Standard colours, as required

1.3.4.3.57. IS Code 3585: 1966 Specification for ready mixed paint, aluminium, brushing, priming, water resistant, for wood work

1.3.4.3.58. IS Code 3678: 1966 Specification for ready mixed paint, thick white, for lettering

1.3.4.3.59. IS Code 9862: 1981 Specification for ready mixed paint, brushing, bituminous black lead free, acid, alkali, water and chlorine resisting

1.3.4.3.60. IS Code 11883: 1986 Specification for ready mixed paint, brushing, red oxide, priming for metals

1.3.4.3.61. IS Code 12744 (Part 1) 2013: Specification for ready mixed paint, air drying, red oxide, zinc phosphate, priming: Part 1 for domestic and decorative applications (fourth revision)

1.3.4.3.62. IS Code 13183: 1991 Specification for aluminium paints, heat resistant

1.3.4.3.63. IS Code 13213: 1991 Specification for poly

1.3.4.3.64.

1.3.4.3.65.

1.3.4.3.66. urethane full gloss enamel (two pack)

1.3.4.3.67. IS Code 13607: 1992 Specification for ready mixed paint, finishing, general purposes, synthetic

1.3.4.3.68. IS Code 13871: 1993 Specification for powder coatings

1.3.4.3.69. IS Code 14506: 1998 Specification for epoxy red oxide zinc phosphate weldable primer, two components

1.3.4.3.70. IS Code 14589: 1999 Specification for zinc priming paint, epoxy based, two pack

1.3.4.4. IS Codes for Ready Mixed Paints, Enamels and Powder Coatings:

1.3.4.4.1. IS Code 82: 1973 Methods of sampling and test for thinners and solvents for paints (first revision)

1.3.4.4.2. IS Code 430: 1972 Paint remover, solvent type, non-flammable (second revision)

1.3.4.4.3. IS Code 431: 1972 Paint remover, solvent type, flammable (second revision)

1.3.4.4.4. IS Code 533: 2007 Specification for gum spirit of turpentine (oil of turpentine) (third revision)

1.3.4.4.5. IS Code 5667: 1970 Specification for thinner for cellulose nitrate based paints and lacquers

1.3.4.4.6. IS Code 14314: 1995 Specification for thinner general purposes for synthetic paints and varnishes

1.3.4.5. IS Codes for Ready Mixed Paints, Enamels and Powder Coatings:

1.3.4.5.1. IS Code 337: 1975 Specification for varnish, finishing, interior (first revision)

1.3.4.5.2. IS Code 344: 1976 Specification for varnish, stoving (first revision)

1.3.4.5.3. IS Code 347: 1975 Specification for varnish, shellac, for general purposes (first revision)

1.3.4.5.4. IS Code 348: 1968 Specification for French polish (first revision)

1.3.4.5.5. IS Code 524: 1983 Specification for varnish, finishing, exterior, synthetic (second revision)

1.3.4.5.6. IS Code 525: 1968 Specification for varnish, finishing, exterior and general purposes (first revision)

1.3.4.5.7. IS Code 642: 1963 Specification for varnish medium for aluminium paint (revised)

1.3.4.5.8. IS Code 10018: 1981 Specification for lacquer, cellulose, nitrate clear, finishing, glossy for wood

1.3.4.5.9. IS Code164: Ready mixed paint for road marking

1.3.4.6. ISO 3233 Methods of Test for Painting.

1.3.4.7. NFPA 101 Life Safety Code.

1.4. Submittals

1.4.1. Technical Data

1.4.2. Comprehensive technical data viz., technical literature, data sheets, product catalogue & brochure, installation instructions, etc. to be provided for Engineer’s approval.

1.4.3. Samples

1.4.3.1. Samples for Initial Selection:

1.4.3.1.1. Provide colour samples for each type of paint/paint system indicated, for Engineer's selection.

1.4.3.2. Samples for Verification:

1.4.3.2.1. Provide samples for verification for each type of paint system and each colour and gloss of topcoat indicated.

1.4.3.2.2. Submit Samples on rigid backing, 300 mm square.

1.4.3.2.3. Step coats on Samples to show each coat required for system.

1.4.3.2.4. Label each coat of each Sample.

1.4.3.2.5. Label each Sample for location and application area.

1.4.4. Shop Drawings

1.4.4.1. Shop drawings shall be submitted to Engineer for approval, prior to start of work at site.

1.4.4.2. Shop drawings shall include all details w.r.t. the material & related product's installation, to the scale & detailing as instructed by the Engineer.

1.4.4.3. Shop drawings shall be developed as per actual measurements from the site & must depict the same in the drawings.

1.4.5. Method Statement

1.4.5.1. Shall include method of construction/installation, Health and Safety management, testing methods, required relative humidity and temperature level and control systems, possible risk associated with the execution works and precautions to be considered, site control system and any other additional requirements as advised by the Engineer for the execution of works.

1.4.5.2. Method statement shall be approved by the Engineer prior to commencement of the works.

1.4.6. Installer Qualifications:

1.4.6.1. Licensed installer specialized in painting shall be employed for the work.

1.4.7. Material/Product Certificates:

1.4.7.1. Comprehensive tests performed by a qualified independent testing agency certifying that the product compiles with specified performance requirements.

1.4.8. Maintenance Data:

1.4.8.1. Details of manufacturer's recommended maintenance procedures for each type of product/material to be included in the operation and maintenance manual.

1.4.9. Spare Materials/Parts

1.4.9.1. Provide the following extra materials that match installed products, packaged for storage with suitable labels for identification.

1.4.9.2. 2% or 50m^2 minimum for each type of paint (product, pattern or colour)

1.5. Warranty / Guarantee

1.5.1. As per the Employer's requirements and general provision of contract, including General, Supplementary and Special Conditions of Contract Documents.

1.6. Performance Requirements

1.6.1. All finishes, materials and components shall comply with requirements of NFPA 101 Life Safety Code, chapter 10 – Interior Finish, Contents and Furnishings with respect to classification of materials, flame spread, smoke generation and other requirements.

1.7. Environment & Sustainability

1.7.1. Recycling Strategy document for the material or product shall be submitted to the Engineer for approval.

1.7.2. LEED Requirements:

1.7.2.1. All the specifications in this document are for USGBC LEED BD+C submission and are applicable to all materials & procedures followed for the project. The specification as stated here should be adhered to the word. In case of a dispute LEED guidelines shall be followed. Brochures, product specification sheets and/or certificates as specified in this document should be submitted along with the material procurement. All contractors, manufactures must submit documents, records and other submittal related to the different credits as mentioned in the specification.

1.7.2.2. Manufacturer product conforming to ISO 14044 or Environmental Product Declarations which conform to ISO 14025, 14040, 14044, and EN 15804 or ISO 21930 and have at least a cradle to gate scope. (If applicable)

1.7.2.3. Manufacturer Material Ingredient Reporting by any of the listed below documents (If applicable):

1.7.2.3.1. Health Product Declaration. The end use product has a published, complete Health Product Declaration with full disclosure of known hazards in compliance with the Health Product Declaration Open Standard.

1.7.2.3.2. Cradle to Cradle Certificate. The end use product has been certified at the Cradle to Cradle v2 Basic level or Cradle to Cradle v3 Bronze level.

1.7.2.3.3. REACH Optimization. End use products and materials that do not contain substances that meet REACH criteria for substances of very high concern.

1.7.2.4. Manufacturer Raw Material Source and Extraction Reporting by any of the listed below activity (If applicable):

1.7.2.4.1. Third-party verified corporate sustainability reports (CSR) which include environmental impacts of extraction operations and activities associated with the manufacturer’s product and the product’s supply chain

1.7.2.4.2. Self-declared reports from manufacturers that have publicly released a report from their raw material suppliers which include raw material supplier extraction locations.

1.7.2.5. Paints & Coating should meet VOC level as per LEED Tender Documents. Manufacturer has to confirm same by providing lab test report.

1.8. Manufacturers

1.8.1. Manufacturer to be accredited and registered under BS EN ISO 9001.

1.8.2. List of preferred or acceptable Manufacturers attached herewith as Annexure 1.

2. Products

2.1. General

2.1.1. Material Compatibility:

2.1.2. Provide materials for use within each paint system that are compatible with one another and substrates indicated, under conditions of service and application as demonstrated by manufacturer, based on testing and field experience.

2.1.3. For each coat in a paint system, provide products recommended in writing by manufacturers of topcoat for use in paint system and on substrate indicated in contract Drawings.

2.2. Paint Systems

2.2.1. General: The following requirements notwithstanding, provide paints, primers, sealers, and fillers as recommended by paint system manufacturer for finish coats. Where manufacturer's recommendations differ from the products specified herein, notify the Engineer in writing prior to application of primer, sealer or filler.

2.2.1.1. Paint with a high gloss level makes cleaning and maintenance easier. For the internal walls and partitions, a minimum gloss level of 30° (Gardner scale) is recommended. Paint for steel staircases is to have a gloss level of 80°. Exposed areas that need to be washable may be specified with a higher gloss.

2.2.1.2. Water based paints should be used wherever possible. The use of solvent based paints is generally not allowed by IKEA.

2.2.1.3. Steel structure should not be painted in addition to a grey primer. The structural steel package contract must include for factory priming and any touch-up repairs on site, following installation.

2.2.1.4. The galvanised or coated inner surfaces of wall cladding or roof lining need no further treatment or painting.

2.2.1.5. Plasterboard walls and any plasterboard lining to the wall cladding, for example office areas, should be painted in accordance with the Room Specifications. Normally, lining is not needed if cladding and structure are well designed.

2.2.1.6. Fair faced concrete or masonry should not be painted or covered unless in highly commercial areas such as the store entrance.

2.2.1.7. External steel structures, for example, escape stairs and canopy supports, are to be galvanised and unpainted. Assembly after galvanising must be a bolted system and shall not be cut nor welded on site.

2.2.1.8. Normally, do not paint any part of the structure, partitions or walls above the open grid ceiling. If anything has to be painted due to an unacceptably low quality unpainted finish, or to conceal mistakes, they should be painted grey to make them inconspicuous. Jointed dry walls shall be painted grey above grid ceiling.

2.2.1.9. Visible services and other installations above open ceiling grid level are not to be painted. In general, all services and installations should be supplied with a galvanised, grey, or black appearance. Sprinkler pipes should be supplied with grey primer. The tender documents states that the sprinkler contractor is responsible for any touching up of the installed pipes on site due to dirt, rust, welding, and so on.

2.2.1.10. Generally, concrete floors should not be painted. However, grey epoxy paint can be used for janitor rooms and some technical rooms. It must not be used where there is forklift traffic, since it does not withstand the wear.

2.2.2. Paint System:

2.2.2.1. Substrate:

2.2.2.1.1. Internal Plaster and Gypsum Plasterboard Painting

2.2.2.2. Base Coat:

2.2.2.2.1. Primer/Sealer/Undercoat: Water Based Acrylic Primer

2.2.2.3. Top Coat:

2.2.2.3.1. Paint: Premium Water Based Acrylic Paint

2.2.3. Dust binder:

2.2.3.1. Substrate:

2.2.3.1.1. Fair Faced Concrete & Block Work Painting

2.2.3.2. Top Coat: Water Based Acrylic Dust Binder

2.2.4. Paint System for Road Marking

2.2.4.1. Used for Roads & Kerbs. use on asphalt and concrete roads, primarily for kerbstones, speed humps.

2.2.4.2. Colour: White, Black & Golden Yellow.

2.2.4.3. Gloss: Matt

2.2.4.4. Thickness:

2.2.4.4.1. DFT = 40-80 microns

2.2.4.4.2. WFT = 110-210 microns

2.2.4.5. Two coat applications are recommended for optimum performance.

2.3. Other Materials

2.3.1. Provide all necessary putty, fillers, sealers, primers, undercoats, and finish materials not specifically described herein for approval by the Engineer prior to application.

2.3.2. Application Equipment: Provide and use only such equipment as recommended by paint manufacturer for application of the paint specified.

2.3.3. Refer Standard attached herewith as Annexure A.03.

2.4. Colours

2.4.1. Colours as per Standard attached herewith as Annexure B.01.

3. Execution

3.1. Examination

3.1.1. Examine substrates and conditions, with Applicator present, for compliance with requirements for maximum moisture content and other conditions affecting performance of work.

3.1.1.1. Verify suitability of substrates, including surface conditions and compatibility with existing finishes and primers.

3.1.1.2. Begin coating application only after unsatisfactory conditions have been corrected and surfaces are dry.

3.1.1.3. Commencement of application constitutes Contractor's acceptance of substrates and conditions.

3.2. Preparation

3.2.1. Comply with manufacturer's written instructions and recommendations in BS 6150 applicable to substrates and paint systems indicated, subject to Engineer’s approval.

3.2.2. Remove plates, machined surfaces, and similar items already in place that are not to be painted. If removal is impractical or impossible because of size or weight of item, provide surface-applied protection before surface preparation and painting.

3.2.2.1. After completing painting operations, use workers skilled in the trades involved to reinstall items that were removed. Remove surface- applied protection if any.

3.2.2.2. Do not paint over labels of independent testing agencies or equipment name, identification, performance rating, or nomenclature plates.

3.2.3. Clean substrates of substances that could impair bond of paints, including dirt, oil, grease, and incompatible paints and encapsulates.

3.2.3.1. Remove incompatible primers and reprime substrate with compatible primers as required to produce paint systems indicated.

3.2.4. Concrete Substrates:

Remove release agents, curing compounds, efflorescence, and chalk. Do not paint surfaces if moisture content or alkalinity of surfaces to be painted exceeds that permitted in manufacturer's written instructions.

3.2.5. Plaster Substrates:

3.2.5.1. Do not begin paint application until plaster is fully cured and dry.

3.3. Application

3.3.1. Apply paints according to manufacturer's written instructions, and as approved by the Engineer.

3.3.1.1. Use applicators and techniques suited for paint and substrate indicated.

3.3.1.2. Paint surfaces behind movable items same as similar exposed surfaces. Before final installation, paint surfaces behind permanently fixed items with prime coat only.

3.3.2. Tint each undercoat a lighter shade to facilitate identification of each coat if multiple coats of same material are to be applied. Tint undercoats to match colour of topcoat, but provide sufficient difference in shade of undercoats to distinguish each separate coat.

3.3.3. If undercoats or other conditions show through topcoat, apply additional coats until cured film has a uniform paint finish, colour, and appearance.

3.3.4. Apply paints to produce surface films without cloudiness, spotting, laps, brush marks, roller tracking, runs, sags, ropiness, or other surface imperfections. Cut in sharp lines and colour breaks.

3.3.5. Miscellaneous Application Procedures

3.3.5.1. Unless otherwise indicated, finish the interior of closets to match the adjacent room; finish back and edges of hinged or removable panels to match the exposed face.

3.3.5.2. Unless otherwise indicated, finish electric panels, access doors, exposed conduit and piping, registers, grilles vents, and similar items to match adjacent wall surfaces.

3.3.5.3. Paint back sides of access panels and removable or hinged covers to match exposed surfaces.

3.3.5.4. Finish interior of field-finished wall and base cabinets and similar casework to match exterior.

3.3.5.5. Finish exterior doors on tops, bottoms, and side edges same as exterior faces.

3.3.5.6. Paint visible surfaces behind grilles, registers, vents, open metal ceilings, and louvers flat black.

3.3.5.7. Size, prime, and paint pipe and duct insulation where exposed to public view. Apply not less than one coat of flat acrylic paint. Remove bands or hangers before painting, and replace after painting is complete.

3.3.5.8. In toilets or other wet areas, add a fungicide recommended by paint manufacturer to paint.

3.4. Workmanship

3.4.1. Workmanship shall be of the highest standard in every respect and tradesmen shall be fully experienced in their field. All work shall be executed neatly and accurately. Products shall be installed according to the manufacturer’s instructions, subject to Engineer’s approval. The work shall be left clean, tidy, and properly finished.

3.5. Cleaning and Protection

3.5.1. At end of each workday, remove rubbish, empty cans, rags, and other discarded materials from Project site.

3.5.2. After completing paint application, clean spattered surfaces. Remove spattered paints by washing, scraping, or other methods. Do not scratch or damage adjacent finished surfaces.

3.5.3. Protect work of other trades against damage from paint application. Correct damage to work of other trades by cleaning, repairing, replacing, and refinishing, as approved by Engineer, and leave in an undamaged condition.

3.5.4. At completion of construction activities of other trades, touch up and restore damaged or defaced painted surfaces.

4. Quality Control

4.1. Delivery, Storage and Handling

4.1.1. All building materials shall be stored on the building site in such a way as to prevent deterioration or the loss or impairment of their structural and other essential properties. Comply to National Building Code (NBC) of India 2016 Volume 1; Part 7 Construction Management, Practices and Safety.

4.2. Source Limitation

4.2.1. All materials & related products shall be from the same manufacturer to ensure material/product integrity & uniformity of colour, texture & quality.

4.3. Mock-ups

4.3.1. Apply sample of each paint system indicated and each colour and finish selected to verify preliminary selections made under sample submittals and to demonstrate aesthetic effects and set quality standards for materials and execution.

4.3.1.1. Engineer will select one surface to represent surfaces and conditions for application of each paint system specified in part 3.

4.3.1.1.1. Vertical and Horizontal surfaces: Provide samples at least 10 Sq.M

4.3.2. Final approval of colour selections will be based on mock-ups.

4.3.2.1. If preliminary colour selections are not approved, apply additional samples of additional colours selected by Engineer at no added cost to owner, until Engineer is satisfied.

4.3.3. Approved mock-ups may be incorporated into the finished work, where directed by the Engineer.

5. Testing and Commissioning

5.1. Inspection

5.1.1. Engineer may direct Contractor to stop applying paints if test results show materials being used do not comply with product requirements. Contractor shall remove non-complying paint materials from Project site, pay for testing, and repaint surfaces painted with rejected materials. Contractor will be required to remove rejected materials from previously painted surfaces if, on repainting with complying materials.

ATK-ARC-004 – Resilient Flooring

1. General

1.1. Introduction

1.1.1. This section specifies supply and installation of Resilient tile/sheet as flooring and skirting, as shown on the tender drawings/schedules as required in completion of the work.

1.2. Related Document

1.2.1. This section to be read in conjunction with Employer's requirements and general provision of contract, including General, Supplementary and Special Conditions of Contract Documents

1.3. Codes and Standards

1.3.1. All materials, workmanship and testing shall be in accordance with the following standards & codes, in their latest revisions. This is not exhaustive and contractor shall verify the listing and supplement with the latest applicable standards & codes.

1.3.2. Comply with the requirements of the reference standards noted herein, except where more stringent requirements are listed herein or otherwise required by the Contract Documents.

1.3.3. Where materials, finishes, fittings, equipment and building components that comply with standards other than those included in these specifications are proposed, the Contractor shall adequately demonstrate that such items are of an equal quality and standard and are fit for their intended purpose, through submission of globally recognized and authorized test methods, technical and certification reports (in compliance with assessment or evaluation criteria based on standards already included in these specifications), to the approval of engineer, prior to use of such items.

1.3.4. Conform to the requirements of the following Standards and Codes of Practice where applicable:

1.3.4.1. National Building Code (NBC) of India 2016 Volume 1 & 2.

1.3.4.2. IS Code 12830: 1989 Rubber based adhesives for fixing PVC tiles to cement.

1.3.4.3. IS Code 3461: 1980 Specification for PVC asbestos floor tiles (first revision).

1.3.4.4. IS Code 3462: 1986 Specification for unbacked flexible PVC flooring (second revision).

1.3.4.5. IS Code 3464: 1986 Methods of test for plastic flooring and wall tiles (second revision).

1.3.4.6. BS 8203 Code of Practice for Installation of Resilient Floor Coverings

1.3.4.7. BS EN 423 Resilient Floor Coverings - Determination of the Resistance to Staining

1.3.4.8. BS EN 425 Resilient and Laminate Floor Coverings.

1.3.4.9. BS EN 426 Resilient floor coverings — Determination of width, length, straightness and flatness of sheet material

1.3.4.10. BS EN 427 Resilient Floor Coverings - Determination of the Side Length, Square ness and Straightness of Tiles

1.3.4.11. BS EN 428 Resilient Floor Coverings - Determination of Overall Thickness

1.3.4.12. BS EN 429 Resilient Floor Coverings - Determination of Thickness of Layers

1.3.4.13. BS EN 430 Resilient Floor Coverings - Determination of Mass Per Unit Area

1.3.4.14. BS EN 433 Resilient Floor Coverings - Determination of Residual Indentation After Static Loading

1.3.4.15. BS EN 434 Resilient Floor Coverings - Determination of Dimensional Stability and Curling After Exposure to Heat

1.3.4.16. BS EN 435 Resilient Floor Coverings - Determination of Flexibility

1.3.4.17. BS EN 436 Resilient Floor Coverings - Determination of Density

1.3.4.18. BS EN 649 Resilient Floor Coverings - Homogeneous and Heterogeneous Polyvinyl Chloride Floor Coverings – Specification.

1.3.4.19. BS EN 684:1996 Resilient Floor Coverings - Determination of Seam Strength

1.3.4.20. BS EN 685 Resilient, textile and laminate floor coverings —Classification

1.3.4.21. BS EN 13553 Resilient Floor Coverings - Polyvinyl Chloride Floor Coverings for Use in Special Wet Areas – Specification

1.3.4.22. EN 660-1 Resilient floor coverings — Determination of wear resistance — Part 1: Stuttgart test.

1.3.4.23. EN 660-2 Resilient floor coverings — Determination of wear resistance — Part 2: Frick-Taber test.

1.3.4.24. NFPA 101 Life Safety Code

1.4. Submittals

1.4.1. Technical Data

1.4.1.1. Comprehensive technical data viz., technical literature, data sheets, product catalogue & brochure, installation instructions, etc. to be provided for Engineer’s approval.

1.4.2. Shop Drawings

1.4.2.1. Shop drawings shall be submitted to Engineer for approval, prior to start of work at site.

1.4.2.2. Shop drawings shall include all details w.r.t. the material & related product’s installation, to the scale & detailing as instructed by the Engineer.

1.4.2.3. Shop drawings shall be developed as per actual measurements from the site & must depict the same in the drawings.

1.4.2.3.1. Shop drawings must comply to BS 1192:2007+A2:2016 Collaborative production of architectural, engineering and construction information & Code of practice.

1.4.3. Samples

1.4.3.1. Samples for Initial Selection: For each type of tile and grout indicated. Include Samples of accessories involving colour selection.

1.4.3.2. Samples for Verification:

1.4.3.2.1. Full-size units of each type and composition of tile and for each colour and finish required.

1.4.3.2.2. Assembled samples with grouted joints for each type and composition of tile and for each colour and finish required, at least 200 mm & 300mm square mounted on rigid panel. Use grout of type and in colour or colours approved for completed work.

1.4.3.2.3. Full-size units of each type of trim and accessory for each colour and finish required.

1.4.3.2.4. Metal edge strips in 150-mm lengths.

1.4.4. Method Statements.

1.4.4.1. Shall include method of construction/installation, Health and Safety management, testing methods, required relative humidity and temperature level and control systems, possible risk associated with the execution works and precautions to be considered, site control system and any other additional requirements as advised by the Engineer for the execution of works.

1.4.4.2. Method statement shall be approved by the Engineer prior to commencement of the works.

1.4.5. Installer Qualifications:

1.4.5.1. Licensed installer specialized in Resilient flooring & skirting installation shall be employed for the work.

1.4.6. Material/Product Certificates:

1.4.6.1. Comprehensive tests performed by a qualified independent testing agency certifying that the product compiles with specified performance requirements.

1.4.7. Maintenance Data:

1.4.7.1. Details of manufacturer's recommended maintenance procedures for each type of product/material to be included in the operation and maintenance manual.

1.4.8. Spare Materials/Parts

1.4.8.1. Provide the following extra materials that match installed products, packaged for storage with suitable labels for identification.

1.4.8.1.1. 2% or 50m^2 minimum for each type of Resilient flooring & skirting (product, pattern or colour)

1.5. Warranty / Guarantee

1.5.1. As per the Employer's requirements and general provision of contract, including General, Supplementary and Special Conditions of Contract Documents.

1.6. Performance Requirements

1.6.1. All finishes, materials and components shall comply with requirements of NFPA 101 Life Safety Code, chapter 10 – Interior Finish, Contents and Furnishings with respect to classification of materials, flame spread, smoke generation and other requirements.

1.7. Environment & Sustainability

1.7.1. Recycling Strategy document for the material or product shall be submitted to the Engineer for approval.

1.7.2. LEED Requirements:

1.7.2.1. All the specifications in this document are for USGBC LEED BD+C submission and are applicable to all materials & procedures followed for the project. The specification as stated here should be adhered to the word. In case of a dispute LEED guidelines shall be followed. Brochures, product specification sheets and/or certificates as specified in this document should be submitted along with the material procurement. All contractors, manufactures must submit documents, records and other submittal related to the different credits as mentioned in the specification.

1.7.2.2. Manufacturer product conforming to ISO 14044 or Environmental Product Declarations which conform to ISO 14025, 14040, 14044, and EN 15804 or ISO 21930 and have at least a cradle to gate scope. (If applicable)

1.7.2.3. Manufacturer Material Ingredient Reporting by any of the listed below documents (If applicable):

1.7.2.3.1. Health Product Declaration. The end use product has a published, complete Health Product Declaration with full disclosure of known hazards in compliance with the Health Product Declaration open Standard.

1.7.2.3.2. Cradle to Cradle Certificate. The end use product has been certified at the Cradle to Cradle v2 Basic level or Cradle to Cradle v3 Bronze level.

1.7.2.3.3. REACH Optimization. End use products and materials that do not contain substances that meet REACH criteria for substances of very high concern.

1.7.2.4. Manufacturer Raw Material Source and Extraction Reporting by any of the listed below activity (If applicable):

1.7.2.4.1. Third-party verified corporate sustainability reports (CSR) which include environmental impacts of extraction operations and activities associated with the manufacturer's product and the product's supply chain

1.7.2.4.2. Self-declared reports from manufacturers that have publicly released a report from their raw material suppliers which include raw material supplier extraction locations.

1.7.2.5. Tile Adhesive should meet VOC level as per LEED Tender Document. Manufacturer has to confirm same by providing lab test report.

1.8. Manufacturer

1.8.1. Manufacturer to be accredited and registered under BS EN ISO 9001.

1.8.2. List of preferred or acceptable Manufacturers attached herewith as Annexure 1.

2. Products

2.1. Materials

2.1.1. Material/Product Specification as per Standard attached herewith as Annexure A.04.

2.2. Colours

2.2.1. Colours as per Standard attached herewith as Annexure B.01.

3. Execution

3.1. Examination

3.1.1. Examine substrates, areas, and conditions where flooring will be installed, with Installer present, for compliance with requirements for installation tolerances and other conditions affecting performance of installed flooring.

3.1.1.1. Verify that substrate is smooth, level with minimum variation from level not to exceed 3 mm in 3m or as recommended by the manufacturer.

3.1.1.2. Verify that installation of grounds, anchors, recessed frames, electrical and mechanical units of work, and similar items located in or behind flooring has been completed before installing flooring.

3.1.1.3. Verify that joints in flooring substrates are coordinated with flooring joint locations; if not coordinated, adjust joint locations in consultation with the Engineer.

3.1.1.4. Verify concrete floors exhibits acceptable moisture emission, negative alkalinity, carbonization or dusting.

3.1.2. Proceed with installation only after unsatisfactory conditions have been corrected.

3.2. Preparation

3.2.1. Remove coatings, including curing compounds and other substances that contain soap, wax, oil, or silicone, that are incompatible with floor-setting materials.

3.2.2. Provide minimum 90 days to cure concrete sub floors in order to achieve acceptable dryness.

3.2.3. Locate material three days prior to installation in order to achieve temperature stability.

3.2.4. Maintain ambient temperature three days prior to installation as recommended by flooring manufacturer.

3.2.5. Fill cracks, holes, and depressions with Trowelable levelling and patching compound according to floor-setting material manufacturer's written instructions. Use product specifically recommended by floor-setting material manufacturer.

3.2.6. Remove protrusions, bumps, and ridges by sanding or grinding.

3.2.7. Blending: For tile exhibiting colour variations within ranges selected during Sample submittals, verify that tile has been factory blended and packaged so tile units taken from one package show same range of colours as those taken from other packages and match approved Samples. If not factory blended, either return to manufacturer or blend tiles at Project site before installing.

3.3. Installation

3.3.1. Installation must be carried out strictly in accordance with written instructions of manufacturer, to ensure product performance and achievement of electrical results outlined in system performance.

3.3.2. Close floor installation area to traffic during the period of installation.

3.3.3. Where demountable partitions and other items are indicated for installation on top of the flooring, install flooring before installing these items.

3.3.4. Do not install flooring over concrete slabs until slabs have cured and are sufficiently dry to bond with adhesive as per manufacturer's recommendations.

3.3.5. Jointing Pattern: Lay tile as per approved shop drawings. Align joints when adjoining tiles on floor and trim are same size. Lay out tile work and centre tile fields in both directions in each space or on each wall area. Adjust to minimize tile cutting. Provide uniform joint widths, unless otherwise indicated.

3.3.6. Expansion Joints: Locate expansion joints and other sealant-filled joints, including control, contraction, and isolation joints, where indicated during installation of setting materials, mortar beds, and sheet/tile. Do not saw- cut joints after installing tiles.

3.3.6.1. Locate joints in sheet/tile surfaces directly above joints in concrete substrates. Prepare joints and apply sealants to comply with requirements shown on the shop drawings.

3.3.7. Do not allow traffic on floors until 4 days after completion of flooring and subsequently permit only light traffic for a further 10 days.

3.4. Workmanship

3.4.1. Thoroughly clean the screed or backing of grease, oil, wax, paint, dirt or dust and fill all cracks, minor holes and services with a suitable filler recommended by the tile or sheeting manufacturer.

3.4.2. Lay all tiles, floor and wall tiles strictly in accordance with the manufacturer’s instructions.

3.4.3. Apply adhesive in a thin film and spread evenly with a notched or plain trowel as recommended by the manufacturer. If a notched trowel is used the notches shall be of the size recommended and shall be renewed periodically to ensure the correct spread of adhesive.

3.4.4. Closely butt joints and cut accurately at right angles to the tiles to an agreed design and with only the minimum necessary jointing. Relay any areas insufficiently bonded or showing an uneven surface.

3.5. Cleaning and Protection

3.5.1. After laying, thoroughly clean the vinyl flooring using detergent recommended by the manufacturer. Apply two coats of non-slip, matt emulsion wax polish. Protect the floors with approved covering.

4. Quality Control

4.1. Delivery, Storage and Handling

4.1.1. All building materials shall be stored on the building site in such a way as to prevent deterioration or the loss or impairment of their structural and other essential properties. Comply to National Building Code (NBC) of India 2016 Volume 1; Part 7 Construction Management, Practices and Safety.

4.2. Source Limitation

4.2.1. All materials & related products shall be from the same manufacturer to ensure material/product integrity & uniformity of colour, texture & quality.

4.3. Mock-up

4.3.1. Build mock-ups for each type of Resilient flooring indicated including all accessories to verify selections made under sample submittals and to demonstrate aesthetic effects and set quality standards for materials and execution.

4.3.2. Size: Minimum 9 Sq.M of typical poured-in place flooring and base condition, in locations selected by Engineer.

4.3.3. Obtain Engineer's approval of design and patterns.

5. Testing and Commissioning

5.1. Inspection

5.1.1. Inspection upon completion of work, all Resilient tile areas shall be carefully tested and examined for faults in application, shall be corrected immediately.

ATK-ARC-005 – Resin Flooring

1. General

1.1. Introduction

1.1.1. This section specifies supply and installation of the Resin flooring & skirting, as shown on the Tender drawings & documents, as required in completion of the work.

1.2. Related Documents

1.2.1. This section to be read in conjunction with Employer's requirements and general provision of contract, including General, Supplementary and Special Conditions of Contract Documents.

1.3. Codes and Standards

1.3.1. All materials, workmanship and testing shall be in accordance with the following documents, in their latest revisions. This is not exhaustive and contractor shall verify the listing and supplement with the latest applicable document or documents.

1.3.2. Comply with the requirements of the reference standards noted herein, except where more stringent requirements are listed herein or otherwise required by the Contract Documents.

1.3.3. Where materials, finishes, fittings, equipment and building components that comply with standards other than those included in these specifications are proposed, the Contractor shall adequately demonstrate that such items are of an equal quality and standard and are fit for their intended purpose, through submission of internationally recognised and authorised test methods, technical and certification reports (in compliance with assessment/evaluation criteria based on standards already included in these specifications), to the approval of engineer, prior to use of such items.

1.3.4. Conform to the requirements of the following British Standards and Codes of Practice where applicable:

1.3.4.1. National Building Code (NBC) of India 2016 Volume 1 & 2.

1.3.4.2. IS Code 9162: 1979 Methods of tests for epoxy resin, hardeners and epoxy resin composition for floor topping.

1.3.4.3. IS Code 9197: 1979 Specification for epoxy resin, hardness and epoxy resin compositions for floor topping.

1.3.4.4. BS 8204-6:2008+A1:2010 Screeds, bases and in situ floorings. Synthetic resin floorings. Code of practice.

1.3.4.5. NFPA 101 Life Safety Code.

1.4. Submittals

1.4.1. Technical Data

1.4.1.1. Comprehensive technical data viz., technical literature, data sheets, product catalogue & brochure, installation instructions, etc. to be provided for Engineer's approval.

1.4.2. Shop Drawings

1.4.2.1. Shop drawings shall be submitted to Engineer for approval, prior to start of work at site.

1.4.2.2. Shop drawings shall include all details w.r.t. the material & related product's installation, to the scale & detailing as instructed by the Engineer.

1.4.2.3. Shop drawings shall be developed as per actual measurements from the site & must depict the same in the drawings.

1.4.2.3.1. Shop drawings must comply to BS 1192:2007+A2:2016 Collaborative production of architectural, engineering and construction information & Code of practice.

1.4.3. Samples

1.4.3.1. Samples for Initial Selection: Provide samples for each type of finish required.

1.4.3.1.1. Samples for Verification: For each resinous flooring system, submit 300 mm square, applied to a rigid backing by Installer.

1.4.4. Method Statements.

1.4.4.1. Shall include method of construction/installation, Health and Safety management, testing methods, required relative humidity and temperature level and control systems, possible risk associated with the execution works and precautions to be considered, site control system and any other additional requirements as advised by the Engineer for the execution of works.

1.4.4.2. Method statement shall be approved by the Engineer prior to commencement of the works.

1.4.5. Installer Qualifications:

1.4.5.1. Licensed installer specialized in Resin flooring & skirting installation shall be employed for the work.

1.4.6. Material/Product Certificates:

1.4.6.1. Comprehensive tests performed by a qualified independent testing agency certifying that the product compiles with specified performance requirements.

1.4.7. Maintenance Data:

1.4.7.1. Details of manufacturer's recommended maintenance procedures for each type of product/material to be included in the operation and maintenance manual.

1.4.8. Spare Materials/Parts

1.4.8.1. Provide the following extra materials that match installed products, packaged for storage with suitable labels for identification.

1.4.8.2. 2% or 50m^2 minimum for each type of Resin flooring & skirting (product, pattern or colour)

1.5. Warranty / Guarantee

1.5.1. As per the Employer's requirements and general provision of contract, including General, Supplementary and Special Conditions of Contract Documents.

1.6. Performance Requirements

1.6.1. All finishes, materials and components shall comply with requirements of NFPA 101 Life Safety Code, chapter 10 – Interior Finish, Contents and Furnishings with respect to classification of materials, flame spread, smoke generation and other requirements.

1.7. Environment & Sustainability

1.7.1. Recycling Strategy document for the material or product shall be submitted to the Engineer for approval.

1.7.2. LEED Requirements:

1.7.2.1. All the specifications in this document are for USGBC LEED BD+C submission and are applicable to all materials & procedures followed for the project. The specification as stated here should be adhered to the word. In case of a dispute LEED guidelines shall be followed. Brochures, product specification sheets and/or certificates as specified in this document should be submitted along with the material procurement. All contractors, manufactures must submit documents, records and other submittal related to the different credits as mentioned in the specification.

1.7.2.2. Manufacturer product conforming to ISO 14044 or Environmental Product Declarations which conform to ISO 14025, 14040, 14044, and EN 15804 or ISO 21930 and have at least a cradle to gate scope. (If applicable).

1.7.2.3. Manufacturer Material Ingredient Reporting by any of the listed below documents (If applicable):

1.7.2.3.1. Health Product Declaration. The end use product has a published, complete Health Product Declaration with full disclosure of known hazards in compliance with the Health Product Declaration open Standard.

1.7.2.3.2. Cradle to Cradle Certificate. The end use product has been certified at the Cradle to Cradle v2 Basic level or Cradle to Cradle v3 Bronze level.

1.7.2.3.3. REACH Optimization. End use products and materials that do not contain substances that meet REACH criteria for substances of very high concern.

1.7.2.4. Manufacturer Raw Material Source and Extraction Reporting by any of the listed below activity (If applicable):

1.7.2.4.1. Third-party verified corporate sustainability reports (CSR) which include environmental impacts of extraction operations and activities associated with the manufacturer’s product and the product’s supply chain

1.7.2.4.2. Self-declared reports from manufacturers that have publicly released a report from their raw material suppliers which include raw material supplier extraction locations

1.8. Manufacturer

1.8.1. Manufacturer to be accredited and registered under BS EN ISO 9001.

1.8.2. List of preferred or acceptable Manufacturers attached herewith as Annexure 1.

2. Products

2.1. Material

2.1.1. Material/Product Specification as per Standard attached herewith as Annexure A.05.

2.2. Colours

2.2.1. Colours as per Standard attached herewith as Annexure B.01.

3. Execution

3.1. Preparation

3.1.1. General: Prepare and clean substrates according to resinous flooring manufacturer's written instructions for substrate indicated, subject to the approval of Engineer. Provide clean, dry, and neutral Ph substrate for resinous flooring application.

3.1.2. Concrete Substrates: Provide sound concrete surfaces free of laitance, glaze, efflorescence, curing compounds, form-release agents, dust, dirt, grease, oil, and other contaminants incompatible with resinous flooring.

3.1.2.1. Repair damaged and deteriorated concrete according to resinous flooring manufacturer's written recommendations. Verify that concrete substrates are dry.

3.1.2.2. Perform additional moisture tests recommended by manufacturer. Proceed with application only after testing the substrates.

3.1.2.3. Verify that concrete substrates have neutral Ph and that resinous flooring will adhere to them. Perform tests recommended by manufacturer. Proceed with application only after testing the substrate.

3.1.3. Resinous Materials: Mix components and prepare materials according to resinous flooring manufacturer's written instructions.

3.1.4. Use patching and fill material to cover holes and depressions in substrates according to manufacturer's written instructions.

3.1.5. Treat control joints and other non-moving substrate cracks to prevent cracks from reflecting through resinous flooring according to manufacturer's written recommendations.

3.2. Application

3.2.1. General: Apply components of Epoxy Resinous Flooring system trowelled or screeded according to manufacturer's written instructions to produce a uniform, monolithic wearing surface of thickness as indicated in the contract drawings/schedules.

3.2.2. Coordinate application of components to provide optimum adhesion of resinous flooring system to substrate, and optimum intercoat adhesion.

3.2.3. Cure resinous flooring components according to manufacturer's written instructions. Prevent contamination during application and curing processes.

3.2.4. At substrate expansion and isolation joints, provide joint in resinous flooring to comply with resinous flooring manufacturer's written recommendations.

3.2.5. Apply primer over prepared substrate at manufacturer's recommended spreading rate.

3.2.6. Integral Cove Base: Apply cove base mix to wall surfaces before applying flooring. Apply according to manufacturer's written instructions and details including those for taping, mixing, priming, trowelling, sanding, and top coating of cove base. Round internal and external corners.

3.2.7. Apply trowelled or screeded body coat(s) in thickness indicated for flooring system. Hand or power trowel and grout to fill voids. When cured, sand to remove trowel marks and roughness.

3.2.8. Apply topcoat(s) in number of coats indicated for flooring system and at spreading rates recommended in writing by manufacturer.

3.3. Workmanship

3.3.1. Workmanship shall be of the highest standard in every respect and tradesmen shall be fully experienced in their field. All work shall be executed neatly and accurately. Products shall be installed according to the manufacturer’s instructions. The work shall be left clean, tidy, and properly finished.

3.4. Cleaning & Protection

3.4.1. Protect resinous flooring from damage and wear during the remainder of construction period. Use protective methods and materials, including temporary covering, recommended in writing by resinous flooring manufacturer.

4. Quality Control

4.1. Delivery, Storage and Handling

4.1.1. All building materials shall be stored on the building site in such a way as to prevent deterioration or the loss or impairment of their structural and other essential properties. Comply to National Building Code (NBC) of India 2016 Volume 1; Part 7 Construction Management, Practices and Safety.

4.2. Source Limitation

4.2.1. All materials & related products shall be from the same manufacturer to ensure material/product integrity & uniformity of colour, texture & quality.

4.3. Mock-ups

4.3.1. Apply mock-ups to verify selections made under sample submittals and to demonstrate aesthetic effects and set quality standards for materials and execution.

4.3.1.1. Apply full-thickness mock-ups on 1200mm square floor area on site as selected by the Engineer

4.3.2. Approved mock-ups may be incorporated into the finished work, where directed by the Engineer.

5. Testing and Commissioning

5.1. Inspection

5.1.1. Inspection upon completion of work, all epoxy-resinous flooring & skirting areas shall be carefully tested and examined for faults in application, shall be corrected immediately.

ATK-ARC-006 – Gypsum Plasterboard Partitions

1. General

1.1. Introduction

1.1.1. This section specifies supply and installation of the Gypsum Plasterboard Partition wall as shown on the Tender drawings & documents, as required in completion of the work.

1.2. Related Documents

1.2.1. This section to be read in conjunction with Employer's requirements and general provision of contract, including General, Supplementary and Special Conditions of Contract Documents.

1.3. Codes and Standards

1.3.1. All materials, workmanship and testing shall be in accordance with the following standards & codes, in their latest revisions. This is not exhaustive and contractor shall verify the listing and supplement with the latest applicable standards & codes.

1.3.2. Comply with the requirements of the reference standards noted herein, except where more stringent requirements are listed herein or otherwise required by the Contract Documents.

1.3.3. Where materials, finishes, fittings, equipment and building components that comply with standards other than those included in these specifications are proposed, the Contractor shall adequately demonstrate that such items are of an equal quality and standard and are fit for their intended purpose, through submission of internationally recognized and authorized test methods, technical and certification reports (in compliance with assessment/evaluation criteria based on standards already included in these specifications), to the approval of engineer, prior to use of such items

1.3.4. Conform to the requirements of the following Standards and Codes of Practice where applicable:

1.3.4.1. National Building Code (NBC) of India 2016 Volume 1 & 2.

1.3.4.2. IS Code 2095 Specification for gypsum plaster boards

1.3.4.2.1. (Part 1): 2011 Plain gypsum plaster boards (third revision)

1.3.4.2.2. (Part 2): 2001 Coated/laminated gypsum plaster boards (second revision)

1.3.4.2.3. (Part 3): 1996 Reinforced gypsum plaster boards (second revision)

1.3.4.3. IS Code 2542 Methods of test for gypsum plaster, concrete and products

1.3.4.3.1. (Part 1/Sec 1) 1978: Plaster and concrete, Section 1 Normal consistency of gypsum plaster (first revision)

1.3.4.3.2. (Part 1/Sec 2) 1978: Plaster and concrete, Section 2 Normal consistency of gypsum concrete (first revision)

1.3.4.3.3. (Part 1/Sec 3) 1978: Plaster and concrete, Section 3 Setting time of plaster and concrete (first revision)

1.3.4.3.4. (Part 1/Sec 4) 1978: Plaster and concrete, Section 4 Transverse strength of gypsum plaster (first revision)

1.3.4.3.5. (Part 1/Sec 5) 1978: Plaster and concrete, Section 5 Compressive strength and dry set density of gypsum plaster (first revision)

1.3.4.3.6. (Part 1/Sec 6) 1978: Plaster and concrete, Section 6 Soundness of gypsum plaster (first revision)

1.3.4.3.7. (Part 1/Sec 7) 1978: Plaster and concrete, Section 7 Mechanical resistance of gypsum plaster by dropping ball test (first revision)

1.3.4.3.8. (Part 1/Sec 8) 1978: Plaster and concrete, Section 8 Freedom from coarse particles (first revision)

1.3.4.3.9. (Part 1/Sec 9) 1978: Plaster and concrete, Section 9 Expansion of plaster (first revision)

1.3.4.3.10. (Part 1/Sec 10) 1978: Plaster and concrete, Section 10 Sand in set plaster (first revision)

1.3.4.3.11. (Part 1/Sec 11) 1978: Plaster and concrete, Section 11 Wood fibre content in gypsum plaster (first revision)

1.3.4.3.12. (Part 1/Sec 12) 1978: Plaster and concrete, Section 12 Dry bulk density (first revision)

1.3.4.3.13. (Part 2/Sec 1) 1981: Gypsum products, Section 1 Measurement of dimensions (first revision)

1.3.4.3.14. (Part 2/Sec 2) 1981: Gypsum products, Section 2 Determination of mass (first revision)

1.3.4.3.15. (Part 2/Sec 3) 1981: Gypsum products, Section 3 Determination of mass and thickness of paper surfacing (first revision)

1.3.4.3.16. (Part 2/Sec 4) 1981: Gypsum products, Section 4 Transverse strength (first revision)

1.3.4.3.17. (Part 2/Sec 5) 1981: Gypsum products, Section 5 Compressive strength (first revision)

1.3.4.3.18. (Part 2/Sec 6) 1981: Gypsum products, Section 6 Water absorption (first revision)

1.3.4.3.19. (Part 2/Sec 7) 1981: Gypsum products, Section 7 Moisture content (first revision)

1.3.4.3.20. (Part 2/Sec 8) 1981: Gypsum products, Section 8 Nail retention of precast reinforced gypsum slabs (first revision)

1.3.4.4. IS Code 2547 Specification for gypsum building plaster

1.3.4.4.1. (Part 1): 1976 Excluding premixed lightweight plaster (first revision)

1.3.4.4.2. (Part 2): 1976 Premixed lightweight plaster (first revision)

1.3.4.5. IS Code 2849: 1983 Specification for non-load bearing gypsum partition blocks (solid and hollow types) (first revision)

1.3.4.6. IS Code 8272: 1984 Specification for gypsum plaster for use in the manufacture of fibrous plasterboards (first revision)

1.3.4.7. IS Code 9498: 1980 Specification for inorganic aggregates for use in gypsum plaster

1.3.4.8. IS Code 12679: 1989 Specification for by-product gypsum for use in plaster, blocks and boards

1.3.4.9. BS 476 Fire tests on building materials and structures.

1.3.4.10. BS 5234 part 1 & 22 Partitions (including matching linings). Specification for performance requirements for strength and robustness including methods of test.

1.3.4.11. BS 8212 Code of Practice for dry lining and partitioning using gypsum plasterboard

1.3.4.12. BS EN 520 Gypsum Plasterboards. Definitions, requirements and test methods.

1.3.4.13. BS EN 10126 Cold Rolled Steel Sections.

1.3.4.14. BS EN 13501 part 1 & 2 Fire classification of construction products and building elements. Classification using test data from reaction to fire tests.

1.3.4.15. BS EN ISO 140-3,4 (also numbered BS 2750-3) Acoustics. Measurement of sound insulation in buildings and of building elements. Laboratory measurement of airborne sound insulation of building elements + AMD 15277

1.3.4.16. BS EN ISO 10848-2,3 (replaces BS EN 20140-9 and BS EN ISO 140-12) Acoustics. Laboratory measurement of the flanking transmission of airborne and impact sound between adjoining rooms. Application to light elements when the junction has a small influence

1.3.4.17. BS EN ISO 14001 (replaced by BS EN ISO 14001 but remains current) Environmental management systems. Specification with guidance for use

1.3.4.18. BS EN 14195 Metal framing components for Gypsum Plasterboard Systems. Definitions, requirements and test methods.

1.3.4.19. NFPA 101 Life Safety Code

1.4. Submittals

1.4.1. Technical Data:

1.4.1.1. Comprehensive technical data viz., technical literature, data sheets, product catalogue & brochure, installation instructions, etc. to be provided for Engineer’s approval.

1.4.2. Shop Drawings:

1.4.2.1. Shop drawings shall be submitted to Engineer for approval, prior to start of work at site.

1.4.2.2. Shop drawings shall include all details w.r.t. the material & related product’s installation, to the scale & detailing as instructed by the Engineer.

1.4.2.3. Shop drawings shall be developed as per actual measurements from the site & must depict the same in the drawings.

1.4.2.4. Shop drawings must comply to BS 1192:2007+A2:2016 Collaborative production of architectural, engineering and construction information & Code of practice.

1.4.3. Samples:

1.4.3.1. Samples for Initial Selection: Provide Gypsum Plasterboard Wall assembly & accessories which are shown or implied by the drawings and required for the work.

1.4.4. Method Statements

1.4.4.1. Shall include method of construction/installation, Health and Safety management, testing methods, required relative humidity and temperature level and control systems, possible risk associated with the execution works and precautions to be considered, site control system and any other additional requirements as advised by the Engineer for the execution of works.

1.4.4.2. Method statement shall be approved by the Engineer prior to commencement of the works.

1.4.5. Installer Qualification:

1.4.5.1. Licensed installer specialized in gypsum plasterboard partition installations, and eligible to receive manufacturer's warranty.

1.4.6. Material Certificates:

1.4.6.1. Comprehensive tests performed by a qualified independent testing agency certifying that the product compiles with specified performance requirements.

1.4.7. Maintenance Data:

1.4.7.1. Details of manufacturer's recommended maintenance procedures for each type of product/material to be included in the operation and maintenance manual.

1.4.8. Spare Materials/Parts

1.4.8.1. Provide the following extra materials that match installed products, packaged for storage with suitable labels for identification.

1.5. Warranty

1.5.1. As per the Employer's requirements and general provision of contract, including General, Supplementary and Special Conditions of Contract Documents.

1.6. Performance Requirements

1.6.1. All finishes, materials and components shall comply with requirements of NFPA 101 Life Safety Code, chapter 10 – Interior Finish, Contents and Furnishings with respect to classification of materials, flame spread, smoke generation and other requirements.

1.6.2. Gypsum Plasterboard Partition Wall Types

Wall Tag	Wall Thickness (MM)	Acoustic Rating (dB)	Fire Rating (Mins.)
C0.0.0	30	00	0
C1.0.0	100	00	0
C1.1.0	100	35	0
C1.2.0	100	40	0
C1.3.0	100	44	0
C1.0.F	100	00	120
C2.0.0	150	00	0
C2.1.0	150	35	0

Wall Tag	Thickness (MM)	Acoustic Rating (dB)	Fire Rating (Mins.)
C2.2.0	150	40	0
C2.3.0	150	44	0
C2.4.0	150	52	0
C2.0.F	150	00	120
C2.1.F	150	35	120
C2.2.F	150	40	120
C2.3.F	150	44	120
C2.4.F	150	52	120
C3.0.0	200	00	0
C3.1.0	200	35	0
C3.2.0	200	40	0
C3.3.0	200	44	0
C3.0.F	200	00	120
C3.1.F	200	35	120
C3.2.F	200	40	120
C3.4.F	200	52	120
C4.0.F	400	00	120
Note: C#.R.# – walls shall be 1mm thick GI perforated sheet over 1st layer of Plasterboard as indicated in the drawings			

1.7. Environment & Sustainability

1.7.1. Recycling Strategy document for the material or product shall be submitted to the Engineer for approval.

1.7.2. LEED Requirements:

1.7.2.1. All the specifications in this document are for USGBC LEED BD+C submission and are applicable to all materials & procedures followed for the project. The specification as stated here should be adhered to the word. In case of a dispute LEED guidelines shall be followed. Brochures, product specification sheets and/or certificates as specified in this document should be submitted along with the material procurement. All contractors, manufactures must submit documents, records and other submittal related to the different credits as mentioned in the specification.

1.7.2.2. Manufacturer product conforming to ISO 14044 or Environmental Product Declarations which conform to ISO 14025, 14040, 14044, and EN 15804 or ISO 21930 and have at least a cradle to gate scope. (If applicable).

1.7.2.3. Manufacturer Material Ingredient Reporting by any of the listed below documents (If applicable):

1.7.2.3.1. Health Product Declaration. The end use product has a published, complete Health Product Declaration with full disclosure of known hazards in compliance with the Health Product Declaration Open Standard.

1.7.2.3.2. Cradle to Cradle Certificate. The end use product has been certified at the Cradle to Cradle v2 Basic level or Cradle to Cradle v3 Bronze level.

1.7.2.3.3. REACH Optimization. End use products and materials that do not contain substances that meet REACH criteria for substances of very high concern.

1.7.2.4. Manufacturer Raw Material Source and Extraction Reporting by any of the listed below activity (If applicable):

1.7.2.4.1. Third-party verified corporate sustainability reports (CSR) which include environmental impacts of extraction operations and activities associated with the manufacturer's product and the product's supply chain.

1.7.2.4.2. Self-declared reports from manufacturers that have publicly released a report from their raw material suppliers which include raw material supplier extraction locations.

1.7.2.5. Dry Wall Adhesive should meet VOC level as per LEED Tender Document. Manufacturer must confirm same by providing lab test report.

1.8. Manufacturers

1.8.1. Manufacturer to be accredited and registered under BS EN ISO 9001.

1.8.2. List of preferred or acceptable Manufacturers attached herewith as Annexure 1.

2. Products

2.1. Materials

2.1.1. Material/Product Specification as per Standard attached herewith as Annexure A.06.

2.1.2. Moisture Resistant Gypsum Plasterboard shall be used for Wet Area & Backside of Water Fountain.

2.2. Colours

2.2.1. Colours as per Standard attached herewith as Annexure B.01.

3. Execution

3.1. Examination

3.1.1. Examine substrates to which gypsum board shaft-wall assemblies attach or abut, with Installer present, and including hollow-metal frames, elevator hoist way doorframes, cast-in anchors, and structural framing. Examine for compliance with requirements for installation tolerances and other conditions affecting performance.

3.1.2. Examine panels before installation. Do not use panels that are wet, moisture damaged, or mould damaged.

3.1.3. Proceed with installation only after unsatisfactory conditions have been corrected.

3.2. Installation

3.2.1. General: Install gypsum board shaft-wall assemblies to comply with requirements of fire-resistance-rated assemblies indicated and manufacturer's written installation instruction. Isolate perimeter of gypsum panels from building structure to prevent cracking of panels, while maintaining continuity of fire-rated construction.

3.2.2. Gypsum Plasterboard Partition shall be provided with required additional channels/members for Openings, Doors, Windows, Louvers, Cut-outs, Lifts/Staircase Walls, Walls with mounted Fixtures, Signage, Equipment, by the Manufacturer/Supplier, ensuring the strength & stability through approved design calculations.

3.2.3. Seal gypsum board shaft walls with acoustical sealant at perimeter of each assembly where it abuts other work and at joints and penetrations within each assembly. Install acoustical sealant to withstand dislocation by air- pressure differential between shaft and external spaces; maintain an airtight and smoke-tight seal; and comply with ASTM C 919 requirements or with manufacturer's written instructions, whichever are more stringent.

3.3. Protection

3.3.1. Protect installed products from damage from weather, condensation, direct sunlight, construction, and other causes during remainder of the construction period. Remove and replace panels that are wet, moisture damaged, or mould damaged, discoloration, sagging, and irregular shape, fuzzy or splotchy surface contamination.

4. Quality Control

4.1. Delivery, Storage and Handling

4.1.1. All building materials shall be stored on the building site in such a way as to prevent deterioration or the loss or impairment of their structural and other essential properties. Comply to National Building Code (NBC) of India 2016 Volume 1; Part 7 Construction Management, Practices and Safety.

4.2. Source Limitation

4.2.1. All materials & related products shall be from the same manufacturer to ensure material/product integrity & uniformity of colour, texture & quality.

4.3. Mock-ups

4.3.1. Build full height mock-ups for each type of panel including all accessories to verify selections made under sample submittals and to demonstrate aesthetic effects and set quality standards for materials and execution.

4.3.2. Approved mock-ups may be incorporated into the finished work, where directed by the Engineer.

5. Testing and Commissioning

5.1. Inspection

5.1.1. Inspection upon completion of work, all Gypsum Plasterboard Wall shall be carefully tested and examined for faults in application, shall be corrected immediately.

ATK-ARC-007 – Toilet Cubicle Partition & Urinal Partition

1. General

1.1. Introduction

1.1.1. This section specifies supply and installation of the Toilet Cubicle & Urinal Partitions as shown on the Tender drawings & documents, as required in completion of the work.

1.2. Related Documents

1.2.1. This section to be read in conjunction with Employer's requirements and general provision of contract, including General, Supplementary and Special Conditions of Contract Documents.

1.3. Codes and Standards

1.3.1. All materials, workmanship and testing shall be in accordance with the following standards & codes, in their latest revisions. This is not exhaustive and contractor shall verify the listing and supplement with the latest applicable standards & codes.

1.3.2. Comply with the requirements of the reference standards noted herein, except where more stringent requirements are listed herein or otherwise required by the Contract Documents.

1.3.3. Where materials, finishes, fittings, equipment and building components that comply with standards other than those included in these specifications are proposed, the Contractor shall adequately demonstrate that such items are of an equal quality and standard and are fit for their intended purpose, through submission of internationally recognized and authorized test methods, technical and certification reports (in compliance with assessment/evaluation criteria based on standards already included in these specifications), to the approval of engineer, prior to use of such items

1.3.4. Conform to the requirements of the following Standards and Codes of Practice where applicable:

1.3.4.1. National Building Code (NBC) of India 2016 Volume 1 & 2.

1.3.4.2. IS Code 2046: 1995 Specification for decorative thermosetting synthetic resin bonded laminated sheets (second revision)

1.3.4.3. BS EN 438-1:2016. High-pressure decorative laminates (HPL). Sheets based on thermosetting resins (usually called laminates). Introduction and general information

1.3.4.4. BS EN 438-2:2016. High-pressure decorative laminates (HPL). Sheets based on thermosetting resins (usually called laminates). Determination of properties

1.3.4.5. BS EN 438-3:2016. High-pressure decorative laminates (HPL). Sheets based on thermosetting resins (usually called laminates). Classification and specifications for laminates less than 2 mm thick intended for bonding to supporting substrates

1.3.4.6. BS EN 438-6:2016. High-pressure decorative laminates (HPL). Sheets based on thermosetting resins (usually called laminates). Classification and specifications for Exterior-grade compact laminates of thickness 2 mm and greater

1.3.4.7. BS EN 438-5:2016. High-pressure decorative laminates (HPL). Sheets based on thermosetting resins (usually called laminates). Classification and specifications for flooring grade laminates less than 2 mm thick intended for bonding to supporting substrates

1.3.4.8. BS EN 438-4:2016. High-pressure decorative laminates (HPL). Sheets based on thermosetting resins (usually called laminates). Classification and specifications for compact laminates of thickness 2 mm and greater

1.3.4.9. ISO 4586-7:2015. High-pressure decorative laminates (HPL, HPDL). Sheets based on thermosetting resins (Usually called Laminates). Classification and specifications for design laminates

1.3.4.10. ISO 4586-8:2015. High-pressure decorative laminates (HPL, HPDL). Sheets based on thermosetting resins (Usually called Laminates). Classification and specifications for alternative core laminates

1.3.4.11. ISO 4586-2:2015. High-pressure decorative laminates (HPL, HPDL). Sheets based on thermosetting resins (Usually called Laminates). Determination of properties

1.3.4.12. ISO 4586-1:2015. High-pressure decorative laminates (HPL, HPDL). Sheets based on thermosetting resins (Usually called Laminates). Introduction and general information

1.3.4.13. ISO 4586-6:2015. High-pressure decorative laminates (HPL, HPDL). Sheets based on thermosetting resins (Usually called Laminates). Classification and specifications for exterior-grade compact laminates of thickness 2 mm and greater

1.3.4.14. ISO 4586-3:2015. High-pressure decorative laminates (HPL, HPDL). Sheets based on thermosetting resins (Usually called Laminates). Classification and specifications for laminates less than 2 mm thick and intended for bonding to supporting substrates

1.3.4.15. ISO 4586-4:2015. High-pressure decorative laminates (HPL, HPDL). Sheets based on thermosetting resins (Usually called Laminates). Classification and specifications for compact laminates of thickness 2 mm and greater

1.3.4.16. ISO 4586-5:2015. High-pressure decorative laminates (HPL, HPDL). Sheets based on thermosetting resins (Usually called Laminates). Classification and specifications for flooring grade laminates less than 2 mm thick intended for bonding to supporting substrates

1.3.4.17. NFPA 101 Life Safety Code

1.4. Submittals

1.4.1. Technical Data:

1.4.1.1. Comprehensive technical data viz., technical literature, data sheets, product catalogue & brochure, installation instructions, etc. to be provided for Engineer’s approval.

1.4.2. Shop Drawings:

1.4.2.1. Shop drawings shall be submitted to Engineer for approval, prior to start of work at site.

1.4.2.2. Shop drawings shall include all details w.r.t. the material & related product’s installation, to the scale & detailing as instructed by the Engineer.

1.4.2.3. Shop drawings shall be developed as per actual measurements from the site & must depict the same in the drawings.

1.4.2.4. Shop drawings must comply to BS 1192:2007+A2:2016 Collaborative production of architectural, engineering and construction information & Code of practice.

1.4.3. Samples:

1.4.3.1. Samples for Initial Selection: Provide Toilet cubicle & Urinal partition assembly & accessories which are shown or implied by the drawings and required for the work.

1.4.4. Method Statements

1.4.4.1. Shall include method of construction/installation, Health and Safety management, testing methods, required relative humidity and temperature level and control systems, possible risk associated with the execution works and precautions to be considered, site control system and any other additional requirements as advised by the Engineer for the execution of works.

1.4.4.2. Method statement shall be approved by the Engineer prior to commencement of the works.

1.4.5. Installer Qualification:

1.4.5.1. Licensed installer specialized in gypsum plasterboard partition installations, and eligible to receive manufacturer's warranty.

1.4.6. Material Certificates:

1.4.6.1. Comprehensive tests performed by a qualified independent testing agency certifying that the product compiles with specified performance requirements.

1.4.7. Maintenance Data:

1.4.7.1. Details of manufacturer's recommended maintenance procedures for each type of product/material to be included in the operation and maintenance manual.

1.4.8. Spare Materials/Parts

1.4.8.1. Provide the following extra materials that match installed products, packaged for storage with suitable labels for identification.

1.5. Warranty

1.5.1. As per the Employer's requirements and general provision of contract, including General, Supplementary and Special Conditions of Contract Documents.

1.6. Performance Requirements

1.6.1. All finishes, materials and components shall comply with requirements of NFPA 101 Life Safety Code, chapter 10 – Interior Finish, Contents and Furnishings with respect to classification of materials, flame spread, smoke generation and other requirements.

1.7. Environment & Sustainability

1.7.1. Recycling Strategy document for the material or product shall be submitted to the Engineer for approval.

1.7.2. LEED Requirements:

1.7.2.1. All the specifications in this document are for USGBC LEED BD+C submission and are applicable to all materials & procedures followed for the project. The specification as stated here should be adhered to the word. In case of a dispute LEED guidelines shall be followed. Brochures, product specification sheets and/or certificates as specified in this document should be submitted along with the material procurement. All contractors, manufactures must submit documents, records and other submittal related to the different credits as mentioned in the specification.

1.7.2.2. Manufacturer product conforming to ISO 14044 or Environmental Product Declarations which conform to ISO 14025, 14040, 14044, and EN 15804 or ISO 21930 and have at least a cradle to gate scope. (If applicable)**.**

1.7.2.3. Manufacturer Material Ingredient Reporting by any of the listed below documents (If applicable):

1.7.2.3.1. Health Product Declaration. The end use product has a published, complete Health Product Declaration with full disclosure of known hazards in compliance with the Health Product Declaration Open Standard.

1.7.2.3.2. Cradle to Cradle Certificate. The end use product has been certified at the Cradle to Cradle v2 Basic level or Cradle to Cradle v3 Bronze level.

1.7.2.3.3. REACH Optimization. End use products and materials that do not contain substances that meet REACH criteria for substances of very high concern.

1.7.2.4. Manufacturer Raw Material Source and Extraction Reporting by any of the listed below activity (If applicable):

1.7.2.4.1. Third-party verified corporate sustainability reports (CSR) which include environmental impacts of extraction operations and activities associated with the manufacturer's product and the product's supply chain**.**

1.7.2.4.2. Self-declared reports from manufacturers that have publicly released a report from their raw material suppliers which include raw material supplier extraction locations.

1.8. Manufacturers

1.8.1. Manufacturer to be accredited and registered under BS EN ISO 9001.

1.8.2. List of preferred or acceptable Manufacturers attached herewith as Annexure 1.

2. Products

2.1. Materials

2.1.1. Material/Product Specification as per Standard attached herewith as Annexure A.07.

2.1.1.1. Toilet Cubicle Partition:

2.1.1.1.1. Frame Material: Brushed Aluminium/Steel

2.1.1.1.2. Panel Material: 18mm Thick Particle Board, Laminated & Square Edged.

2.1.1.1.3. Hardware Material: Brushed Aluminium/Steel

2.1.1.2. Urinal Cubicle Partition:

2.1.1.2.1. Frame Material: Brushed Aluminium/Steel

2.1.1.2.2. Panel Material: 18mm Thick Particle Board, Laminated & Square Edged.

2.1.1.2.3. Hardware Material: Brushed Aluminium/Steel

2.2. Colours

2.2.1. Colours as per Standard attached herewith as Annexure B.01.

2.2.1.1. Toilet Cubicle Partition:

2.2.1.1.1. Panel Colour: Light Grey, RAL 7035

2.2.1.2. Urinal Cubicle Partition:

2.2.1.2.1. Panel Colour: Light Grey, RAL 7035

3. Execution

3.1. Examination

3.1.1. Examine installation area for compliance with all manufacturer's project environmental requirements and ensure uninstalled products have been stored, handled and acclimatized properly prior to commencing installation. Inspect all substrates for completion and quality of work to ensure the surfaces are level, plumb, clean, dry and completely cured from water or solvent evaporation. Do not commence installation if the structural capacity of the substrate is questionable or inadequate.

3.2. Preparation

3.2.1. Review methods and procedures of Toilet cubicle & urinal partition installation, including manufacturer's written instructions. Method statement shall be approved by the Engineer prior to commencement of installation.

3.3. Fabrication

3.3.1. All components of Toilet cubicle & urinal partition system (including accessories) shall be factory finished to manufacturer's standard procedures and processes, as necessary to fulfil the performance requirements, stated in this specification.

3.4. Installation

3.4.1. General: Install composite Toilet cubicle & urinal partition systems according to manufacturer's written instructions in orientation, sizes, and locations as indicated on Contract Drawings or approved shop drawings. Anchor panels and other components of the Work securely in place, with provisions for thermal and structural movement in proprietary retaining channels or direct mounting to substructure, as directed by the Engineer.

4. Quality Control

4.1. Delivery, Storage and Handling

4.1.1. All building materials shall be stored on the building site in such a way as to prevent deterioration or the loss or impairment of their structural and other essential properties. Comply to National Building Code (NBC) of India 2016 Volume 1; Part 7 Construction Management, Practices and Safety.

4.2. Source Limitation

4.2.1. All materials & related products shall be from the same manufacturer to ensure material/product integrity & uniformity of colour, texture & quality.

4.3. Mock-ups

4.3.1. Build full height mock-ups for each type of Toilet cubicle & urinal partition including all accessories to verify selections made under sample submittals and to demonstrate aesthetic effects and set quality standards for materials and execution.

4.3.2. Approved mock-ups may be incorporated into the finished work, where directed by the Engineer.

5. Testing and Commissioning

5.1. Inspection

5.1.1. Inspection upon completion of work, all Toilet Cubicle & Urinal Partition shall be carefully tested and examined for faults in application, shall be corrected immediately.

ATK-ARC-008 – Ceiling

1. General

1.1. Introduction

1.1.1. This section specifies supply and installation of metal ceiling systems, as indicated in the Tender drawings and as required in completion of the construction works.

1.2. Related Documents

1.2.1. This section to be read in conjunction with Employer's requirements and general provision of contract, including General, Supplementary and Special Conditions of Tender Documents.

1.3. Codes and Standards

1.3.1. All materials, workmanship and testing shall be in accordance with the following documents, in their latest revisions. This is not exhaustive and contractor shall verify the listing and supplement with the latest applicable document or documents.

1.3.2. Comply with the requirements of the reference standards noted herein, except where more stringent requirements are listed herein or otherwise required by the Tender Documents.

1.3.3. Where materials, finishes, fittings, equipment and building components that comply with standards other than those included in these specifications are proposed, the Contractor shall adequately demonstrate that such items are of an equal quality and standard and are fit for their intended purpose, through submission of internationally recognized and authorized test methods, technical and certification reports (in compliance with assessment/evaluation criteria based on standards already included in these specifications), to the approval of engineer, prior to use of such items.

1.3.4. Conform to the requirements of the following Standards and Codes of Practice where applicable:

1.3.4.1. National Building Code (NBC) of India 2016 Volume 1 & 2.

1.3.4.2. IS Code 733: 1983 Specification for wrought aluminium and aluminium alloys bars, rods and sections for general engineering purposes (third revision)

1.3.4.3. IS Code 734: 1975 Wrought aluminium and aluminium alloys forging stock and forgings (for general engineering purposes) (second revision)

1.3.4.4. IS Code 736: 1986 Specification for wrought aluminium and aluminium alloys plate for general engineering purposes (third revision)

1.3.4.5. IS Code 737: 2008 Specification for wrought aluminium and aluminium alloys sheet and strip for general engineering purposes (fourth revision)

1.3.4.6. IS Code 738: 1994 Specification for wrought aluminium and aluminium alloys drawn tube for general engineering purposes (third revision)

1.3.4.7. IS Code 739: 1992 Specification for wrought aluminium and aluminium alloys - Wire for general engineering purposes (third revision)

1.3.4.8. IS Code 740: 1977 Specification for wrought aluminium and aluminium alloys rivet stock for general engineering purposes (second revision)

1.3.4.9. IS Code 1254: 2007 Corrugated aluminium sheet (fourth revision)

1.3.4.10. IS Code 1284: 1975 Wrought aluminium alloys bolt and screw stock for general engineering purposes (second revision)

1.3.4.11. IS Code 1285: 2002 Specification for wrought aluminium and aluminium alloys – Extruded round tube and hollow sections for general engineering purposes (third revision)

1.3.4.12. IS Code 2479: 1981 Colour code for the identification of aluminium and aluminium alloys for general engineering purposes (second revision)

1.3.4.13. IS Code 2676: 1981 Dimensions for wrought aluminium and aluminium alloy sheet and strip

1.3.4.14. IS Code 2677: 1979 Dimensions for wrought aluminium and aluminium alloys, plates and hot rolled sheets

1.3.4.15. IS Code 3577: 1992 Wrought aluminium and its alloys - Rivet, bolt and screw stock - Dimensions and tolerances (first revision)

1.3.4.16. IS Code 3965: 1981 Dimensions for wrought aluminium and aluminium alloys bar, rod and section (first revision)

1.3.4.17. IS Code 6477: 1983 Dimensions for wrought aluminium and aluminium alloys extruded hollow sections

1.3.4.18. IS Code 14712: 1999 Specification for wrought aluminium and its alloys - Chequered/tread sheets for general engineering purposes

1.3.4.19. IS Code 15965: 2012 Pre-painted aluminium zinc alloy metallic coated steel strip and sheet (Plain)

1.3.4.20. IS Code 1030: 1998 Carbon steel castings for general engineering purposes (fifth revision)

1.3.4.21. IS Code 1136: 2008 Preferred sizes for wrought metal products (third revision)

1.3.4.22. IS Code 1762 (Part 1) 1974: Code for designation of steels: Part 1 Based on letter symbols (first revision)

1.3.4.23. IS Code 2049: 1978 Colour code for the identification of wrought steel for general engineering purposes (first revision)

1.3.4.24. IS Code 2644: 1994 High tensile steel castings (fourth revision)

1.3.4.25. IS Code 2708: 1993 Specification for 1.5 percent manganese steel castings for general engineering purposes (third revision)

1.3.4.26. IS Code 7598: 1990 Classification of steels (first revision)

1.3.4.27. IS Code 10461 Resistance to inter-granular corrosion of austenitic stainless steels - Method for determination

1.3.4.27.1. (Part 1): 1994 Corrosion test in nitric acid medium by measurement of loss in mass (Huey test) (first revision)

1.3.4.27.2. (Part 2): 1994 Corrosion test in a sulphuric acid/ copper sulphate medium in the presence of copper turnings (Monypenny Strauss Test) (first revision)

1.3.4.28. IS Code 2062: 2011 Specification for hot rolled medium and high tensile structural steel (seventh revision)

1.3.4.29. IS Code 2830: 2012 Specification for carbon steel cast billet ingots, billets, blooms and slabs for re-rolling into steel for general structural purposes (third revision)

1.3.4.30. IS Code 2831: 2012 Specification for carbon steel cast billets ingots, blooms and slabs for re-rolling into structural steel (Ordinary quality) (fourth revision)

1.3.4.31. IS Code 8052: 2006 Specification for steel ingots, billets and blooms for the production of springs, rivets and screws for general engineering applications (second revision)

1.3.4.32. IS Code 8952: 1995 Steel ingots, blooms and billets for production of mild steel wire rods for general engineering purposes (first revision)

1.3.4.33. IS Code 11587: 1986 Specification for structural weather resistant steels

1.3.4.34. IS Code 15103: 2002 Fire resistant steel

1.3.4.35. IS Code 15911: 2010 Structural steel (Ordinary quality)

1.3.4.36. IS Code 15962: 2012 Structural steels for buildings and structures with improved seismic resistance

1.3.4.37. IS Code 277: 2003 Specification for galvanized steel sheets (plain and corrugated) (sixth revision)

1.3.4.38. IS Code 412: 1975 Specification for expanded metal steel sheets for general purposes (second revision)

1.3.4.39. IS Code 513: 2008 Specification for cold reduced low carbon steel sheets and strips (fifth revision)

1.3.4.40. IS Code 1079: 2009 Specification for hot rolled carbon steel sheet and strip (sixth revision)

1.3.4.41. IS Code 6911: 1992 Stainless steel plate, sheet and strip (first revision)

1.3.4.42. IS Code 7226: 1974 Specification for cold rolled medium, high carbon and low alloy steel strip for general engineering purposes

1.3.4.43. IS Code 14246: 2013 Specification for continuously pre- painted galvanized steel sheets and coils (first revision)

1.3.4.44. IS Code 15961: 2012 & IS Code 15965: 2012 Hot-dip Pre-painted aluminium zinc alloy metallic coated steel strip and sheet (plain)

1.3.4.45. IS Code 12313: 1988 Specification for hot-dip tern coated carbon steel sheets

1.3.4.46. IS Code 2095 Specification for gypsum plaster boards

1.3.4.46.1. (Part 1): 2011 Plain gypsum plaster boards (third revision)

1.3.4.46.2. (Part 2): 2001 Coated/laminated gypsum plaster boards (second revision)

1.3.4.46.3. (Part 3): 1996 Reinforced gypsum plaster boards (second revision)

1.3.4.47. IS Code 2542 Methods of test for gypsum plaster, concrete and products

1.3.4.47.1. (Part 1/Sec 1) 1978: Plaster and concrete, Section 1 Normal consistency of gypsum plaster (first revision)

1.3.4.47.2. (Part 1/Sec 2) 1978: Plaster and concrete, Section 2 Normal consistency of gypsum concrete (first revision)

1.3.4.47.3. (Part 1/Sec 3) 1978: Plaster and concrete, Section 3 Setting time of plaster and concrete (first revision)

1.3.4.47.4. (Part 1/Sec 4) 1978: Plaster and concrete, Section 4 Transverse strength of gypsum plaster (first revision)

1.3.4.47.5. (Part 1/Sec 5) 1978: Plaster and concrete, Section 5 Compressive strength and dry set density of gypsum plaster (first revision)

1.3.4.47.6. (Part 1/Sec 6) 1978: Plaster and concrete, Section 6 Soundness of gypsum plaster (first revision)

1.3.4.47.7. (Part 1/Sec 7) 1978: Plaster and concrete, Section 7 Mechanical resistance of gypsum plaster by dropping ball test (first revision)

1.3.4.47.8. (Part 1/Sec 8) 1978: Plaster and concrete, Section 8 Freedom from coarse particles (first revision)

1.3.4.47.9. (Part 1/Sec 9) 1978: Plaster and concrete, Section 9 Expansion of plaster (first revision)

1.3.4.47.10. (Part 1/Sec 10) 1978: Plaster and concrete, Section 10 Sand in set plaster (first revision)

1.3.4.47.11. (Part 1/Sec 11) 1978: Plaster and concrete, Section 11 Wood fibre content in gypsum plaster (first revision)

1.3.4.47.12. (Part 1/Sec 12) 1978: Plaster and concrete, Section 12 Dry bulk density (first revision)

1.3.4.47.13. (Part 2/Sec 1) 1981: Gypsum products, Section 1 Measurement of dimensions (first revision)

1.3.4.47.14. (Part 2/Sec 2) 1981: Gypsum products, Section 2 Determination of mass (first revision)

1.3.4.47.15. (Part 2/Sec 3) 1981: Gypsum products, Section 3 Determination of mass and thickness of paper surfacing (first revision)

1.3.4.47.16. (Part 2/Sec 4) 1981: Gypsum products, Section 4 Transverse strength (first revision)

1.3.4.47.17. (Part 2/Sec 5) 1981: Gypsum products, Section 5 Compressive strength (first revision)

1.3.4.47.18. (Part 2/Sec 6) 1981: Gypsum products, Section 6 Water absorption (first revision)

1.3.4.47.19. (Part 2/Sec 7) 1981: Gypsum products, Section 7 Moisture content (first revision)

1.3.4.47.20. (Part 2/Sec 8) 1981: Gypsum products, Section 8 Nail retention of precast reinforced gypsum slabs (first revision)

1.3.4.48. IS Code 2547 Specification for gypsum building plaster

1.3.4.48.1. (Part 1): 1976 Excluding premixed lightweight plaster (first revision)

1.3.4.48.2. (Part 2): 1976 Premixed lightweight plaster (first revision)

1.3.4.49. IS Code 2849: 1983 Specification for non-load bearing gypsum partition blocks (solid and hollow types) (first revision)

1.3.4.50. IS Code 8272: 1984 Specification for gypsum plaster for use in the manufacture of fibrous plasterboards (first revision)

1.3.4.51. IS Code 9498: 1980 Specification for inorganic aggregates for use in gypsum plaster

1.3.4.52. IS Code 12679: 1989 Specification for by-product gypsum for use in plaster, blocks and boards

1.3.4.53. NFPA 101 Life Safety Code.

1.4. Submittals

1.4.1. Technical Data

1.4.1.1. Comprehensive technical data viz., technical literature, data sheets, product catalogue & brochure, installation instructions, etc. to be provided for Engineer's approval.

1.4.2. Shop drawings

1.4.2.1. Shop drawings shall be submitted to Engineer for approval, prior to start of work at site.

1.4.2.2. Shop drawings shall include all details w.r.t. the material & related product's installation, to the scale & detailing as instructed by the Engineer.

1.4.2.3. Shop drawings shall be developed as per actual measurements from the site & must depict the same in the drawings.

1.4.2.4. Shop drawings must comply to BS 1192:2007+A2:2016 Collaborative production of architectural, engineering and construction information & Code of practice.

1.4.3. Samples

1.4.3.1. Samples for Initial Selection: For each type of ceiling as indicated. Include Samples of accessories involving colour selection.

1.4.4. Method Statement

1.4.4.1. Shall include method of construction/installation, Health and Safety management, testing methods, required relative humidity and temperature level and control systems, possible risk associated with the execution works and precautions to be considered, site control system and any other additional requirements as advised by the Engineer for the execution of works.

1.4.4.2. Method statement shall be approved by the Engineer prior to commencement of the works.

1.4.5. Installer Qualifications

1.4.5.1. Licensed installer specialized in fabricating and installing each type of ceiling, eligible to receive manufacturer's warranty.

1.4.6. Material/Product Certificates

1.4.6.1. Comprehensive tests performed by a qualified independent testing agency certifying that the product compiles with specified performance requirements.

1.4.7. Maintenance Data

1.4.7.1. Details of manufacturer's recommended maintenance procedures for each type of product/material to be included in the operation and maintenance manual.

1.4.8. Spare Materials/Parts

1.4.8.1. Provide the following extra materials that match installed products, packaged for storage with suitable labels for identification.

1.4.8.1.1. 2% or 50m^2 minimum for each type of metal ceiling (product, pattern or colour).

1.5. Warranty / Guarantee

1.5.1. As per the Employer's requirements and general provision of contract, including General, Supplementary and Special Conditions of Contract Documents.

1.6. Performance Requirements

1.6.1. All finishes, materials and components shall comply with requirements of NFPA 101 Life Safety Code, chapter 10 – Interior Finish, Contents and Furnishings with respect to classification of materials, flame spread, smoke generation and other requirements.

1.7. Environment & Sustainability

1.7.1. Recycling Strategy document for the material or product shall be submitted to the Engineer for approval.

1.7.2. LEED Requirements:

1.7.2.1. All the specifications in this document are for USGBC LEED BD+C submission and are applicable to all materials & procedures followed for the project. The specification as stated here should be adhered to the word. In case of a dispute LEED guidelines shall be followed. Brochures, product specification sheets and/or certificates as specified in this document should be submitted along with the material procurement. All contractors, manufactures must submit documents, records and other submittal related to the different credits as mentioned in the specification.

1.7.2.2. Manufacturer product conforming to ISO 14044 or Environmental Product Declarations which conform to ISO 14025, 14040, 14044, and EN 15804 or ISO 21930 and have at least a cradle to gate scope. (If applicable)**.**

1.7.2.3. Manufacturer Material Ingredient Reporting by any of the listed below documents (If applicable):

1.7.2.3.1. Health Product Declaration. The end use product has a published, complete Health Product Declaration with full disclosure of known hazards in compliance with the Health Product Declaration Open Standard.

1.7.2.3.2. Cradle to Cradle Certificate. The end use product has been certified at the Cradle to Cradle v2 Basic level or Cradle to Cradle v3 Bronze level.

1.7.2.3.3. REACH Optimization. End use products and materials that do not contain substances that meet REACH criteria for substances of very high concern.

1.7.2.4. Manufacturer Raw Material Source and Extraction Reporting by any of the listed below activity (If applicable):

1.7.2.4.1. Third-party verified corporate sustainability reports (CSR) which include environmental impacts of extraction operations and activities associated with the manufacturer's product and the product's supply chain**.**

1.7.2.5. Self-declared reports from manufacturers that have publicly released a report from their raw material suppliers which include raw material supplier extraction locations

1.8. Manufacturers

1.8.1. Manufacturer to be accredited and registered under BS EN ISO 9001.

1.8.2. List of preferred or acceptable Manufacturers attached herewith as Annexure 1.

2. Products

2.1. Materials

2.1.1. Material/Product Specification as per Standard attached herewith as Annexure A.08.

2.1.1.1. Ceiling Type:

2.1.1.1.1. Suspended Ceiling - Acoustic

2.1.1.1.2. Suspended Ceiling – Hygienic

2.1.1.1.3. Suspended Ceiling – Fire to BS EN 520 Type F

2.1.1.2. Ceiling Tile Material: Metal

2.2. Colours

2.2.1. Colours as per Standard attached herewith as Annexure B.01.

2.2.2. In Smaland area, Colour of Ceiling shall be Dark Blue NCS S 7020-R90.

2.2.3. In other areas, Colour of Ceiling shall be White NCS S 0500-N.

3. Execution

3.1. Preparation

3.1.1. Review methods and procedures of Ceiling installation, including manufacturer's written instructions. Method statement shall be approved by the Engineer prior to commencement of installation.

3.2. Installation

3.2.1. Install the proprietary concealed ceiling systems complete and securely fix all components all in accordance with the manufacturer's written instructions.

3.3. Cleaning and Protection

3.3.1. Ensure the completed ceilings are thoroughly clean and to the satisfaction of the Engineer before handover. Comply with manufacturer’s written instructions for cleaning and touch-up of minor finish damage. Remove and replace tiles and other ceiling components that cannot be successfully cleaned and repaired to permanently eliminate evidence of damage.

4. Quality Control

4.1. Delivery, Storage and Handling

4.1.1. All building materials shall be stored on the building site in such a way as to prevent deterioration or the loss or impairment of their structural and other essential properties. Comply to National Building Code (NBC) of India 2016 Volume 1; Part 7 Construction Management, Practices and Safety.

4.2. Source Limitation

4.2.1. All materials & related products shall be from the same manufacturer to ensure material/product integrity & uniformity of colour, texture & quality.

4.3. Mock-ups

4.3.1. Build mock-ups of each ceiling type to verify selections made under sample submittals and to demonstrate aesthetic effects and set quality standards for materials and execution to the Engineer's satisfaction. Approved mock-ups may be incorporated into the finished work, where directed by the Engineer.

5. Testing and Commissioning

5.1. Inspection

5.1.1. Inspection upon completion of work, all Ceiling Works shall be carefully tested and examined for faults in application, shall be corrected immediately.

ATK-ARC-009 – Steel Doors & Louvers

1. General

1.1. Introduction

1.1.1. This section specifies supply and installation of proprietary Hollow Steel Doors, Steel Louvers, Fire Rated Doors & Doors Assemblies, Acoustic Rated Doors & Door Assemblies, as shown on the Tender drawings & documents required in completion of the work.

1.2. Related documents

1.2.1. This section to be read in conjunction with Employer's requirements and general provision of contract, including General, Supplementary and Special Conditions of Contract Documents

1.3. Codes and Standards

1.3.1. All materials, workmanship and testing shall be in accordance with the following standards & codes, in their latest revisions. This is not exhaustive and contractor shall verify the listing and supplement with the latest applicable standards & codes.

1.3.2. Comply with the requirements of the reference standards noted herein, except where more stringent requirements are listed herein or otherwise required by the Contract Documents.

1.3.3. Where materials, finishes, fittings, equipment and building components that comply with standards other than those included in these specifications are proposed, the Contractor shall adequately demonstrate that such items are of an equal quality and standard and are fit for their intended purpose, through submission of globally recognized and authorized test methods, technical and certification reports (in compliance with assessment or evaluation criteria based on standards already included in these specifications), to the approval of engineer, prior to use of such items.

1.3.4. Conform to the requirements of the following Standards and Codes of Practice where applicable.

1.3.4.1. National Building Code (NBC) of India 2016 Volume 1 & 2.

1.3.4.2. IS Code 1038: 1983 Specification for steel doors, windows and ventilators (third revision)

1.3.4.3. IS Code 1361: 1978 Specification for steel windows for industrial buildings (first revision)

1.3.4.4. IS Code 1948: 1961 Specification for aluminium doors, windows and ventilators

1.3.4.5. IS Code 1949: 1961 Specification for aluminium windows for industrial buildings

1.3.4.6. IS Code 4351: 2003 Specification for steel door frames (second revision)

1.3.4.7. IS Code 6248: 1979 Specification for metal rolling shutters and rolling grills (first revision)

1.3.4.8. IS Code 7452: 1990 Specification for hot rolled steel sections for doors, windows and ventilators (second revision)

1.3.4.9. IS Code 10451: 1983 Specification for steel sliding shutters (top hung type)

1.3.4.10. IS Code 10521: 1983 Specification for collapsible gates

1.3.4.11. IS Code 14856: 2000 Specification for glass fibre reinforced (GRP) panel type door shutters for internal use

1.3.4.12. IS Code 15380: 2003 Specification for moulded raised high density fibre (HDF) panel doors

1.3.4.13. IS Code 15931: 2012 Specification for solid panel foam UPVC door shutters

1.3.4.14. 3614 Specification for fire-check doors

1.3.4.14.1. (Part 1): 1966 Plate, metal covered and rolling type

1.3.4.14.2. (Part 2): 1992 Metallic and non-metallic fire check doors - Resistance test and performance criteria

1.3.4.15. IS Code 1568: 1970 Specification for wire cloth for general purposes (first revision)

1.3.4.16. IS Code 3150: 1982 Specification for hexagonal wire netting for general purposes (second revision)

1.3.4.17. IS Code 11199: 1985 Specification for HDPE mono- filament twine door nets

1.3.4.18. IS Code 2046: 1995 Specification for decorative thermosetting synthetic resin bonded laminated sheets (second revision)

1.3.4.19. BS EN 438-1:2016. High-pressure decorative laminates (HPL). Sheets based on thermosetting resins (usually called laminates). Introduction and general information

1.3.4.20. BS EN 438-2:2016. High-pressure decorative laminates (HPL). Sheets based on thermosetting resins (usually called laminates). Determination of properties

1.3.4.21. BS EN 438-3:2016. High-pressure decorative laminates (HPL). Sheets based on thermosetting resins (usually called laminates). Classification and specifications for laminates less than 2 mm thick intended for bonding to supporting substrates

1.3.4.22. BS EN 438-6:2016. High-pressure decorative laminates (HPL). Sheets based on thermosetting resins (usually called laminates). Classification and specifications for Exterior-grade compact laminates of thickness 2 mm and greater

1.3.4.23. BS EN 438-5:2016. High-pressure decorative laminates (HPL). Sheets based on thermosetting resins (usually called laminates). Classification and specifications for flooring grade laminates less than 2 mm thick intended for bonding to supporting substrates

1.3.4.24. BS EN 438-4:2016. High-pressure decorative laminates (HPL). Sheets based on thermosetting resins (usually called laminates). Classification and specifications for compact laminates of thickness 2 mm and greater

1.3.4.25. ISO 4586-7:2015. High-pressure decorative laminates (HPL, HPDL). Sheets based on thermosetting resins (Usually called Laminates). Classification and specifications for design laminates

1.3.4.26. ISO 4586-8:2015. High-pressure decorative laminates (HPL, HPDL). Sheets based on thermosetting resins (Usually called Laminates). Classification and specifications for alternative core laminates

1.3.4.27. ISO 4586-2:2015. High-pressure decorative laminates (HPL, HPDL). Sheets based on thermosetting resins (Usually called Laminates). Determination of properties

1.3.4.28. ISO 4586-1:2015. High-pressure decorative laminates (HPL, HPDL). Sheets based on thermosetting resins (Usually called Laminates). Introduction and general information

1.3.4.29. ISO 4586-6:2015. High-pressure decorative laminates (HPL, HPDL). Sheets based on thermosetting resins (Usually called Laminates). Classification and specifications for exterior-grade compact laminates of thickness 2 mm and greater

1.3.4.30. ISO 4586-3:2015. High-pressure decorative laminates (HPL, HPDL). Sheets based on thermosetting resins (Usually called Laminates). Classification and specifications for laminates less than 2 mm thick and intended for bonding to supporting substrates

1.3.4.31. ISO 4586-4:2015. High-pressure decorative laminates (HPL, HPDL). Sheets based on thermosetting resins (Usually called Laminates). Classification and specifications for compact laminates of thickness 2 mm and greater

1.3.4.32. ISO 4586-5:2015. High-pressure decorative laminates (HPL, HPDL). Sheets based on thermosetting resins (Usually called Laminates). Classification and specifications for flooring grade laminates less than 2 mm thick intended for bonding to supporting substrates

1.3.4.33. ASTM D 1187 Standard Specification for Asphalt-Base Emulsions for Use as Protective Coatings for Metal

1.3.4.34. ASTM E90 Acoustic Testing for Door Sets

1.3.4.35. BS 4 Structural Steel Sections

1.3.4.36. BS 476 Fire tests on building materials and structures

1.3.4.37. BS 1245 Metal Door Frames (Steel)

1.3.4.38. BS 3987 Specification for anodic oxidation coatings on wrought aluminium for external architectural applications + AMD 10944

1.3.4.39. BS 5234 Slam test for steel door set.

1.3.4.40. BS 5588 Part 0 to 11 Fire precaution in the design construction and use of building.

1.3.4.41. BS 5950 Structural use of steelwork in building. Code of practice for design. Rolled and welded sections + AMD 13199

1.3.4.42. BS 6496 Specification for powder organic coatings for application and stoving to aluminium alloy extrusions, sheet and preformed sections for external architectural purposes, and for the finish on aluminium alloy extrusions, sheet and preformed sections coated with powder organic coatings + AMD 7182

1.3.4.43. BS 6510 Steel-framed windows and glazed doors

1.3.4.44. BS 6949 Bitumen-Based Coatings for Cold Application, Excluding Use in Contact with Potable Water-AMD 7287:

1.3.4.45. BS 7371 Part 1-11 Coatings on Metal Fasteners

1.3.4.46. BS 8300 Glass Apertures

1.3.4.47. BS EN 1026 Windows and Doors Air Permeability Test Methods

1.3.4.48. BS EN 1027 Windows and Doors Water Tightness Test Methods

1.3.4.49. BS EN 1192 Doors. Classification of strength requirements.

1.3.4.50. BS EN 1993 Euro code 3. Design of steel structures. General rules and rules for buildings

1.3.4.51. BS EN 10025 Hot rolled products of structural steels

1.3.4.52. BS EN 10130 Cold rolled low carbon steel flat products for cold forming Technical delivery conditions

1.3.4.53. BS EN 10131 Cold rolled uncoated and zinc or zinc-nickel electrolytically coated low carbon and high yield strength steel flat products for cold forming Tolerances on dimensions and shape

1.3.4.54. BS EN 10143 Continuously hot-dip coated steel sheet and strip Tolerances on dimensions and shape

1.3.4.55. BS EN 10162 Cold rolled steel sections - Technical delivery conditions - Dimensional and cross-sectional tolerances- Supersedes BS 2994: 1976;

1.3.4.56. BS EN 10326 Continuously hot-dip coated strip and sheet of structural steels Technical delivery conditions-Supersedes BS EN 10147:2000, BS EN 10154:2002, BS EN 10214:1995 and BS EN 10215:1995

1.3.4.57. BS EN 10327 Continuously hot-dip coated strip and sheet of low carbon steels for cold forming Technical delivery conditions- Supersedes BS EN 10142:2000, and together with BS EN 10326:2006, supersedes BS EN 10154:2004:1995 and BS EN

1.3.4.58. 10215:1995

1.3.4.59. BS EN 12114 Thermal Performance of Buildings – Air permeability of building components and building elements. Laboratory test methods.

1.3.4.60. BS EN 12211 Resistant to wind load , Test Methods dd) BS EN 12217 Doors. Operating forces requirements & classification.

1.3.4.61. BS EN 12664 Thermal performance of building material and products.

1.3.4.62. BS EN 12667 Thermal performance of building material and products.

1.3.4.63. BS EN 13162 Thermal insulation products for buildings factory made mineral wool.

1.3.4.64. BS EN 13187 Thermal Performance of Buildings – Qualitative detection of thermal properties in building envelopes – Infrared method.

1.3.4.65. BS EN 13829 Thermal Performance of Buildings – Determination of air permeability of buildings – Fan pressurization method.

1.3.4.66. BS EN ISO 1461 Hot Dip Galvanized Coatings on Fabricated Iron and Steel Articles - Specifications and Test Methods.

1.3.4.67. BS EN ISO 4042 Fasteners - Electroplated Coatings.

1.3.4.68. BS EN ISO 8501, BS 7079-A1 Preparation of steel substrates before application of paints and related products. Visual assessment of surface cleanliness. Rust grades and preparation grades of uncoated steel substrates and of steel substrates after overall removal of previous coatings.

1.3.4.69. BS EN ISO 12412-2 Determination of Thermal Transmittance by hot box method – Part 2 – Frames.

1.3.4.70. BS EN ISO 12567-1 Thermal Performance of Windows, Doors and Shutters – Determination of Thermal Transmittance by hot box method – Part 1 – Windows and Doors.

1.3.4.71. BS EN ISO 12944 (partially replaces BS 5493) Paints and varnishes. Corrosion protection of steel structures by protective paint systems. Types of surface and surface preparation

1.3.4.72. Laminate tested to ASTM D256 Izod impact test.

1.3.4.73. ASTM D790 Flexural Strength3.ASTM D638 Tensile strength.

1.3.4.74. ASTM D543 Evaluation of Plastics to chemical reagents.

1.3.4.75. ASTM D570 Water absorption, ASTM D1308 effect of chemicals.

1.3.4.76. Acoustic doors tested to BS EN ISO 140-3.

1.3.4.77. Hinges certified to BS EN 1935 which include testing for Static load, shear, Endurance, Corrosion, Fire door and Burglary. ANSI 157-1 & 7, UL10C & cUL10C.

1.3.4.78. Locks, cylinder, latch and Lever certified to SS 332:2007 durability testing to 200’000 test cycles to BS EN 12209.

1.3.4.79. Door Closers tested to EN1154 & BS 476 part 22, production surveillance ISO9002 &CERTIFIRE and SS 332:2007 durability testing 500’000 test cycles.

1.3.4.80. PolyIsoCyanurate CFC/HCFC free rigid foam insulation has been tested to ASTM E 84 and has a flame spread index of under 35 AAMA 1503-09 Thermal Performance.

1.3.4.81. Resin is tested to Class 1 fire rating BS476 Part 7 & part 6 and M1 rating also French M1rating to NFP 92-501 fire testing. Tensile strength / Modulus and Elongation are tested to BS EN ISO 527-4:1997, Flexible strength / modulus is tested to BS EN ISO 14125:1998: Notched Izod Impact is tested to BS EN ISO 180:2001. Heat deflection temperature to BSEN ISO 75 and 306. Compression strength and Modulus BS EN ISO 604. BARCOL - ASTMD-2583.

1.3.4.82. Fire rated doors and frame construction conforms to fire standards of American UL10C,British BS476:22, European EN1634-1

1.3.4.83. NFPA 80 Standard for Fire Doors and Other Opening Protective

1.3.4.84. NFPA 101 Life Safety Code

1.3.4.85. NFPA 251 Standard Methods of Tests of Fire Resistance of Building Construction and Materials-2006 Edition

1.3.4.86. NFPA 257 Standard on Fire Test for Window and Glass Block Assemblies-2007 Edition

1.4. Submittals

1.4.1. Technical data

1.4.1.1. Comprehensive technical data viz., technical literature, data sheets, product catalogue & brochure, installation instructions, etc. to be provided for Engineer’s approval.

1.4.2. Samples

1.4.2.1. Samples for Initial Selection

1.4.2.1.1. Provide samples for initial selection of units with factory-applied colour finishes.

1.4.2.1.2. For the following items, prepared on Samples about [300 by 300 mm] to demonstrate compliance with requirements for quality of materials and construction:

1.4.2.1.3. Doors: Show vertical-edge, top, and bottom construction; core construction; and hinge and other applied hardware reinforcement. Include separate section showing glazing if applicable.

1.4.2.1.4. Frames: Show profile, corner joint, floor and wall anchors, and silencers. Include separate section showing fixed hollow metal panels and glazing if applicable.

1.4.3. Shop Drawings

1.4.3.1. Shop drawings shall be submitted to Engineer for approval, prior to start of work at site.

1.4.3.2. Shop drawings shall include all details w.r.t. the material & related product's installation, to the scale & detailing as instructed by the Engineer.

1.4.3.3. Shop drawings shall be developed as per actual measurements from the site & must depict the same in the drawings.

1.4.4. Method Statement

1.4.4.1. Shall include method of construction/installation, Health and Safety management, testing methods, required relative humidity and temperature level and control systems, possible risk associated with the execution works and precautions to be considered, site control system and any other additional requirements as advised by the Engineer for the execution of works.

1.4.4.2. Method statement shall be approved by the Engineer prior to commencement of the works.

1.4.5. Installer Qualifications:

1.4.6. Licensed installer specialized in painting shall be employed for the work.

1.4.7. Material Certificates

1.4.7.1. Comprehensive tests performed by a qualified independent testing agency certifying that the product compiles with specified performance requirements.

1.4.7.2. Fire Door Certificates: From an approved manufacturer that is certified for chain of custody recognized certification body, or independent testing laboratory approved by the engineer. Indicate in the certification, applicable fire test rating for the door constructions.

1.4.8. Maintenance Data

1.4.8.1. Details of manufacturer's recommended maintenance procedures for each type of product/material to be included in the operation and maintenance manual.

1.4.9. Spare Materials/Parts

1.4.9.1. Provide the following extra materials that match installed products, packaged for storage with suitable labels for identification.

1.5. Warranty / Guarantee

1.5.1. As per the Employer's requirements and general provision of contract, including General, Supplementary and Special Conditions of Contract Documents.

1.6. Performance Requirements

1.6.1. All finishes, materials and components shall comply with requirements of NFPA 101 Life Safety Code, chapter 10 – Interior Finish, Contents and Furnishings with respect to classification of materials, flame spread, smoke generation and other requirements.

1.6.2. All Hollow Metal Doors, Louvers and Frames shall be heavy-duty to BS EN 1192 and BS EN 12217.

1.7. Manufacturers

1.7.1. Manufacturer to be accredited and registered under BS EN ISO 9001.

1.7.2. List of preferred or acceptable Manufacturers attached herewith as Annexure 1.

2. Products

2.1. Materials

2.1.1. Material/Product Specification as per Standard attached herewith as Annexure A.09.

2.1.2. Proprietary Hollow Metal Frames

2.1.2.1. General: Fabricated from Cold rolled steel with a single/double rebate to suit the wall thickness, performance parameters & door leaf.

2.1.2.2. Frame Material:

2.1.2.2.1. External: Minimum 1mm Galvanized Steel with 60-micron polyester powder coated finish.

2.1.2.2.2. Internal: Minimum 1mm Galvanized Steel with 60-micron polyester powder coated finish.

2.1.2.2.3. Kitchen/IFS Areas: Minimum 1mm Stainless Steel (Brushed Finish).

2.1.3. Proprietary Steel Leaf

2.1.3.1. Face Material:

2.1.3.1.1. External: Minimum 1mm Galvanized Steel with 60-micron polyester powder coated finish.

2.1.3.1.2. Internal: Minimum 1mm Galvanized Steel with 60-micron polyester powder coated finish.

2.1.3.2. Core Material:

2.1.3.2.1. External/Fire Rated/Acoustic Fire Rated: Mineral wool or equivalent infill to achieve minimum performance parameters as per drawings/schedules.

2.1.3.2.2. Internal/Non-Fire Rated/Non-Acoustic Rated: Honeycomb Infill.

2.1.3.3. Leaf Thickness: Minimum 40mm.

2.1.3.3.1. Vision Panel: Clear Safety Glass (Internal/Non-Fire Rated); Clear Toughened Glass (External/Fire Rated)

2.1.4. Proprietary Solid Particle Board Leaf

2.1.4.1. Face Material: 0.8mm minimum thick high-pressure laminate.

2.1.4.1.1. Core Material: Solid Particle Board. Low formaldehyde content.

2.1.4.1.2. Edge Material: Hardwood edges or equivalent, fixed on bottom and leading edge as 8mm minimum.

2.1.4.1.3. Vision Panel: Clear Safety Glass (Internal/Non-Fire Rated); Clear Toughened Glass (External/Fire Rated)

2.1.5. Proprietary Glass Reinforced Polyester Leaf (Hygienic Doors for Kitchen)

2.1.5.1. Face Material: Minimum 2.0mm thick Moulded Fiberglass Composite Laminate.

2.1.5.1.1. Core Material: PolyIsoCyanurate infill for non-fire rated doors & mineral wool or equivalent for fire rated doors.

2.1.5.1.2. Leaf Thickness: Minimum 40mm.

2.1.5.1.3. Vision Panel: Clear Safety Glass (Internal/Non-Fire Rated); Clear Toughened Glass (External/Fire Rated)

2.1.6. Metal Louvers

2.1.6.1. Louver: Z-shaped blades or Inverted V- shaped or Y-shaped blades.

2.1.6.2. Mesh/Screen: Louver shall have Stainless Steel Insect Mesh, to Engineer's Approval.

2.1.7. Accessories (but not limited to):

2.1.7.1. Mullions and Transom Bars (for support/strengthening)

2.1.7.2. Grout Guards

2.1.7.3. Ironmongery

2.1.7.4. Inserts, Bolts and Fasteners

2.1.7.5. Grout

2.1.7.6. Smoke & Acoustic Seals

2.2. Colours

2.2.1. Colours as per Standard attached herewith as Annexure B.01.

2.2.2. As per National Building Code (NBC) of India 2016; Volume 1 Part 4 Annex-H, H3 (g), All fire exit doors from the car parking to exits shall be painted green and shall display exit signage.

3. Execution

3.1. Examination

3.1.1. Examine substrates, areas, and conditions, with Installer present, for compliance with requirements for installation tolerances and other conditions affecting performance of the Work.

3.2. Fabrications

3.2.1. Fabricate hollow metal work to be rigid seamless and free of defects, warp, or buckle. Accurately form metal to required sizes and profiles, with minimum radius for thickness of metal. Where practical, fit and assemble units in manufacturer's plant. To ensure proper assembly at Project site, clearly identify work that cannot be permanently factory assembled before shipment.

3.3. Installation

3.3.1. General: Install hollow metal work plumb, rigid, properly aligned, and securely fastened in place; comply with Drawings and manufacturer's written instructions. Prior to installation, adjust and securely brace welded hollow metal frames for squareness twist, and plumbness in proper adjustment with surrounding construction. Unless otherwise indicated, set face of frames finish with wall or partition finish on hinge side, to allow full 180º swing of doors.

3.3.2. A gap of 20mm shall be maintained for Door Leaf bottom & Finish Floor Level.

4. Quality Control

4.1. Delivery, Storage and Handling

4.1.1. All building materials shall be stored on the building site in such a way as to prevent deterioration or the loss or impairment of their structural and other essential properties. Comply to National Building Code (NBC) of India 2016 Volume 1; Part 7 Construction Management, Practices and Safety.

4.2. Source Limitation

4.2.1. All materials & related products shall be from the same manufacturer to ensure material/product integrity & uniformity of colour, texture & quality.

4.3. Mock-ups

4.3.1. Build mock-ups of each door & louver type to verify selections made under sample submittals and to demonstrate aesthetic effects and set quality standards for materials and execution to the Engineer's satisfaction. Approved mock-ups may be incorporated into the finished work, where directed by the Engineer.

5. Testing and Commissioning

5.1. Inspection

5.1.1. Inspection upon completion of work, all doors & louvers shall be carefully tested and examined for faults in application, shall be corrected immediately.

ATK-ARC-010 – Ironmongery

1. General

1.1. Introduction

1.1.1. This Section specifies the supply and installation of all door hardware and accessories as required for internal, external doors and the like as shown in the Tender drawings/schedules, as required in completion of the work.

1.1.2. Ironmongery:

1.1.2.1. Hinges

1.1.2.2. Door Closures

1.1.2.3. Lockset & Cylinder

1.1.2.4. Lever Handle

1.1.2.5. Pull Handle

1.1.2.6. Panic Device

1.1.2.7. Panic Trim

1.1.2.8. Door Coordinator

1.1.2.9. Push Plate

1.1.2.10. Flush Bolt

1.1.2.11. Dust Proof Strike

1.1.2.12. Wall Stop

1.1.2.13. Floor Stop

1.1.2.14. Mop Plate

1.1.2.15. Kick Plate

1.1.2.16. Sign Plate

1.1.2.17. Fire Seal

1.1.2.18. Acoustic Seal

1.1.2.19. Astragal Seal

1.1.2.20. Head & Jamb Seal

1.1.2.21. Thresholds

1.1.2.22. Sliding Door hardware

1.1.2.23. Disabled Toilet Door Hardware

1.2. Related Documents

1.2.1. This section to be read in conjunction with Employer's requirements and general provision of contract, including General, Supplementary and Special Conditions of Contract Documents

1.3. Codes and Standards

1.3.1. All materials, workmanship and testing shall be in accordance with the following standards & codes, in their latest revisions. This is not exhaustive and contractor shall verify the listing and supplement with the latest applicable standards & codes.

1.3.2. Comply with the requirements of the reference standards noted herein, except where more stringent requirements are listed herein or otherwise required by the Contract Documents.

1.3.3. Where materials, finishes, fittings, equipment and building components that comply with standards other than those included in these specifications are proposed, the Contractor shall adequately demonstrate that such items are of an equal quality and standard and are fit for their intended purpose, through submission of globally recognized and authorized test methods, technical and certification reports (in compliance with assessment or evaluation criteria based on standards already included in these specifications), to the approval of engineer, prior to use of such items.

1.3.4. Conform to the requirements of the following Standards and Codes of Practice where applicable:

1.3.4.1. National Building Code (NBC) of India 2016 Volume 1 & 2.

1.3.4.2. IS Code 204 Specification for tower bolts

1.3.4.2.1. (Part 1): 1991 Ferrous metals (fifth revision)

1.3.4.2.2. (Part 2): 1992 Non-ferrous metals (fifth revision)

1.3.4.3. IS Code 205: 1992 Specification for non-ferrous metal butt hinges (fourth revision)

1.3.4.4. IS Code 206: 2010 Specification for tee and strap hinges (fifth revision)

1.3.4.5. IS Code 208: 1996 Specification for door handles (fifth revision)

1.3.4.6. IS Code 281: 2009 Specification for mild steel sliding door bolts for use with padlocks (fourth revision)

1.3.4.7. IS Code 362: 1991 Specification for parliament hinges (fifth revision)

1.3.4.8. IS Code 363: 1993 Specification for hasps and staples (fourth revision)

1.3.4.9. IS Code 364: 1993 Specification for fanlight catch (third revision)

1.3.4.10. IS Code 452: 1973 Specification for door springs, rat-tail type (second revision)

1.3.4.11. IS Code 453: 1993 Specification for double-acting spring hinges (third revision)

1.3.4.12. IS Code 729: 1979 Specification for drawer locks, cupboard locks and box locks (third revision)

1.3.4.13. IS Code 1019: 1974 Specification for rim latches (second revision)

1.3.4.14. IS Code 1341: 1992 Specification for steel butt hinges (sixth revision)

1.3.4.15. IS Code 1823: 1980 Specification for floor door stoppers (third revision)

1.3.4.16. IS Code 1837: 1966 Specification for fanlight pivots (first revision)

1.3.4.17. IS Code 2209: 1976 Specification for mortice locks (vertical type) (third revision)

1.3.4.18. IS Code 2681: 1993 Specification for nonferrous metal sliding door bolts for use with padlocks (third revision)

1.3.4.19. IS Code 3564: 1995 Specification for door closers (hydraulically regulated) (second revision)

1.3.4.20. IS Code 3818: 1992 Specification for continuous (piano) hinges (third revision)

1.3.4.21. IS Code 3828: 1966 Specification for ventilator chains

1.3.4.22. IS Code 3843: 1995 Specification for steel back flap hinges (first revision)

1.3.4.23. IS Code 3847: 1992 Specification for mortice night latches (second revision)

1.3.4.24. IS Code 4621: 1975 Specification for indicating bolts for use in public baths and lavatories (first revision)

1.3.4.25. IS Code 4948: 2002 Specification for welded steel wire fabric for general use (second revision)

1.3.4.26. IS Code 4992: 1975 Specification for door handles for mortice locks (vertical type) (first revision)

1.3.4.27. IS Code 5187: 1972 Specification for flush bolts (first revision)

1.3.4.28. IS Code 5899: 1970 Specification for bathroom latches

1.3.4.29. IS Code 5930: 1970 Specification for mortice latch (vertical type)

1.3.4.30. IS Code 6315: 1992 Specification for floor springs (hydraulically regulated) for heavy doors (second revision)

1.3.4.31. IS Code 6318: 1971 Specification for plastic window stays and fasteners

1.3.4.32. IS Code 6343: 1982 Specification for door closers (pneumatically regulated) for light door weighing up to 40 kg (first revision)

1.3.4.33. IS Code 6607: 1972 Specification for rebated mortice locks (vertical type)

1.3.4.34. IS Code 7196: 1974 Specification for hold fast

1.3.4.35. IS Code 7197: 1974 Specification for double action floor springs (without oil check) for heavy doors

1.3.4.36. IS Code 7534: 1985 Specification for sliding locking bolts for use with padlocks (first revision)

1.3.4.37. IS Code 8756: 1978 Specification for ball catches for use in wooden almirah

1.3.4.38. IS Code 9106: 1979 Specification for rising butt hinges

1.3.4.39. IS Code 9131: 1979 Specification for rim locks

1.3.4.40. IS Code 9460: 1980 Specification for flush drop handle for drawer

1.3.4.41. IS Code 9899: 1981 Specification for hat coat and wardrobe hooks

1.3.4.42. IS Code 10019: 1981 Specification for mild steel stays and fasteners

1.3.4.43. IS Code 10090: 1982 Specification for numerical

1.3.4.44. IS Code 10342: 1982 Specification for curtain rail system

1.3.4.45. IS Code 12817: 2013 Specification for stainless steel butt hinges (second revision)

1.3.4.46. IS Code 12867: 1989 Specification for PVC hand rails covers

1.3.4.47. IS Code 14912: 2001 Specification for door closers concealed type (hydraulically regulated)

1.3.4.48. IS Code 15833: 2009 Specification for stainless steel tower bolts

1.3.4.49. IS Code 15834: 2008 Specification for stainless steel sliding door bolts for use with padlocks

1.3.4.50. IS Code 16015: 2013 Specification for mortice locks with lever mechanism (vertical type, sliding door locks and dead locks)

1.3.4.51. IS Code 16016: 2013 Specification for cylindrical locks with pin tumbler mechanism

1.3.4.52. ANSI A156.2 Bored & Preassembled Locks and Latches-Standard establishes performance requirements for bored and preassembled locks and latches, and includes dimensional criteria, operational tests, strength tests, cycle tests, security tests, material evaluation tests, and finish tests.

1.3.4.53. ASTM D 2000 Standard classification system for rubber products in automotive application.

1.3.4.54. ASTM E 283 Standard test methods for determining rate of air leakage through exterior windows, curtain walls and doors under specified pressure differences across the specimen.

1.3.4.55. ASTM E 1408 Standard test method for laboratory measurement of sound transmission.

1.3.4.56. BS 4787 Internal and external wood door sets, door leaves and frames. Specification for dimensional requirements.

1.3.4.57. BS 7036 Part 1 to 5 Safety at powered doors for pedestrian use.

1.3.4.58. BS 8300 Design of buildings and their approaches to meet the needs of disabled people. Code of practice.

1.3.4.59. BS 8424 Building hardware pull handles.

1.3.4.60. BS EN 179 Building hardware - Emergency exit devices operated by a lever handle or push pad - Requirements and test methods.

1.3.4.61. BS EN 1125 Building hardware. Panic exit devices operated by a horizontal bar. Requirements and test methods

1.3.4.62. BS EN 1154 Building hardware. Controlled door closing devices. Requirements and test methods

1.3.4.63. BS EN 1634 - Fire resistance and smoke control tests for door, shutter and, openable window assemblies and elements of building hardware. Fire resistance tests for doors, shutters and openable window

1.4. Submittals

1.4.1. Technical Data

1.4.1.1. Comprehensive technical data viz., technical literature, data sheets, product catalogue & brochure, installation instructions, etc. to be provided for Engineer’s approval.

1.4.2. Shop Drawings

1.4.2.1. Shop drawings shall be submitted to Engineer for approval, prior to start of work at site.

1.4.2.2. Shop drawings shall include all details w.r.t. the material & related product's installation, to the scale & detailing as instructed by the Engineer.

1.4.2.3. Shop drawings shall be developed as per actual measurements from the site & must depict the same in the drawings.

1.4.2.4. Shop drawings must comply to BS 1192:2007+A2:2016 Collaborative production of architectural, engineering and construction information & Code of practice.

1.4.3. Samples

1.4.3.1. Samples for Initial Selection

1.4.3.1.1. For each finish, colour, and texture required for each type of door hardware as per Tender drawings & documents.

1.4.3.2. Samples for Verification:

1.4.3.2.1. Complete range of Engineer's approved hardware & accessories.

1.4.4. Method Statement

1.4.4.1. Shall include method of construction/installation, Health and Safety management, testing methods, required relative humidity and temperature level and control systems, possible risk associated with the execution works and precautions to be considered, site control system and any other additional requirements as advised by the Engineer for the execution of works.

1.4.4.2. Method statement shall be approved by the Engineer prior to commencement of the works.

1.4.5. Installer Qualifications:

1.4.5.1. Licensed installer specialized in painting shall be employed for the work.

1.4.6. Material Certificates

1.4.6.1. Comprehensive tests performed by a qualified independent testing agency certifying that the product compiles with specified performance requirements.

1.4.7. Installer Qualifications

1.4.7.1. Licensed installer specialized in hardware installation shall be employed for the work.

1.4.8. Maintenance Data

1.4.8.1. Details of manufacturer's recommended maintenance procedures for each type of product/material to be included in the operation and maintenance manual.

1.4.9. Spare Materials/Parts

1.4.9.1. Provide the following extra materials that match installed products, packaged for storage with suitable labels for identification.

1.4.9.1.1. 2% minimum for each type of ironmongery item (product or pattern).

1.5. Warranty / Guarantee

1.5.1. As per the Employer's requirements and general provision of contract, including General, Supplementary and Special Conditions of Contract Documents.

1.6. Performance Requirements

1.6.1. All finishes, materials and components shall comply with requirements of NFPA 101 Life Safety Code, chapter 10 – Interior Finish, Contents and Furnishings with respect to classification of materials, flame spread, smoke generation and other requirements.

1.7. Environment & Sustainability

1.7.1. Recycling Strategy document for the material or product shall be submitted to the Engineer for approval.

1.7.2. LEED Requirements:

1.7.2.1. Manufacturer product conforming to ISO 14044 or Environmental Product Declarations which conform to ISO 14025, 14040, 14044, and EN 15804 or ISO 21930 and have at least a cradle to gate scope. (If applicable)

1.7.2.2. Manufacturer Material Ingredient Reporting by any of the listed below documents (If applicable):

1.7.2.2.1. Health Product Declaration. The end use product has a published, complete Health Product Declaration with full disclosure of known hazards in compliance with the Health Product Declaration open Standard.

1.7.2.2.2. Cradle to Cradle Certificate. The end use product has been certified at the Cradle to Cradle v2 Basic level or Cradle to Cradle v3 Bronze level.

1.7.2.2.3. REACH Optimization. End use products and materials that do not contain substances that meet REACH criteria for substances of very high concern.

1.7.2.3. Manufacturer Raw Material Source and Extraction Reporting by any of the listed below activity (If applicable):

1.7.2.3.1. Third-party verified corporate sustainability reports (CSR) which include environmental impacts of extraction operations and activities associated with the manufacturer's product and the product's supply chain

1.7.2.3.2. Self-declared reports from manufacturers that have publicly released a report from their raw material suppliers which include raw material supplier extraction locations

1.8. Manufacturer

1.8.1. Manufacturer to be accredited and registered under BS EN ISO 9001.

1.8.2. List of preferred or acceptable Manufacturers attached herewith as Annexure 1.

2. Products

2.1. Material

2.1.1. Material/Product Specification as per Standard attached herewith as Annexure A.10.

2.2. Colours

2.2.1. Colours as per Standard attached herewith as Annexure B.01.

3. Execution

3.1. Examination

3.1.1. Examine doors and frames, with Installer present, for compliance with requirements for installation tolerances, labelled fire door assembly construction, wall and floor construction, and other conditions affecting performance.

3.2. Fabrication

3.2.1. Base Metals: Produce door hardware units of base metal, fabricated by forming method indicated, using manufacturer's standard metal alloy, composition, temper, and hardness. Furnish metals of a quality equal to or greater than that of specified door hardware units. Do not furnish manufacturer's standard materials or forming methods if different from specified standard.

3.2.2. Fasteners: Provide door hardware manufactured to comply with published templates generally prepared for machine, wood, and sheet metal screws. Provide screws according to commercially recognized industry standards for application intended, except aluminium fasteners are not permitted. Provide Phillips flat-head screws with finished heads to match surface of door hardware, unless otherwise indicated.

3.2.3. Concealed Fasteners: For door hardware units that are exposed when door is closed, except for units already specified with concealed fasteners. Do not use through bolts for installation where bolt head or nut on opposite face is exposed unless it is the only means of securely attaching the door hardware. Where through bolts are used on hollow door and frame construction, provide sleeves for each through bolt.

3.3. Installation

3.3.1. Install each door hardware item to comply with manufacturer's written instructions. Where cutting and fitting are required to install door hardware onto or into surfaces that are later to be painted or finished in another way, coordinate removal, storage, and reinstallation of surface protective trim units with finishing work. Do not install surface-mounted items until finishes have been completed on substrates involved.

3.4. Workmanship

3.4.1. Workmanship shall be of the highest standard in every respect and tradesmen shall be fully experienced in their field. All work shall be executed neatly and accurately. Products shall be installed according to the manufacturer's instructions. The work shall be left clean, tidy, and properly finished. Movable parts shall be lubricated as necessary and left in good working order.

4. Quality Control

4.1. Delivery, Storage and Handling

4.1.1. All building materials shall be stored on the building site in such a way as to prevent deterioration or the loss or impairment of their structural and other essential properties. Comply to National Building Code (NBC) of India 2016 Volume 1; Part 7 Construction Management, Practices and Safety.

4.2. Source Limitation

4.2.1. All materials & related products shall be from the same manufacturer to ensure material/product integrity & uniformity of colour, texture & quality.

4.3. Mock-ups

4.3.1. Build mock-ups of each hardware type to verify selections made under sample submittals and to demonstrate aesthetic effects and set quality standards for materials and execution to the Engineer's satisfaction. Approved mock-ups may be incorporated into the finished work, where directed by the Engineer.

5. Testing and Commissioning

5.1. Inspection

5.1.1. Inspection upon completion of works all ironmongery shall be carefully tested and examine for faults in application, shall be corrected immediately.

ATK-ARC-011 – Glazing

1. General

1.1. Introduction

1.1.1. This Section specifies supply and installation of Glazing, Glazing assemblies & accessories, as shown on the tender drawings/schedules as required in completion of the work.

1.1.2. External Curtain Wall Glazing

1.1.3. Internal Curtain Wall Glazing

1.1.4. External Windows

1.1.5. Internal Windows

1.1.6. Glazed Partition

1.1.7. Sliding Doors

1.2. Related Documents

1.2.1. This section to be read in conjunction with Employer's requirements and general provision of contract, including General, Supplementary and Special Conditions of Contract Documents

1.3. Codes and Standards

1.3.1. All materials, workmanship and testing shall be in accordance with the following standards & codes, in their latest revisions. This is not exhaustive and contractor shall verify the listing and supplement with the latest applicable standards & codes.

1.3.2. Comply with the requirements of the reference standards noted herein, except where more stringent requirements are listed herein or otherwise required by the Contract Documents.

1.3.3. Where materials, finishes, fittings, equipment and building components that comply with standards other than those included in these specifications are proposed, the Contractor shall adequately demonstrate that such items are of an equal quality and standard and are fit for their intended purpose, through submission of globally recognized and authorized test methods, technical and certification reports (in compliance with assessment or evaluation criteria based on standards already included in these specifications), to the approval of engineer, prior to use of such items.

1.3.4. Conform to the requirements of the following Standards and Codes of Practice where applicable:

1.3.4.1. National Building Code (NBC) of India 2016 Volume 1 & 2.

1.3.4.2. IS Code 2553 (Part 1) 1990: Specification for safety glass: Part 1 General purpose (third revision)

1.3.4.3. IS Code 2835: 1987 Specification for flat transparent sheet glass (third revision)

1.3.4.4. IS Code 3438: 1994 Specification for silvered glass mirrors for general purposes (second revision)

1.3.4.5. IS Code 5437: 1994 Specification for figured rolled and wired glass (first revision)

1.3.4.6. IS Code 14900: 2000 Specification for transparent float glass

1.3.4.7. IS Code 343: 1994 - CHD 10: Silvered Glass Mirrors for General Purposes

1.3.4.8. IS Code 1200: Part XIV: Method of measurement of building and civil engineering work glazing

1.3.4.9. IS 1081 Code of practice for fixing and glazing of metal (Steel and aluminium) doors, windows and ventilators.

1.3.4.10. ANSI Z 97.1-1975 Safety performance specifications and methods of test for safety glazing material used in buildings

1.3.4.11. ANSI 297 Safety glazing materials used in building.

1.3.4.12. BS 544 Linseed oil putty for use in wooden frames.

1.3.4.13. BS 952 Glass for glazing.

1.3.4.14. BS 4255 Rubber used in preformed gaskets for weather exclusion from buildings. Specification for non-cellular gaskets

1.3.4.15. BS 5516 Patent glazing and sloping glazing for buildings. Code of practice for design and installation of sloping and vertical patent glazing.

1.3.4.16. BS 6206 Specification for impact performance requirements for flat safety glass and safety plastics for use in buildings.

1.3.4.17. BS 6262 Code of practice for glazing for buildings.

1.3.4.18. BS 6375 Performance of windows.

1.3.4.19. BS 6399 Part 1 to 3 Loading for buildings.

1.3.4.20. BS EN 572-1 to 9 Glass in building

1.3.4.21. BS EN 1096-1,2,3 Glass in building. Coated glass.

1.3.4.22. BS EN 1279 -1,2,3,4,5,6 Glass in building. Insulating glass units.

1.3.4.23. BS EN 1748 Part 1-1, 1-2 & 2-1 Glass in building. Borosilicate glass.

1.3.4.24. BS EN 1863 Glass in building. Heat strengthened soda lime silicate glass. Definition and description

1.3.4.25. BS EN 12150 Glass in building. Thermally toughened soda lime silicate safety glass. Definition and description

1.3.4.26. BS EN 12337 Part 1 & 2 Glass in building. Chemically strengthened soda lime silicate glass.

1.3.4.27. BS EN 13541 Glass in building. Security glazing.

1.3.4.28. BS EN ISO 12543-1,3,4,5,6:1998 Glass in building. Laminated glass and laminated safety glass. Definitions and description of component parts.

1.3.4.29. BS EN 14179-1 Glass in building. Heat-soaked thermally- toughened soda lime silicate safety glass. Definition and description.

1.3.4.30. NFPA 101 Life Safety Code.

1.4. Submittals

1.4.1. Technical Data

1.4.1.1. Comprehensive technical data viz., technical literature, data sheets, product catalogue & brochure, installation instructions, etc. to be provided for Engineer's approval.

1.4.2. Shop Drawings

1.4.2.1. Shop drawings shall be submitted to Engineer for approval, prior to start of work at site.

1.4.2.2. Shop drawings shall include all details w.r.t. the material & related product's installation, to the scale & detailing as instructed by the Engineer.

1.4.2.3. Shop drawings shall be developed as per actual measurements from the site & must depict the same in the drawings.

Shop drawings must comply to BS 1192:2007+A2:2016 Collaborative production of architectural, engineering and construction information & Code of practice.

1.4.3. Samples

1.4.3.1. Samples for Initial Selection

1.4.3.1.1. For each type of Glazing indicated. Include Samples of accessories involving colour selection.

1.4.4. Method Statement

1.4.4.1. Shall include method of construction/installation, Health and Safety management, testing methods, required relative humidity and temperature level and control systems, possible risk associated with the execution works and precautions to be considered, site control system and any other additional requirements as advised by the Engineer for the execution of works.

1.4.4.2. Method statement shall be approved by the Engineer prior to commencement of the works.

1.4.5. Installer Qualifications:

1.4.5.1. Licensed installer specialized in Resilient flooring & skirting installation shall be employed for the work.

1.4.6. Material/Product Certificates:

1.4.6.1. Comprehensive tests performed by a qualified independent testing agency certifying that the product compiles with specified performance requirements.

1.4.7. Maintenance Data:

1.4.7.1. Details of manufacturer's recommended maintenance procedures for each type of product/material to be included in the operation and maintenance manual.

1.4.8. Spare Materials/Parts

1.4.8.1. Provide the following extra materials that match installed products, packaged for storage with suitable labels for identification.

1.4.8.2. 2% or 50m^2 minimum for each type of Glazing (product, pattern or colour)

1.5. Warranty / Guarantee

1.5.1. As per the Employer's requirements and general provision of contract, including General, Supplementary and Special Conditions of Contract Documents.

1.6. Environment & Sustainability

1.6.1. Recycling Strategy document for the material or product shall be submitted to the Engineer for approval.

1.6.2. LEED Requirements:

1.6.2.1. Manufacturer product conforming to ISO 14044 or Environmental Product Declarations which conform to ISO 14025, 14040, 14044, and EN 15804 or ISO 21930 and have at least a cradle to gate scope (If applicable).

1.6.2.2. Manufacturer Material Ingredient Reporting by any of the listed below documents (If applicable):

1.6.2.2.1. Health Product Declaration. The end use product has a published, complete Health Product Declaration with full disclosure of known hazards in compliance with the Health Product Declaration Open Standard.

1.6.2.2.2. Cradle to Cradle Certificate. The end use product has been certified at the Cradle to Cradle v2 Basic level or Cradle to Cradle v3 Bronze level.

1.6.2.2.3. REACH Optimization. End use products and materials that do not contain substances that meet REACH criteria for substances of very high concern.

1.6.2.3. Manufacturer Raw Material Source and Extraction Reporting by any of the listed below activity (If applicable):

1.6.2.3.1. Third-party verified corporate sustainability reports (CSR) which include environmental impacts of extraction operations and activities associated with the manufacturer's product and the product's supply chain.

1.6.2.3.2. Self-declared reports from manufacturers that have publicly released a report from their raw material suppliers which include raw material supplier extraction locations

1.6.2.4. External facade glazing should comply with listed below properties:

1.6.2.4.1. U value <= 1.1 W/sq M K

1.6.2.4.2. SHGC (Solar Heat Gain Coefficient) <= 0.25

1.6.2.4.3. VLT >= 30%

1.6.2.5. All Sealants and Adhesive should meet VOC level as mentioned below. Manufacturer has to confirm same by providing lab test report.

Sealants & Sealant Primers	**VOC Limit (g/L less water)**
Architectural	250
Non-membrane roof	300
Roadway	250
Single-ply roof membrane	450
Other	420
Architectural Applications	**VOC Limit (g/L less water)**
Structural glazing adhesives	100

1.7. Manufacturers

1.7.1. Manufacturer to be accredited and registered under BS EN ISO 9001.

1.7.2. List of preferred or acceptable Manufacturers attached herewith as Annexure 1.

2. Products

2.1. Materials

2.1.1. Material/Product Specification as per Standard attached herewith as Annexure A.11.

2.1.1.1. Glass Unit for External Glazing & Windows shall comply with the following:

2.1.1.1.1. U value <= 1.1 W/Sq.M.K

2.1.1.1.2. SHGC (Solar Heat Gain Coefficient) <= 0.25

2.1.1.1.3. VLT (Visible Light Transmission) >= 30%

2.1.1.2. Glass Unit for Internal Glazing & Windows shall be Toughened Glass complying with the Performance parameters as per schedule.

2.1.1.3. Glass Unit for Sliding Doors shall be Toughened Glass complying with the Performance parameters as per schedule.

2.2. Colours

2.2.1. Colours as per Standard attached herewith as Annexure B.01.

3. Execution

3.1. Examination

3.1.1. Verification of Conditions: Examine areas and conditions under which the work of this Section will be performed. Do not proceed with the work until unsatisfactory conditions have been corrected. Commencement of work implies acceptance of all areas and conditions.

3.1.2. Examine framing, glazing channels, and stops, with Installer present, for compliance with the following:

3.1.2.1. Manufacturing and installation tolerances, including those for size, square, and offsets at corners.

3.1.2.2. Presence and functioning of weep systems.

3.1.2.3. Minimum required face and edge clearances.

3.1.2.4. Effective sealing between joints of glass-framing members.

3.2. Preparation

3.2.1. Substrate Preparation

3.2.2. Clean glazing channels, stops and rebates to receive glazing materials. Remove obstructions and substances which would impair the work or interfere with bond of sealants and adhesive. Solvents used for cleaning must be compatible with finished aluminium, glass and glazing materials.

3.2.3. Prime surfaces as recommended by sealant or adhesive manufacturer. Mask or otherwise protect surfaces adjacent to installation of sealants.

3.2.4. Preparation of Framing

3.2.4.1. Before beginning work, the contractor shall measure and inspect frames to receive glazing to determine that other trades have completed preparatory work and that sash and frames are ready to receive his materials.

3.2.4.2. Frames and sash shall be adjusted, plumbed and squared. All rivets, screws, bolts nail heads, welds, and other projections shall be finished flush in the glazing rebates. All corners and intersections shall be sealed and weather tight.

3.2.4.3. Opening sashes shall be fastened and kept closed until glazing compounds, except non-setting type, have cured or set.

3.2.4.4. Surfaces to receive glazing materials shall be free of dirt, dust, grease, oil and other foreign materials, and shall be painted or sealed, before glazing work is begun.

3.2.4.5. Do not begin glazing work until all cleaning and repairing of surfaces has been completed.

3.2.4.6. Aluminium glazed frames requiring dry glazing method shall be glazed in accordance with the manufacturer’s instructions.

3.2.4.7. Cut glass accurately to fit openings. Sizes of glass indicated on drawings are approximate only, and the actual sizes required shall be determined by measuring on site the frames to receive the glass. Size glass to permit the required clearances around the full perimeter of the glass.

3.2.5. Preparation of Glazing

3.2.5.1. The glass shall be cut of sizes with proper clearance and shall be well back puttied, sprigged for timber rebates or pegged for metal rebates and neatly front puttied. Putty must not appear above the sight lines. Glass to all windows and doors (except toilets/bathrooms) shall be clear sheet glass or as specified. Glass to toilets/bathrooms shall be an approved obscured glass or as specified.

3.2.5.2. Glass shall be fitted in accordance with BS 6262:1982.

3.2.5.3. Glass panes over 0.1m2 in metal windows for putty glazing shall be secured by means of spring glazing clips fixed in the performed holes.

3.2.5.4. Georgian wired glass shall have the wiring aligned vertically and horizontally one or both ways to adjacent glazed panels.

3.2.5.5. The provision of glazing compounds and putties and springs, clips and other sundry fixings shall be included in the cost of glazing and of fixing of mirrors.

3.2.5.6. Distance pieces and setting blocks in appropriate materials shall be provided in accordance with good glazing practice and they shall be included in the cost of glazing.

3.3. Installation

3.3.1. General

3.3.1.1. Inspect glass before installation. Do not install glass which does not conform to specification. Replace glass which is broken or damaged.

3.3.1.2. Defer glazing of openings which are obstructed during construction. Glaze such openings when obstructions are removed.

3.3.1.3. Provide minimum nominal glass bite of 12.5mm. Where joint movement will result in variable glass bite, increase normal bite to provide 9.5mm minimum bite and 6.4mm minimum edge clearance.

3.3.1.4. Set glazing to accurately fit frames and provide equal bearing along entire length of each pane. Secure glazing with clips unless detailed with beads or stops. When glazing is to be placed with metal stops, set glazing in full bed of glazing putty or vinyl strip.

3.3.1.5. Place manufacturing waves, if apparent in panes, in horizontal orientation.

3.3.1.6. Replace glass broken due to improper settings.

3.3.1.7. Notwithstanding the glass types and locations indicated on the contract drawings, provide safety glass at all locations required by codes or glazing ordinances.

3.3.1.8. Glazing details shall permit glass replacement after initial construction, shall permit reuse of original gaskets, shall permit replacement glass of the same nominal size as original glass, and shall not require cutting of framing members or removal of interior finishes. Vision glass in conventional frames shall be replaceable from the interior. Spandrel glass shall be replaceable from the exterior. Silicone supported vision glass shall be replaceable from the exterior and/or interior.

3.3.2. Setting and Side Blocks

3.3.2.1. Provide setting blocks of size to fit sill rebate. Locate ¼ of glass width from each corner. Set in thin course of heel-bead compound, if any.

3.3.2.2. Locate side blocks between mid-height and top corner of glass.

3.3.2.3. Side blocks, setting blocks and chairs shall be positively retained in position.

3.3.3. Spacers

3.3.3.1. Except where continuous gaskets are used for glazing glass sizes larger than 1200mm cumulative (length plus width), provide spacers on both faces of glass at maximum 600mm intervals and no less than 300mm from corners, inside and out, of thickness equal to sealant width. Place spacers opposite one another. Provide 6mm minimum bite of spacers on glazing.

3.3.4. Gasketing

3.3.4.1. Install pressurized tapes and gaskets to protrude slightly out of channel to eliminate dirt and moisture pockets.

3.3.4.2. Install gaskets in continuous lengths without splicing. Miter cut and bond ends together at corners in manner to prevent pull out.

3.3.4.3. Field vulcanize or heat weld corners of exterior glazing gaskets.

3.3.5. Sealants

3.3.5.1. Remove and replace stops and apply sealants as required for complete glass installation.

3.3.5.2. To prevent exudation of sealant, provide filler rods at heels of glazing channels.

3.3.6. Fixing Mirrors:

3.3.6.1. Avoid distorting or stressing mirrors during fixing. Prevent any contact with irregularities in the background. Secure to block work or concrete using proprietary expanding wall plugs with chromium plated screws with detachable dome heads. Provide compressible polyethylene sleeves and washers to isolate mirror from screws. Provide additional support to bottom edge of mirrors over 1m2 (e.g. supplementary clips or a ledge). Fix mirrors with 3mm air space at back in areas subject to humidity. Fix adjacent mirrors in the same plane to eliminate distortion.

3.3.6.2. Mirrors to be fixed on the masonry wall directly & not on the tiles.

3.4. Workmanship

3.4.1. Workmanship shall be of the highest standard in every respect and tradesmen shall be fully experienced in their field. All work shall be executed neatly and accurately. Products shall be installed according to the manufacturer's instructions. The work shall be left clean, tidy and properly finished. Movable parts shall be lubricated as necessary and left in good working order.

4. Quality Control

4.1. Delivery, Storage and Handling

4.1.1. All building materials shall be stored on the building site in such a way as to prevent deterioration or the loss or impairment of their structural and other essential properties. Comply to National Building Code (NBC) of India 2016 Volume 1; Part 7 Construction Management, Practices and Safety.

4.2. Source Limitation

4.2.1. All materials & related products shall be from the same manufacturer to ensure material/product integrity & uniformity of colour, texture & quality.

4.3. Mock-ups

4.3.1. Build full height mock-ups for each type of panel including all accessories to verify selections made under sample submittals and to demonstrate aesthetic effects and set quality standards for materials and execution.

4.3.2. Approved mock-ups may be incorporated into the finished work, where directed by the Engineer.

5. Testing and Commissioning

5.1. Inspection

5.1.1. Inspection upon completion of work, all Glazing shall be carefully tested and examined for faults in application, shall be corrected immediately.

ATK-ARC-012 – Skylights

1. General

1.1. Introduction

1.1.1. This section specifies supply and installation of Skylight, as indicated in the Tender drawings and as required in completion of the construction works.

1.2. Related Documents

1.2.1. This section to be read in conjunction with Employer's requirements and general provision of contract, including General, Supplementary and Special Conditions of Tender Documents.

1.3. Codes and Standards

1.3.1. All materials, workmanship and testing shall be in accordance with the following documents, in their latest revisions. This is not exhaustive, and contractor shall verify the listing and supplement with the latest applicable document or documents.

1.3.2. Comply with the requirements of the reference standards noted herein, except where more stringent requirements are listed herein or otherwise required by the Tender Documents.

1.3.3. Where materials, finishes, fittings, equipment and building components that comply with standards other than those included in these specifications are proposed, the Contractor shall adequately demonstrate that such items are of an equal quality and standard and are fit for their intended purpose, through submission of internationally recognized and authorized test methods, technical and certification reports (in compliance with assessment/evaluation criteria based on standards already included in these specifications), to the approval of engineer, prior to use of such items.

1.3.4. Conform to the requirements of the following Standards and Codes of Practice where applicable:

1.3.4.1. National Building Code (NBC) of India 2016 Volume 1 & 2.

1.3.4.2. ASTM E2268 - 04(2016). Standard Test Method for Water Penetration of Exterior Windows, Skylights, and Doors by Rapid Pulsed Air Pressure.

1.3.4.3. ASTM E547 - 00(2016). Standard Test Method for Water Penetration of Exterior Windows, Skylights, Doors, and Curtain Walls by Cyclic Static Air Pressure Difference.

1.3.4.4. ASTM E331 - 00(2016). Standard Test Method for Water Penetration of Exterior Windows, Skylights, Doors, and Curtain Walls by Uniform Static Air Pressure Difference.

1.3.4.5. ASTM E2112 - 07(2016). Standard Practice for Installation of Exterior Windows, Doors and Skylights.

1.3.4.6. ASTM E1105 - 15. Standard Test Method for Field Determination of Water Penetration of Installed Exterior Windows, Skylights, Doors, and Curtain Walls, by Uniform or Cyclic Static Air Pressure Difference.

1.3.4.7. ASTM G197 - 14. Standard Table for Reference Solar Spectral Distributions: Direct and Diffuse on 20$0D Tilted and Vertical Surfaces.

1.3.4.8. ASTM E1425 - 14. Standard Practice for Determining the Acoustical Performance of Windows, Doors, Skylight, and Glazed Wall Systems.

1.3.4.9. ASTM E1423 - 14. Standard Practice for Determining Steady State Thermal Transmittance of Fenestration Systems.

1.3.4.10. ASTM E1233/E1233M - 14. Standard Test Method for Structural Performance of Exterior Windows, Doors, Skylights, and Curtain Walls by Cyclic Air Pressure Differential.

1.3.4.11. ASTM E330/E330M - 14. Standard Test Method for Structural Performance of Exterior Windows, Doors, Skylights and Curtain Walls by Uniform Static Air Pressure Difference.

1.3.4.12. ASTM E971 – 11: Standard Practice for Calculation of Photometric Transmittance and Reflectance of Materials to Solar Radiation.

1.3.4.13. ASTM E2692 - 10. Standard Test Method for Structural Performance of Thermal Barriers in Fenestration Products.

1.3.4.14. BS EN 12833:2001. Skylight and conservatory roller shutters. Resistance to snow load. Test method.

1.3.4.15. NFPA 101 Life Safety Code.

1.4. Submittals

1.4.1. Technical Data

1.4.1.1. Comprehensive technical data viz., technical literature, data sheets, product catalogue & brochure, installation instructions, etc. to be provided for Engineer's approval.

1.4.2. Shop drawings

1.4.2.1. Shop drawings shall be submitted to Engineer for approval, prior to start of work at site.

1.4.2.2. Shop drawings shall include all details w.r.t. the material & related product's installation, to the scale & detailing as instructed by the Engineer.

1.4.2.3. Shop drawings shall be developed as per actual measurements from the site & must depict the same in the drawings.

1.4.2.4. Shop drawings must comply to BS 1192:2007+A2:2016 Collaborative production of architectural, engineering and construction information & Code of practice.

1.4.3. Samples

1.4.3.1. Samples for Initial Selection: For each type of Skylight assembly or accessories as indicated. Include Samples of accessories involving colour selection.

1.4.4. Method Statement

1.4.4.1. Shall include method of construction/installation, Health and Safety management, testing methods, required relative humidity and temperature level and control systems, possible risk associated with the execution works and precautions to be considered, site control system and any other additional requirements as advised by the Engineer for the execution of works.

1.4.4.2. Method statement shall be approved by the Engineer prior to commencement of the works.

1.4.5. Installer Qualifications

1.4.5.1. Licensed installer specialized in fabricating and installing each type of ceiling, eligible to receive manufacturer's warranty.

1.4.6. Material/Product Certificates

1.4.6.1. Comprehensive tests performed by a qualified independent testing agency certifying that the product compiles with specified performance requirements.

1.4.7. Maintenance Data

1.4.7.1. Details of manufacturer's recommended maintenance procedures for each type of product/material to be included in the operation and maintenance manual.

1.4.8. Spare Materials/Parts

1.4.8.1. Provide the following extra materials that match installed products, packaged for storage with suitable labels for identification.

1.5. Warranty / Guarantee

1.5.1. As per the Employer's requirements and general provision of contract, including General, Supplementary and Special Conditions of Contract Documents.

1.6. Performance Requirements

1.6.1. All finishes, materials and components shall comply with requirements of NFPA 101 Life Safety Code, chapter 10 – Interior Finish, Contents and Furnishings with respect to classification of materials, flame spread, smoke generation and other requirements.

1.7. Environment & Sustainability

1.7.1. Recycling Strategy document for the material or product shall be submitted to the Engineer for approval.

1.7.2. LEED Requirements:

1.7.2.1. All the specifications in this document are for USGBC LEED BD+C submission and are applicable to all materials & procedures followed for the project. The specification as stated here should be adhered to the word. In case of a dispute LEED guidelines shall be followed. Brochures, product specification sheets and/or certificates as specified in this document should be submitted along with the material procurement. All contractors, manufactures must submit documents, records and other submittal related to the different credits as mentioned in the specification.

1.7.2.2. Manufacturer product conforming to ISO 14044 or Environmental Product Declarations which conform to ISO 14025, 14040, 14044, and EN 15804 or ISO 21930 and have at least a cradle to gate scope. (If applicable)**.**

1.7.2.3. Manufacturer Material Ingredient Reporting by any of the listed below documents (If applicable):

1.7.2.3.1. Health Product Declaration. The end use product has a published, complete Health Product Declaration with full disclosure of known hazards in compliance with the Health Product Declaration Open Standard.

1.7.2.3.2. Cradle to Cradle Certificate. The end use product has been certified at the Cradle to Cradle v2 Basic level or Cradle to Cradle v3 Bronze level.

1.7.2.3.3. REACH Optimization. End use products and materials that do not contain substances that meet REACH criteria for substances of very high concern.

1.7.2.4. Manufacturer Raw Material Source and Extraction Reporting by any of the listed below activity (If applicable):

1.7.2.4.1. Third-party verified corporate sustainability reports (CSR) which include environmental impacts of extraction operations and activities associated with the manufacturer's product and the product's supply chain**.**

1.7.3. Self-declared reports from manufacturers that have publicly released a report from their raw material suppliers which include raw material supplier extraction locations

1.8. Manufacturers

1.8.1. Manufacturer to be accredited and registered under BS EN ISO 9001.

1.8.2. List of preferred or acceptable Manufacturers attached herewith as Annexure 1.

2. Products

2.1. Materials

2.1.1. Material/Product Specification as per Standard attached herewith as Annexure A.12.

2.2. Colours

2.2.1. Colours as per Standard attached herewith as Annexure B.01.

3. Execution

3.1. Examination

3.1.1. Verification of existing conditions before starting work. Installer must examine the location and advise the Contractor of any site conditions unacceptable for proper installation of product. These conditions include but are not limited to the following:

3.2. Preparation

3.2.1. Coordinate dimensions, tolerance, and method of attachment with other work.

3.2.2. Manufacturer shall offer a Factory Acceptance Test to the Employer with fully functioning door prior to shipment.

3.3. Fabrication

3.3.1. General: Factory fabricated skylight components to designs, profiles, sizes, and thicknesses indicated and to comply with indicated standards. Provide skylights as prefabricated assemblies. Complete fabrication, assembly, finishing, hardware application, and other work before shipment to Project site.

3.4. Installation

3.4.1. Install skylights in accordance with manufacturer's printed instructions. Set units' level, plumb, and with uniform hairline joints. Anchor securely into place. Use only factory-trained installers.

4. Quality Control

4.1. Delivery, Storage and Handling

4.1.1. All building materials shall be stored on the building site in such a way as to prevent deterioration or the loss or impairment of their structural and other essential properties. Comply to National Building Code (NBC) of India 2016 Volume 1; Part 7 Construction Management, Practices and Safety.

4.2. Source Limitation

4.2.1. All materials & related products shall be from the same manufacturer to ensure material/product integrity & uniformity of colour, texture & quality.

5. Testing & Commissioning

5.1. Factory Inspection & Tests

5.1.1. Arrange for Employer and Employer's representative visits (minimum of two visits) for material/product/prototype reviews or tests to factories or sites as agreed with the Engineer.

5.1.2. The contractor shall inform the Engineer of test/inspection date, time and location at least 4 weeks in advance. The cost of travel, accommodation and other related expenses shall be as included in tender.

5.2. Site Inspection

5.2.1. Inspection upon completion of work, Overhead Sectional Doors shall be carefully tested and examined for faults in operation, shall be corrected immediately.

ATK-ARC-013 – Sliding Doors & Folding Partition

1. General

1.1. Introduction

1.1.1.1. This section specifies supply and installation of Automatic Sliding Doors & Folding Partition, as indicated in the Tender drawings and as required in completion of the construction works.

1.2. Related Documents

1.2.1. This section to be read in conjunction with Employer's requirements and general provision of contract, including General, Supplementary and Special Conditions of Tender Documents.

1.3. Codes and Standards

1.3.1. All materials, workmanship and testing shall be in accordance with the following documents, in their latest revisions. This is not exhaustive and contractor shall verify the listing and supplement with the latest applicable document or documents.

1.3.2. Comply with the requirements of the reference standards noted herein, except where more stringent requirements are listed herein or otherwise required by the Tender Documents.

1.3.3. Where materials, finishes, fittings, equipment and building components that comply with standards other than those included in these specifications are proposed, the Contractor shall adequately demonstrate that such items are of an equal quality and standard and are fit for their intended purpose, through submission of internationally recognized and authorized test methods, technical and certification reports (in compliance with assessment/evaluation criteria based on standards already included in these specifications), to the approval of engineer, prior to use of such items.

1.3.4. Conform to the requirements of the following Standards and Codes of Practice where applicable:

1.3.4.1. National Building Code (NBC) of India 2016 Volume 1 & 2.

1.3.4.2. IS 277: Specification for galvanized steel sheets (Plain and corrugated)

1.3.4.3. IS 1038: Steel doors, windows and ventilators

1.3.4.4. IS 1081: Code of practice for fixing and glazing of metal (Steel and aluminium) doors, windows and ventilators

1.3.4.5. IS 4351: Specification for steel door frames

1.3.4.6. IS 7452: Specification for hot rolled steel sections for doors, windows and ventilators.

1.3.4.7. BS 2994 Specification for cold rolled steel sections

1.3.4.8. BS 3987 Specification for anodic oxidation coatings on wrought aluminium for external architectural applications.

1.3.4.9. BS 4800 Schedule of paint colours for building purposes

1.3.4.10. BS 4842 Specification for liquid organic coatings for application to aluminium alloy extrusions, sheet and preformed sections for external architectural purposes, and for the finish on aluminium alloy extrusions, sheet and preformed sections coated with liquid organic coatings

1.3.4.11. BS 6150 Painting of buildings. Code of practice

1.3.4.12. BS 6180 Barriers in and about buildings. Code of practice.

1.3.4.13. BS 6206 Specification for impact performance requirements for flat safety glass and safety plastics for use in buildings.

1.3.4.14. BS 6262 (partially replaced by BS 6262-1, -2, -3, -4, -6 and - 7:2005) Code of practice for glazing for buildings

1.3.4.15. BS 6496 Specification for powder organic coatings for application and stoving to aluminium alloy extrusions, sheet and preformed sections for external architectural purposes, and for the finish on aluminium alloy extrusions, sheet and preformed sections coated with powder organic coatings

1.3.4.16. BS 7036-1-5 Code of practice for safety at powered doors for pedestrian use.

1.3.4.17. BS 7036-1 Code of practice for safety at powered doors for pedestrian use. General

1.3.4.18. BS 7036-2 Code of practice for safety at powered doors for pedestrian use. Straight and curved sliding doors and prismatic and folding doors.

1.3.4.19. BS 7773 Code of practice for cleaning and preparation of metal surfaces.

1.3.4.20. BS EN 755-9 Aluminium and Aluminium alloys, extruded rod/bar, tubes & profiles.

1.3.4.21. BS EN 10088 Stainless steels. Technical delivery conditions for semi-finished products, bars, rods, wire, sections and bright products of corrosion resisting steels for general purposes

1.3.4.22. BS EN 10111 Continuously hot rolled low carbon steel sheet and strip for cold forming. Technical delivery conditions

1.3.4.23. BS EN 10162 Cold rolled steel sections. Technical delivery conditions. Dimensional and cross-sectional tolerances

1.3.4.24. BS EN 13888 Grouts for tiles. Definitions and specifications

1.3.4.25. BS EN ISO 12944 Paints and varnishes. Corrosion protection of steel structures by protective paint systems. Protective paint systems

1.3.4.26. Laminate tested to ASTM D256 Izod impact test.

1.3.4.27. ASTM D790 Flexural Strength3.ASTM D638 Tensile strength.

1.3.4.28. ASTM D543 Evaluation of Plastics to chemical reagents.

1.3.4.29. ASTM D570 Water absorption, ASTM D1308 effect of chemicals.

1.3.4.30. Acoustic doors tested to BS EN ISO 140-3.

1.3.4.31. Hinges certified to BS EN 1935 which include testing for Static load, shear, Endurance, Corrosion, Fire door and Burglary. ANSI 157-1 & 7, UL10C & cUL10C.

1.3.4.32. Locks, cylinder, latch and Lever certified to SS 332:2007 durability testing to 200’000 test cycles to BS EN 12209.

1.3.4.33. Door Closers tested to EN1154 & BS 476 part 22, production surveillance ISO9002 &CERTIFIRE and SS 332:2007 durability testing 500’000 test cycles.

1.3.4.34. PolyIsoCyanurate CFC/HCFC free rigid foam insulation has been tested to ASTM E 84 and has a flame spread index of under 35 AAMA 1503-09 Thermal Performance.

1.3.4.35. Resin is tested to Class 1 fire rating BS476 Part 7 & part 6 and M1 rating also French M1rating to NFP 92-501 fire testing. Tensile strength / Modulus and Elongation are tested to BS EN ISO 527-4:1997, Flexible strength / modulus is tested to BS EN ISO 14125:1998: Notched Izod Impact is tested to BS EN ISO 180:2001. Heat deflection temperature to BSEN ISO 75 and 306. Compression strength and Modulus BS EN ISO 604. BARCOL - ASTMD-2583.

1.3.4.36. NFPA 101 Life Safety Code.

1.4. Submittals

1.4.1. Technical Data

1.4.1.1. Comprehensive technical data viz., technical literature, data sheets, product catalogue & brochure, installation instructions, etc. to be provided for Engineer’s approval.

1.4.2. Shop drawings

1.4.2.1. Shop drawings shall be submitted to Engineer for approval, prior to start of work at site.

1.4.2.2. Shop drawings shall include all details w.r.t. the material & related product’s installation, to the scale & detailing as instructed by the Engineer.

1.4.2.3. Shop drawings shall be developed as per actual measurements from the site & must depict the same in the drawings.

1.4.2.4. Shop drawings must comply to BS 1192:2007+A2:2016 Collaborative production of architectural, engineering and construction information & Code of practice.

1.4.3. Samples

1.4.3.1. Samples for Initial Selection: For Sliding door assembly as indicated. Include Samples of accessories involving colour selection.

1.4.4. Method Statement

1.4.4.1. Shall include method of construction/installation, Health and Safety management, testing methods, required relative humidity and temperature level and control systems, possible risk associated with the execution works and precautions to be considered, site control system and any other additional requirements as advised by the Engineer for the execution of works.

1.4.4.2. Method statement shall be approved by the Engineer prior to commencement of the works.

1.4.5. Installer Qualifications

1.4.5.1. Licensed installer specialized in fabricating and installing each type of ceiling, eligible to receive manufacturer’s warranty.

1.4.6. Material/Product Certificates

1.4.6.1. Comprehensive tests performed by a qualified independent testing agency certifying that the product compiles with specified performance requirements.

1.4.7. Maintenance Data

1.4.7.1. Details of manufacturer's recommended maintenance procedures for each type of product/material to be included in the operation and maintenance manual.

1.4.8. Spare Materials/Parts

1.4.8.1. Provide the following extra materials that match installed products, packaged for storage with suitable labels for identification.

1.5. Warranty / Guarantee

1.5.1. As per the Employer's requirements and general provision of contract, including General, Supplementary and Special Conditions of Contract Documents.

1.6. Performance Requirements

1.6.1. All finishes, materials and components shall comply with requirements of NFPA 101 Life Safety Code, chapter 10 – Interior Finish, Contents and Furnishings with respect to classification of materials, flame spread, smoke generation and other requirements.

1.7. Environment & Sustainability

1.7.1. Recycling Strategy document for the material or product shall be submitted to the Engineer for approval.

1.7.2. LEED Requirements:

1.7.2.1. All the specifications in this document are for USGBC LEED BD+C submission and are applicable to all materials & procedures followed for the project. The specification as stated here should be adhered to the word. In case of a dispute LEED guidelines shall be followed. Brochures, product specification sheets and/or certificates as specified in this document should be submitted along with the material procurement. All contractors, manufactures must submit documents, records and other submittal related to the different credits as mentioned in the specification.

1.7.2.2. Manufacturer product conforming to ISO 14044 or Environmental Product Declarations which conform to ISO 14025, 14040, 14044, and EN 15804 or ISO 21930 and have at least a cradle to gate scope. (If applicable)**.**

1.7.2.3. Manufacturer Material Ingredient Reporting by any of the listed below documents (If applicable):

1.7.2.3.1. Health Product Declaration. The end use product has a published, complete Health Product Declaration with full disclosure of known hazards in compliance with the Health Product Declaration Open Standard.

1.7.2.3.2. Cradle to Cradle Certificate. The end use product has been certified at the Cradle to Cradle v2 Basic level or Cradle to Cradle v3 Bronze level.

1.7.2.3.3. REACH Optimization. End use products and materials that do not contain substances that meet REACH criteria for substances of very high concern.

1.7.2.4. Manufacturer Raw Material Source and Extraction Reporting by any of the listed below activity (If applicable):

1.7.2.4.1. Third-party verified corporate sustainability reports (CSR) which include environmental impacts of extraction operations and activities associated with the manufacturer’s product and the product’s supply chain**.**

1.7.3. Self-declared reports from manufacturers that have publicly released a report from their raw material suppliers which include raw material supplier extraction locations

1.8. Manufacturers

1.8.1. Manufacturer to be accredited and registered under BS EN ISO 9001.

1.8.2. List of preferred or acceptable Manufacturers attached herewith as Annexure 1.

2. Products

2.1. Materials

2.1.1. Material/Product Specification as per Standard attached herewith as Annexure A.13.

2.1.2. Proprietary Steel Leaf

2.1.2.1. Face Material:

2.1.2.1.1. External: Minimum 1mm galvanized steel with 60-micron polyester powder coated finish.

2.1.2.1.2. Internal: Minimum 1mm mild steel with 60-micron polyester powder coated finish.

2.1.2.2. Core Material: Hollow or filled with mineral wool or equal depending on the function. Minimum U-Value is 2.0 W/m²K for exterior doors.

2.1.3. Leaf Thickness: Minimum 40mm.

2.1.4. Proprietary Solid Particle Board Leaf

2.1.4.1. Face Material: 0.8mm minimum thick high pressure laminate.

2.1.4.2. Core Material: Solid Particle Board. Low formaldehyde content.

2.1.4.3. Edge Material: Hardwood edges or equivalent, fixed on bottom and leading edge as 8mm minimum.

2.1.5. Proprietary Glass Reinforced Polyester Leaf (Hygienic Sliding Doors for Kitchen)

2.1.5.1. Face Material: Minimum 2.0mm thick Moulded Fiberglass Composite Laminate.

2.1.5.2. Core Material: Polyisocyanurate infill for non-fire rated doors & mineral wool or equivalent for fire rated doors.

2.1.5.3. Leaf Thickness: Minimum 40mm.

2.2. Colours

2.2.1. Colours as per Standard attached herewith as Annexure B.01.

3. Execution

3.1. Examination

3.1.1. Examine conditions, with Installer present, for compliance with requirements for installation tolerances, header support, and other conditions affecting performance of the installation.

3.1.2. Proceed with installation only after unsatisfactory conditions have been corrected.

3.2. Fabrication

3.2.1. General: Factory fabricated door components to designs, profiles, sizes, and thicknesses indicated and to comply with indicated standards.

3.2.2. Framing: Provide power operated glazed doors as prefabricated assemblies. Complete fabrication, assembly, finishing, hardware application, and other work before shipment to Project site.

3.2.3. Doors: Factory fabricated and assembled in profiles indicated. Reinforce as required to support imposed loads and for installing hardware.

3.2.4. Door Operators: Factory fabricated and installed in headers, including adjusting and testing.

3.2.5. Glazing: Fabricate framing with minimum glazing edge clearances for thickness and type of glazing indicated, according to British standards.

3.2.6. Laminated: Laminated Particle Board leaf as per the specified Material.

3.2.7. Hardware: Factory install hardware to greatest extent possible; remove only as required for final finishing operation and for delivery to and installation at Project site. Cut, drill, and tap for factory-installed hardware before applying finishes.

3.2.8. Aluminium Finishes

3.2.8.1. High-Performance Organic Finish: 3-coat fluoropolymer finish complying with BS 4694 and containing not less than 70 percent PVDF resin by weight in both colour coat and clear topcoat. Prepare, pre-treat, and apply coating to exposed metal surfaces to comply with coating and resin manufacturers' written instructions.

3.3. Installation

3.3.1.1. General: Do not install damaged components. Fit frame joints to produce hairline joints free of burrs and distortion. Rigidly secure non-movement joints. Seal joints watertight. Where aluminium will contact dissimilar metals, protect against galvanic action by painting contact surfaces with primer or by applying sealant or tape recommended by manufacturer for this purpose. Where aluminium will contact concrete or masonry, protect against corrosion by painting contact surfaces with bituminous paint.

3.3.2. Entrances: Install power operated doors plumb and true in alignment with established lines and grades without warp or rack of framing members and doors. Anchor securely in place.

3.3.3. Door Operators: Connect door operators to electrical power distribution system as specified in the MEP Specifications.

3.3.4. Activation and Safety Devices: Install and adjust devices to provide detection field and functions indicated.

3.3.5. Guide Rails: Install rails according to BS 7036 and manufacturer's written instructions unless otherwise indicated.

3.3.6. Sealants: Comply with requirements specified Section "Joint Sealants" to provide weathertight installation.

3.3.7. Signage: Apply signage on both sides of each door [and breakaway sidelight] as required by referenced door standards.

3.3.8. Wiring within power operated enclosures: Bundle, lace, and train conductors to terminal points with no excess and without exceeding manufacturer's written limitations on bending radii. Provide and use lacing bars and distribution spools.

4. Quality Control

4.1. Delivery, Storage and Handling

4.1.1. All building materials shall be stored on the building site in such a way as to prevent deterioration or the loss or impairment of their structural and other essential properties. Comply to National Building Code (NBC) of India 2016 Volume 1; Part 7 Construction Management, Practices and Safety.

4.2. Source Limitation

4.2.1. All materials & related products shall be from the same manufacturer to ensure material/product integrity & uniformity of colour, texture & quality.

4.3. Mock-ups

4.3.1. Build mock-ups of sliding door to verify selections made under sample submittals and to demonstrate aesthetic effects and set quality standards for materials and execution to the Engineer's satisfaction. Approved mock-ups may be incorporated into the finished work, where directed by the Engineer.

5. Testing & Commissioning

5.1. Factory Inspection & Tests

5.1.1. Arrange for Employer and Employer’s representative visits (minimum of two visits) for material/product/prototype reviews or tests to factories or sites as agreed with the Engineer.

5.1.2. The contractor shall inform the Engineer of test/inspection date, time and location at least 4 weeks in advance. The cost of travel, accommodation and other related expenses shall be as included in tender.

5.2. Site Inspection

5.2.1. Inspection upon completion of work, all Sliding doors shall be carefully tested and examined for faults in application, shall be corrected immediately.

ATK-ARC-014 – Overhead Sectional Doors

1. General

1.1. Introduction

1.1.1. This section specifies supply and installation of Overhead Sectional Doors, as indicated in the Tender drawings and as required in completion of the construction works.

1.2. Related Documents

1.2.1. This section to be read in conjunction with Employer's requirements and general provision of contract, including General, Supplementary and Special Conditions of Tender Documents.

1.3. Codes and Standards

1.3.1. All materials, workmanship and testing shall be in accordance with the following documents, in their latest revisions. This is not exhaustive and contractor shall verify the listing and supplement with the latest applicable document or documents.

1.3.2. Comply with the requirements of the reference standards noted herein, except where more stringent requirements are listed herein or otherwise required by the Tender Documents.

1.3.3. Where materials, finishes, fittings, equipment and building components that comply with standards other than those included in these specifications are proposed, the Contractor shall adequately demonstrate that such items are of an equal quality and standard and are fit for their intended purpose, through submission of internationally recognized and authorized test methods, technical and certification reports (in compliance with assessment/evaluation criteria based on standards already included in these specifications), to the approval of engineer, prior to use of such items.

1.3.4. Conform to the requirements of the following Standards and Codes of Practice where applicable:

1.3.4.1. National Building Code (NBC) of India 2016 Volume 1 & 2.

1.3.4.2. IS 277: Specification for galvanized steel sheets (Plain and corrugated)

1.3.4.3. IS 1038: Steel doors, windows and ventilators

1.3.4.4. IS 1081: Code of practice for fixing and glazing of metal (Steel and aluminium) doors, windows and ventilators

1.3.4.5. IS 4351: Specification for steel door frames

1.3.4.6. IS 7452: Specification for hot rolled steel sections for doors, windows and ventilators.

1.3.4.7. BS EN 12433-1:2000: Industrial, commercial, and garage doors and gates. Terminology. Types of doors

1.3.4.8. BS EN 12433-2:2000: Industrial, commercial, and garage doors and gates. Terminology. Parts of doors

1.3.4.9. BS EN 12425:2000: Industrial, commercial and garage doors and gates. Resistance to water penetration. Classification

1.3.4.10. BS EN 12489:2000: Industrial, commercial and garage doors and gates. Resistance to water penetration. Test method

1.3.4.11. BS EN 12426:2000: Industrial, commercial and garage doors and gates. Air permeability. Classification

1.3.4.12. BS EN 12427:2000: Industrial, commercial and garage doors and gates. Air permeability. Test method

1.3.4.13. BS EN ISO 717-1:2013: Acoustics. Rating of sound insulation in buildings and of building elements

1.3.4.14. BS EN 12424:2000: Industrial, commercial and garage doors and gates. Resistance to wind load. Classification

1.3.4.15. BS EN 12428:2013: Industrial, commercial and garage doors. Thermal transmittance. Requirements for the calculation

1.3.4.16. BS EN 13241:2003+A2:2016: Industrial, commercial, garage doors and gates. Product standard, performance characteristics

1.3.4.17. NFPA 101 Life Safety Code.

1.4. Submittals

1.4.1. Technical Data

1.4.1.1. Comprehensive technical data viz., technical literature, data sheets, product catalogue & brochure, installation instructions, etc. to be provided for Engineer's approval.

1.4.2. Shop drawings

1.4.2.1. Shop drawings shall be submitted to Engineer for approval, prior to start of work at site.

1.4.2.2. Shop drawings shall include all details w.r.t. the material & related product's installation, to the scale & detailing as instructed by the Engineer.

1.4.2.3. Shop drawings shall be developed as per actual measurements from the site & must depict the same in the drawings.

1.4.2.4. Shop drawings must comply to BS 1192:2007+A2:2016 Collaborative production of architectural, engineering and construction information & Code of practice.

1.4.3. Samples

1.4.3.1. Samples for Initial Selection: For each type of Overhead sectional door assembly or accessories as indicated. Include Samples of accessories involving colour selection.

1.4.4. Method Statement

1.4.4.1. Shall include method of construction/installation, Health and Safety management, testing methods, required relative humidity and temperature level and control systems, possible risk associated with the execution works and precautions to be considered, site control system and any other additional requirements as advised by the Engineer for the execution of works.

1.4.4.2. Method statement shall be approved by the Engineer prior to commencement of the works.

1.4.5. Installer Qualifications

1.4.5.1. Licensed installer specialized in fabricating and installing each type of ceiling, eligible to receive manufacturer's warranty.

1.4.6. Material/Product Certificates

1.4.6.1. Comprehensive tests performed by a qualified independent testing agency certifying that the product compiles with specified performance requirements.

1.4.7. Maintenance Data

1.4.7.1. Details of manufacturer's recommended maintenance procedures for each type of product/material to be included in the operation and maintenance manual.

1.4.8. Spare Materials/Parts

1.4.8.1. Provide the following extra materials that match installed products, packaged for storage with suitable labels for identification.

1.5. Warranty / Guarantee

1.5.1. As per the Employer's requirements and general provision of contract, including General, Supplementary and Special Conditions of Contract Documents.

1.6. Performance Requirements

1.6.1. All finishes, materials and components shall comply with requirements of NFPA 101 Life Safety Code, chapter 10 – Interior Finish, Contents and Furnishings with respect to classification of materials, flame spread, smoke generation and other requirements.

1.7. Environment & Sustainability

1.7.1. Recycling Strategy document for the material or product shall be submitted to the Engineer for approval.

1.7.2. LEED Requirements:

1.7.2.1. All the specifications in this document are for USGBC LEED BD+C submission and are applicable to all materials & procedures followed for the project. The specification as stated here should be adhered to the word. In case of a dispute LEED guidelines shall be followed. Brochures, product specification sheets and/or certificates as specified in this document should be submitted along with the material procurement. All contractors, manufactures must submit documents, records and other submittal related to the different credits as mentioned in the specification.

1.7.2.2. Manufacturer product conforming to ISO 14044 or Environmental Product Declarations which conform to ISO 14025, 14040, 14044, and EN 15804 or ISO 21930 and have at least a cradle to gate scope. (If applicable)**.**

1.7.2.3. Manufacturer Material Ingredient Reporting by any of the listed below documents (If applicable):

1.7.2.3.1. Health Product Declaration. The end use product has a published, complete Health Product Declaration with full disclosure of known hazards in compliance with the Health Product Declaration Open Standard.

1.7.2.3.2. Cradle to Cradle Certificate. The end use product has been certified at the Cradle to Cradle v2 Basic level or Cradle to Cradle v3 Bronze level.

1.7.2.3.3. REACH Optimization. End use products and materials that do not contain substances that meet REACH criteria for substances of very high concern.

1.7.2.4. Manufacturer Raw Material Source and Extraction Reporting by any of the listed below activity (If applicable):

1.7.2.4.1. Third-party verified corporate sustainability reports (CSR) which include environmental impacts of extraction operations and activities associated with the manufacturer's product and the product's supply chain**.**

1.7.3. Self-declared reports from manufacturers that have publicly released a report from their raw material suppliers which include raw material supplier extraction locations

1.8. Manufacturers

1.8.1. Manufacturer to be accredited and registered under BS EN ISO 9001.

1.8.2. List of preferred or acceptable Manufacturers attached herewith as Annexure 1.

2. Products

2.1. Materials

2.1.1. Material/Product Specification as per Standard attached herewith as Annexure A.14.

2.2. Colours

2.2.1. Colours as per Standard attached herewith as Annexure B.01.

3. Execution

3.1. Examination

3.1.1. Verification of existing conditions before starting work. Installer must examine the location and advise the Contractor of any site conditions unacceptable for proper installation of product. These conditions include but are not limited to the following:

3.2. Preparation

3.2.1. Coordinate dimensions, tolerance, and method of attachment with other work.

3.2.2. Manufacturer shall offer a Factory Acceptance Test to the Employer with fully functioning door prior to shipment.

3.3. Fabrication

3.3.1. General: Factory fabricated door components to designs, profiles, sizes, and thicknesses indicated and to comply with indicated standards. Provide Overhead sectional doors as prefabricated assemblies. Complete fabrication, assembly, finishing, hardware application, and other work before shipment to Project site.

3.4. Installation

3.4.1. Install Overhead Sectional Doors in accordance with manufacturer's printed instructions. Set units' level, plumb, and with uniform hairline joints. Anchor securely into place. Use only factory-trained installers. All Operating Panels & Supports shall be installed by Vendor.

4. Quality Control

4.1. Delivery, Storage and Handling

4.1.1. All building materials shall be stored on the building site in such a way as to prevent deterioration or the loss or impairment of their structural and other essential properties. Comply to National Building Code (NBC) of India 2016 Volume 1; Part 7 Construction Management, Practices and Safety.

4.2. Source Limitation

4.2.1. All materials & related products shall be from the same manufacturer to ensure material/product integrity & uniformity of colour, texture & quality.

5. Testing & Commissioning

5.1. Factory Inspection & Tests

5.1.1. Arrange for Employer and Employer's representative visits (minimum of two visits) for material/product/prototype reviews or tests to factories or sites as agreed with the Engineer.

5.1.2. The contractor shall inform the Engineer of test/inspection date, time and location at least 4 weeks in advance. The cost of travel, accommodation and other related expenses shall be as included in tender.

5.2. Site Inspection

5.2.1. Inspection upon completion of work, Overhead Sectional Doors shall be carefully tested and examined for faults in operation, shall be corrected immediately.

ATK-ARC-015 – Roller Shutter

1. General

1.1. Introduction

1.1.1. This section specifies supply and installation of Roller Shutter; High Speed Roller Shutter, Fire & Smoke Curtain, as indicated in the Tender drawings and as required in completion of the construction works.

1.2. Related Documents

1.2.1. This section to be read in conjunction with Employer's requirements and general provision of contract, including General, Supplementary and Special Conditions of Tender Documents.

1.3. Codes and Standards

1.3.1. All materials, workmanship and testing shall be in accordance with the following documents, in their latest revisions. This is not exhaustive and contractor shall verify the listing and supplement with the latest applicable document or documents.

1.3.2. Comply with the requirements of the reference standards noted herein, except where more stringent requirements are listed herein or otherwise required by the Tender Documents.

1.3.3. Where materials, finishes, fittings, equipment and building components that comply with standards other than those included in these specifications are proposed, the Contractor shall adequately demonstrate that such items are of an equal quality and standard and are fit for their intended purpose, through submission of internationally recognized and authorized test methods, technical and certification reports (in compliance with assessment/evaluation criteria based on standards already included in these specifications), to the approval of engineer, prior to use of such items.

1.3.4. Conform to the requirements of the following Standards and Codes of Practice where applicable:

1.3.4.1. National Building Code (NBC) of India 2016 Volume 1 & 2.

1.3.4.2. IS 277: Specification for galvanized steel sheets (Plain and corrugated)

1.3.4.3. IS 1038: Steel doors, windows and ventilators

1.3.4.4. IS 1081: Code of practice for fixing and glazing of metal (Steel and aluminium) doors, windows and ventilators

1.3.4.5. IS 4351: Specification for steel door frames

1.3.4.6. IS 7452: Specification for hot rolled steel sections for doors, windows and ventilators.

1.3.4.7. ASTM E119 - 16a Standard Test Methods for Fire Tests of Building Construction and Materials

1.3.4.8. ASTM E84 – 17 Standard Test Method for Surface Burning Characteristics of Building Materials

1.3.4.9. UL 1784 Standard for Air Leakage Tests of Door Assemblies and Other Opening Protectives

1.3.4.10. UL 10D Standard for Fire Tests of Fire Protective Curtain Assemblies

1.3.4.11. BS EN 12433-1:2000: Industrial, commercial, and garage doors and gates. Terminology. Types of doors

1.3.4.12. BS EN 12433-2:2000: Industrial, commercial, and garage doors and gates. Terminology. Parts of doors

1.3.4.13. BS EN 12425:2000: Industrial, commercial and garage doors and gates. Resistance to water penetration. Classification

1.3.4.14. BS EN 12489:2000: Industrial, commercial and garage doors and gates. Resistance to water penetration. Test method

1.3.4.15. BS EN 12426:2000: Industrial, commercial and garage doors and gates. Air permeability. Classification

1.3.4.16. BS EN 12427:2000: Industrial, commercial and garage doors and gates. Air permeability. Test method

1.3.4.17. BS EN ISO 717-1:2013: Acoustics. Rating of sound insulation in buildings and of building elements

1.3.4.18. BS EN 12424:2000: Industrial, commercial and garage doors and gates. Resistance to wind load. Classification

1.3.4.19. BS EN 12428:2013: Industrial, commercial and garage doors. Thermal transmittance. Requirements for the calculation

1.3.4.20. BS EN 13241:2003+A2:2016: Industrial, commercial, garage doors and gates. Product standard, performance characteristics

1.3.4.21. BS: 476 Fire tests on building materials and structures. Method for determination of the fire resistance of non-loadbearing elements of construction

1.3.4.22. BS EN 16034:2014 Pedestrian doorsets, industrial, commercial, garage doors and openable windows. Product standard, performance characteristics. Fire resisting and/or smoke control characteristics

1.3.4.23. NFPA 701 Standard Methods of Fire Tests for Flame Propagation of Textiles and Films

1.3.4.24. NFPA 101 Life Safety Code.

1.4. Submittals

1.4.1. Technical Data

1.4.1.1. Comprehensive technical data viz., technical literature, data sheets, product catalogue & brochure, installation instructions, etc. to be provided for Engineer’s approval.

1.4.2. Shop drawings

1.4.2.1. Shop drawings shall be submitted to Engineer for approval, prior to start of work at site.

1.4.2.2. Shop drawings shall include all details w.r.t. the material & related product’s installation, to the scale & detailing as instructed by the Engineer.

1.4.2.3. Shop drawings shall be developed as per actual measurements from the site & must depict the same in the drawings.

1.4.2.4. Shop drawings must comply to BS 1192:2007+A2:2016 Collaborative production of architectural, engineering and construction information & Code of practice.

1.4.3. Samples

1.4.3.1. Samples for Initial Selection: For each type of Overhead sectional door assembly or accessories as indicated. Include Samples of accessories involving colour selection.

1.4.4. Method Statement

1.4.4.1. Shall include method of construction/installation, Health and Safety management, testing methods, required relative humidity and temperature level and control systems, possible risk associated with the execution works and precautions to be considered, site control system and any other additional requirements as advised by the Engineer for the execution of works.

1.4.4.2. Method statement shall be approved by the Engineer prior to commencement of the works.

1.4.5. Installer Qualifications

1.4.5.1. Licensed installer specialized in fabricating and installing each type of ceiling, eligible to receive manufacturer's warranty.

1.4.6. Material/Product Certificates

1.4.6.1. Comprehensive tests performed by a qualified independent testing agency certifying that the product compiles with specified performance requirements.

1.4.7. Maintenance Data

1.4.7.1. Details of manufacturer's recommended maintenance procedures for each type of product/material to be included in the operation and maintenance manual.

1.4.8. Spare Materials/Parts

1.4.8.1. Provide the following extra materials that match installed products, packaged for storage with suitable labels for identification.

1.5. Warranty / Guarantee

1.5.1. As per the Employer's requirements and general provision of contract, including General, Supplementary and Special Conditions of Contract Documents.

1.6. Performance Requirements

1.6.1. All finishes, materials and components shall comply with requirements of NFPA 101 Life Safety Code, chapter 10 – Interior Finish, Contents and Furnishings with respect to classification of materials, flame spread, smoke generation and other requirements.

1.6.2. Fire & Smoke Curtain:

1.6.2.1. Audio-Visual: An audio siren and flashing beacon offer a warning that the fire curtain is about to descend. This warning will continue until the fire curtain is in its fire-ready position (closed).

1.6.2.2. Battery Back-Up: If the power supply fails, the in-built battery back-up system maintains operation of the fire curtain for up to 48 hours. If the power isn't restored, the fire curtain will gravity fall to its fire-ready position and alerting the user of the power failure.

1.6.2.3. Delay Before Descent: When a fire alarm is activated, fire curtains can delay their descent for up to 3 minutes. The fire curtain will remain in its retracted position, allowing safe evacuation of a building. After a set period of time the fire curtain will automatically close.

1.6.2.4. Split Drop Delay: Once the alarm is activated, the fire curtain will descend to its split drop position. The curtain can remain in this position for up to 3 minutes. It also acts as a smoke curtain, stopping the spread of smoke and allowing safe evacuation before descent.

1.6.2.5. Emergency Retract: If the fire curtain has been activated and people are trapped, the emergency retract switch can be activated which raises the fire curtain allow people to escape. The fire curtain will once again descend giving full fire protection.

1.6.2.6. Double Knock Fire Activation: The Fire Control Panel can operate a double knock fire alarm activation. For instance, upon alarm activation, the fire curtain will descend to the split drop delay position and act solely as a smoke curtain. The fire curtain will only fall to the floor once a second signal is received, either from a second fire signal or a standalone relay based smoke or heat detector.

1.6.2.7. Building Management System: The fire control panel allows fire curtains to be connected to a building management system. This means that the control room in a large building complex can monitor designated areas to see the position of all the fire curtains.

1.6.2.8. Obstacle Detection: A photocell with a flashing beacon and sounder is connected to the Fire Control Panel. If an object is under the fire curtain, the alarm will activate until the object is removed.

1.7. Environment & Sustainability

1.7.1. Recycling Strategy document for the material or product shall be submitted to the Engineer for approval.

1.7.2. LEED Requirements:

1.7.2.1. All the specifications in this document are for USGBC LEED BD+C submission and are applicable to all materials & procedures followed for the project. The specification as stated here should be adhered to the word. In case of a dispute LEED guidelines shall be followed. Brochures, product specification sheets and/or certificates as specified in this document should be submitted along with the material procurement. All contractors, manufactures must submit documents, records and other submittal related to the different credits as mentioned in the specification.

1.7.2.2. Manufacturer product conforming to ISO 14044 or Environmental Product Declarations which conform to ISO 14025, 14040, 14044, and EN 15804 or ISO 21930 and have at least a cradle to gate scope. (If applicable)**.**

1.7.2.3. Manufacturer Material Ingredient Reporting by any of the listed below documents (If applicable):

1.7.2.3.1. Health Product Declaration. The end use product has a published, complete Health Product Declaration with full disclosure of known hazards in compliance with the Health Product Declaration Open Standard.

1.7.2.3.2. Cradle to Cradle Certificate. The end use product has been certified at the Cradle to Cradle v2 Basic level or Cradle to Cradle v3 Bronze level.

1.7.2.3.3. REACH Optimization. End use products and materials that do not contain substances that meet REACH criteria for substances of very high concern.

1.7.2.4. Manufacturer Raw Material Source and Extraction Reporting by any of the listed below activity (If applicable):

1.7.2.4.1. Third-party verified corporate sustainability reports (CSR) which include environmental impacts of extraction operations and activities associated with the manufacturer’s product and the product’s supply chain**.**

1.7.3. Self-declared reports from manufacturers that have publicly released a report from their raw material suppliers which include raw material supplier extraction locations

1.8. Manufacturers

1.8.1. Manufacturer to be accredited and registered under BS EN ISO 9001.

1.8.2. List of preferred or acceptable Manufacturers attached herewith as Annexure 1.

2. Products

2.1. Materials

2.1.1. Material/Product Specification as per Standard attached herewith as Annexure A.15.

2.1.2. Fire & Smoke Curtain:

2.1.2.1. Fire Curtain to withstand direct flame at temperatures up to 1065°C for periods of up to 2 hours with no flame penetration and low thermal shrinkage.

2.1.2.2. Fire curtain to be strong, flexible and with no toxic gassing or surface flaming when exposed to fire. It will also provide an insulative barrier between the fire and unexposed side.

2.2. Colours

2.2.1. Colours as per Standard attached herewith as Annexure B.01.

3. Execution

3.1. Examination

3.1.1. Verification of existing conditions before starting work. Installer must examine the location and advise the Contractor of any site conditions unacceptable for proper installation of product. These conditions include but are not limited to the following:

3.2. Preparation

3.2.1. Coordinate dimensions, tolerance, and method of attachment with other work.

3.2.2. Manufacturer shall offer a Factory Acceptance Test to the Employer with fully functioning door prior to shipment.

3.3. Fabrication

3.3.1. General: Factory fabricated door components to designs, profiles, sizes, and thicknesses indicated and to comply with indicated standards. Provide Roller shutter as prefabricated assemblies. Complete fabrication, assembly, finishing, hardware application, and other work before shipment to Project site.

3.4. Installation

3.4.1. Install Roller Shutter in accordance with manufacturer's printed instructions. Set units' level, plumb, and with uniform hairline joints. Anchor securely into place. Use only factory-trained installers. All Operating Panels & Supports shall be installed by the Vendor.

4. Quality Control

4.1. Delivery, Storage and Handling

4.1.1. All building materials shall be stored on the building site in such a way as to prevent deterioration or the loss or impairment of their structural and other essential properties. Comply to National Building Code (NBC) of India 2016 Volume 1; Part 7 Construction Management, Practices and Safety.

4.2. Source Limitation

4.2.1. All materials & related products shall be from the same manufacturer to ensure material/product integrity & uniformity of colour, texture & quality.

5. Testing & Commissioning

5.1. Factory Inspection & Tests

5.1.1. Arrange for Employer and Employer's representative visits (minimum of two visits) for material/product/prototype reviews or tests to factories or sites as agreed with the Engineer.

5.1.2. The contractor shall inform the Engineer of test/inspection date, time and location at least 4 weeks in advance. The cost of travel, accommodation and other related expenses shall be as included in tender.

5.2. Site Inspection

5.2.1. Inspection upon completion of work, Roller Shutter shall be carefully tested and examined for faults in operation, shall be corrected immediately.

ATK-ARC-016 – Loading Dock Assembly

1. General

1.1. Introduction

1.1.1. This section specifies supply and installation of Loading Dock systems, as indicated in the Tender drawings and as required in completion of the construction works.

1.1.1.1. Climate Dock Shelter

1.1.1.2. Dock Leveller

1.1.1.3. Loading Dock House

1.2. Related Documents

1.2.1. This section to be read in conjunction with Employer's requirements and general provision of contract, including General, Supplementary and Special Conditions of Tender Documents.

1.3. Codes and Standards

1.3.1. All materials, workmanship and testing shall be in accordance with the following documents, in their latest revisions. This is not exhaustive and contractor shall verify the listing and supplement with the latest applicable document or documents.

1.3.2. Comply with the requirements of the reference standards noted herein, except where more stringent requirements are listed herein or otherwise required by the Tender Documents.

1.3.3. Where materials, finishes, fittings, equipment and building components that comply with standards other than those included in these specifications are proposed, the Contractor shall adequately demonstrate that such items are of an equal quality and standard and are fit for their intended purpose, through submission of internationally recognized and authorized test methods, technical and certification reports (in compliance with assessment/evaluation criteria based on standards already included in these specifications), to the approval of engineer, prior to use of such items.

1.3.4. Conform to the requirements of the following Standards and Codes of Practice where applicable:

1.3.4.1. National Building Code (NBC) of India 2016 Volume 1 & 2.

1.3.4.2. ANSI MH14.1-1987 - Loading Dock Levellers and Dock boards

1.3.4.3. ANSI MH29.1 - Industrial Scissor Lifts.

1.3.4.4. ANSI MH30.1 - Industrial Loading Dock Boards (Ramps).

1.3.4.5. ANSI MH30.3 - Vehicle Restraining Devices (Safety, Performance, and Testing).

1.3.4.6. ASTM D6413 / D6413M Standard Test Method for Flame Resistance of Textiles

1.3.4.7. FED-STD-191 A Textile Test Methods

1.3.4.8. EN 1398 CEE machine

1.3.4.9. directive

1.3.4.10. EN 605529 Safety

1.3.4.11. IKEA Engineering and construction manual Section 4.1.3.

1.3.4.12. IKEA Standard 04-0140 Climate Dock Shelter, Dock Leveller and Exterior Loading Dock House.

1.3.4.13. BS EN ISO 9001: Quality management systems. Requirements

1.3.4.14. NFPA 101 Life Safety Code.

1.4. Submittals

1.4.1. Technical Data

1.4.1.1. Comprehensive technical data viz., technical literature, data sheets, product catalogue & brochure, installation instructions, etc. to be provided for Engineer's approval.

1.4.2. Shop drawings

1.4.2.1. Shop drawings shall be submitted to Engineer for approval, prior to start of work at site.

1.4.2.2. Shop drawings shall include all details w.r.t. the material & related product's installation, to the scale & detailing as instructed by the Engineer.

1.4.2.3. Shop drawings shall be developed as per actual measurements from the site & must depict the same in the drawings.

1.4.2.4. Shop drawings must comply to BS 1192:2007+A2:2016 Collaborative production of architectural, engineering and construction information & Code of practice.

1.4.3. Samples

1.4.3.1. Samples for Initial Selection

1.4.3.1.1. For loading dock assembly as indicated. Include Samples of accessories involving colour selection.

1.4.3.2. Samples for Verification

1.4.3.2.1. Full-size units of loading dock equipment and for each colour and finish required.

1.4.4. Method Statement

1.4.4.1. Shall include method of construction/installation, Health and Safety management, testing methods, required relative humidity and temperature level and control systems, possible risk associated with the execution works and precautions to be considered, site control system and any other additional requirements as advised by the Engineer for the execution of works.

1.4.4.2. Method statement shall be approved by the Engineer prior to commencement of the works.

1.4.5. Installer Qualifications

1.4.5.1. Licensed installer specialized in fabricating and installing each type of ceiling, eligible to receive manufacturer's warranty.

1.4.6. Material/Product Certificates

1.4.6.1. Comprehensive tests performed by a qualified independent testing agency certifying that the product compiles with specified performance requirements.

1.4.7. Maintenance Data

1.4.7.1. Details of manufacturer's recommended maintenance procedures for each type of product/material to be included in the operation and maintenance manual.

1.4.8. Spare Materials/Parts

Provide the following extra materials that match installed products, packaged for storage with suitable labels for identification.

1.5. Warranty / Guarantee

1.5.1.1. As per the Employer's requirements and general provision of contract, including General, Supplementary and Special Conditions of Contract Documents.

1.6. Performance Requirements

1.6.1. All finishes, materials and components shall comply with requirements of NFPA 101 Life Safety Code, chapter 10 – Interior Finish, Contents and Furnishings with respect to classification of materials, flame spread, smoke generation and other requirements.

1.7. Environment & Sustainability

1.7.1. Recycling Strategy document for the material or product shall be submitted to the Engineer for approval.

1.7.2. LEED Requirements:

1.7.2.1. All the specifications in this document are for USGBC LEED BD+C submission and are applicable to all materials & procedures followed for the project. The specification as stated here should be adhered to the word. In case of a dispute LEED guidelines shall be followed. Brochures, product specification sheets and/or certificates as specified in this document should be submitted along with the material procurement. All contractors, manufactures must submit documents, records and other submittal related to the different credits as mentioned in the specification.

1.7.2.2. Manufacturer product conforming to ISO 14044 or Environmental Product Declarations which conform to ISO 14025, 14040, 14044, and EN 15804 or ISO 21930 and have at least a cradle to gate scope. (If applicable)**.**

1.7.2.3. Manufacturer Material Ingredient Reporting by any of the listed below documents (If applicable):

1.7.2.3.1. Health Product Declaration. The end use product has a published, complete Health Product Declaration with full disclosure of known hazards in compliance with the Health Product Declaration Open Standard.

1.7.2.3.2. Cradle to Cradle Certificate. The end use product has been certified at the Cradle to Cradle v2 Basic level or Cradle to Cradle v3 Bronze level.

1.7.2.3.3. REACH Optimization. End use products and materials that do not contain substances that meet REACH criteria for substances of very high concern.

1.7.2.4. Manufacturer Raw Material Source and Extraction Reporting by any of the listed below activity (If applicable):

1.7.2.4.1. Third-party verified corporate sustainability reports (CSR) which include environmental impacts of extraction operations and activities associated with the manufacturer's product and the product's supply chain**.**

1.7.3. Self-declared reports from manufacturers that have publicly released a report from their raw material suppliers which include raw material supplier extraction locations

1.8. Manufacturers

1.8.1. Manufacturer to be accredited and registered under BS EN ISO 9001.

1.8.2. List of preferred or acceptable Manufacturers attached herewith as Annexure 1.

2. Products

2.1. Materials

2.1.1. Material/Product Specification as per Standard attached herewith as Annexure A.16.

2.1.2. Dock house shall be provided with Traffic Light for operation monitoring & control.

2.2. Colours

2.2.1. Colours as per Standard attached herewith as Annexure B.01.

3. Execution

3.1. Examination

3.1.1. Verification of Conditions: Examine areas and conditions under which the work of this Section will be performed. Do not proceed with the work until unsatisfactory conditions have been corrected. Commencement of work implies acceptance of all areas and conditions.

3.2. Installation

3.2.1. Install products in prepared opening in accordance with manufacturer's instructions. Set square and level. Anchor unit securely, flush with dock. Weld back of dock leveller to pit frame. Touch-up welds with matching paint.

4. Quality Control

4.1. Delivery, Storage and Handling

4.1.1. All building materials shall be stored on the building site in such a way as to prevent deterioration or the loss or impairment of their structural and other essential properties. Comply to National Building Code (NBC) of India 2016 Volume 1; Part 7 Construction Management, Practices and Safety.

4.2. Source Limitation

4.2.1. All materials & related products shall be from the same manufacturer to ensure material/product integrity & uniformity of colour, texture & quality.

4.3. Mock-ups

Build mock-ups of each ceiling type to verify selections made under sample submittals and to demonstrate aesthetic effects and set quality standards for materials and execution to the Engineer's satisfaction. Approved mock-ups may be incorporated into the finished work, where directed by the Engineer.

5. Testing and Commissioning

5.1. Factory Inspection & Tests

5.1.1. Arrange for Employer and Employer’s representative visits (minimum of two visits) for material/product/prototype reviews or tests to factories or sites as agreed with the Engineer.

5.1.2. The contractor shall inform the Engineer of test/inspection date, time and location at least 4 weeks in advance. The cost of travel, accommodation and other related expenses shall be as included in tender.

5.2. Site Inspection

5.2.1. Inspection upon completion of work, all Ceiling Works shall be carefully tested and examined for faults in application, shall be corrected immediately.

ATK-ARC-017 – Wall, Column & Edge Protection

1. General

1.1. Introduction

1.1.1. This section specifies supply and installation of Wall & Edge Protection, as indicated in the Tender drawings and as required in completion of the construction works.

1.1.2. Railings & Balustrades

1.1.3. Wall Protection Systems

1.1.4. Edge Protection Systems

1.2. Related Documents

1.2.1. This section to be read in conjunction with Employer's requirements and general provision of contract, including General, Supplementary and Special Conditions of Tender Documents.

1.3. Codes and Standards

1.3.1. All materials, workmanship and testing shall be in accordance with the following documents, in their latest revisions. This is not exhaustive and contractor shall verify the listing and supplement with the latest applicable document or documents.

1.3.2. Comply with the requirements of the reference standards noted herein, except where more stringent requirements are listed herein or otherwise required by the Tender Documents.

1.3.3. Where materials, finishes, fittings, equipment and building components that comply with standards other than those included in these specifications are proposed, the Contractor shall adequately demonstrate that such items are of an equal quality and standard and are fit for their intended purpose, through submission of internationally recognized and authorized test methods, technical and certification reports (in compliance with assessment/evaluation criteria based on standards already included in these specifications), to the approval of engineer, prior to use of such items.

1.3.4. Conform to the requirements of the following Standards and Codes of Practice where applicable:

1.3.4.1. National Building Code (NBC) of India 2016 Volume 1 & 2.

1.3.4.2. IS 7537 - Road traffic signals

1.3.4.3. BS 873-1:1983 - Road traffic signs and internally illuminated bollards. Methods of test.

1.3.4.4. BS 8442:2015 - Miscellaneous road traffic signs and devices. Requirements and test methods.

1.3.4.5. BS 8081 - Code of practice for grouted anchors.

1.3.4.6. BS EN 1090-5:2017. Execution of steel structures and aluminium structures.

1.3.4.7. BS EN 17104 - Thermoplastics rigid protective wallcovering panels.

1.3.4.8. EN ISO 11925-2:2010, EN 13823:2010 - Reaction to fire tests

1.3.4.9. EN 13501-1: Fire Test to Building Material

1.3.4.10. BS EN ISO 175:2010 - Plastics. Methods of test for the determination of the effects of immersion in liquid chemicals

1.3.4.11. ISO 22196:2011 - Measurement of antibacterial activity on plastics and other non-porous surfaces.

1.3.4.12. ISO 846:1997 - Plastics - Evaluation of the action of microorganisms

1.3.4.13. BS EN 573-3:2013 - Aluminium and aluminium alloys

1.3.4.14. ASTM F1141/F1141M - Standard Specification for Wall Coverings.

1.3.4.15. ASTM F2656/F2656M - Standard Test Method for Crash Testing of Vehicle Security Barriers.

1.3.4.16. ASTM E84 – 16 - Standard Test Method for Surface Burning Characteristics of Building Materials.

1.3.4.17. ASTM B221-14 - Aluminium and Aluminium-Alloy Extruded Bars, Rods, Wire, Shapes, and Tubes

1.3.4.18. ASTM B221M-13 - Aluminium and Aluminium-Alloy Extruded Bars, Rods, Wire, Shapes, and Tubes (Metric)

1.3.4.19. ASTM D256-10 - Impact Resistance of Plastics

1.3.4.20. ASTM D635-10 - Rate of Burning and/or Extent and Time of Burning of Self-Supporting Plastics in a Horizontal Position

1.3.4.21. ASTM E84-14 - Surface Burning Characteristics of Building Materials

1.3.4.22. NFPA 101 Life Safety Code.

1.4. Technical Submittals

1.4.1. Technical Data

1.4.1.1. Comprehensive technical data viz., technical literature, data sheets, product catalogue & brochure, installation instructions, etc. to be provided for Engineer's approval.

1.4.2. Shop drawings

1.4.2.1. Shop drawings shall be submitted to Engineer for approval, prior to start of work at site.

1.4.2.2. Shop drawings shall include all details w.r.t. the material & related product's installation, to the scale & detailing as instructed by the Engineer.

1.4.2.3. Shop drawings shall be developed as per actual measurements from the site & must depict the same in the drawings.

1.4.2.4. Shop drawings must comply to BS 1192:2007+A2:2016 Collaborative production of architectural, engineering and construction information & Code of practice.

1.4.3. Samples

1.4.3.1. Samples for Initial Selection: For each type of wall & edge protection, as indicated. Include Samples of accessories involving colour selection.

1.4.3.2. Samples for Verification:

1.4.3.3. Full-size units of each wall & edge protection and for each colour and finish required.

1.4.4. Method Statement

1.4.4.1. Shall include method of construction/installation, Health and Safety management, testing methods, required relative humidity and temperature level and control systems, possible risk associated with the execution works and precautions to be considered, site control system and any other additional requirements as advised by the Engineer for the execution of works.

1.4.4.2. Method statement shall be approved by the Engineer prior to commencement of the works.

1.4.5. Installer Qualifications

1.4.5.1. Licensed installer specialized in fabricating and installing each type of ceiling, eligible to receive manufacturer's warranty.

1.4.6. Material/Product Certificates

1.4.6.1. Comprehensive tests performed by a qualified independent testing agency certifying that the product compiles with specified performance requirements.

1.4.7. Maintenance Data

1.4.7.1. Details of manufacturer's recommended maintenance procedures for each type of product/material to be included in the operation and maintenance manual.

1.4.8. Spare Materials/Parts

Provide the following extra materials that match installed products, packaged for storage with suitable labels for identification.

1.5. Warranty / Guarantee

1.5.1. As per the Employer's requirements and general provision of contract, including General, Supplementary and Special Conditions of Contract Documents.

1.6. Performance Requirements

1.6.1. All finishes, materials and components shall comply with requirements of NFPA 101 Life Safety Code, chapter 10 – Interior Finish, Contents and Furnishings with respect to classification of materials, flame spread, smoke generation and other requirements.

1.7. Environment & Sustainability

1.7.1. Recycling Strategy document for the material or product shall be submitted to the Engineer for approval.

1.7.2. LEED Requirements:

1.7.2.1. All the specifications in this document are for USGBC LEED BD+C submission and are applicable to all materials & procedures followed for the project. The specification as stated here should be adhered to the word. In case of a dispute LEED guidelines shall be followed. Brochures, product specification sheets and/or certificates as specified in this document should be submitted along with the material procurement. All contractors, manufactures must submit documents, records and other submittal related to the different credits as mentioned in the specification.

1.7.2.2. Manufacturer product conforming to ISO 14044 or Environmental Product Declarations which conform to ISO 14025, 14040, 14044, and EN 15804 or ISO 21930 and have at least a cradle to gate scope. (If applicable)**.**

1.7.2.3. Manufacturer Material Ingredient Reporting by any of the listed below documents (If applicable):

1.7.2.3.1. Health Product Declaration. The end use product has a published, complete Health Product Declaration with full disclosure of known hazards in compliance with the Health Product Declaration Open Standard. Cradle to Cradle Certificate. The end use product has been certified at the Cradle to Cradle v2 Basic level or Cradle to Cradle v3 Bronze level. REACH Optimization. End use products and materials that do not contain substances that meet REACH criteria for substances of very high concern.

1.7.2.4. Manufacturer Raw Material Source and Extraction Reporting by any of the listed below activity (If applicable):

1.7.2.5. Third-party verified corporate sustainability reports (CSR) which include environmental impacts of extraction operations and activities associated with the manufacturer's product and the product's supply chain**.** Self-declared reports from manufacturers that have publicly released a report from their raw material suppliers which include raw material supplier extraction locations

1.8. Manufacturers

1.8.1. Manufacturer to be accredited and registered under BS EN ISO 9001.

1.8.2. List of preferred or acceptable Manufacturers attached herewith as Annexure 1.

2. Products

2.1. Railings & Balustrades

2.1.1. Material/Product Specification as per 04-0130 of IKEA Standards attached herewith as Annexure 2.

2.1.2. External Staircase Railings & Balustrades (Type – 1)

2.1.2.1. Tread/Stringer Mounted for External Staircases

2.1.2.2. Height: 1000mm

2.1.2.3. Primary Balustrade (Post): Circular GI Pipe of 40mm Dia.

2.1.2.4. Secondary Balustrade: Circular GI Pipe of 20mm Dia.

2.1.2.5. Handrail Profile: Circular GI Pipe 40mm Dia.

2.1.2.6. Upper Guardrail Profile: Circular GI Pipe 40mm Dia.

2.1.2.7. Lower Guardrail Profile: Circular GI Pipe 30mm Dia.

2.1.3. Internal Staircase Railings & Balustrades (Type – 2)

2.1.3.1. Tread/Stringer Mounted for Internal Staircases

2.1.3.2. Height: 900mm (at Stair flight) & 1100mm (at Landing)

2.1.3.3. Primary Balustrade (Post): Circular GI Pipe of 40mm Dia.

2.1.3.4. Secondary Balustrade: Circular GI Pipe of 20mm Dia.

2.1.3.5. Handrail Profile: Circular GI Pipe 40mm Dia.

2.1.3.6. Upper Guardrail Profile: Circular GI Pipe 40mm Dia.

2.1.3.7. Lower Guardrail Profile: Circular GI Pipe 30mm Dia.

2.1.4. Internal Staircase Railings & Balustrades (Type – 3)

2.1.4.1. Wall Mounted for Internal Staircases

2.1.4.2. Height: 900mm (at Stair flight) & 1100mm (at Landing)

2.1.4.3. Handrail Profile: Circular GI Pipe 50mm diameter.

2.1.5. Internal Staircase Railings & Balustrades (Type – 4)

2.1.5.1. Tread/Stringer Mounted for Internal Customer Staircases

2.1.5.2. Top Rail: Height: 1500mm Profile: 40mm x 40mm GI L-Section.

2.1.5.3. Primary Rail: Height: 900mm Profile: Circular Wooden (Hardwood) of 50mm diameter.

2.1.5.4. Secondary Rail (Child): Height: 600mm Profile: Circular Wooden (Hardwood) of 30mm diameter.

2.1.5.5. Primary Balustrades: Profile: 40mm x 40mm GI L-Section.

2.1.5.6. Infill Panel: GI Perforated Sheet 3mm thick.

2.2. Wall Protection

2.2.1. Wall Protection Against Costumer Trolleys **(1)**

Particle Board with Laminate Finish

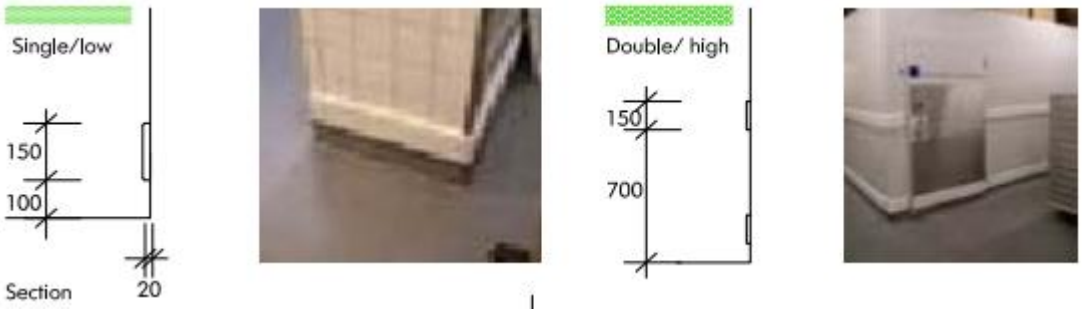


Type 1.1: Single/low, Height: 150mm, Thickness: 20mm, Hight from base: 100mm

Type 1.2: Double/high, Height: 150mm, Thickness: 20mm, Hight from base: 100mm and 700mm

2.2.2. Wall Protection in Kitchen Area, Washable **(2) – Refer IFS**

Particle Board with Laminate Finish



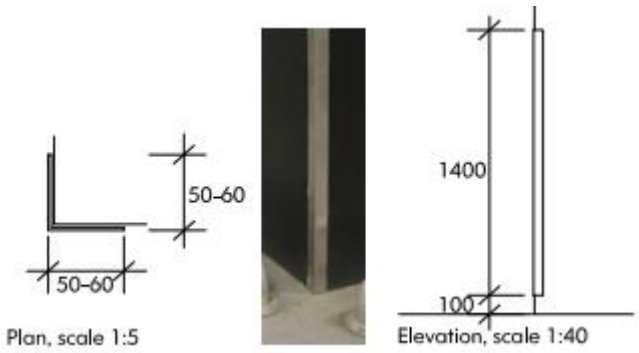
Type 2.1: Single/low, Height: 150mm, Thickness: 20mm, Hight from base: 100mm

Type 2.2: Double/high, Height: 150mm, Thickness: 20mm, Hight from base: 100mm and 700mm

2.3. Corner Protection

2.3.1. Corner Protection on all Outstanding Corners, Stainless Steel **(3.1)**

One-piece, surface mount corner guards. For hygienic, durable protection.



Material: Stainless Steel.

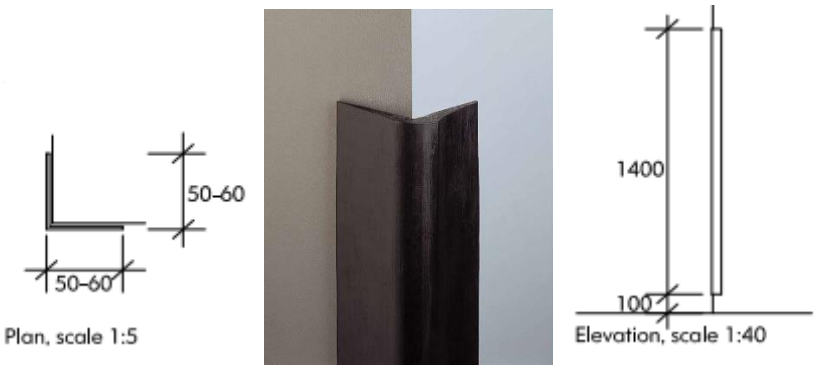
Type 3.1.1: Height: 1400mm, Thickness: 2mm, Hight from base: 100mm, Length: 60mm, Width: 60mm

Type 3.1.2: Height: 450mm, Thickness: 2mm, Hight from base: 350mm, Length: 60mm, Width: 60mm

Type 3.1.3: Height: 1150mm, Thickness: 2mm, Hight from base: 350mm, Length: 60mm, Width: 60mm

2.3.2. Corner Protection in Mechanical Areas **(3.2)**

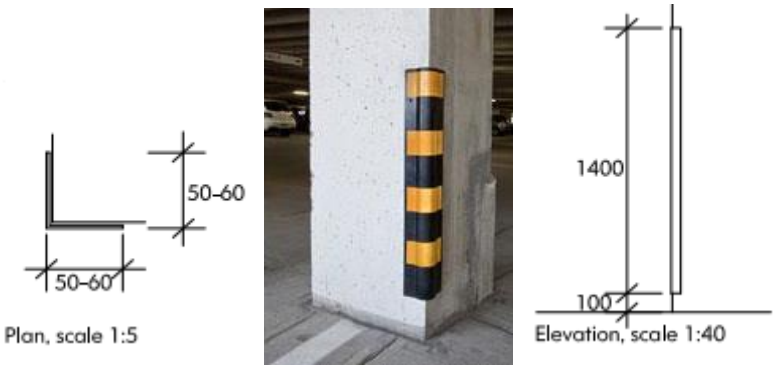
One-piece, surface mount corner guards incorporating impact absorption pads. Highly visible reflectors to alert vehicle users to potential hazards.



Type 3.2: Material: Rubber, Height: 1150mm, Thickness: 2mm, Hight from base: 350mm, Length: 60mm, Width: 60mm

2.3.3. Corner Protection at Parking areas **(3.3)**

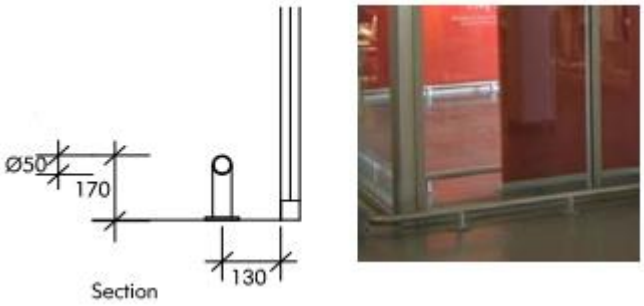
One-piece, surface mount corner guards incorporating impact absorption pads. Highly visible reflectors to alert vehicle users to potential hazards.



Type 3.3: Material: Heavy duty rubber/vinyl, Reflector Colour: Black/Yellow, Height: 1400mm, Hight from base: 100mm, Length: 60mm, Width: 60mm

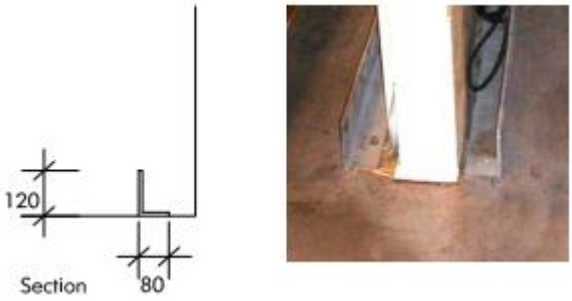
2.4. **Safety Barrier**

2.4.1. Window and wall Protection against Trolleys **(4.1)**



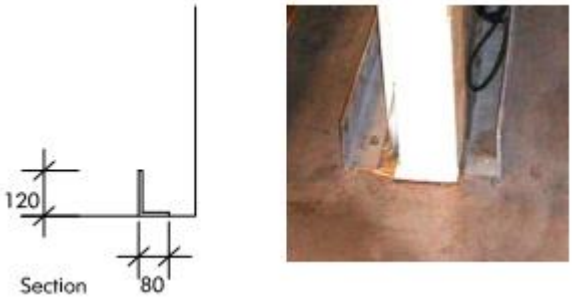
Type 4.1: Material: Stainless Steel, Dia: 50mm, Hight from base: 170mm, Vertical support spacing: 1200mm

2.4.2. Protection for Sliding Doors **(4.2)**



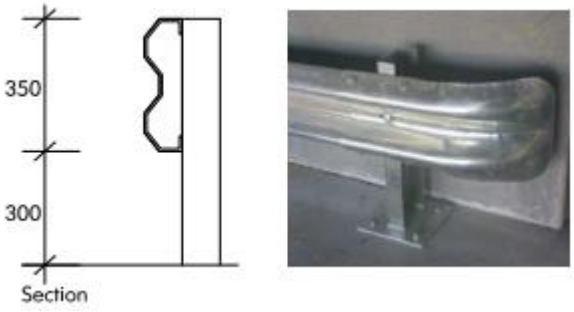
Type 4.2: Material: Minimum 8mm GI Sheet, Profile: L Section, Height: 120mm, Length: 80mm, Thickness: 10mm

2.4.3. Protection against Forklifts **(4.3)**



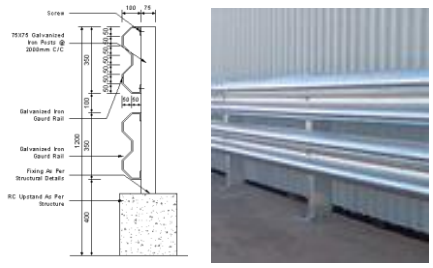
Type 4.3: Material: Minimum 8mm GI Sheet, Profile: L Section, Height: 120mm, Length: 80mm, Thickness: 10mm

2.4.4. Protection against Forklifts, Non-Costumer Areas **(4.4)**



Type 4.4: Material: Galvanized Steel, Profile: Height: 350mm, Thickness: 100mm, Hight from base: 300mm, Width: 100mm, Vertical support profile: 75mmx75mm, Vertical support spacing: 2000mm with L Channel at floor level.

2.4.5. Protection against Vehicles Parking Areas **(4.5)**



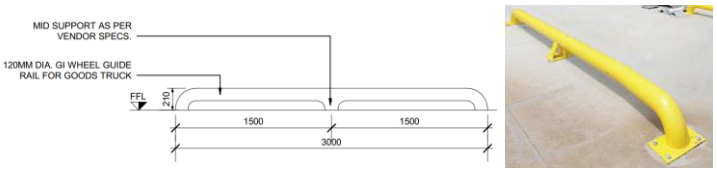
Type 4.5: Material: Galvanized Steel

Profile 1: Height: 400mm, Thickness: 100mm, Hight from base: 300mm, Width: 100mm

Profile 2: Height: 750mm, Thickness: 100mm, Hight from base: 300mm, Width: 100mm

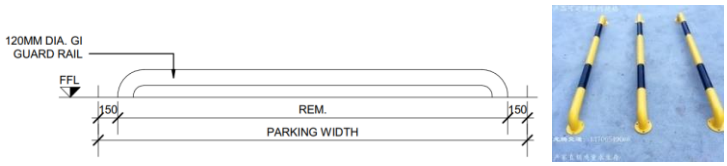
Vertical support profile: 75mmx75mm, Vertical support spacing: 2000mm

2.4.6. Wheel Guides for Goods Truck **(4.5)**



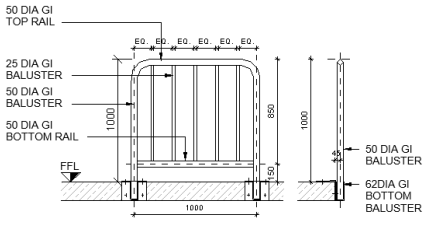
Type 4.5: Material: Galvanized Steel with Black/Yellow Reflective Coating, Dia.: 120mm, Hight from base: 150mm, Vertical support spacing: 1500mm

2.4.7. Wheel Stop for Parking **(4.6)**



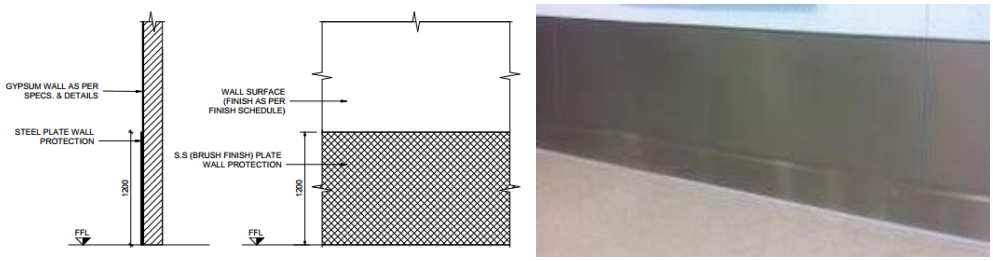
Type 4.6: Material: Galvanized Steel with Black/Yellow Reflective Coating, Dia.: 120mm, Hight from base: 150mm, Length Varies (As per wall protection details).

2.4.8 Removable Protection **(4.7)**



2.5. Wall Protective Sheet

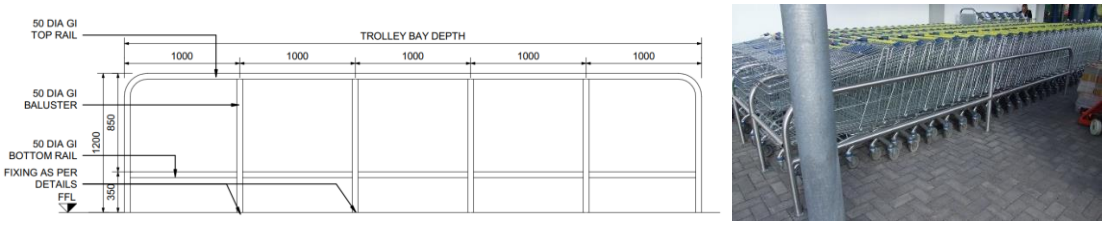
2.5.1. Chequered Plate Wall Protection **(5.1)**



Type 5.1: Material: Stainless Steel (Brush Finish), Height: 1200mm, Thickness: 6mm

2.6. Trolley Protection Rail

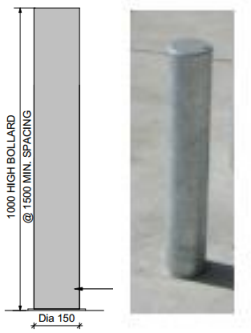
2.6.1. Protection against Vehicles at Trolley Bays in Parking Areas **(6.1)**



Type 6.1: Galvanized Steel Guard Rail Height: 1200mm, Dia: 50mm, Bottom Rail Height: 3500mm, Dia: 50mm, Vertical support profile: Dia: 50mm, Vertical support spacing: 1000mm

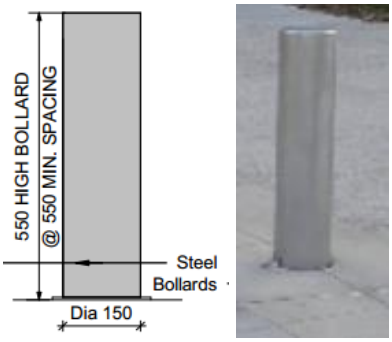
2.7. Safety and Protection System

2.7.1. Bollards for Exterior Areas (Pick Up Area Along Driving Lane) **(7.1)**



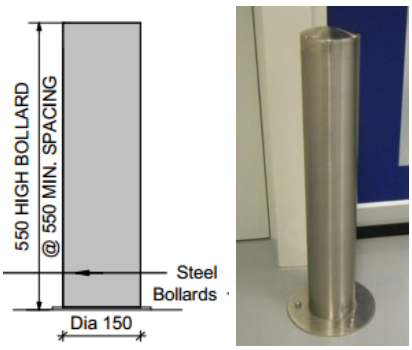
Type 7.1: Material: Galvanized Steel, Height: 1000mm, Dia: 150mm, Spacing: 1500mm. With Reflective

2.7.2. Bollards for Exterior Areas **(7.2)**



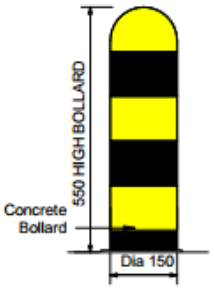
Type 7.2: Material: Galvanized Steel, Height: 500mm, Dia: 150mm, Spacing: minimum 550mm With

2.7.3. Bollards for Interior Areas **(7.3)**



Type 7.3: Material: Stainless Steel (Brush Finish), Height: 550mm, Dia: 150mm, Spacing: minimum 1300mm

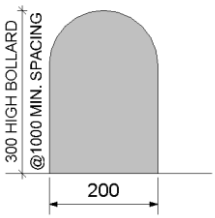
2.7.4. Bollards for Parking Areas with Reflective Finish **(7.4)**





Type 7.2: Material: Galvanized Steel, Reflector Colour: Black/Yellow, Height: 550mm, Dia: 150mm

2.7.5. Heavy Duty Steel Bollard for Loading Bay Parking **(7.5)**



Type 7.3: Material: Galvanized Steel, Height: 450mm, Dia: 300mm, Spacing: minimum 1000mm

2.8. Colours

2.8.1. Colours as per 04-0600 of IKEA Standards attached herewith as Annexure 2.

3. Execution

3.1. Examination

3.1.1. Examine walls for availability of backing to ensure secure attachment of wall protection products.

3.1.2. Verify that HVAC is operating, and installation area temperature has been maintained between 65°-75°F (18°-24°C) for at least 72 hours prior to installation.

3.1.3. Verify application of wall finishes has been completed in accordance with project finish plan.

3.2. Fabrication

3.2.1. Comply with requirements indicated for design, dimensions, details, finish, and member sizes, including wall thicknesses of components.

3.2.2. Shop-assemble components to the greatest extent possible. Disassemble only as necessary for shipping and handling.

3.2.3. Fabricate components with tight seams and joints with exposed edges rolled. Provide surfaces free of evidence of wrinkling, chipping, uneven coloration, dents, and other imperfections. Fabricate members and fittings to produce flush, smooth, and rigid hairline joints.

3.2.4. Brackets, Flanges, Fittings, and Anchors: Provide wall brackets, flanges, miscellaneous fittings, and anchors for interconnection of members to other construction.

3.3. Preparation

3.3.1. Surface Preparation: Clean substrate to remove dust and debris.

3.4. Installation

3.4.1. Acclimate materials to building conditions for at least 24 hours prior to installation.

3.4.2. Install wall protection products in accordance with manufacturer’s installation instructions.

3.4.3. Install wall surface protection units’ plumb, level, and true to line without distortions.

3.4.4. Do not use materials with chips, cracks, voids, stains, or other defects that might be visible in the finished work.

3.4.5. Install aluminium retainers, mounting brackets, and other accessories in strict accordance with the manufacturer's instructions.

3.4.6. Where splices occur in horizontal runs of over 20 feet (6 m), splice aluminium retainer and plastic cover at same locations along the run.

3.5. Workmanship

3.5.1. Workmanship shall be of the highest standard in every respect and tradesmen shall be fully experienced in their field. All work shall be executed neatly and accurately. Products shall be installed according to the manufacturer’s instructions. The work shall be left clean, tidy and properly finished. Movable parts shall be lubricated as necessary and left in good working order.

4. Quality Control

4.1. Delivery, Storage and Handling

4.1.1. All building materials shall be stored on the building site in such a way as to prevent deterioration or the loss or impairment of their structural and other essential properties. Comply to National Building Code (NBC) of India 2016 Volume 1; Part 7 Construction Management, Practices and Safety.

4.2. Source Limitation

4.2.1. All materials & related products shall be from the same manufacturer to ensure material/product integrity & uniformity of colour, texture & quality.

4.3. Mock-ups

4.3.1. Build mock-ups of each Wall & Edge Protection type to verify selections made under sample submittals and to demonstrate aesthetic effects and set quality standards for materials and execution to the Engineer's satisfaction. Approved mock-ups may be incorporated into the finished work, where directed by the Engineer.

5. Testing and Commissioning

5.1. Inspection

5.1.1. Inspection upon completion of work, all Ceiling Works shall be carefully tested and examined for faults in application, shall be corrected immediately.

ATK-ARC-018 – Fire Sealant & Protection

1. General

1.1. Introduction

1.1.1. This section specifies supply and installation of Fire Sealant & Protection products, as indicated in the Tender drawings and as required in completion of the construction works.

1.1.2. Fire Sealant & Protection Products as follows:

1.1.2.1. For Through Penetrations

1.1.2.2. For Construction Joints

1.1.2.3. For Flexible Wrap Systems

1.1.2.4. For General Firestopping

1.2. Related Documents

1.2.1. This section to be read in conjunction with Employer's requirements and general provision of contract, including General, Supplementary and Special Conditions of Tender Documents.

1.3. Codes and Standards

1.3.1. All materials, workmanship and testing shall be in accordance with the following documents, in their latest revisions. This is not exhaustive and contractor shall verify the listing and supplement with the latest applicable document or documents.

1.3.2. Comply with the requirements of the reference standards noted herein, except where more stringent requirements are listed herein or otherwise required by the Tender Documents.

1.3.3. Where materials, finishes, fittings, equipment and building components that comply with standards other than those included in these specifications are proposed, the Contractor shall adequately demonstrate that such items are of an equal quality and standard and are fit for their intended purpose, through submission of internationally recognized and authorized test methods, technical and certification reports (in compliance with assessment/evaluation criteria based on standards already included in these specifications), to the approval of engineer, prior to use of such items.

1.3.4. Conform to the requirements of the following Standards and Codes of Practice where applicable:

1.3.4.1. National Building Code (NBC) of India 2016 Volume 1 & 2.

1.3.4.2. IS 7537 - Road traffic signals

1.3.4.3. American National Standards Institute (ANSI):

1.3.4.3.1. ANSI/UL 263 - Fire Tests of Building Construction and Materials.

1.3.4.3.2. ANSI/UL 723 - Surface Burning Characteristics of Building Materials.

1.3.4.3.3. ANSI/UL 1479 - Standard for Fire Tests of Through-Penetration Firestops.

1.3.4.3.4. ANSI/UL 2079 - Tests for Fire Resistance of Building Joint Systems.

1.3.4.3.5. American Society for Testing and Materials(ASTM):

1.3.4.3.6. ASTM E 84 - Standard Test Method for Surface Burning Characteristics of Building Materials.

1.3.4.3.7. ASTM E 119 - Standard Test Methods for Fire Tests of Building Construction and Materials.

1.3.4.3.8. ASTM E 814 - Standard Test Method for Fire Tests of Through-Penetration Firestops.

1.3.4.3.9. ASTM E 1399 - Standard Test Method for Cyclic Movement and Measuring the Minimum and Maximum Joint Widths of Architectural Joint Systems.

1.3.4.3.10. ASTM E 1966 - Standard Test Method for Fire Resistive Joint Systems.

1.3.4.3.11. ASTM E 2174 - Standard Practice for On-Site Inspection of Installed Firestops.

1.3.4.3.12. ASTM E 2307 - Fire Tests of Perimeter Fire Barrier Systems Using Intermediate Scale, Multi-Story Test Apparatus.

1.3.4.3.13. ASTM E 2393 - Standard Practice for On-Site Inspection of Installed Fire Resistive Joint Systems and Perimeter Fire Barriers.

1.3.4.4. National Fire Protection Association (NFPA):

1.3.4.4.1. NFPA 70 - National Electrical Code.

1.3.4.4.2. NFPA 80 - Standard for Fire Doors and Other Opening Protectives.

1.3.4.4.3. NFPA 96 - Standard for Ventilation Control and Fire Protection of Commercial Cooking Operations.

1.3.4.4.4. NFPA 101 - Life Safety Code.

1.3.4.4.5. NFPA 5000 - Building Construction and Safety Code.

1.3.4.5. International Standards Organization (ISO):

1.3.4.5.1. ISO 6944 Fire containment -- Elements of building construction

1.3.4.5.2. ISO 10295-1: 2007 Fire tests for building elements and components -- Fire testing of service installations -- Part 1: Penetration seals

1.3.4.5.3. ISO 10295-2: 2009 Fire tests for building elements and components -- Fire testing of service installations -- Part 2: Linear joint (gap) seals

1.3.4.5.4. ISO 10295-3: 2012 Fire tests for building elements and components -- Fire testing of service installations -- Part 3: Single component penetration seals -- Guidance on the construction and use of test configurations and simulated services to characterized sealing materials

1.4. Technical Submittals

1.4.1. Technical Data

1.4.1.1. Comprehensive technical data viz., technical literature, data sheets, product catalogue & brochure, installation instructions, etc. to be provided for Engineer's approval.

1.4.2. Shop drawings

1.4.2.1. Shop drawings shall be submitted to Engineer for approval, prior to start of work at site.

1.4.2.2. Shop drawings shall include all details w.r.t. the material & related product's installation, to the scale & detailing as instructed by the Engineer.

1.4.2.3. Shop drawings shall be developed as per actual measurements from the site & must depict the same in the drawings.

1.4.2.4. Shop drawings must comply to BS 1192:2007+A2:2016 Collaborative production of architectural, engineering and construction information & Code of practice.

1.4.3. Samples

1.4.3.1. Samples for Initial Selection: For each type of Fire sealant & protection product, as indicated. Include Samples of accessories involving colour selection.

1.4.4. Method Statement

1.4.4.1. Shall include method of construction/installation, Health and Safety management, testing methods, required relative humidity and temperature level and control systems, possible risk associated with the execution works and precautions to be considered, site control system and any other additional requirements as advised by the Engineer for the execution of works.

1.4.4.2. Method statement shall be approved by the Engineer prior to commencement of the works.

1.4.5. Installer Qualifications

1.4.5.1. Licensed installer specialized in fabricating and installing each type of ceiling, eligible to receive manufacturer's warranty.

1.4.5.2. Installer must be licensed in Maharashtra. Names of approved vendors is given on Maharashtra government website.

1.4.6. Material/Product Certificates

1.4.6.1. Comprehensive tests performed by a qualified independent testing agency certifying that the product compiles with specified performance requirements.

1.4.7. Maintenance Data

1.4.7.1. Details of manufacturer's recommended maintenance procedures for each type of product/material to be included in the operation and maintenance manual.

1.4.8. Spare Materials/Parts

Provide the following extra materials that match installed products, packaged for storage with suitable labels for identification.

1.5. Warranty / Guarantee

1.5.1. As per the Employer's requirements and general provision of contract, including General, Supplementary and Special Conditions of Contract Documents.

1.6. Performance Requirements

1.6.1. All finishes, materials and components shall comply with requirements of NFPA 101 Life Safety Code, chapter 10 – Interior Finish, Contents and Furnishings with respect to classification of materials, flame spread, smoke generation and other requirements.

1.7. Environment & Sustainability

1.7.1. Recycling Strategy document for the material or product shall be submitted to the Engineer for approval.

1.7.2. LEED Requirements:

1.7.2.1. All the specifications in this document are for USGBC LEED BD+C submission and are applicable to all materials & procedures followed for the project. The specification as stated here should be adhered to the word. In case of a dispute LEED guidelines shall be followed. Brochures, product specification sheets and/or certificates as specified in this document should be submitted along with the material procurement. All contractors, manufactures must submit documents, records and other submittal related to the different credits as mentioned in the specification.

1.7.2.2. Manufacturer product conforming to ISO 14044 or Environmental Product Declarations which conform to ISO 14025, 14040, 14044, and EN 15804 or ISO 21930 and have at least a cradle to gate scope. (If applicable)**.**

1.7.2.3. Manufacturer Material Ingredient Reporting by any of the listed below documents (If applicable):

1.7.2.3.1. Health Product Declaration. The end use product has a published, complete Health Product Declaration with full disclosure of known hazards in compliance with the Health Product Declaration Open Standard.

1.7.2.3.2. Cradle to Cradle Certificate. The end use product has been certified at the Cradle to Cradle v2 Basic level or Cradle to Cradle v3 Bronze level.

1.7.2.3.3. REACH Optimization. End use products and materials that do not contain substances that meet REACH criteria for substances of very high concern.

1.7.2.4. Manufacturer Raw Material Source and Extraction Reporting by any of the listed below activity (If applicable):

1.7.2.4.1. Third-party verified corporate sustainability reports (CSR) which include environmental impacts of extraction operations and activities associated with the manufacturer's product and the product's supply chain**.**

1.7.2.5. Self-declared reports from manufacturers that have publicly released a report from their raw material suppliers which include raw material supplier extraction locations

1.8. Manufacturers

1.8.1. Manufacturer to be accredited and registered under BS EN ISO 9001.

1.8.2. List of preferred or acceptable Manufacturers attached herewith as Annexure 1.

2. Products

2.1. Fire Sealant & Protection Products

2.1.1. Fire Stopping Cast-in-Place Devices: Firestopping device for use prior to a concrete pour. Adjustable height with pull tabs, straight edge design for close placement to walls and adjacent devices.

2.1.2. Fire Stopping Devices: One piece metal collar assembly encasing intumescent material for firestopping of pipes and cables through rated walls and floors.

2.1.3. Fire Stopping Plastic Pipe Device: Intumescent device for firestopping of plastic pipe and cables through rated walls and floors.

2.1.4. Fire Stopping Sealant: High-performance, intumescent, water-based sealant. No-sag, fast drying, paintable, red in colour. Versatile firestop sealant for pipes (not for use with CPVC), cables, cable tray, blank opening and other penetrations along with mineral wool or other fire-rated assembly products.

2.1.5. Fire Stopping Sealant: Single-part, non-slump elastomeric silicone firestop sealant. Sag-resistant, low VOC. Light grey colour. Used in mechanical, electrical and plumbing applications to firestop openings and penetrations through fire-rated floor or wall assemblies. Typical penetrants include: metallic pipe, non-metallic pipe (FGG/BM system CPVC compatible), conduit, power and communication cable and telephone or electrical wiring.

2.1.6. Fire Stopping Water Tight Sealant: Single-part, water-tight, intumescent silicone firestop sealant for filling voids in concrete gypsum, metal, plastic, wood and insulation. Light grey colour with black flecks. Meets UL Water Leakage Test, W Rating – Class 1 requirements.

2.1.7. Fire Stopping Mouldable Putty: One-part, 100 percent solids intumescent firestop. Remains pliable, flexible and easily re-enterable. Non-toxic synthetic formula.

3. Execution

3.1. Examination

3.1.1. Conduct tests according to manufacturer's written recommendations to verify that substrates are free of oil, grease, rolling compounds, incompatible primers, loose mill scale, dirt and other foreign substances capable of impairing bond of firestopping.

3.1.2. Verify that openings and adjacent areas are not obstructed by construction that would interfere with installation of firestopping, including ducts, piping, equipment, and other suspended construction. Verify that environmental conditions are safe and suitable for installation of firestopping.

3.2. Preparation

3.2.1. Prepare substrates in accordance with manufacturer's instructions and recommendations.

3.2.2. Install masking and temporary coverings as required to prevent contamination or defacement of adjacent surfaces due to firestopping installation.

3.3. Installation

3.3.1. Install in strict accordance with manufacturer's detailed installation instructions and procedures.

3.3.2. Install so that openings are completely filled and material is securely adhered.

3.3.3. Where firestopping surface will be exposed to view, finish to a smooth, uniform surface flush with adjacent surfaces.

3.3.4. After installation is complete, remove combustible forming materials and accessories that are not part of the listed system.

3.3.5. Repair or replace defective installations in accordance with manufacturer’s recommendations, listed systems details and applicable code requirements.

3.3.6. At each through penetration or fire-resistive joint system, attach identification labels on both sides in location where label will be visible to anyone seeking to remove penetrating items or firestopping.

3.3.7. Clean firestop materials off surfaces adjacent to openings as work progresses, using methods and cleaning materials approved in writing by firestop system manufacturer and which will not damage the surfaces being cleaned.

3.3.8. Notify Authority Having Jurisdiction when firestopping installation is ready for inspection; obtain advance approval of anticipated inspection dates and phasing, if any, required to allow subsequent construction to proceed.

3.3.9. Do not cover firestopping with other construction until approval of authority having jurisdiction has been received.

3.4. Workmanship

3.4.1. Workmanship shall be of the highest standard in every respect and tradesmen shall be fully experienced in their field. All work shall be executed neatly and accurately. Products shall be installed according to the manufacturer's instructions. The work shall be left clean, tidy and properly finished. Movable parts shall be lubricated as necessary and left in good working order.

4. Quality Control

4.1. Delivery, Storage and Handling

4.1.1. All building materials shall be stored on the building site in such a way as to prevent deterioration or the loss or impairment of their structural and other essential properties. Comply to National Building Code (NBC) of India 2016 Volume 1; Part 7 Construction Management, Practices and Safety.

4.2. Source Limitation

4.2.1. All materials & related products shall be from the same manufacturer to ensure material/product integrity & uniformity of colour, texture & quality.

4.3. Mock-ups

4.3.1. Build mock-ups of each type of Fire Stopping to verify selections made under sample submittals and to demonstrate aesthetic effects and set quality standards for materials and execution to the Engineer's satisfaction. Approved mock-ups may be incorporated into the finished work, where directed by the Engineer.

5. Testing and Commissioning

5.1. Inspection

5.1.1. Inspection upon completion of work, all Fire Sealant & Protection shall be carefully tested and examined for faults in application, shall be corrected immediately.

ATK-ARC-019 – Specialities

1. General

1.1. Introduction

1.1.1. This section specifies supply and installation of the following materials/products, as indicated in the Tender drawings and as required in completion of the construction works.

1.1.1.1. Canopies (Galvanised Steel Fascia & Rain Screen)

1.1.1.2. Stone (Visitor Staircase Tread)

1.1.1.3. SS Strips (Anti-slip) for Cast In-Situ Concrete Internal Staircases.

1.1.1.4. Fire & Emergency Signage

1.2. Related Documents

1.2.1. This section to be read in conjunction with Employer's requirements and general provision of contract, including General, Supplementary and Special Conditions of Tender Documents.

1.3. Codes and Standards

1.3.1. All materials, workmanship and testing shall be in accordance with the following documents, in their latest revisions. This is not exhaustive and contractor shall verify the listing and supplement with the latest applicable document or documents.

1.3.2. Comply with the requirements of the reference standards noted herein, except where more stringent requirements are listed herein or otherwise required by the Tender Documents.

1.3.3. Where materials, finishes, fittings, equipment and building components that comply with standards other than those included in these specifications are proposed, the Contractor shall adequately demonstrate that such items are of an equal quality and standard and are fit for their intended purpose, through submission of internationally recognized and authorized test methods, technical and certification reports (in compliance with assessment/evaluation criteria based on standards already included in these specifications), to the approval of engineer, prior to use of such items.

1.3.4. Conform to the requirements of the following Standards and Codes of Practice where applicable:

1.3.4.1. National Building Code (NBC) of India 2016 Volume 1 & 2.

1.3.4.2. NFPA 101 - Life Safety Code.

1.4. Technical Submittals

1.4.1. Technical Data

1.4.1.1. Comprehensive technical data viz., technical literature, data sheets, product catalogue & brochure, installation instructions, etc. to be provided for Engineer's approval.

1.4.2. Shop drawings

1.4.2.1. Shop drawings shall be submitted to Engineer for approval, prior to start of work at site.

1.4.2.2. Shop drawings shall include all details w.r.t. the material & related product's installation, to the scale & detailing as instructed by the Engineer.

1.4.2.3. Shop drawings shall be developed as per actual measurements from the site & must depict the same in the drawings.

1.4.2.4. Shop drawings must comply to BS 1192:2007+A2:2016 Collaborative production of architectural, engineering and construction information & Code of practice.

1.4.3. Samples

1.4.3.1. Samples for Initial Selection: For each type of Fire sealant & protection product, as indicated. Include Samples of accessories involving colour selection.

1.4.4. Method Statement

1.4.4.1. Shall include method of construction/installation, Health and Safety management, testing methods, required relative humidity and temperature level and control systems, possible risk associated with the execution works and precautions to be considered, site control system and any other additional requirements as advised by the Engineer for the execution of works.

1.4.4.2. Method statement shall be approved by the Engineer prior to commencement of the works.

1.4.5. Installer Qualifications

1.4.5.1. Licensed installer specialized in fabricating and installing each type of material/product, eligible to receive manufacturer's warranty.

1.4.5.2. Installer must be licensed in Maharashtra. Names of approved vendors is given on Maharashtra government website.

1.4.6. Material/Product Certificates

1.4.6.1. Comprehensive tests performed by a qualified independent testing agency certifying that the product compiles with specified performance requirements.

1.4.7. Maintenance Data

1.4.7.1. Details of manufacturer's recommended maintenance procedures for each type of product/material to be included in the operation and maintenance manual.

1.4.8. Spare Materials/Parts

Provide the following extra materials that match installed products, packaged for storage with suitable labels for identification.

1.5. Warranty / Guarantee

1.5.1. As per the Employer's requirements and general provision of contract, including General, Supplementary and Special Conditions of Contract Documents.

1.6. Performance Requirements

1.6.1. All finishes, materials and components shall comply with requirements of NFPA 101 Life Safety Code, chapter 10 – Interior Finish, Contents and Furnishings with respect to classification of materials, flame spread, smoke generation and other requirements.

1.7. Environment & Sustainability

1.7.1. Recycling Strategy document for the material or product shall be submitted to the Engineer for approval.

1.7.2. LEED Requirements:

1.7.2.1. All the specifications in this document are for USGBC LEED BD+C submission and are applicable to all materials & procedures followed for the project. The specification as stated here should be adhered to the word. In case of a dispute LEED guidelines shall be followed. Brochures, product specification sheets and/or certificates as specified in this document should be submitted along with the material procurement. All contractors, manufactures must submit documents, records and other submittal related to the different credits as mentioned in the specification.

1.7.2.2. Manufacturer product conforming to ISO 14044 or Environmental Product Declarations which conform to ISO 14025, 14040, 14044, and EN 15804 or ISO 21930 and have at least a cradle to gate scope. (If applicable)**.**

1.7.2.3. Manufacturer Material Ingredient Reporting by any of the listed below documents (If applicable):

1.7.2.3.1. Health Product Declaration. The end use product has a published, complete Health Product Declaration with full disclosure of known hazards in compliance with the Health Product Declaration Open Standard.

1.7.2.3.2. Cradle to Cradle Certificate. The end use product has been certified at the Cradle to Cradle v2 Basic level or Cradle to Cradle v3 Bronze level.

1.7.2.3.3. REACH Optimization. End use products and materials that do not contain substances that meet REACH criteria for substances of very high concern.

1.7.2.4. Manufacturer Raw Material Source and Extraction Reporting by any of the listed below activity (If applicable):

1.7.2.4.1. Third-party verified corporate sustainability reports (CSR) which include environmental impacts of extraction operations and activities associated with the manufacturer's product and the product's supply chain**.**

1.7.2.5. Self-declared reports from manufacturers that have publicly released a report from their raw material suppliers which include raw material supplier extraction locations

1.8. Manufacturers

1.8.1. Manufacturer to be accredited and registered under BS EN ISO 9001.

1.8.2. List of preferred or acceptable Manufacturers attached herewith as Annexure 1.

2. Products

2.1. Canopy

2.1.1. Material/Product Specification as per 04-0130 of IKEA Standards attached herewith as Annexure 17.

2.2. Stone Tread

2.2.1. 30mm Thick Stone Tread (Granite); Colour & Finish to Engineer's approval, to full width of the Staircase, laid in Structural Steel sectional bracket.

2.3. SS Strips (Anti-slip)

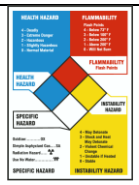
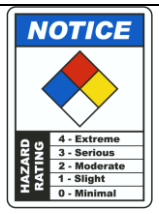
2.3.1. 2 Nos. Stainless Steel Strips, 5mm Brush Finish, Inserts in Tread to full width of the Staircase, laid 30mm c/c.

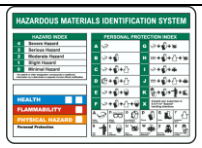
2.4. Signage

2.4.1. Fire & Emergency Signage: Location/Application as per Fire Strategy/Fire Fighting & Protection/Fire Alarm & Detection Drawings.

Fire/Emergency Sign Type	Graphic (Indicative)	Dimension (W x H) (mm)	Specification
Emergency Exit Sign		35 x 25	Glow-in-the-Dark Adhesive-Backed Polyester 4 mils thick glow-in-the-dark polyester film with a LEED compliant adhesive, suitable for indoor and outdoor applications. Not resistant to contact with chemicals. This material will glow for up to 10 hours when it is charged with a strong light source for at least 60 minutes. Daylight color is soft-yellow to eggshell white. Color in darkness is photoluminescent yellowish.
This Is Not An Exit Sign		35 x 25	Glow-in-the-Dark Adhesive-Backed Polyester 4 mils thick glow-in-the-dark polyester film with a LEED compliant adhesive, suitable for indoor and outdoor applications. Not resistant to contact with chemicals. This material will glow for up to 10 hours when it is charged with a strong light source for at least 60 minutes. Daylight color is soft-yellow to eggshell white. Color in darkness is photoluminescent yellowish.
Emergency Exit Sign		25 x 35	Adhesive-Backed Vinyl Outdoor grade high gloss calendared vinyl film, and coated with aggressive LEED compliant acrylic adhesive formulated to adhere to a variety of surfaces, ensuring durability in most commercial and industrial environments. Resistant to fading and mild chemicals.
Assembly Point Sign		25 x 35	Adhesive-Backed Vinyl Outdoor grade high gloss calendared vinyl film, and coated with aggressive LEED compliant acrylic adhesive formulated to adhere to a variety of surfaces, ensuring durability in most commercial and industrial environments. Resistant to fading and mild chemicals.

Accessible Refuge Area Sign		25 x 35	Adhesive-Backed Vinyl Outdoor grade high gloss calendared vinyl film, and coated with aggressive LEED compliant acrylic adhesive formulated to adhere to a variety of surfaces, ensuring durability in most commercial and industrial environments. Resistant to fading and mild chemicals.
Break Glass In Case Of Emergency Sign		25 x 35	Adhesive-Backed Vinyl Outdoor grade high gloss calendared vinyl film, and coated with aggressive LEED compliant acrylic adhesive formulated to adhere to a variety of surfaces, ensuring durability in most commercial and industrial environments. Resistant to fading and mild chemicals.
First Aid Station Sign		25 x 35	Adhesive-Backed Vinyl Outdoor grade high gloss calendared vinyl film, and coated with aggressive LEED compliant acrylic adhesive formulated to adhere to a variety of surfaces, ensuring durability in most commercial and industrial environments. Resistant to fading and mild chemicals.
Fire Department Standpipe Connection Sign		30 x 30	Adhesive-Backed Vinyl Outdoor grade high gloss calendared vinyl film, and coated with aggressive LEED compliant acrylic adhesive formulated to adhere to a variety of surfaces, ensuring durability in most commercial and industrial environments. Resistant to fading and mild chemicals.
Automatic Sprinkler Control Valve Sign		30 x 30	Adhesive-Backed Vinyl Outdoor grade high gloss calendared vinyl film, and coated with aggressive LEED compliant acrylic adhesive formulated to adhere to a variety of surfaces, ensuring durability in most commercial and industrial environments. Resistant to fading and mild chemicals.
Automatic Sprinkler and Standpipe Connection Sign		30 x 30	Adhesive-Backed Vinyl Outdoor grade high gloss calendared vinyl film, and coated with aggressive LEED compliant acrylic adhesive formulated to adhere to a variety of surfaces, ensuring durability in most commercial and industrial environments. Resistant to fading and mild chemicals.
Fire Hose or Standpipe Outlet Sign		30 x 30	Adhesive-Backed Vinyl Outdoor grade high gloss calendared vinyl film, and coated with aggressive LEED compliant acrylic adhesive formulated to adhere to a variety of surfaces, ensuring durability in most commercial and industrial environments. Resistant to fading and mild chemicals.
Fire Hydrant Sign		30 x 30	Adhesive-Backed Vinyl Outdoor grade high gloss calendared vinyl film, and coated with aggressive LEED compliant acrylic adhesive formulated to adhere to a variety of surfaces, ensuring durability in most commercial and industrial environments. Resistant to fading and mild chemicals.
Do Not Use Elevator In Case Of Fire Sign		30 x 30	Adhesive-Backed Vinyl Outdoor grade high gloss calendared vinyl film, and coated with aggressive LEED compliant acrylic adhesive formulated to adhere to a variety of surfaces, ensuring durability in most commercial and industrial environments. Resistant to fading and mild chemicals.
Use Stairs In Case of Fire Sign		30 x 30	Adhesive-Backed Vinyl Outdoor grade high gloss calendared vinyl film, and coated with aggressive LEED compliant acrylic adhesive formulated to adhere to a variety of surfaces, ensuring durability in most commercial and industrial environments. Resistant to fading and mild chemicals.
Use Stairs In Case of Fire Sign		30 x 30	Adhesive-Backed Vinyl Outdoor grade high gloss calendared vinyl film, and coated with aggressive LEED compliant acrylic adhesive formulated to adhere to a variety of surfaces,

			ensuring durability in most commercial and industrial environments. Resistant to fading and mild chemicals.
No Smoking Symbol Sign		30 x 30	Adhesive-Backed Vinyl Outdoor grade high gloss calendared vinyl film, and coated with aggressive LEED compliant acrylic adhesive formulated to adhere to a variety of surfaces, ensuring durability in most commercial and industrial environments. Resistant to fading and mild chemicals.
No Cooking Symbol Sign		30 x 30	Adhesive-Backed Vinyl Outdoor grade high gloss calendared vinyl film, and coated with aggressive LEED compliant acrylic adhesive formulated to adhere to a variety of surfaces, ensuring durability in most commercial and industrial environments. Resistant to fading and mild chemicals.
International Symbol of Accessibility Sign		30 x 30	Adhesive-Backed Vinyl Outdoor grade high gloss calendared vinyl film, and coated with aggressive LEED compliant acrylic adhesive formulated to adhere to a variety of surfaces, ensuring durability in most commercial and industrial environments. Resistant to fading and mild chemicals.
Electric Panel or Electric Shutoff Sign		30 x 30	Adhesive-Backed Vinyl Outdoor grade high gloss calendared vinyl film, and coated with aggressive LEED compliant acrylic adhesive formulated to adhere to a variety of surfaces, ensuring durability in most commercial and industrial environments. Resistant to fading and mild chemicals.
Press For Fire Alarm Sign		25 x 35	Glow-in-the-Dark Adhesive-Backed Polyester 4 mils thick glow-in-the-dark polyester film with a LEED compliant adhesive, suitable for indoor and outdoor applications. Not resistant to contact with chemicals. This material will glow for up to 10 hours when it is charged with a strong light source for at least 60 minutes. Daylight color is soft-yellow to eggshell white. Color in darkness is photoluminescent yellowish.
Danger Hazardous Chemicals Sign		25 x 17.5	Adhesive-Backed Vinyl Outdoor grade high gloss calendared vinyl film, and coated with aggressive LEED compliant acrylic adhesive formulated to adhere to a variety of surfaces, ensuring durability in most commercial and industrial environments. Resistant to fading and mild chemicals.
Spill Response Equipment Sign		25 x 35	Adhesive-Backed Vinyl Outdoor grade high gloss calendared vinyl film, and coated with aggressive LEED compliant acrylic adhesive formulated to adhere to a variety of surfaces, ensuring durability in most commercial and industrial environments. Resistant to fading and mild chemicals.
NFPA Reference Chart		25 x 35	Adhesive-Backed Vinyl Outdoor grade high gloss calendared vinyl film, and coated with aggressive LEED compliant acrylic adhesive formulated to adhere to a variety of surfaces, ensuring durability in most commercial and industrial environments. Resistant to fading and mild chemicals.
Notice NFPA Hazard Rating Sign		25 x 35	Glow-in-the-Dark Adhesive-Backed Polyester 4 mils thick glow-in-the-dark polyester film with a LEED compliant adhesive, suitable for indoor and outdoor applications. Not resistant to contact with chemicals. This material will glow for up to 10 hours when it is charged with a strong light source for at least 60 minutes. Daylight color is soft-yellow to eggshell white. Color in darkness is photoluminescent yellowish.

NFPA Diesel Fuel 1-2-0 Sign		25 x 35	Glow-in-the-Dark Adhesive-Backed Polyester 4 mils thick glow-in-the-dark polyester film with a LEED compliant adhesive, suitable for indoor and outdoor applications. Not resistant to contact with chemicals. This material will glow for up to 10 hours when it is charged with a strong light source for at least 60 minutes. Daylight color is soft-yellow to eggshell white. Color in darkness is photoluminescent yellowish.
Hazardous Materials Identification System		25 x 17.5	Adhesive-Backed Vinyl Outdoor grade high gloss calendared vinyl film, and coated with aggressive LEED compliant acrylic adhesive formulated to adhere to a variety of surfaces, ensuring durability in most commercial and industrial environments. Resistant to fading and mild chemicals.
Fire Extinguisher 3-Way Sign		Two 10 × 30 Panels with 2.5 Flanges on Sides	Glow-in-the-Dark Plastic .046" photoluminous plastic will not rust, is resistant to chips and cracks, and will easily bend to certain surfaces without causing material fractures. It is designed for indoor use only. This material can be charged by any light source in as little as 15 minutes, and will glow for up to 10 hours when it is charged with a strong light source for at least 60 minutes. Cool white light sources such as fluorescent lighting and sunshine are excellent activators. Soft warm lighting such as incandescent light bulbs are not as effective.
3-Way Emergency Shower Sign		Two 25 × 17.5 Panels with 2.5 Flanges on Sides	Polystyrene Plastic Polystyrene is a low weight, high impact resistant plastic that is rust-free, chip and crack resistant, and is ideal for many applications.

2.4.2. CO2 Display Sensor (Wall Mounted): One Each for all Fire Compartment Zones.

2.4.2.1. Compliance: EMC directive 89/336/EEC. RoHS directive 2002/95/EG

2.4.2.2. Operating Temperature Range: 0 - 50 °C

2.4.2.3. Storage Temperature Range: -40 to +70 °C (display model -D: -20 to +70 °C)

2.4.2.4. Operating Humidity Range: 0 to 95% RH (non-condensing)

2.4.2.5. Sensor Life Expectancy: > 15 years

2.4.2.6. Sensing Method: Gold-plated infrared (NDIR) waveguide technology with passive gas diffusion (no moving parts)

2.4.2.7. Calibration: Automatic Background Calibration (ABC)

2.4.2.8. Response Time (T1/e): < 10 sec. @ 30 cc/min. flow rate, < 3 min. diffusion time

2.4.2.9. Repeatability: ± 20 ppm ± 1 % of reading

2.4.2.10. Accuracy: ±30 ppm ±3% of reading

2.4.2.11. Annual Zero Drift: < ± 10 ppm

2.4.2.12. Pressure Dependence: 1.6% of reading per kPa

2.4.2.13. Measurement range: 0 - 6,000 ppm

2.4.2.14. Power Input: 24 VAC/VDC ±20%, 50 Hz (half-wave rectifier input)

2.4.2.15. Analog Output: 0 -10 VDC

2.4.2.16. Audible alarm: 94 Db Buzzer with Mute button, D/A resolution 10 bits, 10 mV

2.4.3. Wayfinding & Customer Signage

2.4.3.1. Nominated Sub-Contract

2.5. Colours

2.5.1. Colours as per 04-0600 of IKEA Standards attached herewith as Annexure 2.

3. Execution

3.1. Examination

3.1.1. Conduct tests according to manufacturer's written recommendations to verify that substrates are free of oil, grease, rolling compounds, incompatible primers, loose mill scale, dirt and other foreign substances capable of impairing bond of firestopping.

3.1.2. Verify that openings and adjacent areas are not obstructed by construction that would interfere with installation of firestopping, including ducts, piping, equipment, and other suspended construction. Verify that environmental conditions are safe and suitable for installation of firestopping.

3.2. Preparation

3.2.1. Prepare substrates in accordance with manufacturer's instructions and recommendations.

3.2.2. Install masking and temporary coverings as required to prevent contamination or defacement of adjacent surfaces due to firestopping installation.

3.3. Installation

3.3.1. Install in strict accordance with manufacturer's detailed installation instructions and procedures.

3.3.2. Install so that openings are completely filled and material is securely adhered.

3.3.3. Where firestopping surface will be exposed to view, finish to a smooth, uniform surface flush with adjacent surfaces.

3.3.4. After installation is complete, remove combustible forming materials and accessories that are not part of the listed system.

3.3.5. Repair or replace defective installations in accordance with manufacturer’s recommendations, listed systems details and applicable code requirements.

3.3.6. At each through penetration or fire-resistive joint system, attach identification labels on both sides in location where label will be visible to anyone seeking to remove penetrating items or firestopping.

3.3.7. Clean firestop materials off surfaces adjacent to openings as work progresses, using methods and cleaning materials approved in writing by firestop system manufacturer and which will not damage the surfaces being cleaned.

3.3.8. Notify Authority Having Jurisdiction when firestopping installation is ready for inspection; obtain advance approval of anticipated inspection dates and phasing, if any, required to allow subsequent construction to proceed.

3.3.9. Do not cover firestopping with other construction until approval of authority having jurisdiction has been received.

3.4. Workmanship

3.4.1. Workmanship shall be of the highest standard in every respect and tradesmen shall be fully experienced in their field. All work shall be executed neatly and accurately. Products shall be installed according to the manufacturer's instructions. The work shall be left clean, tidy and properly finished. Movable parts shall be lubricated as necessary and left in good working order.

4. Quality Control

4.1. Delivery, Storage and Handling

4.1.1. All building materials shall be stored on the building site in such a way as to prevent deterioration or the loss or impairment of their structural and other essential properties. Comply to National Building Code (NBC) of India 2016 Volume 1; Part 7 Construction Management, Practices and Safety.

4.2. Source Limitation

4.2.1. All materials & related products shall be from the same manufacturer to ensure material/product integrity & uniformity of colour, texture & quality.

4.3. Mock-ups

4.3.1. Build mock-ups of each type of Fire Stopping to verify selections made under sample submittals and to demonstrate aesthetic effects and set quality standards for materials and execution to the Engineer's satisfaction. Approved mock-ups may be incorporated into the finished work, where directed by the Engineer.

5. Testing and Commissioning

5.1. Inspection

5.1.1. Inspection upon completion of work, all Fire Sealant & Protection shall be carefully tested and examined for faults in application, shall be corrected immediately.

Annexure A.01 – List of Acceptable Manufacturers/Suppliers

DESCRIPTION	ACCEPTABLE MANUFACTURERS
Ceramic Tile	Johnson Tiles NITCO Tiles Kajaria Tiles
Painting	Asian Paints Ltd. Berger Paints Ltd. Jotun Paints Ltd.
Resilient Flooring	Armstrong Flooring Asian Flooring India Pvt. Ltd. Wonderfloor
Resin Flooring	Flowcrete India Pvt. Ltd. Sika India Pvt. Ltd. Dubond Products India Pvt. Ltd. Cipy Polyurethanes Pvt. Ltd
Gypsum Plasterboard Partitions	USG Boral Ltd. Gyproc Saint Gobain Ltd.
Toilet Cubicle & Urinal Partition	T Line Pvt. Ltd. Merino Besco Ltd. Greenlam Sturdo Ltd
Metal Ceiling	Techno Ceilings Armstrong Ceilings Rockfon Ceilings Durlum Ceilings
Steel Doors & Louvers	Steel Frame Steel/Glass Leaf Door: Shakti Hormann Pvt. Ltd. Assa Abloy India Private Ltd. Metaflex Doors India Pvt. Ltd. Steel Frame Particle Board Leaf Door: Greenply Plywood Merino Group Assa Abloy India Private Ltd. Century Plywood India Limited Steel Frame GRP Door (Hygienic Door): Metaflex Doors India Pvt. Ltd. Assa Abloy India Private Ltd. Integrated Clean Room Technologies Pvt. Ltd. Doortek Pvt. Ltd. Glass: Saint-Gobain India Pvt Ltd. Schott AG Asahi India Glass Ltd.
Ironmongery	Häfele India Pvt Ltd. DORMA India Pvt. Ltd. Assa Abloy India Private Ltd.*
Glazing	Kalco Alu-Systems Pvt. Ltd. Aluplex India Ltd. Alupro Building Systems (P) Ltd. Fenesta DCM Shriram Group Ltd. Glass: Saint-Gobain India Pvt Ltd. Schott AG

	Asahi India Glass Ltd.
Skylights	Acros Pvt. Ltd. Skyshade Daylights Pvt. Ltd. Palram India Pvt. Ltd.
Sliding Doors/Folding Partition	DORMA India Pvt. Ltd Häfele India Pvt Ltd.
Loading Dock Assembly, Overhead Sectional Door, Automatic High-Speed Roller Shutter	Gandhi Automations Pvt. Ltd. Shakti Hormann Pvt. Ltd. Metaflex Doors India Pvt. Ltd. Assa Abloy India Private Ltd. Avians
Fire Curtain	Orient Fire Curtains India Pvt. Ltd. NECO Electrical Company Limited GreenInnovior Technosolutions Pvt. Ltd.
Wall & Edge Protection	Imperio Railing Systems Pvt. Ltd. Frontier Ploymers Ltd. Kiran Rubber Industries Pvt. Ltd. Kantaflex India Pvt. Ltd. WallPro Protections Pvt. Ltd.
Fire Sealant & Protection	Hilti India Pvt. Ltd. 3M India Ltd
Fire & Emergency Signages	Prolite Usha Fire Safety Equipments Pvt. Ltd. Indra Glowtech Private Limited

Annexure A.02 - Ceramic tiles



150 x 150mm ceramic floor tiles

		Remark
Description of Standard		
This standard describes ceramicfloor tiles to be used in stores on ground (OG), lifted (LS) and garaged stores (GS). Tiles be installed on smooth structural surfaces which have been prepared for the installation.		Refer to left side of page
Function		
The tiled surface shall constitute the finished wearing surface in the object area and shall comply with the requirements set on performance regarding aesthetic appearance, wearing ability, hygiene and low maintenance cost. All functions to correspond with the usage of each particular area.		Refer to left side of page
Validity		
This standard is valid within IKEA in Europe		Region 1
Standard Specification		
Type of wall tiling:	Medium size ceramic tiles applied to horisontal surface with cementitious joints.	Refer to left side of page
Substrate:	Concrete ground floor slab, raised concrete slab or screed. Power float or trowelled finish.	
Material:	Dry-pressed ceramic tiles manufactured of fine stoneware or stoneware.	EN 176 Slightly rough surface
Size:	Tiles in squares of 150x150 or 200x200 mm. Exact size to allow joint of 3 mm. Maximum difference of single tile to average size: L<12cm/L>12cm max +/ 0,75/0,50 %.	Maximum thickness 8 mm
Tolerances in %	Straightness +/-0,3 – Right angle level +/- 0,5 - Centre/side camber +0,5/-0,3 – Distortion +/- 0,5 - Thickness +- 0,8 mm or supplier stand.	
Bonding:	In cementitious glue as per manufacturers re-commendation. Quantity and thickness to be recommended.	
Jointing:	Joints to be 3 mm wide and flush finished of material compatible with tiles. Resistance to chemical staining or urine	
Skirting:	Coved base of same type as floor	Supplier system, corner details
Colour and gloss:	Grey with satin gloss finish (gloss 55)	Grey from suppliers range to be approved by IKEA
Water absorption:	Average 0,5%>E>3%, Single E<3,0%	
Slip resistance:	Value 30 in polished/unpolished CEN test	Colour difference between tiles not visible from watching floor from 2 m
Breaking module:	Average > 27 MPa	
Maintenance, cleaning:	Proven manufactures manual to be followed.	
Environmental:	Documented recycling strategy required	
Transportation/storage:	Suppliers recommendations required.	
Conformity:	EN 176, GB/T4100, EN ISO 101545	
Warranty:	5 years on material.	
Options		
Large size tiles for, restaurant etc.		300,400,450 modular size For IKEA approval
Related documents		
Laying pattern to be agreed in advance of laying		
Other		
Moistures testing to be agreed and documented.		
Best Buy		
Supplier / installer availability		See intranet BB site for detail

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150 x 150mm ceramic floor tiles

Remark

Description of Standard
This standard describes ceramic floor tiles to be used in stores on ground (OG), lifted (LS) and garaged stores (GS). Tiles are installed on smooth structural surfaces which have been prepared for the installation.

Refer to left side of page

Function
The tiled surface shall constitute the finished wearing surface in the object area and shall comply with the requirements set on performance regarding aesthetic appearance, wearing ability, hygiene and low maintenance cost. All functions to correspond with the usage of each particular area.

Refer to left side of page

Validity
This standard is valid within IKEA in Asia & Pacific

Region 3

Standard Specification

Type of floor tiling:	Medium size ceramic tiles applied to horizontal surface with cementitious joints.	Refer to left side of page
Substrate:	Concrete ground floor slab, raised concrete slab or screed. Power float or trowelled finish.	
Material:	Dry-pressed ceramic tiles manufactured of fine stoneware or stoneware.	Slightly rough surface
Size:	Tiles in squares of 150x150 or 200x200 mm. Exact size to allow joint of 3 mm. Maximum difference of single tile to average size: L<12cm,max+0,75%, L>12cm,max+0,5%	Maximum thickness 8 mm
Tolerances in %	Straightness +/-0,3 Right angle level +/- 0,5 Centre/side camber +0,5/-0,3 Distortion +/- 0,5 Thickness +/- 0,8 mm or supplier stand.	
Bonding:	In cementitious glue as per manufacturer's re-commendation. Quantity and thickness to be recommended.	
Jointing:	Joints to be 3 mm wide and flush finished of material compatible with tiles. Resistance to chemical staining or urine	
Skirting:	Coved base of same type as floor	Supplier system, corner details
Colour:	Grey with satin gloss finish	Grey from suppliers range to be approved by IKEA
Gloss:	55 Gardner at 60º +/- 5%	
Water absorption:	Average 0,5%<E<3%,	Colour difference between tiles not visible from watching floor from 2 m
Slip resistance:	Acc. to norms, both polished/unpolished	
Breaking module:	Average > 27 MPa	
Maintenance, cleaning:	Proven manufactures manual to be followed.	
Transportation/storage:	Suppliers recommendations required.	
Conformity:	National/local norms, ev. British Standard	
Warranty:	5 years on material.	

Options
Large size tiles for, restaurant etc.

300,400,450 modular size
For IKEA approval

Related documents
Laying pattern to be agreed in advance of laying

Other
Moistures testing to be agreed and documented.

Best Buy
Supplier / installer availability

See intranet BB site for detail

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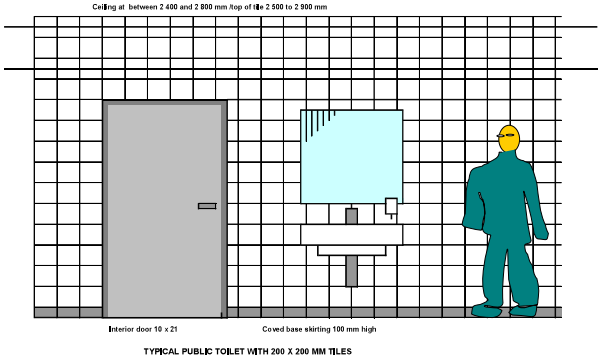
150 x 150mm wall tiles



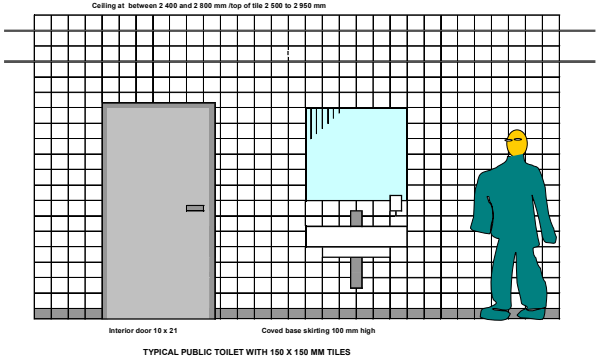
200 x 200mm wall tiles



200 x 200mm wall tiles



TYPICAL PUBLIC TOILET WITH 200 X 200 MM TILES



TYPICAL PUBLIC TOILET WITH 150 X 150 MM TILES

Remark

Description of Standard
This standard describes ceramic wall tiles to be used in stores on ground (OG), lifted (LS) and garaged stores (GS). Tiles be installed on smooth vertical surfaces which have been prepared for the installation.

Refer to left side of page

Function
The tiled surface shall constitute the finished surface in the object area and shall comply with the requirements set on performance regarding aesthetic appearance, wearing ability, hygiene and low maintenance cost. All functions to correspond with the usage of each particular area.

Refer to left side of page

Validity
This standard is valid within IKEA in Europe.

Region 1

Standard Specification

Type of wall tiling:	Medium size ceramic tiles applied to vertical surface with cementitious joints.	Refer to left side of page
Substrate:	Gypsum board, plastered block wall finished to within required tolerances. Water proofing membrane if required.	
Material:	Dry-pressed ceramic tiles manufactured of fine stoneware or stoneware with glazed surface	
Size:	Tiles in squares of 150x150, 100x200 or 200x200 mm. Exact size to allow joint of 3 mm. Maximum difference of single tile to average size: L<12cm/L>12cm max +/ 0,75/0,50 %.	
Tolerances in %	Straightness +/-0,3 – Right angle level +/- 0,5 - Centre/side camber +0,5/-0,3 – Distortion +/- 0,5 - Thickness +- 0,8 mm	
Bonding:	In cementitious glue as per manufacturers re-commendation. Quantity and thickness to be recommended.	
Jointing:	Joints to be 3 mm wide and flush finished of material compatible with tiles.	
Colour and gloss:	White with satin gloss finish (gloss 55)	NCS S0500-N, colour difference between tiles not visible from watching wall from 2 m
Water absorption:	Average 0,5%>E>3%, Single E<3,0%	
Breaking mould:	Average > 27 MPa	
Maintenance, cleaning:	Proven manufactures manual to be followed.	
Environmental:	Documented recycling strategy required	
Transportation/storage:	Suppliers recommendations required.	
Conformity:	EN 176, GB/T4100, ASTM	
Warranty:	2 years on material.	

Options
Skirtings or recommended uplift detail at walls.
Accent colour for feature details

Related documents
Laying pattern to be agreed in advance of laying

Other
Moistures testing to be agreed and documented.

Best Buy
Supplier / installer availability

See intranet BB site for detail

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Annexure A.03 - Paints

Remark

Description of standard
This standard describes the typical type of paints being used for construction of IKEA stores and similar projects.

Function
The functions of paints in this standard are to constitute a finished surface on floors, walls, ceilings and other building element (excluding pre-treatment like wallpaper or plastering) in order to meet the minimum functional requirements as per project design and IKEA Roombook.

Validity
This standard is valid worldwide within IKEA. Invalidity due to local demands to be treated separately.

Standard Specification
Paintmaterial shall be of various types like emulsion, latex or solvent based as per manufacturers standard and shall be suitable to meet the requirements in this specification.

Substrate 1:	Walls: Smooth surfaces of concrete, masonry or drywall. Textured surface to be specifically specified
Substrate 2:	Floors: Trowelled concrete
Substrate 3:	Ceilings: Smooth drywall or concrete /steel
Substrate 4:	Wood, plastic, primed steel members
Application:	Brush, roller or spraygun depending on function.
Coverage:	Staining, light cover or full cover depending function of room and substrate type.
Number of coats:	As per suppliers recommencation, application type and specific requirement
Gloss:	30-80 degree Gardner as required.
Material base:	Water or solvent ?
Thinning:	Only when required
Colour:	White or off white Accent colours to own specification
Environmental:	Strictest health and safety requirement or local demand
Durability:	
Warranty:	5 years

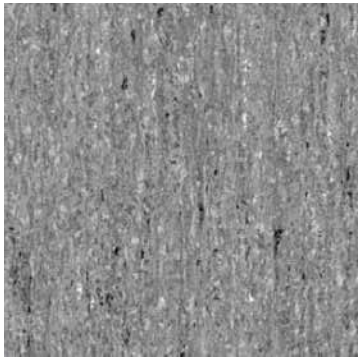
IKEA NCS ?

Regional requirements.
Europe:
North America:
Asia & Pacific:

Options

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Annexure A.04 - Flooring - Resilient



Type A
Homogeneous PVC



Type C
Vinyl Composition Tile

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Remark

Description of Standard
This standard describes the resilient flooring to be used in stores on ground (OG), lifted (LS) and garaged stores (GS) to be installed on rough structural surfaces which have been prepared for the installation.

Refer to left side of page

Function
The flooring shall constitute the finished surface in the object area and shall comply with the requirements set on performance regarding comfort, aesthetic appearance, wearing ability and low maintenance cost. All functions to correspond with the usage of each particular area. In sales areas electrical transport vehicles may be used extensively.

Refer to left side of page

Validity
This standard is valid within IKEA worldwide.

Region 0

Standard Specification

Type of flooring:	Single sheet in tiles or sheeting	
Substrate:	Trowelled concrete or levelling compound Moisture content to max. supplier requirement	Refer to left side of page
Material:	VCT, PVC, or equal material in thicknesses of 2.0 for PVC, 2.5 or 3.2mm for VCT.	
Size:	Tiles in squares of 300 or 500mm Sheet flooring in rolls 1800-2000mm wide .typically, and 18000 – 25000mm long	
Bonding:	Fully glued in adhesive recommended by the supplier. Use of recommended spreader.	
Jointing:	Tiles to be butt jointed, max. joint width 0.5mm Sheet flooring to have welded joints. Stretching ability to guarantee no joint failure.	
Colour and pattern:	Typically 2 shades of grey to IKEA approval. Fabrication batch number to avoid patchiness. Gloss of surface to be satin	
Surface:	Smooth, non-slip, hard to stain or damage.	
Resistance:	Product to be resistant to fat, mineral oil, acids and alkalis.	
Abrasion:	Tested to pr EN 660-1 & -2, Stuttgart & Frick-Taber tests.	Refer to technical specification.
Sound:	Step noise reduction 4dBA	
Maintenance, cleaning:	Proven manufactures manual to be followed.	
Environmental:	Documented recycling strategy required	
Transportation/storage:	Suppliers recommendations required.	
Conformity:	EN or BS, ASTM depending type and material.	Refer to technical specification.
Warranty:	2 years on material.	

Options
Pattern as feature floor
Skirtings or recommended uplift detail at walls.
Alternative thicknesses as per technical specification

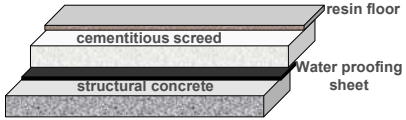
Related documents
Laying pattern to be agreed in advance of laying.
IKEA Technical specification — Link to come

Other
Moistures testing to be agreed and documented.

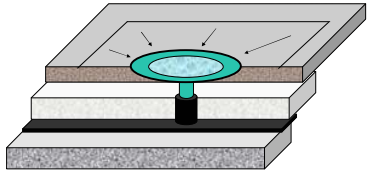
Best Buy
Supplier / installer availability — See intranet BB site for detail

Annexure A.05 - Flooring - Resin

Typical build up of kitchen floor



Typical floor drain connection



Description of standard — **Remark**

This standard describes a thin coat resin floor to be used mainly in food preparation and other wet areas in IKEA stores and other building with similar demands.

Function

The flooring here described shall constitute the finished surface in a hygienic area and shall have water repellent properties and also be anti-skid safe and easy to clean and maintain. Together with an under lying water proof sheet this flooring will constitute the complete composite solution.

Validity

This standard is valid worldwide within IKEA. — Region 0

Standard Specification

This resin material shall be of a chemical composition as per the supplier's own specification and be fit for the purpose and meet requirement here within.

Substrate:	Sand cement, light weight screed on a depressed waterproofed mezzanine structure. Surface to be flat with a tolerance of +/- 5 mm in 4 metres (3/16" in 12 feet). Moisture content to be maximum 5% at time of laying floor.	Refer to illustration
Substrate detail:	Area to include a limited number of floor drain penetrations where a local fall shall be provided.	Refer to IKEA Construction manual
Material:	Acrylic or epoxy resin including sand aggregate to supplier's specification.	
Colour:	Grey base with grey/black/white aggregate	To be agreed
Thickness:	Minimum 4 mm (5/32") at any point.	
Finish:	Slightly rough, non slip finish. Local approval required.	See technical specification
Resistance:	Floor must be resistant to water up to 100°C and be suitable in refrigerated areas. Oil/grease and acid resistant in kitchen environment.	
Flexibility:	Generally joints shall be avoided. Concrete structure may create movements and the flooring material must have a stretching ability or method to avoid cracking.	
Interface details:	Compatibility with substrate for bonding and waterproofing of drain penetrations to be provided for system guarantee.	
Skirting.	The flooring shall be wrapped up vertically 100 mm (4") on to main block work or gypsum walls.	
Installation:	The installation shall be made by a skilled worker approved by the supplier in order to obtain a complete guarantee.	See options
Acceptance:	Before laying the installer shall accept substrate.	
Maintenance:	Repair and stain removal manual to be provided. References shall be required.	
Environmental:	Recycling plan required	
Warranty:	10 years	

Regional requirements.

The product shall comply with the local requirement where it is going to be installed. Hygienic certificates where applicable.

Europe: EN

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North America: ASTM
Asia & Pacific: Local national rules. Japanese certificate required.

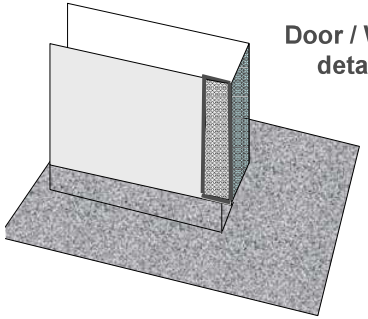
Options
If needed installation to be done others approved as above by supplier
Other material base than specified that will meet requirement

Related documents (on available on demand)
IKEA Roombook
IKEA Construction manual
IKEA technical tender specification
IKEA standard layout drawing
Construction drawing (project)

Other

Best Buy

Supplier / installer available		See intranet BB site
Feed back	Standard form	With BB



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Annexure A.06 – Dry wall Partition

	Remark
Description of standard This Standard describes a typical internal plasterboard partition in an IKEA store or distribution centre. The partition is to include boards, studs, movement joints and sealant at head, fixings, and metal plaster beads on external corners. **Function** To provide an enclosure around an internal space for the purpose of fire protection, sound reduction, security, privacy or other. The partitions are to be lightweight, and installed or remodelled without the need for traditional 'wet trades'. **Validity** This specification is valid within IKEA worldwide. **Standard Specification** **Standard partition** Partition height: Approx 5000mm typically to underside of soffit or roof above (or just above suspended ceiling height, 3000mm approximately, if a closed suspended ceiling both sides). Partition depth: 115mm minimum Framing: Metal studs fabricated from galvanised mild steel sheet. Size and centres to Manufacturer's design to meet design parameters. Head restraint: Studs extended to underside of floor slab or roof above, or otherwise restrained to Manufacturer's design. Head restraint to accommodate vertical deflection of floor slab or roof as advised by Structural Engineer, up to 35mm. Lining: Minimum single layer 12mm thickness standard plasterboard, paper-faced with rebated edges. Number of layers, thickness and size to manufacturer's design to meet design parameters. Provide sufficient screws to fix boards. Accessories: Provide metal corner plaster beads for all external corners. **Acoustic partition** As standard partition above, except partition to achieve a minimum of 45R_w dB 100-3150Hz. Insulation: Continuous layer of 25Kg/m³ minimum density mineral wool insulation hung between studs with no air gaps. Thickness to Manufacturer's design. Accessories: Provide sufficient sealant for all junctions with adjacent materials. **60min fire rated partition** As standard partition above, except fire rated to achieve EI60 (60 minutes integrity and insulation). Lining and frame upgraded as necessary in accordance with Manufacturer's tested systems. Plasterboard to be type DFI to EN 520. Insulation: Continuous layer of mineral wool insulation hung between studs with no air gaps. Density and thickness to Manufacturer's design.	EN 520 or equivalent local code

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Moisture resistant partition
As standard partition above, except suitable for use in high humidity areas (shower rooms, etc). Plasterboard to be type DH1I to EN 520.

Design parameters

Design life:	15 years.	
Warranty:	5 years minimum, insurance backed.	Refer to IKEA standard terms.
Structural design:	Maximum deflection L/240 at 200Pa. Where applicable, the wall must be designed to resist seismic loads in accordance with EN 1998 (Eurocode 8).	EN 1998 or equivalent local code

Options

1. Clerestory glazing:	As standard partition above, except f*raming to accommodate 400mm deep continuous strip of glazed screens by others. Architraves by others.* Framing sections continue past glazed screens at 1200mm centres.	

Related documents

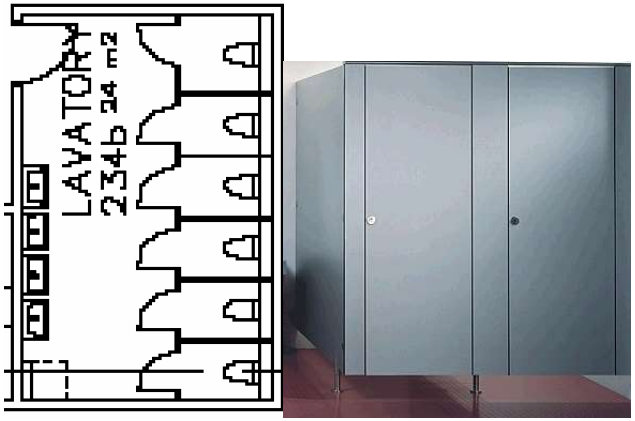
IKEA concept drawings	Refer to project specific Intranet/Web when project commences.
Construction drawings	
Construction details	
IKEA standard warranty terms	

Best Buy

Supplier / installer available		See intranet BB site for details
Feed back	Standard form	With BB

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Annexure A.07 – Sanitaryware, Fixture and Cubicle



Remark

Description of Standard
This standard describes modular demountable toilet cubicle systems for use in toilets/washrooms in both Customer and Staff areas of an IKEA store or DC. The store may be on ground (OG), lifted (LS), or with garage (GS).

Function
Provide a degree of privacy to users of water closets, while permitting air flow for ventilation, allowing ease of cleaning, economy of construction, and a minimum of space requirement. Comply with all code requirements, especially relating to disabled access provisions.

.

Validity
This standard is valid within IKEA in Europe — Region 1

Standard Specification

Cubicle type	Outward opening door	
Cubicle width	900 mm	typical
Cubicle depth	1400 mm	
Panel material	18 mm particle board, laminate covered, square edged	water resistant
Laminate colour	Light grey, RAL 7035	Easy clean
Frame material	Brushed aluminium, adjustable, header rail continuous across top of door	Sag resistant system
Attachment	Floor mounted, feet designed for ease of cleaning. Attached to back wall	Compatible with tiles or IPS
Gap below	100 mm min, 200 mm max	
Height	2000 mm	Approx
Door hardware	Brushed aluminium	Robust
Hinges	Self closing design	Gravity
Disabled access	One cubicle per washroom to comply with code requirements for ambulant disabled use, if required by local code.	Cubicle size, accessories and compatibility with required grab bars etc
Accessories	Coat hook	

Options

Cubicle type and details	Manufacturer proposal	Cost relevant

Related documents

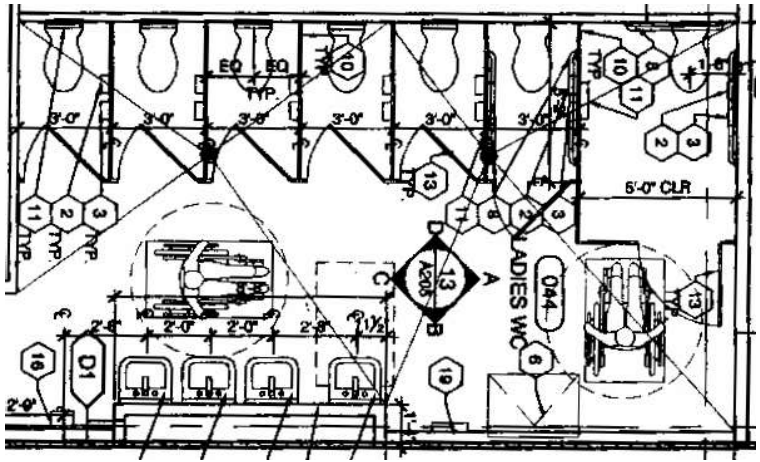
Technical Specification	link
IKEA concept drawings	See intranet/web
Construction Drawings	For co-ordination

Other

Best Buy

Supplier / installer available		See intranet BB for details
Feed back	Standard form	With BB

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Remark

Description of Standard
This standard describes modular demountable toilet cubicle systems for use in toilets/washrooms in both Customer and Staff areas of an IKEA store or DC. The store may be on ground (OG), lifted (LS), or with garage (GS).

Function
Provide a degree of privacy to users of water closets, while permitting air flow for ventilation, allowing ease of cleaning, economy of construction, and a minimum of space requirement. Comply with all code requirements, especially relating to disabled access provisions.

.

Validity
This standard is valid within IKEA in North America — Region 2

Standard Specification

Cubicle type	Inward opening	
Cubicle width	3'	typical
Cubicle depth	5' 2"	
Panel material	Galvanized steel, factory painted/powder coated, stretcher levelled	water resistant Easy clean (graffiti)
Panel colour	Light grey, RAL 7035	Sag resistant system
Frame material	Brushed aluminium or stainless steel, adjustable, header rail continuous across top of door	
Attachment	Floor mounted, feet designed for ease of cleaning. Attached to back wall	Compatible with tiles
Gap below	4" min, 10" max	Approx
Height	6' 6"	Robust
Door hardware	Brushed aluminium or stainless steel	
Locks	Sliding	
Hinges	Self closing design	Gravity
Handicapped access	One cubicle per washroom to comply with code requirements for handicapped use, as required by code.	Cubicle size, door width, accessories and compatibility with required grab bars etc
Accessories	Coat hook	
Urinal vision screens	Standard wall hung screens in same materials as cubicles	

Options

Cubicle type and details	Manufacturer proposal	Cost relevant

Related documents

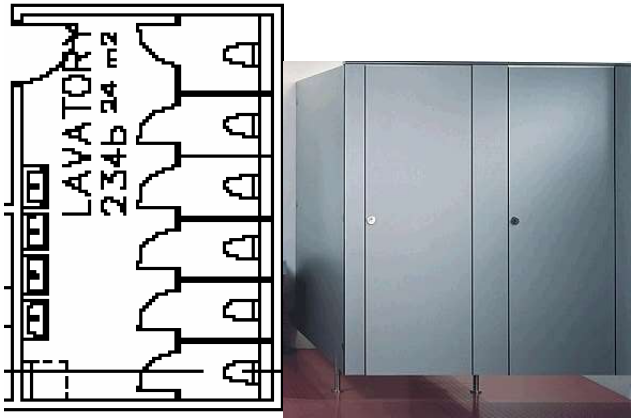
Technical Specification	link
IKEA concept drawings	See intranet/web
Construction Drawings	For co-ordination

Other

Best Buy

Supplier / installer available		See intranet BB for details
Feed back	Standard form	With BB

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Remark

Description of Standard
This standard describes modular demountable toilet cubicle systems for use in toilets/washrooms in both Customer and Staff areas of an IKEA store or DC. The store may be on ground (OG), lifted (LS), or with garage (GS).

Function
Provide a degree of privacy to users of water closets, while permitting air flow for ventilation, allowing ease of cleaning, economy of construction, and a minimum of space requirement. Comply with all code requirements, especially relating to disabled access provisions.

.

Validity
This standard is valid within IKEA in Asia & Pacific — Region 3

Standard Specification

Cubicle type	Outward opening door	
Cubicle width	900 mm	typical
Cubicle depth	1400 mm	
Panel material	18 mm particle board, laminate covered, square edged	water resistant
Laminate colour	Light grey, RAL 7035	Easy clean (graffiti)
Frame material	Brushed aluminium, adjustable, header rail continuous across top of door	Sag resistant system
Attachment	Floor mounted, feet designed for ease of cleaning. Attached to back wall	Compatible with tiles or IPS
Gap below	100 mm min, 200 mm max	
Height	2000 mm	Approx
Door hardware	Brushed aluminium	Robust
Locks	Sliding	Not gravity-related
Hinges	Self closing design	Gravity
Disabled access	One cubicle per washroom to comply with code requirements for ambulant disabled use, if required by local code.	Cubicle size, accessories and compatibility with required grab bars etc
Accessories	Coat hook Hardware to fix toilet paper dispenser	

Options

Cubicle type and details	Manufacturer proposal	Cost relevant

Related documents

Technical Specification	link
IKEA concept drawings	See intranet/web
Construction Drawings	For co-ordination

Other

Best Buy

Supplier / installer available		See intranet BB for details
Feed back	Standard form	With BB

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1. Description of Standard

1.1 Short description

This standard describes general sanitary ware and plumbing fixtures used in IKEA stores or distribution warehouses, with the exception of food service areas.

1.1.1 Function

The fittings are to perform the required sanitary function in a hygienic and user friendly way, with good durability, maintenance, environmental and aesthetic qualities.

1.2 Validity

This IKEA Standard is valid worldwide.
IKEA Standards describe IKEA minimum requirements for a product/component.

Products/components must always comply with regional/Local laws, norms, authority requirements etc. if these are higher than IKEA Standards.
To be recognized and fulfilled by suppliers, contractors, consultants etc.

1.3 Conceptual, Mandatories

2. Technical Specification

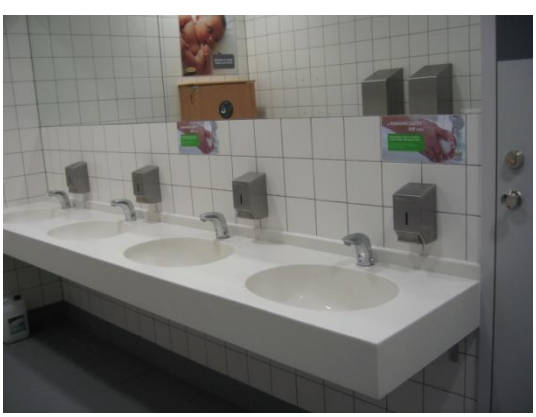
2.1 General

This description describes only basic requirements for plumbing fixtures.

2.2 Washbasins in customer and staff toilets

2.2.1 Vanity units

Made of solid homogeneous acrylic polymer/mineral composite counter with integral wash basins.
Size is depending on space.



Example of vanity unit

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2.2.2 Pedestal washbasin

Made of solid vitreous china
Approximately size WxD = 500x400 mm



Example of wash basin

2.3 Taps

2.3.1 Automatic sensor taps in customer and staff toilets

Made of chrome plated brass, with thermostatic blending valve, and integral pop up waste.
The electronics can be hardwired or battery-operated. Hardwired products are harder and more expensive to install but require less maintenance than battery-operated devices. Most battery-operated automated systems use standard batteries available and rechargeable batteries can also be used.



Example of tap with automatic censor

2.3.2 Push button water taps in customer and staff toilets

Made of chrome plated brass, with thermostatic blending valve, and integral pop up waste.



Example of tap with push button

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2.3.3 For utilities in door

Chrome plated brass, wall mounted, 12 mm inclined bib taps. Pipe work is preferably exposed but may be concealed.



Example of wall mounted taps

2.3.4 For utilities out door and in customer areas

Chrome plated brass, wall mounted, with detachable key. Pipe work is preferably exposed but should be concealed if placed out door.



Example of tap with key

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2.4 Water closets

2.4.1 With exposed water cistern

Solid vitreous china, wall mounted on concealed purpose made bracket, Wash down water closet with air trap and outlet concealed .Exposed water cistern with capacity of 6 litres and water saving feature to max . 4 litres. One top mounted flush button with toggle max/min flushing choice. Closet to be without cover lid but with possibility to add. A variation suitable for use by disabled persons is to be included.

Note that this type of exposed water cistern can be vulnerable to vandalism.



Example of water closet with exposed water cistern

2.4.2 With concealed water cistern

Solid vitreous china, wall mounted on concealed purpose made bracket, Wash down water closet with air trap and outlet concealed .Concealed water cistern with capacity of 6 litres and water saving feature to min. 4 litres. Flush button with toggle max/min flushing choice. Closet to be without cover lid but with possibility to add. A variation suitable for use by disabled persons is to be included.

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Example of concealed water tank, inkl. frame for water closet

2.4.3 Water closet connected to the water main.

Solid vitreous china, wall mounted on concealed purpose made bracket, Wash down water closet with air trap and outlet concealed . Closet to be without cover lid but with possibility to add. A variation suitable for use by disabled persons is to be included. The flushing valve can be manual or as in picture below with automatic sensor.The electronics can be hardwired or battery-operated. Hardwired products are harder and more expensive to install but require less maintenance than battery-operated devices. Most battery-operated automated systems use standard batteries available and rechargeable batteries can also be used.



Example of water closet connected to the water main

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2.5 Urinals

In customer and staff toilets

Made of solid vitreous china, wall mounted and divider in same style as cubicle partitions (laminate) and 32mm bottle traps with 75mm seal.



Example of urinals and divider

2.6 Utility sinks

To be in heavy duty stainless steel, sink with integral splash back, approx. 500x400mm. Cleaner's sinks to also have stainless steel bucket grating to put the bucket on.



Example of stainless steel sinks.

2.7 Compliance

Sanitary and plumbing installation is to comply with all relevant regional or national codes, cisterns, overflow and flushing system to be revised if necessary.
Fitting are to comply with specific Local Norms relevant to each item.

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3. Related Products / Accessories / Options

3.1 Sundry items

Showers, shower trays, baby change tables, baby seats, grab rails, soap dispensers, etc are to be included. Refer to the Room Specification for details.

3.2 Options

3.2.1 Utility sink

Heavy duty fireclay sinks with integral splash back, approx. 500x400mm. Cleaner's sinks to also have stainless steel bucket grating and hardwood pad.



Example of heavy duty fire clay utility sink

4. Social & Environmental

4.1 Code of conduct

All manufacturing and supply of products must comply with IKEA Services AB:
"Code of Conduct – for supply of Property Services"
"IWAY Compliance Commitment"
Latest revision

4.2 Environmental

4.2.1 Materials

Legally forbidden materials/substances are not accepted.
A recycling plan must be available for the product.

4.2.2 Water saving

All taps in customer and staff toilets should be equipped with water saving devices type aerators.

5. References and Related documents

IKEA Engineering and Construction Manual
IKEA Roombook
Project specific: IKEA concept drawings, construction drawings & details, and sanitary ware schedule

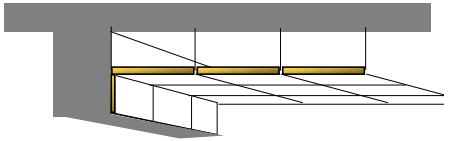
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Annexure A.08 - False ceiling

Type A Acoustic ceiling



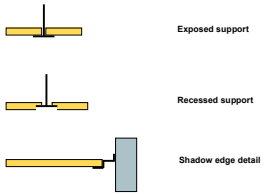
Ceiling Type D / insolation at parking -1



Type C Metal ceiling



Typical support details



Description of standard — **Remark**

This standard describes suspended tiled ceilings used in an IKEA store, distribution centres and offices.

Function

The function of the suspended ceiling is to reduce the visual height, conceal services above and/or reduce acoustic reverberation depending on the area of installation. In certain areas it shall act as hygienic, cleanable shield. When part of building envelope have or be part of thermal insulation.

Validity

This standard is valid worldwide within IKEA. — Region 0

Standard Specification

The tiled ceiling shall be of lay-in type resting on special profiles suspended from mezzanine slabs or roof decks and connected to adjacent walls. Tiles shall be easy to remove, replace and to clean.

Substrates: Concrete, sheet metal profile horizontally
Block work, gypsum walls vertically

Vertical support: Steel wires, chains or similar members according to suppliers system design. Non corrosive material. Capacity to carry a general load of 10 kg/m^2 from installations without visible deflection..

Secondary support: Aluminium T-shape track for lateral support of individual tiles according to manufacturer's design. Visible part to be white lacquered. Additional shall be maximum allowed by manufacturer's design and support detail (clip).

Tile material: Mineral fibres, metal or other incombustible material

Fire rating. None, material shall not ignite when exposed to open flame. — Refer to local requirement

Acoustic property: To meet requirements according to Roombook

Thermal conductivity: Only for external tiles

Moisture resistance: Types B and C shall be able to withstand 100% relative humidity at 40°C with sustained form stability in wet and exterior areas.

Fixing: Where uplift forces occur, fixing clips shall be used.

Ceiling types:

Type A: Regular acoustical ceiling of mineral fibres in thickness of approximately 20 mm.
Tile size to be 600 x 1200 mm (2 x 4 foot)
Finish white smooth surface with slightly fissured pattern

Type B: Washable hygienic tile for kitchen areas. Tile size 600 x 1200
Material to be metal or coated mineral fibre and be installed with fixing clips. The butt ends of tiles to have a return of the same type as the surface material. White smooth finish.

Type C: Metal tile for areas exposed to wind and moisture and no sound requirement. Tiles to be in size 600 x 600 mm and to have white smooth lacquered soffit. Installed with fixing clips.

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Type D: Thick insulating tile in mineral wool for exterior use in lifted store.
Tile size to be 600 x 1200 mm and thickness 50 mm.
Flame spread rating minimum class 1 for surface materials.
Good sound absorption properties. Surface to be white or natural with good light reflection and possible to clean.

Refer to left side

Warranty: 5 years.

Regional requirements.

The material shall comply with all local requirement regarding fire, environmental requirement and hygienic regulations.

Europe:	prEN 13501, EN ISO 717,11654 and 140
North America:	ASTM, UL, ULC, installation according to seismic demands.
Asia & Pacific:	Local demands Japanese approval required. PSE certificate

Options

Type D ceiling to be installed direct to concrete
Thickness of tile type D to be 70 mm.

Related documents (only available on demand)

IKEA Roombook
IKEA technical tender specification
IKEA standard layout drawings

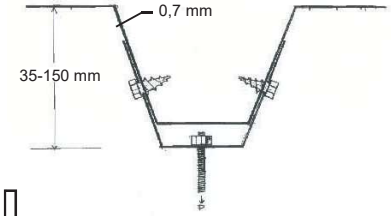
Other

Best Buy

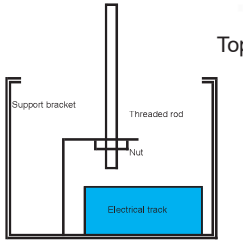
Supplier / installer available — See intranet BB site

Feed back — Standard form — With BB

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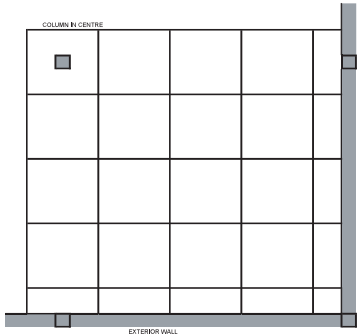


Top bracket detail

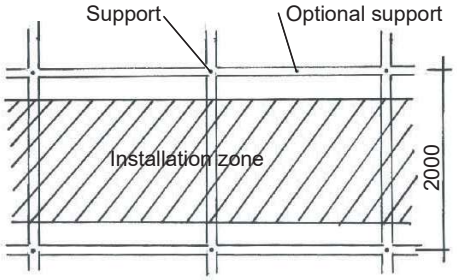


Cross section of 80 x 100 mm channel

Torsion under load not accepted



STANDARD LAYOUT OF 2 x 2 M GRID ALONG PERIMETER



Coordination building installation



Wall detail to be adapted per project



General view of grid

Description of standard
This standard describes an open metal grid ceiling that is installed in IKEA store on ground (OG), lifted stores (LS) and stores with garage (GS). Ceiling to be installed in sales areas, restaurant and offices possible.

Remarks
Refer to left side of this page

Function
The grid ceiling shall function as a visual break and to support lighting fixtures and commercial items (signs and various infills) as well as being an electrical raceway with a high level of flexibility for the power/data distribution. Power distribution to be either track or cable.
In certain areas a higher load capacity is required.

Refer to left side of this page

Validity
This standard is valid worldwide within IKEA.
Invalidity due to national, regional or other other demands to be treated as option or other standard.

Regions 0

Standard specification

Ceiling type	¤Open grid		
Grid measurements	2000x2000 mm		centre to centre
Beam measurements HxW	100x80 mm		
Support height roof to ceiling	2500 / 7000 mm		Approximate
Face material and finish	¤Steel powder coated gloss		White white NCS 0500/RAL 9010 30°5 (60E)
Fire requirement	¤non combustible		
Sound requirement	¤none		
Support capacity	¤live load	20 kg/m² (average)	See Roombook
	¤point load	50 kg	max one adjacent beam
	¤no torsion or buckling allowed in grid		
Fixing to structure	¤threaded rod	at 2000 centre appr.	Max deflection 1/250
	¤top fixing to concrete or metal sheet deck typically 130 mm deep profile with wall thickness of 0,7 mm		depending roof deck/purlins
	¤seismic forces		Local code
Transport and mounting			By IKEA, see option below
Warranty	¤2 years on material		

Options

Transport	Possible by supplier	
Perimeter wall detail/length of channel	Adapt to project	Refer to left side of this page
Bulkhead detail / restaurant / checkout	Additional	Refer to left side of this page
Ceiling accessories	Retail requirements	See BB link/intranet
Extra support ¤Heavy loads		Local decision

Related documents

Detailed ceiling drawing and fittings		By manufacturer
IKEA Technical specifications	Includes accessories	Link
Installation drawings		For co-ordination
IKEA layout drawings		For co-ordination
Building main sectional drawings		For co-ordination

Other
Coordination between ventilation ducts and support before exection

Refer to left side of this page

Best Buy

Detailed technical description ITS		See intranet BB for details
Supplier/installer availability		See intranet BB for details
Feedback	Standard form	With Best Buy

*) Footnote 0,00/AAAA/000/0 = = Cost Acc # / brief descr / item # / region #

0 World, 1 EU, 2 NA, 3 AP

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Annexure A.09 - Door Made of Steel

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1. Description of Standard

1.1 Short description

This standard describes the **minimum** requirements for all different types and sizes of **exterior and interior** steel doors in IKEA Stores and Warehouses.

1.1.1 Function

The doors shall provide a closure in order to separate rooms to rooms and rooms to outside. The basic functions vary with the location and are expressed as requirements regarding appearance, fire and smoke rating, sound reduction ability, thermal insulating property, operation and resistance to wear and tear. The complete function of the door will be seen in conjunction with ironmongery forming part of a related standard.

1.2 Validity

This IKEA Standard is valid worldwide.
IKEA Standards describe IKEA minimum requirements for a product/component.

Products/components must always comply with regional/Local laws, norms, authority requirements etc. if these are higher than IKEA Standards.
Specific door types and sizes may be valid only for certain regions.

To be recognized and fulfilled by suppliers, contractors, consultants etc.

1.3 Conceptual, Mandatories

Door colours should follow recommendations from Inter IKEA Systems B.V, IKEA standards colours.
Sizes may vary due to local rules.

2. Technical Specification

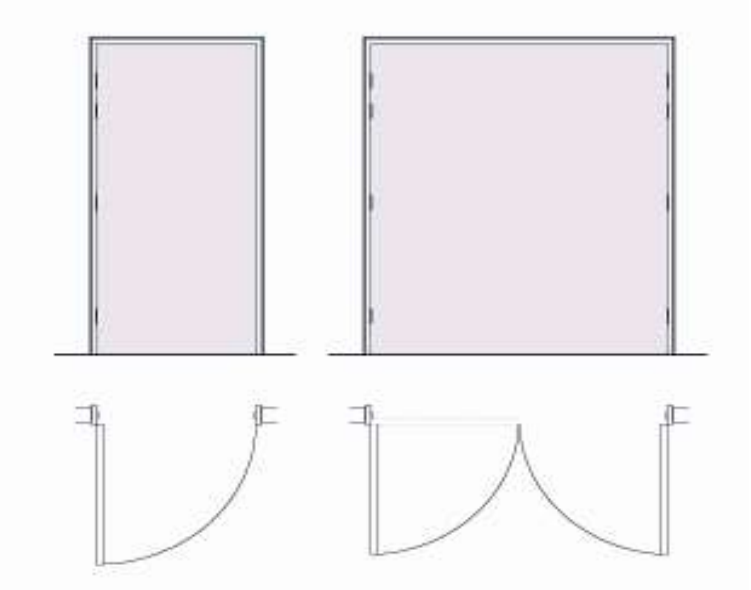
2.1 General

This specification describes basic requirements only.

2.2 Door action

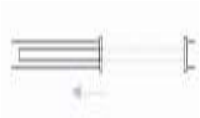
Depending on door location and use, following actions can be chosen:

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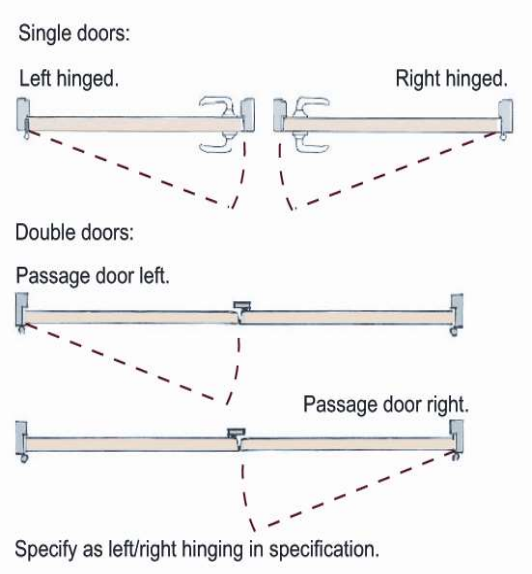


Single leaf
Single swing

Double leaf
Single swing



Single leaf
Sliding



2.3 Door leaf

2.3.1 Face material

External: Minimum 1 mm galvanized steel with 60 micron polyester powder coated finish.
Internal: Minimum 1 mm mild steel with 60 micron polyester powder coated finish.

2.3.2 Colour

External: NCS S 3000-N (Grey).
Internal: NCS S 0500-N (White), NCS S 2500-N (Grey) or NCS S 3000-N (Grey).

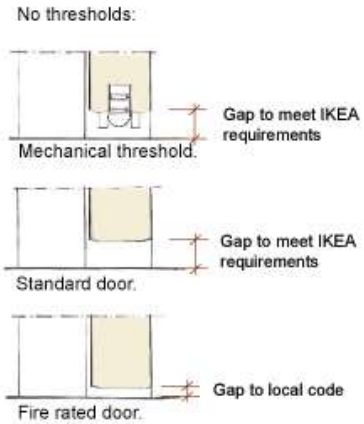
2.3.3 Core material

Hollow or filled with mineral wool or equal depending on the function.
Minimum U-value: 2.0 W/m²K for exterior doors.

2.3.4 Leaf thickness

Minimum 40 mm (1 ¾")

2.4 Threshold details



2.5 Door frame

Frames shall be included for all doors. They shall be made of cold rolled steel with a double rebate to suit the door leaf and form an architrave on one side

2.5.1 Frame material

External: Galvanized steel with 60 micron polyester powder coated finish.
Internal: Standard is steel with 60 micron polyester powder coated finish.
In wet areas such as kitchen, dishwashing room etc. Stainless steel frames are to be chosen to reduce maintenance costs.

2.5.2 Measurements

To supplier's standard. Minimum 30 mm architrave.
Material thickness minimum 1,5 mm (20 gauge).
To suite wall thicknesses 70 – 200 mm.

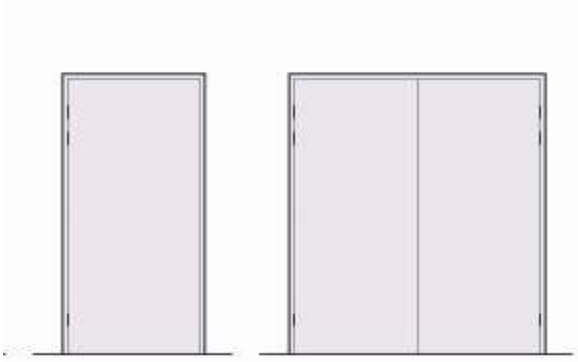
2.5.3 Colour

Same colour as door leaf except when stainless steel, then it is brushed stainless steel.

2.5.4 Mounting

Pre-assembled with sleeved adjustable fixing bolts

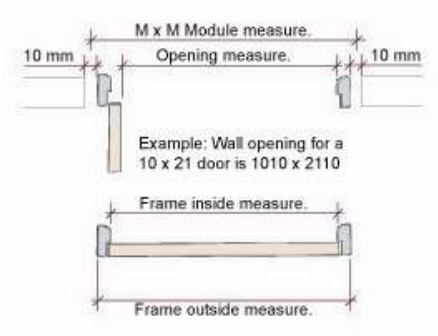
2.6 Door size



W X H
appr.1.0 m x 2.1 m

W X H
appr. 2.1 m x 2.1 m

Examples of door opening measures:



Example of sliding steel door:



Structural opening sizes to be checked and confirmed with supplier of door

2.7 Basic ironmongery

All doors supplied with **minimum 3 no** of heavy duty butt hinges per leaf.
All leaf to receive mortised lock cases. Refer to ironmongery schedule and ironmongery standard (04-0230) for other preparation required.

2.8 Others

2.8.1 Fire and smoke rating

Depending on location in building the following ratings applies:
None, EI30, EI60, or EI120 minutes

Smoke seals to be installed if required.

2.8.2 Sound reduction

Depending on location in building the following ratings applies:
None and / or Rw=30dB

2.8.3 Labels

All doors shall have certification and ID-number.

2.8.4 Glazing

Glazing not allowed in fire doors

2.8.5 Norms applicable

Local, National, EU (EN 13162, 1935, 1634-1) or British Standard.

3. Related Products / Accessories/Options

Other sizes than mentioned above can occur due to legal or operational demands.

4. Social & Environmental

4.1 Code of conduct

All manufacturing and supply of products must comply with IKEA Services AB:
"Code of Conduct – for supply of Property Services"
"IWAY Compliance Commitment"
Latest revision

4.2 Environmental

4.2.1 Materials

Legally forbidden materials/substances are not accepted.
A recycling plan must be available for the product.

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1. Description of Standard

1.1 Short description

This standard describes the **minimum** requirements for all different types and sizes of interior wooden doors in IKEA Stores and Warehouses.

1.1.1 Function

The basic functions of a door are to act as a closure in an opening. Doors are either opened manually or automatically. The door type is determined by its usage and is identified in the standard specification Chp 2, and in diagram below.

1.2 Validity

This IKEA Standard is valid worldwide.
IKEA Standards describe IKEA minimum requirements for a product/component.

Products/components must always comply with regional/Local laws, norms, authority requirements etc. if these are higher than IKEA Standards.
Specific door types and sizes may be valid only for certain regions.
To be recognized and fulfilled by suppliers, contractors, consultants etc.

1.3 Conceptual, Mandatories

Door colours should follow recommendations from Inter IKEA Systems B.V, IKEA standards colours.
Sizes may vary due to local rules.

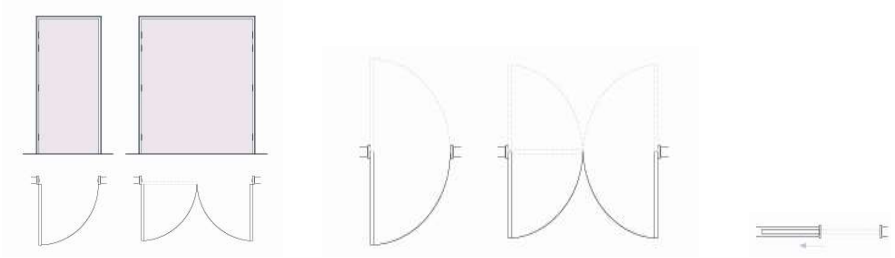
2. Technical Specification

2.1 General

Subject to door type, location and use the door will be required to fulfil any of these basic requirements.
Alterations and/or additions to the doors to be found under options Chp 3

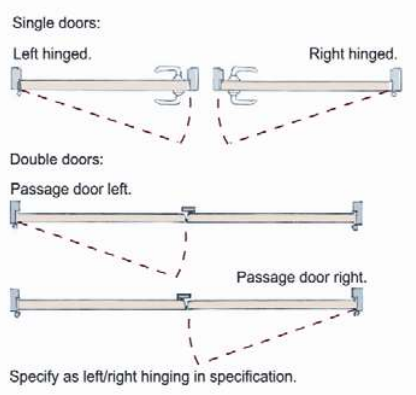
2.2 Door action

Depending on door location and use, following actions can be chosen:



Single leaf Single swing | Double leaf Single swing | Single leaf Double swing | Double leaf Double swing | Single leaf Sliding

V. 2005



2.3 Door leaf

2.3.1 Face material

0.8mm minimum thick high pressure laminate

2.3.2 Colour

NCS S 0500-N (White), NCS S 2500-N (Grey), NCS S 3000-N (Grey).

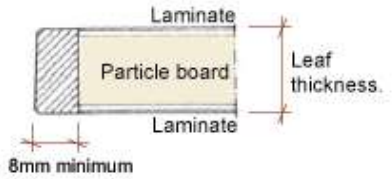
2.3.3 Core material

Solid particle board. Low formaldehyde content

2.3.4 Edge material

Hardwood edges or equivalent, fixed on bottom and leading edge as a **minimum.**
Use only wood with approval from world forestry. See example below

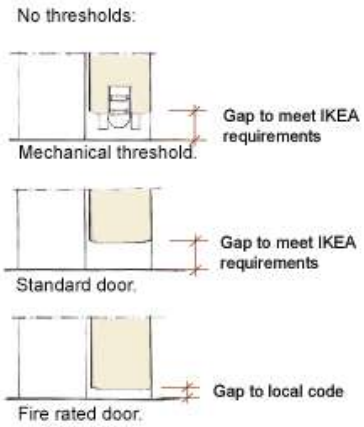




2.3.5 Leaf thickness

Minimum 40 mm

2.3.6 Threshold details



2.4 Door frame

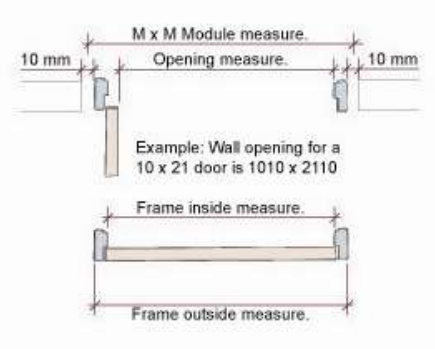
2.4.1 Frame material

Standard is powder coated steel.
In wet areas such as kitchen, dishwashing room etc. Stainless steel frames are to be chosen to reduce maintenance costs.

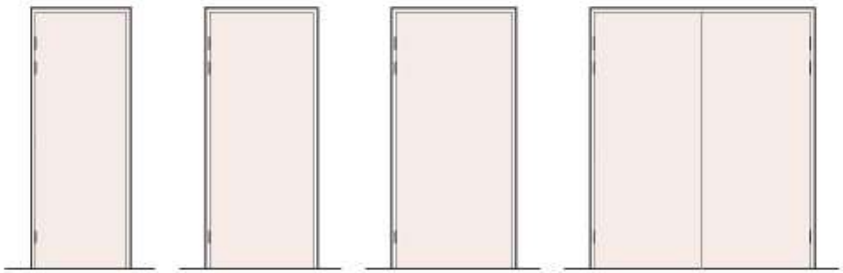
2.4.2 Colour

Same colour as door leaf except when stainless steel, then it is brushed stainless steel.

2.5 Door sizes



Door opening Sizes

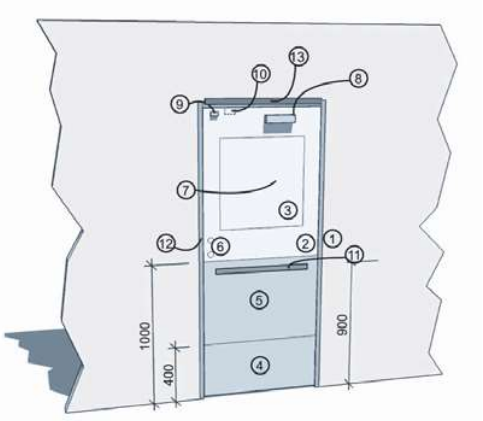


Door sizes W x H:

0.8 x 2.1 m	0.9 x 2.1 m	1.0 x 2.1 m	2.1 x 2.1 m Swing and sliding

2.6 Basic ironmongery

All doors supplied with minimum 3 no of heavy duty butt hinges per leaf.
All leaf to receive mortised lock cases. Refer to ironmongery schedule and ironmongery standard (04-0230).



1. FRAME (laquered steel)
2. LEAF (massive, laminated)
3. WINDOW (option, laminated or tempered glass)
4. KICKPLATE 1 (option, stainless steel , ~1mm)
5. KICKPLATE 2 (option, stainless steel, ~1mm) - see Roombook
6. HANDLE/KNOB
7. SIGN (option)
8. DOOR PUMP (option)
9. DOOR OPENHOLDER MAGNET (option)
10. ALARM MAGNET (option, recessed)
11. PUSHBAR
12. LOCK
13. DOOR COORDINATOR

Refer to ironmongery standard 04-0230

2.7 Others

2.7.1 Fire and smoke rating

Depending on location in building the following ratings applies:
None, EI 30, EI 60, or EI120 minutes

Smoke seals to be installed if required.

2.7.2 Sound reduction

Depending on location in building the following ratings applies:
None, Rw=35dB and Rw=45Db

2.7.3 Labels

All doors shall have certification and ID-number.

2.7.4 Glazing

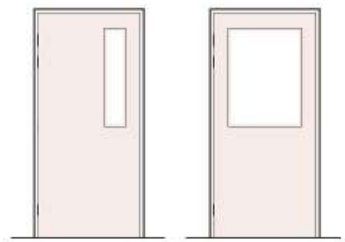
See options below

2.7.5 Norms applicable

Local, National, EU or British Standard.

3. Related Products / Accessories/Options

3.1.1 Vision panels



Single vision panel

Half glazed

3.1.2 Hollow core particle board

To save weight, but please observe that these kind of doors have limited wear and tear.

4. Social & Environmental

4.1 Code of conduct

All manufacturing and supply of products must comply with IKEA Services AB:
"Code of Conduct – for supply of Property Services"
"IWAY Compliance Commitment"
Latest revision

4.2 Environmental

4.2.1 Materials

Legally forbidden materials/substances are not accepted.
A recycling plan must be available for the product.

Annexure A.10 - Ironmongery

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1. Description of Standard

1.1 Short description

This standard describes general ironmongery to be used on doors in IKEA stores, DC: s and other project with same requirements.

1.1.1 Function

In conjunction with the door the different items of ironmongery shall contribute to the required operation of the door with regard to normal opening and closing, automatic closing, sequential closing, safety opening and protection of doors with heavy usage. Further on, the locks shall accommodate a variety of cylinder and privacy locks

1.2 Validity

This IKEA Standard is valid worldwide.
IKEA Standards describe IKEA minimum requirements for a product/component.

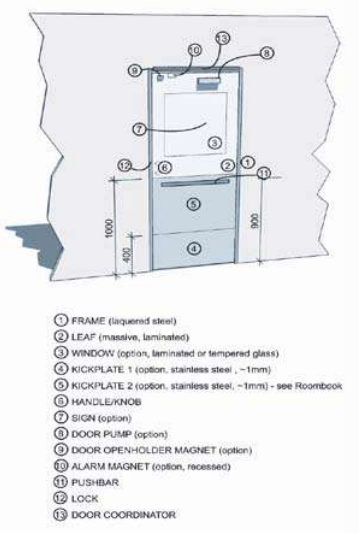
Products/components must always comply with regional/Local laws, norms, authority requirements etc. if these are higher than IKEA Standards.
To be recognized and fulfilled by suppliers, contractors, consultants etc.

1.3 Conceptual, Mandatories

2. Technical Specification

2.1 General

The ironmongery items included shall be by proven design and be suited to all standard door leaves within a store. The performance shall be as per supplier's design and as a minimum as per this specification. All necessary details to be included for full installation.



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2.2 Lock case

Mortise lock case with cold-rolled steel chassis with spring operated latch bolt or dead bolt to fit striking plate in stainless steel. The lock case shall be operated via one or two side door lever. When cylinder on only one side the opposite side shall have thumb turn. **The unit shall be made for heavy duty usage**



Example of lock case

2.3 Door lever

Double sided connected with one square bolt to suite industry standard lock cases. Lever shall be matt chrome plated steel and be straight design without end return. Circular cross section with a minimum diameter of 15 mm. Roses and escutcheons to suit lever and door.

2.4 Door closer

Shall be of heavy duty type, fully hydraulic, full rack and pinion action with high strength material. Tested for10 million loads cycles. Spring power to be adjustable. No seasonal adjustment to be necessary. When fire rated, closer shall meet local requirements and have test certificate.

2.5 Panic bar

This is for doors with special evacuation requirements (single or double). Material to be of brushed aluminium or of coated steel. Assembly shall be tested for its intended use and certified to operate with a minimum of 50 N applied force in emergency situations. Design can be made for surface mounting or concealed mounting. Compatibility required for installation together with all industry standard doors.

2.6 Door coordinator

The door-coordinator shall control a self closing double door in order that the leaves will close in the correct order to leave doors in closed position. Design to be heavy duty in galvanized steel, include fixing bolts and be compatible with all industry standard doors.

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Picture showing a door with door closer, door coordinator and panic bars installed

2.7 Emergency exit handle

For doors with no special evacuation requirements (non public areas). One action handle to activate the lock case to enable exiting through a door in an emergency situation. Chrome plated steel. Handle or turn knob to be secured to avoid misuse in non emergency situations.

2.8 Kick plates

Plates in 1 mm stainless steel with pre-drilled holes for counter sunk screws.
Edges are to be tapered. Sizes to fit doors and to be approximate 800 and 1000 mm wide and 400 and 1000 mm high

2.9 Regional requirements and standards.

2.9.1 Europe

EN179, 1125 (escape), EN 1154, 1634 (Fire), ISO 9001

2.9.2 North America

ANSI A156.2, CSA, UL, ULC, NFPA Standards no. 80

2.9.3 Asia & Pacific

Local or National Standards.
Ev. British Standards

3. Related Products / Accessories / Options

Panic bar in slim line design

4. Social & Environmental

4.1 Code of conduct

All manufacturing and supply of products must comply with IKEA Services AB:
“Code of Conduct – for supply of Property Services”
“IWAY Compliance Commitment”
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4.2 Environmental

4.2.1 Materials

Legally forbidden materials/substances are not accepted.
A recycling plan must be available for the product.

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Annexure A.11 - Sunshade, Windows and Curtain wall & Automatic sliding doors

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1. Description of Standard

1.1 Short description

This standard describes Sunshades/brise soleil over windows or in front of curtain walling on an IKEA store or Warehouse, including associated fixings and supports.
The store may be on ground (OG), lifted slab (LS), or with garage (GS).
The sunshades are to be fixed **outdoors** on south elevations and, subject to solar gain and glare calculations, on east and west elevations.
Two types of sunshades are addressed: small horizontal blades in a horizontal plane over strip windows, and larger horizontal blades in a vertical plane in front of windows and curtain walling.

1.1.1 Function

Provide a degree of solar gain control and glare control to the glazed element below or behind.
The degree of control will vary subject to the latitude of the store location and the store orientation

1.1.2 Recommendations

Small window areas:
Use lamellas placed above windows and perpendicular to the façade (type A).

Large window areas:
Use the horizontal lamellas placed in front of the glass as described in this standard (type B).

1.2 Validity

This IKEA Standard is valid worldwide.
IKEA Standards describe IKEA minimum requirements for a product/component.

Products/components must always comply with regional/Local laws, norms, authority requirements etc. if these are higher than IKEA Standards.
To be recognized and fulfilled by suppliers, contractors, consultants etc.

1.3 Conceptual, Mandatories

The sunshade shall be subordinated to the rest of the façade/trade dress and shall not be perceived as an extra designed feature.

Sunshade on an IKEA façade must always be done in a way that secures transparency inside/out and outside/in.

For type B sunshade design approval from IISSAB is mandatory.

2. Technical Specification

2.1 General

This specification describes basic requirements only

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2.2 Material and Finish

Extruded (natural) anodised aluminium.

2.3 Type A – Horizontal

Horizontal louvre blades to span horizontally between horizontal cantilever brackets, to form a sunshade in a horizontal plane above strip windows. Diagonal supports to provide additional bracing if necessary (Example, see picture below)



2.3.1 Louvre blades

100mm wide flat profile horizontal blades (louvers), at 45°and 125mm centres

2.3.2 Size

Sunshade to extend 1200 mm from window

2.3.3 Support brackets

T shaped brackets at 2000mm centres to match the glazing module, with no matching framing element along outside edge. Bracket depth to manufacturer's design to meet all design parameters

2.3.4 Diagonal support

Preferably none but if necessary they should be as slender as possible, maximum 25 mm in diameter.

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2.4 Type B – Vertical

Horizontal louvre blades to span horizontally between vertical columns, to form a sunshade in a vertical plane in front of curtain walling and windows. (Example, see picture below)



2.4.1 Louvre blades

300 mm wide 'aerofoil' profile horizontal blades (louvers), at approx 15°and 700mm vertical centres, to match glazing module

2.4.2 Supports

Slender square or circular hollow section columns placed at 2 m or 4 m centres, to coordinate with the glazing module. Columns to be fixed at base and head, and braced back to curtain wall as necessary

2.4.3 Loads

Wind and snow loading as local code requirements

2.5 Warranty

5 years

3. Related Products / Accessories / Options

3.1 Finish

Polyester powder coated, IKEA standard colour Dark grey NCS S 6502-B.

3.2 Design

If the manufacturer has an alternative design it must be to be presented, and needs IKEA approval.

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4. Social & Environmental

4.1 Code of conduct

All manufacturing and supply of products must comply with IKEA Services AB:
"Code of Conduct – for supply of Property Services"
"IWAY Compliance Commitment"
Latest revision

4.2 Environmental

4.2.1 Materials

Legally forbidden materials/substances are not accepted.
A recycling plan must be available for the product.

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1. Description of Standard

1.1 Short description

This standard describes Aluminium Window and Curtain Wall glazing systems in an IKEA store or DC façade.
The store may be on ground (OG), lifted (LS), or with garage (GS)

1.1.1 Function

To form a part of the façade, creating an architectural feature (part of the IKEA trade dress) which conveys an impression of openness and provides visual communication between exterior and interior and allows natural light into staff and customer areas.

1.2 Validity

This IKEA Standard is valid worldwide.
IKEA Standards describe IKEA minimum requirements for a product/component.

Products/components must always comply with regional/Local laws, norms, authority requirements etc. if these are higher than IKEA Standards.
To be recognized and fulfilled by suppliers, contractors, consultants etc.

1.3 Conceptual, Mandatory

Conceptual are the external colour of the frame.

2. Technical Specification

2.1 General

This standard is divided into 3 parts describing Window Frames, Curtain Wall Frame and Glazing

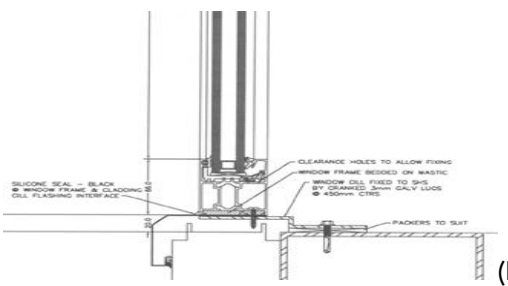
2.2 Window Frame

2.2.1 Orientation

Vertical flat plane

2.2.2 Supporting structure

Primary structural frame



(Example of fixing in façade)

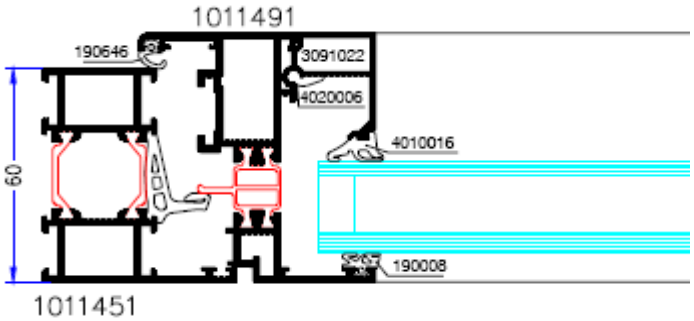
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2.2.3 Span

Vertical, varies up to 2800 mm in units of 700 mm.

2.2.4 Type

Standard Aluminium thermally broken system, drained, ventilated and pressure equalised.



(Example of window profile thermally broken)

2.2.5 Dimensions

Slender, approx 50 mm width (viewed from front)

2.2.6 Material

Extruded aluminium (EN 755)

2.2.7 External finish

Frame in powder coated aluminium
Colour: IKEA Dark Grey NCS (National colour system) S 6502-B with a satin finish gloss, 20 Gardner, +10, -5 at 60 degrees angle. Minimum thickness shall be 40 microns.

2.2.8 Internal finish

Frame in powder coated aluminium
Colour: IKEA Dark Grey NCS (National colour system) S 6502-B with a satin finish gloss, 20 Gardner, +10, -5 at 60 degrees angle. Minimum thickness shall be 40 microns.

Options (consider the extra costs for this before deciding)
Frame in powder coated aluminium
Colour: IKEA White NCS (National colour system) S0500-N with a satin finish gloss, 20 Gardner, +10, -5 at 60 degrees angle. Minimum thickness shall be 40 microns.
OR
Frame in powder coated aluminium
Colour: IKEA Grey NCS (National colour system) S 3000-N with a satin finish gloss, 20 Gardner, +10, -5 at 60 degrees angle. Minimum thickness shall be 40 microns.
OR
Frame in powder coated aluminium
Colour: IKEA Grey NCS (National colour system) S 2500-N with a satin finish gloss, 20 Gardner, +10, -5 at 60 degrees angle. Minimum thickness shall be 40 microns.

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2.2.9 Accessories

Include all fixing, interfaces, seals and gaskets for complete installation.

2.2.10 Glass unit type

See 2.4 below

2.2.11 Wind loading

Check and follow local code requirements.

2.2.12 Horizontal module

2000 mm horizontal spacing of vertical members

2.2.13 Vertical module

700 mm, or multiples of 700 mm (up to 2800 mm)

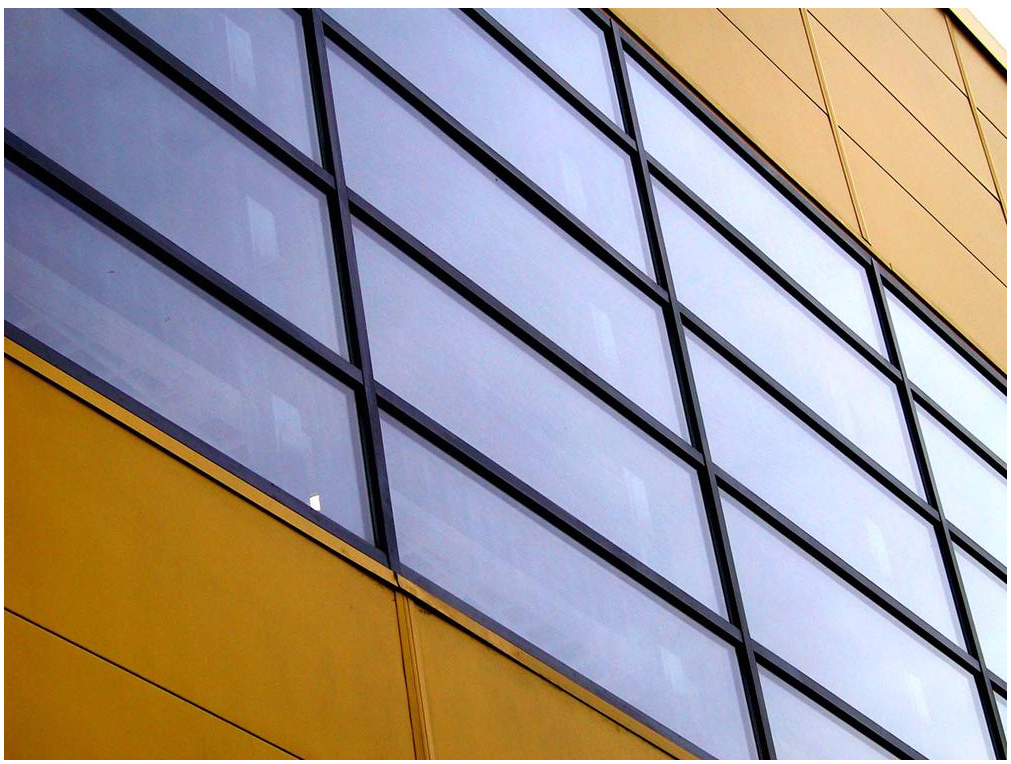
2.2.14 Air Permeability

Max 1, 5 m3/hr/m2 at 600 Pa, total installation.

2.3 Curtain wall frame

2.3.1 Orientation

Vertical flat plane



2.3.2 Supporting structure

Structurally self supporting, requiring a minimum of additional supporting structural members, attaches to primary structural frame, concrete floor /mounting base.

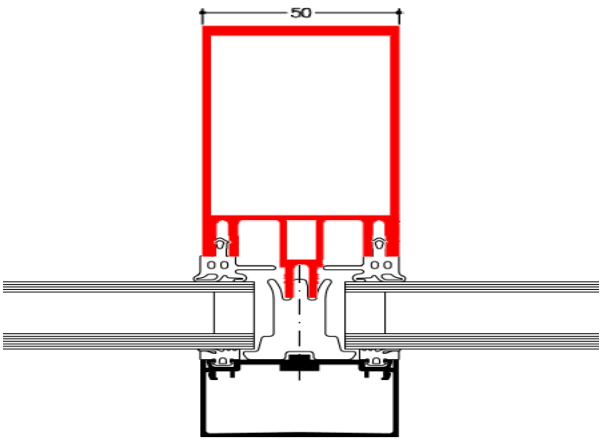
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2.3.3 Span

Vertical, 9 m, lateral structural support at 5 m

2.3.4 Type

Standard Aluminium thermally broken system, drained, ventilated and pressure equalised.



(Example of profile for curtain wall, thermally broken, high insulating)

2.3.5 Dimensions

Slender, approx 50 mm width (viewed from front)

2.3.6 Material

Extruded aluminium (EN 755)

2.3.7 Reinforcing

Galvanized steel insert, internal concealed, if necessary. To be calculated by manufacturer.

2.3.8 External finish

Frame in powder coated aluminium
Colour: IKEA Dark Grey NCS (National colour system) S 6502-B with a satin finish gloss, 20 Gardner, +10, -5 at 60 degrees angle. Minimum thickness shall be 40 microns.

2.3.9 Internal finish

Frame in powder coated aluminium
Colour: IKEA Dark Grey NCS (National colour system) S 6502-B with a satin finish gloss, 20 Gardner, +10, -5 at 60 degrees angle. Minimum thickness shall be 40 microns.

Options (consider the extra costs for this before deciding)
Frame in powder coated aluminium
Colour: IKEA White NCS (National colour system) S0500-N with a satin finish gloss, 20 Gardner, +10, -5 at 60 degrees angle. Minimum thickness shall be 40 microns.
OR
Frame in powder coated aluminium
Colour: IKEA Grey NCS (National colour system) S 3000-N with a satin finish gloss, 20 Gardner, +10, -5 at 60 degrees angle. Minimum thickness shall be 40 microns.
OR

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Frame in powder coated aluminium
Colour: IKEA Grey NCS (National colour system) S 2500-N with a satin finish gloss, 20 Gardner, +10, -5 at 60 degrees angle. Minimum thickness shall be 40 microns.

2.3.10 Accessories

Include all fixing, interfaces, seals and gaskets for complete installation.

2.3.11 Glass unit type

See 2.4 below

2.3.12 Wind loading

Check and follow local code requirements.

2.3.13 Horizontal module

2000 mm horizontal spacing of vertical members

2.3.14 Vertical module

700 mm, or multiples of 700 mm

2.3.15 Air Permeability

Max 1, 5 $m^3/hr/m2$ at 600 Pa, total installation.

2.4 Glass

2.4.1 Type

Fixed light, double glazed insulated unit, hermetically sealed.

2.4.2 Performance

U - Value 1.1 W/m²K, (maximum in centre of glass).
Need normal Low Emission glass (e<0, 0024) inner pane. EN 1096

G – Value (g) > 35 %
This is the solar transparency factor of the glass, the higher value the more heat from the sun will be transmitted through the glass.

LT – Value (LT) > 65%
This is the daylight transparency factor, the higher number the more transparent the glass will be.

2.4.3 Horizontal dimensions

2000 mm c/c of frames

2.4.4 Vertical dimensions

A – 700 c/c frames
B – 1400 c/c frames
C – 2100 c/c frames
D – 2800 c/c frames

2.4.5 Inner pane

6 mm clear glass (minimum), EN 572-2

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2.4.6 Outer pane

4 mm (for A&B) / 6mm (for C&D) clear glass (minimum), EN 572-2

2.4.7 Cavity

16 mm (minimum). Supplier to increase if necessary to meet strength or thermal requirements.

2.4.8 Spandrel panels

Accommodate spandrel panel at edge of 2nd floor.
Glass fronted, dark grey tinted, sandwich type, size 2000 by 700 mm.
Detail to be approved by IKEA

2.5 Warranty

5 years minimum on frame, 10 years minimum on glass.

3. Related Products / Accessories/Options

3.1.1 Outer pane

Solar protective Low E glass: Light transmission/reflection (It~15%), g-factor < 0, 05.
Colour of glass to be approved by IISSAB.

3.1.2 Inner and outer pane

When combination low E and Neutral energy saving glass on outer pane, then low E should be on the inner pane.
Supplier to provide details.

3.1.3 Safety glass

Toughened, only in combination with standard glass. Must comply with Local and EN code requirements.

3.1.4 Cleaning

Dirt repelling glass

3.1.5 Horizontal dimensioning

1000 c/c frames. Possible detail adj. doors etc.

3.1.6 Security glass

Only to be installed in very special applications. EN 365

4. Social & Environmental

4.1 Code of conduct

All manufacturing and supply of products must comply with IKEA Services AB:
"Code of Conduct – for supply of Property Services"
"IWAY Compliance Commitment"
Latest revision

4.2 Environmental

4.2.1 Materials

Legally forbidden materials/substances are not accepted.

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A recycling plan must be available for the product.

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1. Description of Standard

1.1 Short description

This standard describes aluminium automatic sliding doors for **exterior and interior** installation in entrances and exits of IKEA stores and warehouses.

1.1.1 Function

To provide a closure to an opening with the basic functions regarding night security, climate control and comfortable exiting for customers with purchases from the whole range on trolleys. Have the possibility to use it as an emergency exit.

1.2 Validity

This IKEA Standard is valid worldwide.
IKEA Standards describe IKEA minimum requirements for a product/component.

Products/components must always comply with regional/Local laws, norms, authority requirements etc. if these are higher than IKEA Standards.
To be recognized and fulfilled by suppliers, contractors, consultants etc.

1.3 Conceptual, Mandatory

Not applicable

2. Technical Specification

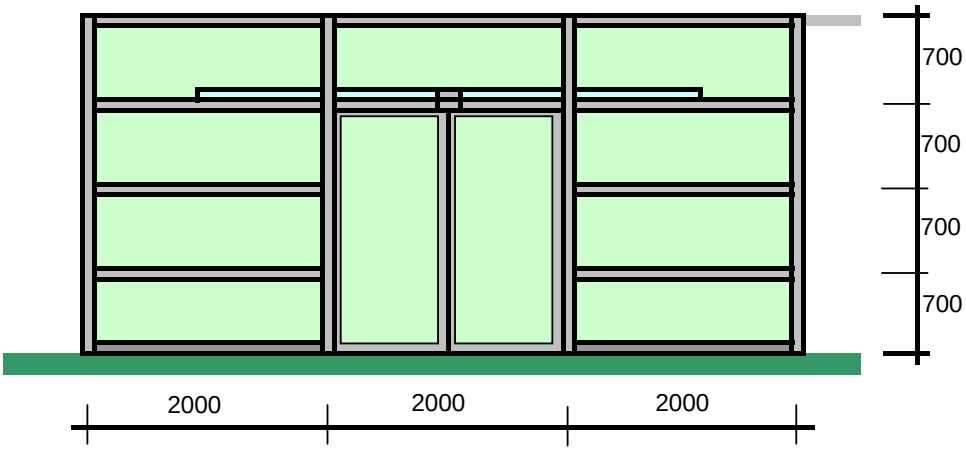
2.1 General

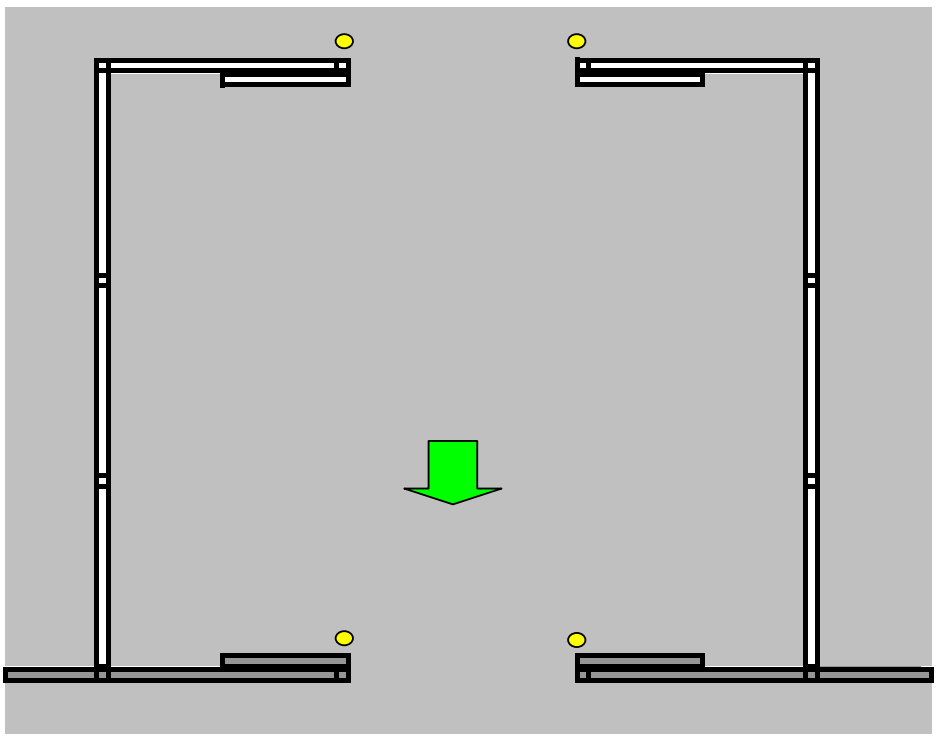
This door shall fit into interior and exterior glazed wall systems based on IKEA standard modular system 2000x700 mm and shall be a double sliding door with centre opening and consist of 2 door leaves. The 2 glazed side panels in the exterior wall and the interior glazed walls for the air lock are often delivered by the curtain wall supplier.
To be coordinated.

Door opening: W X H=2000mm x 2100mm.

Positioning: Interior and exterior window assembly, (see drawing below).
Exterior door ESGD 2000 x 2100 mm and Interior door ISGD 2000 x 2100 mm. Air-lock size approximately 6000 x 6000 mm. Ceiling at 3000mm

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2.2 Door action

Automatic sliding doors, overhead sliding mechanism

2.2.1 Driving mechanism

Electrical motor (120, 200 or 230V, 50 or 60 Hz depending on country), with belt transmission for continuous operation. Aluminium cover and support beam, same finish as door and frame.

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2.2.2 Control

Microprocessor control. Speed adjustment up to 1.4 m/sec
Opening width adjustment. Winter /summer mode settings.
Key switch box/programme selector, remote positioned maximum 5 m.

2.2.3 Activation

Individually controlled PIR or Radar sensors on both sides of the doors, with fully adjustable spread.

2.3 Material / Colour

2.3.1 Exterior door

Frame in powder coated aluminium, to match with exterior windows/curtain wall
Colour: IKEA Grey NCS (National colour system) S 6502-B with a satin finish gloss, 20 Gardner, +10, -5 at 60 degrees angle. Minimum thickness shall be 40 microns.

2.3.2 Interior door

Frame in powder coated aluminium, to match with interior curtain wall.
Colour: IKEA White NCS (National colour system) S 0500-N with a satin finish gloss 20 Gardne, +10, -5 at 60 degrees angle. Minimum thickness shall be 40 microns.

2.3.3 Glass filling

Laminated clear safety glass. Glass thickness minimum 6 mm.
No requirements on thermal insulation for the doors and internal glazed walls for the airlock.

2.4 Safety

Photocells with reverse action, 2 per door,
Ability to open the doors, either manually or automatically. in case of fire alarm.
Finger safety feature must be included.
(If the sliding door moves past a vertical member there is a potential risk for a finger trap. This should be avoided by design or else protected against)

2.5 Security

Mechanical latch lock for sliding door, operated by interior cylinder only. Inter-locked with door operation.

2.6 Warranty

Minimum 2 years on mechanical parts. 5 years on material.

2.7 Regional requirements

2.7.1 Europe

European norms, EN 12650-2; BS 70356
Local or national codes (ref. to technical spec.)

2.7.2 North America

ASTM, ANSI, NEMA, Local and national codes.

2.7.3 Asia & Pacific

Local requirements or British Standard whichever is more onerous

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3. Related Products / Accessories/Options

Double telescopic door for greater width (over 2400) and speed, depending on space and cost.
Emergency “break-away” function of doors, if required by local code.
Emergency backup battery for opening, if required by local code.

4. Social & Environmental

4.1 Code of conduct

All manufacturing and supply of products must comply with IKEA Services AB:
“Code of Conduct – for supply of Property Services”
“IWAY Compliance Commitment”
Latest revision

4.2 Environmental

4.2.1 Materials

Legally forbidden materials/substances are not accepted.
A recycling plan must be available for the product.

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1. Description of Standard

1.1 Short description

This standard describes window assemblies fitted in interior separation walls in IKEA stores and warehouses.
They shall be designed as a whole together with internal doors and walls. The design shall be co-ordinated and checked to work together with windows in the façade. The internal window shall blend in, in the total design and act neutral

1.1.1 Function

To create visual connection between inside and outside of a room and to add daylight into room. Window shall maintain overall technical standard set for the room and be structurally self supporting.

1.2 Validity

This IKEA Standard is valid worldwide.
IKEA Standards describe IKEA minimum requirements for a product/component.

Products/components must always comply with regional/Local laws, norms, authority requirements etc. if these are higher than IKEA Standards.
To be recognized and fulfilled by suppliers, contractors, consultants etc.

1.3 Conceptual, Mandatories

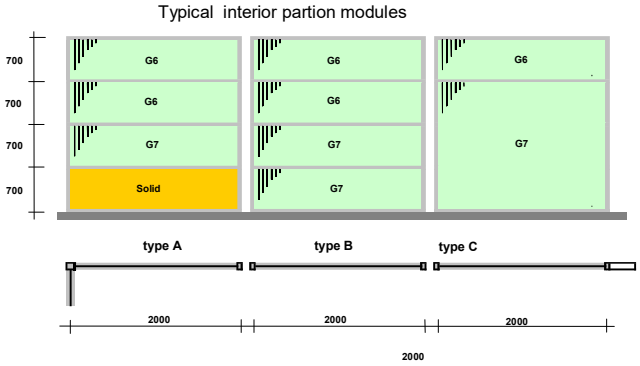
2. Technical Specification

2.1 General

Window assemblies for single glazing to be sectional divided into standard modules and mounted vertically.
Substrates: Stud frame in gypsum wall or block work wall with concrete lintel.

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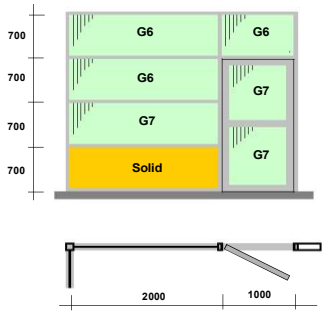
2.2 Typical interior partition modules



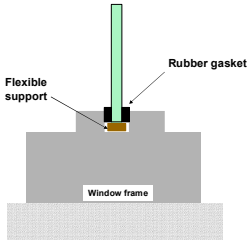
Legend to Glass types
Minimum requirements

G 6 = Regular clear float glass

G 7 = Safety glass



Frame detail



2.3 Module size

L x H = 2000 x 700 mm

2.4 Span (vertical)

Varies, up to 2800 mm

2.5 Measurement

Overall size and type according to type schedule

2.6 Material

Wood or Extruded aluminium profile.

2.7 System type

One piece or two piece box profiles.
Vertical mullions typically at 2000 mm spacing
Transom (horizontal) devision at 700 mm centres or multiple thereof.
Outside to be slender approximately 50 mm wide

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2.8 Finish

Painted in IKEA std. colours. NCS S 0500-N (White), NCS S 2500-N (Grey) or NCS S 3000-N (Grey)

2.9 Glass unit

Single clear float glass, minimum 4 mm
Safety glass as required by local code or IKEA type schedule.
Minimum requirements: Lowest module with safety glass

2.10 Sound

Sound reduction as per location and window type schedule

2.11 Regional requirements

2.11.1 Europe

EN 755, EN 12373-1

2.11.2 North America

As demanded by Local or National codes.

2.11.3 Asia & Pacific

As demanded by Local or National codes.

2.12 Warranty

5 years minimum

3. Related Products / Accessories / Options

3.1 Accessories

Include all fixings, interfaces, seals and gaskets for a complete installation.

3.2 Options

Toughened glass to be used as safety glass.
Fire rating only when required.
Småland window assembly in wood.

4. Social & Environmental

4.1 Code of conduct

All manufacturing and supply of products must comply with IKEA Services AB:
“Code of Conduct – for supply of Property Services”
“IWAY Compliance Commitment”
Latest revision

4.2 Environmental

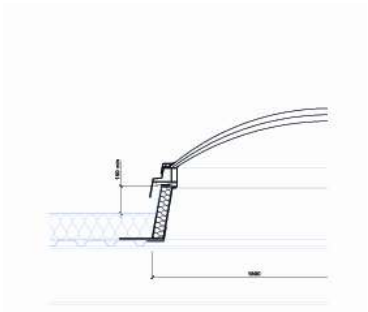
4.2.1 Materials

Legally forbidden materials/substances are not accepted.
A recycling plan must be available for the product.

ems B.V. 2005

Annexure A .12 - Skylight





Remark

Description of Standard
This standard describes a translucent, multi layered, fixed plastic skylight on an insulated upstand kerb.

The rooflight is to be installed on a roof base and insulation by others on an IKEA store on ground (OG), lifted (LS), or with garage (GS), or a distribution center (DC).

Refer to left side of page

Function
Provide secure, tamper proof, impact resistant, waterproof, and weather resistant skylight with good light transmitting qualities, suitable for all conditions in which the proposed building is to be located. The skylight is to resist all weather and other local environmental conditions, and resist the force of a falling person. The design is to be compatible with ESFR sprinklers.

Refer to diagrams adjacent.

Heat resistance
Height above sprinkler head

Validity
This standard is valid for all stores within IKEA in Europe

Region 1

Standard Specification

Skylight type	Domed or ridge, multi-layer skylight on an integrated, insulated kerb.	Comply with prEN1873
Material	Polycarbonate with an acrylic coating Acrylic GRP	
Colour	Clear	
Size	1.8m x 1.8m	Or similar standard size
Thermal insulation	U-value 1.7 W/m2K max	
Light transmittance	65% minimum	
Air permeability	max 5 mcu/hr at 50 Pa	EN12153
Fire standards	Meet national codes of all European countries Tested for 5 minutes at 300 deg C	
Ventilation	None	
Upward load resist	UL 1500	prEN1873
Downward load resist	DL 1125	
Kerb material	Galvanized steel or anodized aluminium	
Kerb insulation	Mineral wool or Polyisocynurate	
Manufacturer's warranty	15 years	Insurance backed
Options		
Thermal insulation	U-value 1.5 W/m2K	Depends local requirements
Upward load resist	UL 3000	
Safety	Fall protection grid	If openable
Related documents		
IKEA concept drawings		See intranet/web
Construction Drawings		For co-ordination
Cross sections and connecting details		For co-ordination
Door and window sections and details		For co-ordination

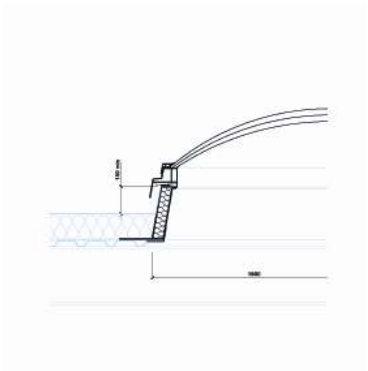
Other
Wind loading — Local condition

Frame Agreement

Supplier / installer available		See intranet FA for details
Feed back	Standard form	With FA

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Remark

Description of Standard
This standard describes a translucent, multi layered, fixed plastic skylight on an insulated upstand kerb.
The rooflight is to be installed on a roof base and insulation by others on an IKEA store on ground (OG), lifted (LS), or with garage (GS), or a distribution center (DC).

Refer to left side of page

Function
Provide secure, tamper proof, impact resistant, waterproof, and weather resistant skylight with good light transmitting qualities, suitable for all conditions in which the proposed building is to be located. The skylight is to resist all weather and other local environmental conditions, and resist the force of a falling person. The design is to be compatible with sprinkler-systems (among others ESFR).

.

Heat resistance, max. volume

Validity
This standard is valid for IKEA in North America

Region 2

Standard Specification

Skylight type	Domed or ridge, multi-layer skylight on an integrated, insulated curb.	
Material	Polycarbonate with an acrylic coating Acrylic Glass fibre Reinforced Polyester	
Colour	Clear	
Size	6' x 6', max. 30 cu.ft volume	Or similar standard size
Thermal insulation	U-value 3.35 BTU/hr.sf.degF max	
Light transmittance	65% minimum	
Air permeability	max 150 cu.ft/hr at 10 psf	
Fire standards	Meet national codes Tested to resist 300 deg C for 5 minutes	
Ventilation	None	
Upward load resist	UL 300 psf	
Downward load resist	DL 200 psf	
Curb material	Galvanized steel or anodized aluminum, with thermal break	
Curb insulation	Mineral wool or Polyisocyanurate	
Manufacturer's warranty	15 years	Insurance backed

Options

Thermal insulation	U-value 3.80 BTU/hr.sf.degF	
Upward load resist	UL 600 psf	
Colour	Opal	
Safety	Fall protection grid	If openable

Related documents

Technical specification	link
IKEA concept drawings	See intranet/web
Construction Drawings	For co-ordination
Cross sections and connecting details	For co-ordination
Door and window sections and details	For co-ordination

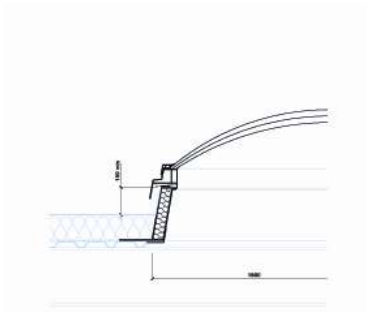
Other
Wind loading — Local condition

Frame Agreement

Supplier / installer available		See intranet FA for details
Feed back	Standard form	With FA

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Remark

Description of Standard
This standard describes a translucent, multi layered, fixed plastic skylight on an insulated upstand kerb.
The rooflight is to be installed on a roof base and insulation by others on an IKEA store on ground (OG), lifted (LS), or with garage (GS), or a distribution centre (DC).

Refer to left side of page

Function
Provide secure, tamper proof, impact resistant, waterproof, and weather resistant skylight with good light transmitting qualities, suitable for all conditions in which the proposed building is to be located. The skylight is to resist all weather and other local environmental conditions, and resist the force of a falling person. The design is to be compatible with sprinkler-systems (among others ESFR).

.

Heat resistance, max. content

Validity
This standard is valid for Asia/Pacific within IKEA

Region 3

Norms
All definitions and requirements mentioned below are based on British Standard norms or local norms, whichever is more onerous

Standard Specification

Skylight type	Domed or ridge, multi-layer skylight on an integrated, insulated curb.	
Material	Polycarbonate with an acrylic coating Acrylic Glass fibre Reinforced Polyester	
Colour	Clear	
Size	1.8m x 1.8m, max. 2.5 m3 content	Or similar standard size
Thermal insulation	U-value 1.7 W/m2K max	
Light transmittance	65% minimum	
Air permeability	max 5 mcu/hr at 50 Pa	
Fire standards	Meet national codes Tested to resist 300 deg C for 5 minutes	
Ventilation	None	
Upward load resist	UL 1500 kg/m2	
Downward load resist	DL 1125 kg/m2	
Curb material	Galvanized steel or anodized aluminium, with thermal break	
Curb insulation	Mineral wool or Polyisocynurate	
Manufacturer's warranty	15 years	Insurance backed

Options

Thermal insulation	U-value 1.5 W/m2K	
Upward load resist	UL 3000 kg/m2	
Colour	Opal	
Safety	Fall protection grid	If openable

Related documents

IKEA concept drawings	See intranet/web
Construction Drawings	For co-ordination
Cross sections and connecting details	For co-ordination
Door and window sections and details	For co-ordination

Other
Wind loading — Local condition

Frame Agreement

Supplier / installer available		See intranet FA for details
Feed back	Standard form	With FA

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Annexure A.13 - Automatic Sliding Doors

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1. Description of Standard

1.1 Short description

This standard describes aluminium automatic sliding doors for **exterior and interior** installation in entrances and exits of IKEA stores and warehouses.

1.1.1 Function

To provide a closure to an opening with the basic functions regarding night security, climate control and comfortable exiting for customers with purchases from the whole range on trolleys. Have the possibility to use it as an emergency exit.

1.2 Validity

This IKEA Standard is valid worldwide.
IKEA Standards describe IKEA minimum requirements for a product/component.

Products/components must always comply with regional/Local laws, norms, authority requirements etc. if these are higher than IKEA Standards.
To be recognized and fulfilled by suppliers, contractors, consultants etc.

1.3 Conceptual, Mandatory

Not applicable

2. Technical Specification

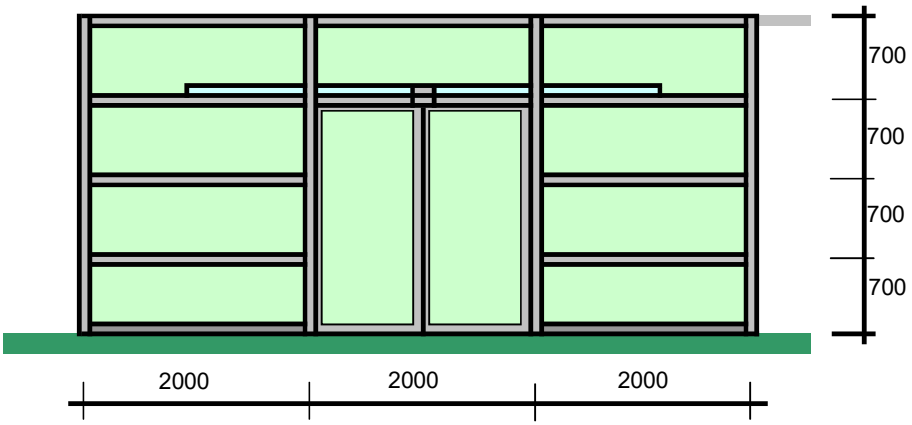
2.1 General

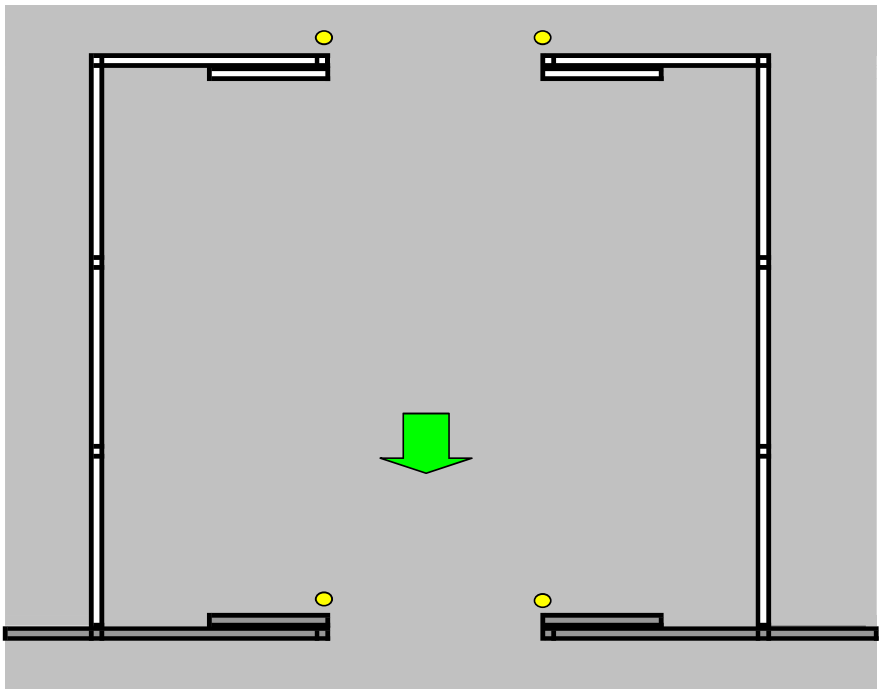
This door shall fit into interior and exterior glazed wall systems based on IKEA standard modular system 2000x700 mm and shall be a double sliding door with centre opening and consist of 2 door leaves. The 2 glazed side panels in the exterior wall and the interior glazed walls for the air lock are often delivered by the curtain wall supplier.
To be coordinated.

Door opening: W X H=2000mm x 2100mm.

Positioning: Interior and exterior window assembly, (see drawing below).
Exterior door ESGD 2000 x 2100 mm and Interior door ISGD 2000 x 2100 mm. Air-lock size approximately 6000 x 6000 mm. Ceiling at 3000mm

, 2005





2.2 Door action

Automatic sliding doors, overhead sliding mechanism

2.2.1 Driving mechanism

Electrical motor (120, 200 or 230V, 50 or 60 Hz depending on country), with belt transmission for continuous operation. Aluminium cover and support beam, same finish as door and frame.

2.2.2 Control

Microprocessor control. Speed adjustment up to 1.4 m/sec
Opening width adjustment. Winter /summer mode settings.
Key switch box/programme selector, remote positioned maximum 5 m.

2.2.3 Activation

Individually controlled PIR or Radar sensors on both sides of the doors, with fully adjustable spread.

2.3 Material / Colour

2.3.1 Exterior door

Frame in powder coated aluminium, to match with exterior windows/curtain wall
Colour: IKEA Grey NCS (National colour system) S 6502-B with a satin finish gloss, 20 Gardner, +10, -5 at 60 degrees angle. Minimum thickness shall be 40 microns.

2.3.2 Interior door

Frame in powder coated aluminium, to match with interior curtain wall.
Colour: IKEA White NCS (National colour system) S 0500-N with a satin finish gloss 20 Gardne, +10, -5 at 60 degrees angle. Minimum thickness shall be 40 microns.

2.3.3 Glass filling

Laminated clear safety glass. Glass thickness minimum 6 mm.
No requirements on thermal insulation for the doors and internal glazed walls for the airlock.

2.4 Safety

Photocells with reverse action, 2 per door,
Ability to open the doors, either manually or automatically. in case of fire alarm.
Finger safety feature must be included.
(If the sliding door moves past a vertical member there is a potential risk for a finger trap. This should be avoided by design or else protected against)

2.5 Security

Mechanical latch lock for sliding door, operated by interior cylinder only. Inter-locked with door operation.

2.6 Warranty

Minimum 2 years on mechanical parts. 5 years on material.

2.7 Regional requirements

2.7.1 Europe

European norms, EN 12650-2; BS 70356
Local or national codes (ref. to technical spec.)

2.7.2 North America

ASTM, ANSI, NEMA, Local and national codes.

2.7.3 Asia & Pacific

Local requirements or British Standard whichever is more onerous

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3. Related Products / Accessories/Options

Double telescopic door for greater width (over 2400) and speed, depending on space and cost.
Emergency "break-away" function of doors, if required by local code.
Emergency backup battery for opening, if required by local code.

4. Social & Environmental

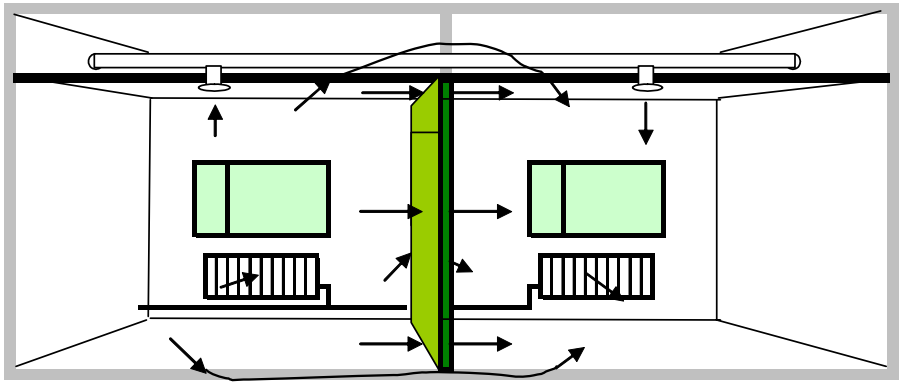
4.1 Code of conduct

All manufacturing and supply of products must comply with IKEA Services AB:
"Code of Conduct – for supply of Property Services"
"IWAY Compliance Commitment"
Latest revision

4.2 Environmental

4.2.1 Materials

Legally forbidden materials/substances are not accepted.
A recycling plan must be available for the product.



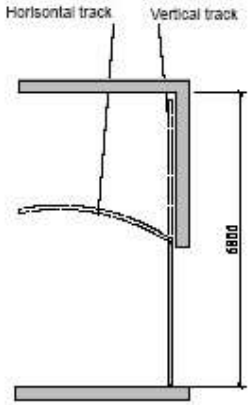
Possible flank transmissions of sound

		Remark
Description of standard		
This standard describes a sliding or folding door / wall that is used in IKEA stores and other project with office compartments.		
Function		
The folding door / wall shall serve as a room divider in a larger room and shall be fully retractable. Where required it shall perform to meet sound or fire demands together with surrounding building elements..		
Validity		
This standard is valid for IKEA projects worldwide. Invalidity due to local rules shall be treated separately.		Region 0
Standard Specification		
The unit shall consist of a proven standard door consisting of foldable elements of a construction meeting the actual requirements. The door shall fit into framework by others.		Sandwich elements or accordion type
Substrate:	The opening consists of concrete floor, wall of gypsum or block and bulkhead of gypsum.	
Fixing base:	Sides and top to metal studs or blockwork Vertical or horizontal bracings to be as demanded by supplier.	
Open size:	Approximately L x H = 6000 x 2200 mm	
Guides:	Top and bottom guide rail	Loads by supplier
Operation.	Manual bi-parting or to one side folding into wall niches.	
Core of element:	As per supplier	
Finish:	Smooth surface in board or pastic Uniform colour light grey	RAL 7035 = NCS S 1502-G
Hardware:	Standard lock, cylinder to fit locksystem Operation handles Wheels to facilitate smooth operation.	
Sound requirement:	Reduction R_w = 35 dB including reduction for nomal gap of 0,005 m2.	
Fire rating:	None or 30 minutes.	
Compatibilty:	Norm for sound test proposed Approval of installation by supplier	Nordtest ACOU 086 EN ISO 140
Installation:	By local installer	
Warranty:	5 years	
Regional requirements		

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Annexure A.14 - Overhead sectional Doors





Description of standard **Remark**

This standard describes the external overhead doors that are used in mainly transport openings in stores and distribution centres. Size and function vary.

Function

The doors shall be part of the outer skin of the building and hence meet requirement of security, thermal insulation and air tightness. The size of the door shall be big enough to accommodate the intended forklift (truck) traffic including goods. Doors must be able to withstand rough handling and frequent use without needing extensive maintenance. Adequate protection must be possible.

Validity

This standard is valid for IKEA stores and DC:s worldwide.

Standard Specification

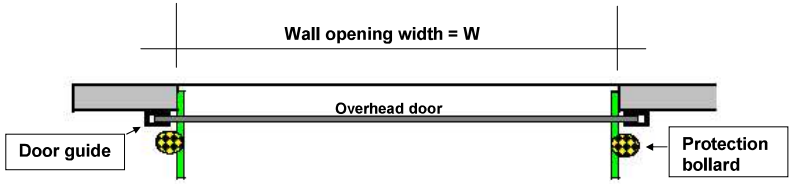
A sectional sandwich panel door opening upwards on vertical or horizontal guide rail (tracks) with following minimum requirements.

Substrate: Concrete or block work wall opening or framed opening in mild steel or wall studs. Door mounted inside of wall.

Main measurements:	Clear wall opening Width (W) x Height (H) Door leaf min. 100 mm wider than opening. Room height 2 x H + 400 for vertical lift.	Refer to Schedule A
Panel size: Height to be minimum 500 mm (20”) to accommodate window and to supplier’s design. Thickness to be minimum 40 mm (1½”)		
Construction:	Sandwich panel with core of PIR/PUR or equal. Face material to be coated steel/ galvanised steel or aluminium. Wind load resistance to local norm.	
Climate properties:	U-value depending design and material. Assembly to be water and wind proof according to local norm. Rubber seal to close gap door to floor.	Minimum 1.0 W/m^{2}K
Window:	Min. 500 x 300 mm in unbreakable double glass or PMMC centred at 1500 mm above floor.	Only loading doors
Finish of panel:	Natural metal (aluminium or galvanized) or High quality coated steel in supplier’s standard grey	
Operation:	Manual chain operated	
Safety:	Spring break safety device. Finger trap protection	
Protection:	Doors need to be protected with bollards on both sides reducing the effective width and saving the door guide rail.	Refer to left side
Locking device:	Lock bolt to be integrated with central lock system.	
Maintenance:	Spare parts must be available within 48 hours.	
Warranty:	5 years o material, 2 years on moving parts	

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Typical overhead door arrangement





Construction:	Sandwich panel with core of PIR/PUR or equal. Face material to be coated steel/ galvanised steel or aluminium. Wind load resistance to local norm.	
Climate properties:	U-value depending design and material. Assembly to be water and wind proof according to local norm. Rubber seal to close gap door to floor.	Minimum 1.0 W/m^{2}K
Window:	Min. 500 x 300 mm in unbreakable double glass or PMMC centred at 1500 mm above floor.	Only loading doors
Finish of panel:	Natural metal (aluminium or galvanized) or High quality coated steel in supplier's standard grey	
Operation:	Manual chain operated	
Safety:	Spring break safety device. Finger trap protection	
Protection:	Doors need to be protected with bollards on both sides reducing the effective width and saving the door guide rail.	Refer to left side
Locking device:	Lock bolt to be integrated with central lock system.	
Maintenance:	Spare parts must be available within 48 hours.	
Warranty:	5 years o material, 2 years on moving parts	

Regional requirements.

Europe:	Properties to EN 13241. Safety to EN 12604 Powered door to CEE machine directive
North America:	ASTM, NEMA
Asia & Pacific:	Local requirements

Options

Rolling overhead door can be used if standard causes conflict with other functions such as air blowers and sprinkler.
A door can be equipped with motor only when it needs to be operated remotely.

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Related documents		
IKEA standard layout drawings		
IKEA roombook		
Project construction details		
Standard protection detail		Refer to left side
Other		
Frame Agreement (FA)		
Supplier / installer available		See intranet FA site
Feed back	Standard form	With FA

Schedule A - OH Doors - types / functions

Type	Name/Position	Region	Function	Operation	Size opening *) Width x Height mm(eng)	Remark
1	Loading doors	EU/AP	Goods receiving doors	Manual	3200/3650 x 3200	The wider door to be used
		NA		Manual	3050 x 3050 (10' x 10')	when dockhouse at 45°
2	Trolley door	EU/AP	Intake for trolley to SS / MP	Manual	1000 x 2100	Option remote control
		NA		Manual	917 x 2134 (3' x 7')	
3	Home transport	EU/AP	Loading for home transport	Manual	2100 x 2800	
		NA		Manual	1829 x 2745 (6' x 9')	
4	Forklift door	EU/AP	Exit for forklift to exterior	Manual	1500 x3500	
		NA		Manual	1524 x 3505 (5' x 11'-5")	

*) Footnote — Door leaf to be opening + 100 mm (4")

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Annexure A.15 - Fire Shutter Doors and Panels

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1. Description of Standard

1.1 Short description

This Standard describes the IKEA requirements for fire shutting doors and panels. His standard replaces standard 04.0120

1.1.1 Function

Fire shutting doors and panels shall automatically close the designated opening by means of approved links to the fire detection systems.

1.2 Validity

This IKEA Standard is valid worldwide.
IKEA Standards describe IKEA minimum requirements for material, workmanship and overall function of the facade.

Products/components must always comply with regional/Local laws, norms, authority requirements etc. if these are higher than IKEA Standards.

This Standard shall be recognized and fulfilled by suppliers, contractors, consultants etc, working with the construction of IKEA stores.

1.3 Conceptual, Mandatories

None

2. Technical Specification

The function can be achieved with either sliding panel doors or overhead rolling shutters. Doors shall close automatically by gravity unless specified otherwise
Doors shall be designed to be closed every night.

2.1 Specifications

Material	Mild steel
Colour	RAL 7035
Material thickness	As per manufacturers specification, but min 1mm
Fire rating	As per local code
Smoke rating	As per local code
Control / Release	By manual switch or by local smoke detectors located on each side of door. To be approved by local authority.
Certificate/label	Approved test certificates as required locally. Visible approval label.
Warranty	5 years

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2.2 Panel sliding door

Total panel leaf in one or more insulated elements to supplier's design. Heavy duty support track fixed to structure, sloped to accommodate closing or equipped with counter weight system
Open hold magnets released by fire alarm system or push button. Shock absorber to retard door when closing
Connections in accordance with approved tested system



Panel Sliding Door

2.3 Overhead Rolling Shutter

Coiled segmented door consisting of hinged segment to manufacturer's approved system. Coil supported by heavy duty steel brackets fixed to building structure. Release mechanism approved by local code.
Heavy duty vertical guides with protection towards mechanical damage. Cover to coil in sheet metal only if part of suppliers' standard



Overhead Rolling Shutter

2.4 Options

Motor operation on shutter only when needed due to door size and remote operation
Door release by fusible link only if required by local authority.

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3. Related Products / Accessories

-

4. Sustainability

4.1 Code of conduct

All manufacturing and supply of products must comply with IKEA's "IWAY Compliance Commitment", Latest revision

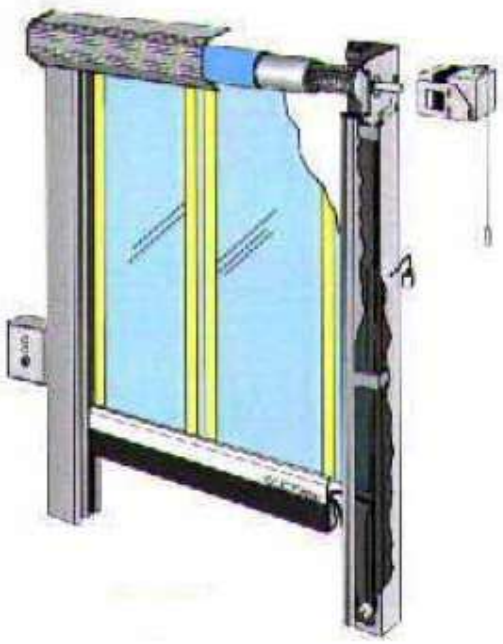
4.2 Materials

Suppliers shall provide full documentation and traceability for all materials.
No Hazardous substances may be used
Legally forbidden substances may not be used

4.3 Life Cycle costs

Suppliers shall submit relevant product information to enable IKEA to produce its Life Cycle Cost calculation

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Description of standard
This standard describes the rapid action roll up door used in IKEA stores and distribution centres.

Function
The function of the rapid roll door is primarily to provide separation between public and goods handling areas for safety reasons, whilst allowing uninterrupted forklift traffic. They shall also prevent nuisance drafts and inclement weather entering into the building.

Validity
This standard is valid for IKEA buildings worldwide. Where local special demands this will be treated separately.

Standard Specification
Internal rapid roll door for internal opening. Surrounding to be block work, gypsum wall or sandwich panes.

		Remark
Frame:	Structural steel frame to be advised by manufacturer	By others
Door action:	Opening speed max 2m/sec with soft start and finish With average speed of 1 m/sec. Closing speed variable, max 2 m/sec	
Opening:	W x W = 3200 x 3200 mm clear opening size	Other size see options
Door panel:	Robust, tear resistant, clear PVC, with vertical reinforcement strips to manufacturers design	Alternatively window
Appearance:	Transparent	
Wind resistance:	Door and track to have devise to prevent wind blow out.	
Track:	Continuous aluminium track with accidental knock	
out	and self repairing door system features.	
Motor:	Concealed drive system, rated IP 54. Motor to be NT-S and to meet local power standards	
Casing:	Coated steel cover to supplier standard, grey	
Activation:	Pull string or induction loop in floor, 2 sides Manual override via crank	
Safety features:	Contact edge. Stop and return to open. Obstacle detection via photocell. Stops descent. Emergency stop button.	
Design parameters:	To technical specification	
Warranty:	5 years	

Variations/Options

Door size:	W x H = 3200 x 3800 mm
External doors	
Warning lights and alarms	
Non-transparent with windows	

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Related documents (On demand)
IKEA Technical specification
IKEA Roombook
Wall details

Other

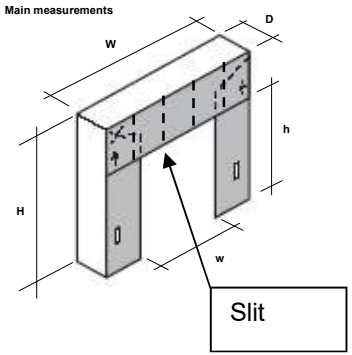
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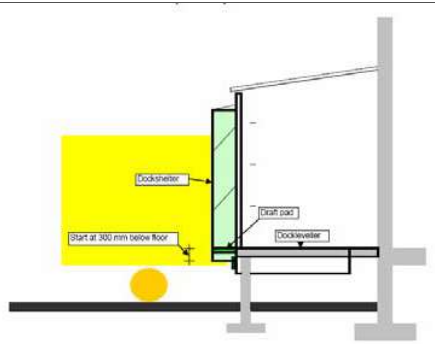
Supplier / installer available		See intranet BB site
Feed back	Standard form	With BB

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Annexure A.16 - Dock Leveller







Dock shelter principle (dock house)

Description of standard
This standard describes the dock shelter to be used in IKEA stores and distribution centres and shall be read in conjunction with the related standards for dock levellers, overhead doors and dock houses.

Remark
Refer to left side

Function
The dock shelter shall reduce draft and energy leakage during the process of unloading trucks and containers when doors must be open. It shall close the gap between vehicle and opening as much as possible and suite most vehicles and withstand rough mechanical abuse.

Validity
This standard is valid worldwide within IKEA.

Standard Specification
This unit is designed as a collapsible box made of galvanized steel profile and covered with reinforced PVC coated sides, top and flaps.
Substrate: Unit to be mounted in front of opening in dock house with steel frame or exterior wall opening in steel frame or concrete when no dock house.
Measurements:
External approximate (min) measurements of unit: W x H x D = 3400x340x550 mm (11x11x2 feet). Refer to left side
Free opening between flaps: w x h = 2000 x 2300 mm (6'6"x7'6")
Ramp height: 1200 mm (NA 3' 8" = 1120 mm)
Mounting height: Underside of shelter at 1000 mm (3'4")

Box cover:	PVC coated reinforced fabric with flexibility to withstand impact from reversing vehicles
Flap material:	Semi rigid reinforced fabric coated with PVC or similar robust design. Flaps to be designed to exert pressure to against truck to improve the seal.
Draft prevention:	Draft pads on both sides to close horizontal gap at floor level. Hinged design to avoid damage.
Colour:	Black
Flap function:	The upper flap shall be slitted (see picture) in order to accommodate higher vehicles
Maintenance:	Spare parts to be available within 48 hours
Warranty:	For all integrated materials warranty shall be 5 years And for any mechanical or moving parts 2 years.

Regional requirements.
Europe:
North America:
Asia & Pacific:

Options
In extreme winter conditions inflatable type shelter can be used

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Related documents (available on demand only)
IKEA Roombook
IKEA standard for dockhouse
IKEA technical specification
IKEA construction drawings

Other
No brand labels to be visible on shelters

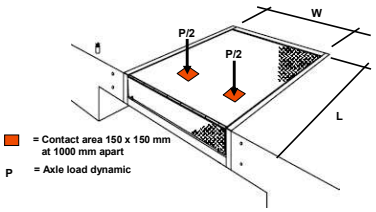
Frame Agreement (FA)

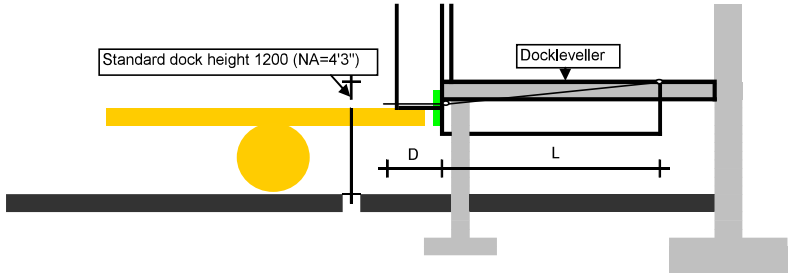
Supplier / installer available		See intranet FA site
Feed back	Standard form	With FA

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Definitions of size and capacity





Description of standard

This standard describes the dock leveller to be used in IKEA stores and distribution centres and shall be read in conjunction with related standards for dock shelters, overhead exterior doors and dock houses.

Function

This equipment shall facilitate the easy unloading of goods from a lorry (truck) / container into the receiving area with forklifts. It shall accommodate as many standard vehicles as possible regarding height and width and have a moderate inclination in its most common operative position. The unit shall withstand tough handling. Cold bridges to be avoided. Jumbo leveller to be used for not standard trucks.

Validity

This standard is valid for IKEA store and DC:s worldwide

Standard Specification

The dock leveller box is a hinged platform made of mild steel with an anti-slip surface (checkered) including a hinged lip. The unit shall be of heavy duty type and to the manufacturer's detailed design. It shall meet the below requirements:

Substrate: In standard store format integrated in exterior dock house steel floor slab.

Standard installed height above ground level 1 200 mm. — NA 3'-8" = 1120 mm

Types: Open pit, closed pit, standing or hanging

Platform sizes: Width W x Length L excluding lip. Standard size and jumbo (extended) — Refer to table left side

Capacity: Minimum loading capacity 6 000 kg (14 000 lbs) dynamic load. Allowed loads shall give not damage any part of the unit. Vertical reach minimum depending type. Max. platform twisting +/- 150 mm (6") — Refer to table left side

Swing lip: Width to be same as platform. Min. length 400 mm (16"). Automatic operation

Operation: 2 Hydraulic pistons. If not used return to neutral position level floor

Climate: The leveller shall be made to operate in temperatures between -30 to + 50 °C. Platform uninsulated but equipped with draught breaks — Arctic oil when required

Safety: Full length toe guards. Hose break valve to keep platform in position. Maintenance safety support. Emergency shut-down button. And no-restart facility.

Power: Local requirement

Protection: Rubber bumpers to be included.

Colour: Light grey to supplier's standard

Compliance: Local regulation and commissioning demands

Compatibility: Dock shelter and dock house construction

Remark

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Warranty: Material 5 years, mechanical parts 2 years

Regional requirements.

Europe: EN 1398, CEE machine directive, EN 605529 regarding safety
Voltage 230 volts

North America: ASTM, control panel NEMA 12.
Voltage 120 volt

Asia & Pacific: Local regulation

Options

Leveller with telescopic lip when standard lip not possible — Unloading regime
Interlock with door via limit switch.
Leveller with thermal insulation for cold climates zones (< -30°C)
Leveller integrated on concrete floor slab if required — Not standard

Related documents (available on demand)

IKEA standard layout drawings
IKEA standard for dock houses
IKEA Roombook

Other

Maintenance by local provider

Frame Agreement (FA)

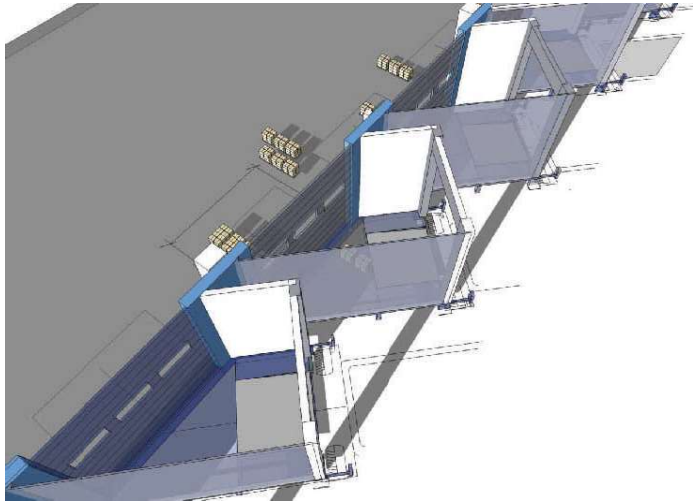
Supplier / installer available — See intranet FA site
Feed back — Standard form — With FA

Standard type docklevellers

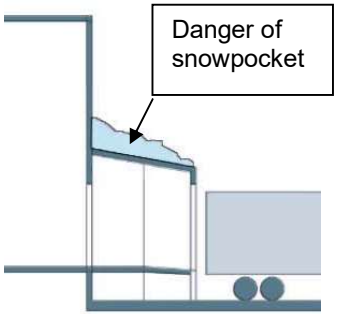
Minimum requirement

Type	Description of type	Platform size				Remark
	Nominal size	Width	Length	Lip		
		mm (eng.)	mm (eng.)	mm (eng.)		
A	European standard size	2100	2400	400		
A1	North America standard size 7x8'	2096 (6'-10")	2108 (6'-11")	457 (18")		
B	European large size	2100	3500	400		Jumbo
B1	North America large size 7x10'	2096 (6'-10")	2718 (8'-11")	457 (18")		

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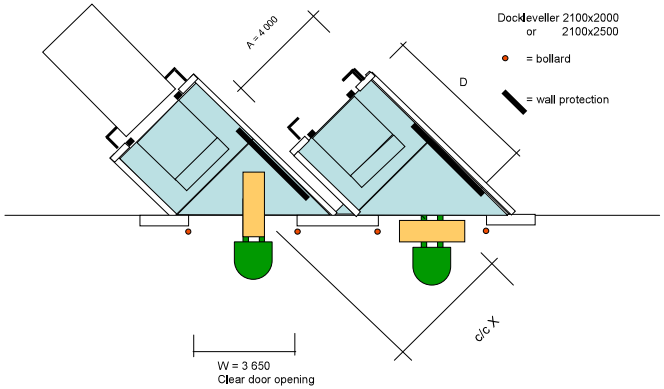


Principle section

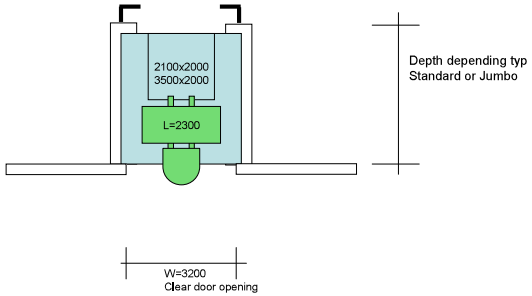
Description of standard	Remark
This standard describes an uninsulated dockhouse attached to the main building of a store on ground (OG), lifted store (LS) or a garaged store (GS). Also applicable to DC:s. Standard refers to other dock equipment.	Refer to picture adjacent See 4.14.1-5
Function	
Act as weather protection to prevent draft into goods receiving. To support dockleveller and shelter/bumper and allow flexibility for all relevant vehicles with different heights and attachments (lifts). Enable smooth off loading and to minimise the maintenance. The dockhouse shall be thermally broken to the main building i.e. no cold bridges.	Note snow loads per zone
Validity	
This standard is valid for IKEA units worldwide.	
Invalidity due to national or other demands to be treated as option.	Region 0
Standard Specification	
House consisting of frame, floor, walls and deck in non-corrosive material.	
Dock house floor in steel supported on columns	Same type as leveller
House at 90° angle to building or as required by size of site	See options below
Spacing of dockhouses depending on angle to building and store design	See principle X mm
Overall approximate measurements house: WxDxH mm /english	3600x2500x3900 / 12'x11'x13'
Opening approximate measurement doorseal: wxh mm/ english	3000x3000 / 10'x10'
Ramp height (ground to floor): RH mm / english	1200 (region 1,3)/ 3'-8"(region 2)
Wall cladding Uninsulated trapezoidal steel	Standard steel grey
Floor in checkered steel plate compatible with leveller 4.141	Standard steel grey
Standard galvanised steel support columns	Supplier standard
Deck in waterproof design sloping at 10° away from building	
Drainage to suppliers design. Rainwater to be led away to connecting point.	Freezing to be avoided.
Snowload and windload	To local conditions
Design responsibility	Supplier
Compatibility with 4.141 leveller & 4.142 shelter	Supplier
Connections and loads to main building	Standard details
Lighting by others	
Protection Integrated heavy duty wall protections in wood	By supplier
Installation	By IKEA
Warranty 5 years on material, 2 years on moving parts	
Variations/Options	
Angle to building to be at 45° to façade line in lieu of standard 90°	For size and design see
Overall approximate measurements house (45°) WxDxH mm/ english	3600x4500x39900 / 12'x15'x13'
Support columns in concrete	Loading as per supplier
Pitched roof to drain water to the sides	If feasible

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Exterior Dockhouse at 45º



Exterior Dockhouse 90°



Maintenance and repair	To be agreed locally
Maintenance spare parts availability	By supplier

Related documents (on demand)

IKEA standard layout drawings	
Model store	For details
Construction drawings facade and foundations	

Other

Protection standard	Additional

Frame Agreement (FA)

Supplier/installer available:		See FA link
Feedback	Mandatory on standard form	See FA link

0.000/AAAA/00/0 = Cost Acc # / brief descr / item # / region #	0 World, 1 EU, 2 NA, 3 AP

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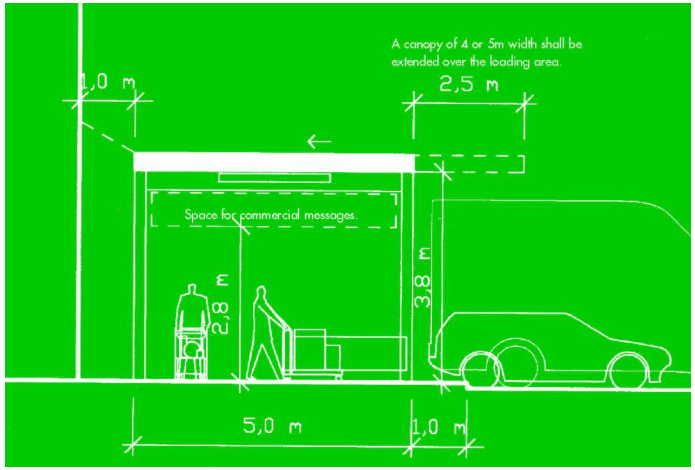
Annexure A.17 - Canopy

Release Notes / Revision Log:

Revision Date	Summary of Changes	Changes marked
2006-05-26	Conversion of European version to new global version intermediate layout	
2005-03-08	European version approved by SSG 5	

Canopy (Exit)

IS IKEA Standard



Description of Standard
This standard describes the Canopy at the customer exit, in front of an IKEA Store.

Remark
Refer technical spec

Function
Provides weather protection to customers exiting the IKEA store, and preparing to transfer their purchases to their vehicle. Forms part of the IKEA Trade Dress.
The Canopy is configured to permit daylight to reach the façade below, while not allowing rain to reach the exiting customer.

Validity
This standard is valid for stores within IKEA in Europe — Region 1

Standard Specification

Front edge	Horizontal, viewed from front. No visible deflection. Simple, clean structure. Slender, approx 200 to 300mm	
Front edge colour	Painted yellow, NCS S 1070 Y10R	
Roof slope	3% slope down from front to back	
Column spacing	8 m in long direction (matching store grid), 5 m in cross direction	
Position	Rear column 1 m (approx) from façade. Gap between roof and store façade generally, with glazed weather protection over doorways only.	
Length	To suit store design, between 60 m and 100 m	Refer architect drwgs
Height	3.8 m from ground to underside front edge	
Overhang	front edge 2.5 m cantilever from front column line	
Materials	Structure: Closed hollow section (circular, square or rectangular), galvanized steel. Top surface of all structural members to be in contact with deck.	Closed to restrict birds perching
	Roof: Galvanized trapezoidal cold deck (sealed), draining to gutter at back. Profile fillers over beams.	No membrane, no insulation
	Gutter: Galvanized sheet metal square/rectangular channel	
	Downspouts: Galvanized steel pipe, fixed to rear columns	Rainwater goods design to suit local code
Glazed rain screen	Polycarbonate, Perspex or toughened glass in aluminium glazing members, sloping, covering gap between façade and canopy	Over doors
Loading	Snow, wind, rain, live loads	Local code
Lighting	Surface mounted fluorescent fixtures, fixed to deck	Outdoor class
Options		

Related documents

Technical Specification	
IKEA concept drawings	See intranet/web
Construction Drawings	For co-ordination
Cross sections and connecting details	For co-ordination
Door and window sections and details	For co-ordination
Best Buy	
Supplier / installer available	See intranet BB for details
Feed back Standard form	With BB

Annexure B.01 - IKEA Colours

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1. Description of Standard

1.1 Short description

This standard describe mandatory and preferred colours, in accordance with NCS colour code, to be used internally and externally in IKEA stores and DC's.

1.1.1 Function

The colours addressed in this standard shall fulfil the trade dress, mandatories and recommendations as stated by Inter IKEA System BV. Colours shall be sustainable over reasonable time.

1.2 Validity

This IKEA Standard is valid worldwide.
IKEA Standards describe IKEA minimum requirements for a product/component.

Products/components must always comply with regional/Local laws, norms, authority requirements etc. if these are higher then IKEA Standards.
To be recognized and fulfilled by suppliers, contractors, consultants etc.

1.3 Conceptual, Mandatories

IKEA Trade dress

2. Technical Specification

2.1 General

The colours described herein make reference solely to the NCS (Natural Colour System) system for determining colours. Any other colour coding system shall always make reference to the NCS system and when necessary, regarding the IKEA blue and yellow, be tested and determined by Inter IKEA Systems Services AB.

2.2 Substrates

Depending on the application, the substrate could be any material common in the construction of an IKEA store or DC. Depending on the substrate the paint material shall be of the type best suited for the purpose (see standard for paint). The majority of applications have standardized designs and well proven materials.

2.3 Materials

By definitions the material is the same as the face material with either an applied or integrated colour coat or pigment. Depending on the actual material the colour will appear different with different properties like texture, gloss and age. Below the most common materials are described:

2.3.1 Concrete

Smooth surface wall, ceiling or floor. Natural or painted

2.3.2 Steel sheet

Flat or profiled cladding or sandwich panels. Galvanized, coated or painted

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2.3.3 Steel structure

Profiled rolled or bent. Galvanized, primed or painted

2.3.4 Aluminium

Flat or profiled, doors and windows. Natural anodized or coated.

2.3.5 Blockwork

Smooth or textured surface with fill joints. Untreated or painted.

2.3.6 Wood

Solid, veneered or trims. Natural, lacquered or painted

2.3.7 Gypsum

Boards, jointed. Untreated, wallpaper (fibre glass) or painted.

2.3.8 Plastic

Laminate, extruded. Pre-coloured or painted.

2.4 Gloss

The ability of a surface to reflect light without scattering is known as Gloss. Gloss is measured by directing a constant power light beam at an angle to the test surface and then by monitoring the amount of reflected light. Different surfaces require different reflective angles.

2.4.1 Values

All colour indications are accompanied by a gloss requirement including the reflecting angle. Common values are 7 Flat (ceilings), 20 Medium (interior walls) and 60 High gloss (joinery/steel details and wet areas).

2.5 Paint codes

The following paint codes are the IKEA TRADE DRESS COLOURS and are either mandatory or recommended and shall be used in the different applications as part of the concept. Various special standards indicated when a certain colour code shall be used.

2.5.1 IKEA Blue and Yellow

Blue	NCS S 4550-R80B
Yellow	NCS S 1070-Y10R

2.5.2 IKEA Complementing colours

Red	NCS S 1580-R
White	NCS S 0500-N
Grey	NCS S 2500-N
Grey	NCS S 3000-N
Dark grey	NCS S 6502-B

2.5.3 IKEA Outdoor parking signage colours

Blue	NCS S 4550-R80B
Yellow	NCS S 1070-Y10R
Red	NCS S 1580-R
Green	NCS S 3065-G10Y

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2.5.4 IKEA Non ground parking signage colours

Blue	NCS S 4550-R80B
Blue	NCS S 2565-R80B
Yellow	NCS S 0580-Y
Red	NCS S 1085-Y90R
Green	NCS S 3065-G10Y
Orange	NCS S 0585-Y50R

2.5.5 Store equipment

Metal racking-white	NCS 0500-N	Gloss 60+/-10
Metal racking-grey silver	NCS 5500-N	Gloss 60+/-10
Wood- white	NCS 0500-N	Gloss 25+/-5
Warehouse racking-grey	NCS 4005-R80B	Gloss 80+/-5
Check out counter Blue	NCS S4550-N	Gloss 60+/-5
Grey silver gates	NCS S2500-N	Gloss80+/-5
Grey silver MDF	NCS S2500-N	Gloss25+/-5

2.6 Test

Testing is only applicable for IKEA blue and yellow.
When a certain application is not pre-approved, or a parameter in the supply process has been altered or the combination of substrate and paint is not frequently used an approval is required.
An application consisting of a formal letter and 3 representative paint samples shall be sent to IKEA Services for the approval of Inter IKEA Systems Services AB.

3. Related Products / Accessories

Standard for Façade sandwich panels	3.640.FCDESSW.01.1,2&3
Standard for Façade yellow	3.642.FCDEYEL.02.1&2
Standard for curtain wall & windows	4.112.CTNWALL & 4.113.WNDW
Standard wood & steel doors	4.221.WDDOOR &.4.120.STDOOR
Standard for paint	04-0610.Paint.pdf

4. Social & Environmental

4.1 Code of conduct

All manufacturing and supply of products must comply with IKEA Services AB:
“Code of Conduct – for supply of Property Services”
“IWAY Compliance Commitment”
Latest revision

4.2 Environmental

Not applicable

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